

Exercise Companion

This is a raw, unabridged transcription of the exercise sets from Joseph Edwards's *A Treatise on the Integral Calculus with Applications, Examples and Problems*, Volume I (Macmillan and Co., 1921), companion to the theory-only formulario. Wording, numbering, and citations are reproduced exactly as printed; nothing has been reworded, abbreviated, or solved. Exercises are grouped under the article ranges they follow in the original text, so each set sits under the theory it tests.

Chapter II: Standard Forms — Exercises

Exercises I — Reversal of Differentiation & the Logarithmic Rule (Arts. 27–42)

Write down the indefinite integrals of:

1. $x^{10}, x^{-10}, 1, 0, x^{\frac{7}{5}}, x^{-\frac{5}{7}}, \sqrt[3]{x^{-2}}, \frac{1}{\sqrt{x}}, x^7 \times x^{-\frac{9}{2}},$
2. $a\sqrt{x} + \frac{b}{\sqrt{x}}, a\sqrt[3]{x} + \frac{b}{\sqrt[3]{x}}, (ax^{\frac{1}{p}} + b)(cx^{\frac{1}{q}} + d), \frac{ax^2 + bx + c}{x^2}.$
3. $\frac{(ax^2 + bx + c)(ax^{-2} + bx^{-1} + c)}{x}, \frac{1}{a - x}, \frac{1}{(a - x)^2}, \frac{1}{(a - x)^p}.$
4. $\frac{1}{a - x} + \frac{1}{a + x}, \frac{x}{a + x}, \frac{1}{(a + x)^2} + \frac{1}{(a - x)^2}, \frac{(x - a)(x + a)(x^2 + a^2)}{x^4}.$
5. Calculate $\int_0^2 x^{0.3} dx; \int_3^5 x^{-\frac{1}{3}} dx; \int_1^2 \frac{1}{2x + 3} dx.$
6. Calculate $\int_2^3 (a + bx^3)^2 3bx^2 dx$ for the values $a = 2, b = 5.$
7. $\int_a^{2a} \left(\sqrt{\frac{a}{x}} + \sqrt{\frac{x}{a}} \right)^2 dx.$
8. If the retardation of a particle be 2 foot-seconds per second, and its initial velocity 10 f.s., when and where will it come to a stop?
9. Given $p v = \text{constant}$, and that $p = 40$ when $v = 10$, calculate $\int_{10}^{20} p dv$. What does this integration mean?
10. Calculate $\int (x - 1)(x - 2)(x - 3)(x - 4) dx$ between limits (a) 0 and 1, (b) 1 and 2, (c) 2 and 3, (d) 3 and 4, (e) 4 and 5. Explain the signs which occur in the results. Illustrate by a graph.
11. Write down the indefinite integrals of:
 - (i) $(ae^x + b)^n e^x,$
 - (ii) $\frac{ce^x}{ae^x + b},$
 - (iii) $\left(ax + \frac{b}{x} + c \right)^n \left(\frac{ax^2 - b}{x^2} \right),$
 - (iv) $(ax^p + be^x)^n (pax^{p-1} + be^x).$
12. Integrate
 - (i) $\int \frac{ae^{ax} + be^{bx}}{e^{ax} + e^{bx}} dx,$
 - (ii) $\int \cot 2x dx,$

- (iii) $\int \tanh x \, dx,$
 (iv) $\int \frac{2ax^n + b}{(ax^{2n} + bx^n + c)^p} x^{n-1} dx.$

13. Integrate

- (i) $\int \frac{dx}{(1+x^2)\tan^{-1}x},$
 (ii) $\int \frac{dx}{(1+x^2)(\tan^{-1}x)^n},$
 (iii) $\int \frac{(\sin^{-1}x)^n}{\sqrt{1-x^2}} dx,$
 (iv) $\int \frac{1}{\sin^{-1}x} \frac{dx}{\sqrt{1-x^2}},$
 (v) $\int \frac{\text{vers}^{-1}x}{\sqrt{2x-x^2}} dx.$

14. Integrate

- (i) $\int \frac{dx}{x \log x},$
 (ii) $\int \frac{1}{x \log x} \frac{1}{\log \log x} dx,$
 (iii) $\int \frac{1}{x \log x} \frac{1}{\log \log x} \frac{1}{(\log \log \log x)^n} dx,$
 (iv) $\int \frac{dx}{x l(x) l^2(x) l^3(x) \dots l^r(x) [l^{r+1}(x)]^n},$

where $l^r x$ represent $\log \log \log \dots \log x$, the log being repeated r times.

Exercises II — The Preliminary Standard-Forms Table (Arts. 43–45)

Write down the indefinite integrals of the following functions:

- $\frac{1}{x+1}, \frac{x-a}{x+a}, \frac{x}{x^2+a^2}, \frac{x+a}{x^2+a^2}, \frac{x^2}{x^3+a^3}, \frac{x^{n-1}}{x^n+a^n}.$
- $2^x, 2x, \frac{2}{x}, x^2, x^3+3^x, a+b^x+c^{2x}+d^{3x}.$
- $\cos^2 \frac{x}{2}, \sin^2 \frac{x}{2}, \cot x + \tan x, \cos x \left(\frac{1}{\sin x} + \frac{1}{\sin^2 x} \right).$
- $\frac{1}{\sqrt{9-x^2}}, \frac{1}{\sqrt{9+x^2}}, \frac{1}{\sqrt{x^2-9}}, \frac{1}{9+x^2}, \frac{1}{9-x^2}, \frac{1}{x^2-9}.$
- $\frac{1}{x\sqrt{x^2-4}}, \frac{x+1}{x\sqrt{x^2-4}}, \frac{ax+b}{\sqrt{c^2-x^2}}, \frac{ax+b}{\sqrt{x^2-c^2}}, \frac{ax+b}{\sqrt{x^2+c^2}}.$
- $\frac{1}{\sqrt{x-x^2}}, \frac{1}{x\sqrt{3x^2-27}}, \frac{1}{\sqrt{27-3x^2}}, \frac{x^2-4}{x^2+4}, \frac{x^2+4}{x^2-4}.$
- Write down the indefinite integrals of:

$$(i) \int \sin^{-3} x \cos x \, dx,$$

$$(ii) \int \cot x \sec^2 x \, dx,$$

$$(iii) \int (e^x + a)^n e^x \, dx,$$

$$(iv) \int (x^3 + a^3)^n x^2 \, dx,$$

$$(v) \int (ax^3 + bx + c)^n (3ax^2 + b) \, dx.$$

8. Write down the indefinite integrals of:

$$(i) \int \frac{1}{1+x^2} \frac{1}{\tan^{-1} x} dx,$$

$$(ii) \int \frac{dx}{\sqrt{1-x^2}(\sin^{-1} x)^2},$$

$$(iii) \int \frac{dx}{x(\log x)^3}.$$

9. Evaluate

$$(i) \int_1^2 \frac{2x+1}{(x^2+x+1)} dx,$$

$$(ii) \int_1^2 \frac{2x+1}{(x^2+x+1)^2} dx.$$

10. Draw the graph of $64(x-2)(x-3)(2x-5)$, and show that the area between $x=2$ and $x=2.5$, bounded by the curve and the x -axis, is $32 [(x-2)^2(x-3)^2]_2^{2.5}$, i.e. = 2 square units. Verify by multiplying out and integrating each term.

11. Write down the values of:

$$(i) \int_0^x \frac{e^{3x} + e^{6x}}{e^x + e^{-x}} dx,$$

$$(ii) \int_0^x \frac{e^{(n+1)x} - e^{(n-1)x}}{\sinh x} dx,$$

$$(iii) \int_0^1 \frac{e^{2x} + e^{4x}}{e^{3x}} dx,$$

$$(iv) \int_a^b \cosh(\log x) \, dx.$$

12. Evaluate

$$(i) \int_0^{\pi/2} \cos x \, dx,$$

$$(ii) \int_0^{\pi/2} \cos^2 x \, dx,$$

$$(iii) \int_0^{\pi/4} \cos 2x \, dx,$$

$$(iv) \int_0^x (\cosh x + \cos x) \, dx.$$

13. Evaluate

(i) $\int_0^{\pi/4n} \sec^2 nx \, dx,$

(ii) $\int_0^{\pi/4} \sec x \tan x \, dx,$

(iii) $\int_1^{\sqrt{3}} \frac{dx}{1+x^2},$

(iv) $\int_0^1 \frac{dx}{\sqrt{1-x^2}}.$

14. Evaluate

(i) $\int \frac{x^n - a^n}{x - a} dx,$

(ii) $\int \frac{x^n}{x - a} dx,$

(iii) $\int \frac{x^5 - 2x + 1}{x - 1} dx,$

(iv) $\int \frac{x^6}{x - 1} dx,$

(v) $\int \frac{x^3 - 6x^2 + 11x - 6}{x^2 - 3x + 2} dx.$

Exercises III — Miscellaneous (Art. 46 onward)

1. Write down the indefinite integrals of:

(1) $\frac{(x+a)(x+b)(x+c)}{x^4},$

(2) $\frac{x^3 + a^3 + b^3 - 3abx}{x + a + b},$

(3) $\sqrt[p-q]{x^{\frac{q+r}{r-p}}} \sqrt[q-r]{x^{\frac{r+p}{p-q}}} \sqrt[r-p]{x^{\frac{p+q}{q-r}}},$

(4) $\frac{a \cos x - b \sin x}{a \sin x + b \cos x + c},$

(5) $e^{a \log x} + e^{x \log a},$

(6) $\frac{\tan^{-1} \frac{x}{3}}{9 + x^2},$

(7) $\frac{1}{\sin x \cos x},$

(8) $\frac{\cos x(1 + \sin x)}{\sin^2 x},$

(9) $\frac{1}{1 - \cos x},$

(10) $\sin \left(x + \frac{\pi}{4} \right),$

$$(11) \frac{x^2 + \sin^2 x}{x^2 + 1} \sec^2 x,$$

$$(12) (1 + \sin x \cos x) \sec^2 x,$$

$$(13) \tan x(1 + \sec x),$$

$$(14) \frac{a \sin^3 x + b \cos^3 x}{\sin^2 x \cos^2 x},$$

$$(15) (\cos x - \sin x)(2 + \sin 2x) \sec^2 x \operatorname{cosec}^2 x,$$

$$(16) (a + \tan x)(b + \tan x) \sec^2 x,$$

$$(17) \frac{1}{x[1 + (\log x)^2]},$$

$$(18) \frac{1}{x} \cos(\log x),$$

$$(19) \frac{x^5 + 1}{x - 1},$$

$$(20) \frac{e^x}{a^2 e^{2x} + 1}.$$

2. (a) If $f(x) = \frac{1}{1-x}$, prove that $\int f f f(x) dx = \frac{x^2}{2}$.

(b) If $f(x) = a+bx$, prove that $\int f^r(x) dx = a \frac{b^r - 1}{b - 1} x + b^r \frac{x^2}{2}$; where $f^r(x)$ means $f f f \dots f(x)$, the functionality symbol f occurring r times.

3. Show by expansion that

$$(a) \int \log(1+x) dx = \frac{x^2}{1.2} - \frac{x^3}{2.3} + \frac{x^4}{3.4} - \frac{x^5}{4.5} + \dots = (1+x) \log(1+x) - x,$$

$$(b) \int \log \frac{1+x+x^2}{1-x+x^2} dx = 2 \left\{ \frac{x^2}{1.2} - \frac{2x^4}{3.4} + \frac{x^6}{5.6} + \frac{x^8}{7.8} - \frac{2x^{10}}{9.10} + \frac{x^{12}}{11.12} + \frac{x^{14}}{13.14} - \frac{2x^{16}}{15.16} + \dots \right\},$$

$$(c) \int (1+x)^n dx = x + \binom{n}{1} \frac{x^2}{2} + \binom{n}{2} \frac{x^3}{3} + \binom{n}{3} \frac{x^4}{4} + \binom{n}{4} \frac{x^5}{5} + \dots,$$

$$\text{where } \binom{n}{r} \equiv \frac{n(n-1)\dots(n-r+1)}{1.2\dots r}.$$

4. Prove by Differentiation or Integration from the Binomial Expansion of $(1+x)^n$, where n is a positive integer,

$$(a) 1C_1 + 2C_2 + 3C_3 + \dots + nC_n = n2^{n-1},$$

$$(b) 1.2C_2 + 2.3C_3 + 3.4C_4 + \dots + (n-1)nC_n = n(n-1)2^{n-2},$$

$$(c) 1C_1 + 3C_3 + 5C_5 + \dots = n2^{n-2},$$

$$(d) 1^2C_1 + 2^2C_2 + 3^2C_3 + \dots + n^2C_n = n(n+1)2^{n-2},$$

$$(e) 1^3C_1 + 2^3C_2x + 3^3C_3x^2 + \dots + n^3C_nx^{n-1} = x\{n^2x^2 + (3n-1)x + 1\}(1+x)^{n-3},$$

$$(f) 1C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n = 2^{n-1}(n+2),$$

$$(g) \frac{C_0}{1} + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1},$$

$$(h) \frac{C_0}{1.2} + \frac{C_1}{2.3} + \frac{C_2}{3.4} + \dots + \frac{C_n}{(n+1)(n+2)} = \frac{2^{n+2} - n - 3}{(n+1)(n+2)},$$

- (i) $\frac{C_0}{1.2.3} - \frac{C_1}{2.3.4} + \frac{C_2}{3.4.5} - \cdots + \frac{(-1)^n C_n}{(n+1)(n+2)(n+3)} = \frac{1}{2(n+3)},$
- (j) $\frac{3.4}{1.2}C_0 - \frac{4.5}{2.3}C_1 + \frac{5.6}{3.4}C_2 - \cdots + (-1)^n \frac{(n+3)(n+4)}{(n+1)(n+2)}C_n = 2 \frac{3n+5}{(n+1)(n+2)}.$

5. Prove from the expansions of $\sin x$ and $\cos x$ in powers of x that $\int_0^x \sin x \, dx = 1 - \cos x$ and that $\int_0^x \cos x \, dx = \sin x.$

6. Prove from the expansion of $\exp x$ that $\int_{-\infty}^x \exp x \, dx = \exp x \quad [\exp x \equiv e^x].$

7. Prove that

(a) $\int_1^x \frac{(1+x)^n}{x} dx = \log x + \binom{n}{1}(x-1) + \binom{n}{2} \frac{x^2-1}{2} + \binom{n}{3} \frac{x^3-1}{3} + \cdots,$

(b) $\int_{2 \dots}^x \frac{(x+1)^n}{x-1} dx = 2^n \log(x-1) + \binom{n}{1} 2^{n-1} [(x-1)-1] + \binom{n}{2} 2^{n-2} \frac{(x-1)^2-1}{2} + \binom{n}{3} 2^{n-3} \frac{(x-1)^3-1}{3} + \cdots$

8. Show that $\int (a+bx)(x+c)^n dx = b \frac{(x+c)^{n+2}}{n+2} + (a-bc) \frac{(x+c)^{n+1}}{n+1}.$

9. Show that $\int_a^b (1+x)^n dx = (b-a) + \binom{n}{1} \frac{b^2-a^2}{2} + \binom{n}{2} \frac{b^3-a^3}{3} + \cdots + \binom{n}{n} \frac{b^{n+1}-a^{n+1}}{n+1}.$

10. If $\phi(x)$ be a rational integral algebraic function of x , show that $\int_c^x \phi(x)(x-c)^n dx = (x-c)^{n+1} \left[\frac{\phi(c)}{n+1} + \frac{\phi'(c)}{n+2} \frac{x-c}{1!} + \frac{\phi''(c)}{n+3} \frac{(x-c)^2}{2!} + \cdots \right].$

11. By considering $\int (x-a)^p (x-b)^q dx$, show that the difference of the series (p and q being positive integers)

$$\begin{aligned} & \frac{(x-b)^{p+q+1}}{p+q+1} + p(b-a) \frac{(x-b)^{p+q}}{p+q} + \frac{p(p-1)}{1.2} (b-a)^2 \frac{(x-b)^{p+q-1}}{p+q-1} + \cdots, \\ & \frac{(x-a)^{p+q+1}}{p+q+1} - q(b-a) \frac{(x-a)^{p+q}}{p+q} + \frac{q(q-1)}{1.2} (b-a)^2 \frac{(x-a)^{p+q-1}}{p+q-1} - \cdots \end{aligned}$$

is independent of x .

12. Verify by differentiation that

(1) $\int \frac{dx}{1+x^4} = \frac{1}{4\sqrt{2}} \log \frac{1+x\sqrt{2}+x^2}{1-x\sqrt{2}+x^2} + \frac{1}{2\sqrt{2}} \tan^{-1} \frac{x\sqrt{2}}{1-x^2},$

(2) $\int \frac{\sqrt{1-x^2}-x}{\sqrt{1-x^2}[1+x\sqrt{1-x^2}]} dx = 2 \tan^{-1}(x+\sqrt{1-x^2}).$

13. If $\phi(x) = A_0 x^n + A_1 x^{n-1} + A_2 x^{n-2} + \cdots + A_n$, prove that

$$\int \frac{\phi(x)}{x-h} dx = B_0 \frac{x^n}{n} + B_1 \frac{x^{n-1}}{n-1} + B_2 \frac{x^{n-2}}{n-2} + \cdots + \phi(h) \log(x-h)$$

where $B_r = hB_{r-1} + A_r$. Write down the values of B_0, B_1, B_2 .

14. Show that $\int \frac{\phi(x)}{x-h} dx$ for rational integral algebraic forms of $\phi(x)$ may also be expressed as

$$\phi(h) \log(x-h) + \phi'(x) \frac{(x-h)}{1!} + \phi''(x) \frac{(x-h)^2}{2!} + \phi'''(x) \frac{(x-h)^3}{3!} + \dots$$

Prove that

$$\int \frac{e^x - e^h}{x-h} dx = e^h \left[\frac{x-h}{1^2 \cdot 1!} + \frac{(x-h)^2}{2^2 \cdot 2!} + \frac{(x-h)^3}{3^2 \cdot 3!} + \frac{(x-h)^4}{4^2 \cdot 3!} + \dots \right].$$

15. Prove that $\int_a^b \frac{\log x}{x} dx = \frac{1}{2} \log(ab) \log\left(\frac{b}{a}\right)$.

16. If

$$J_n(x) = \frac{x^n}{2^n n!} \left[1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2 \cdot 4(2n+2)(2n+4)} - \dots \right]$$

(i.e. Bessel's function), prove that

$$(1) \int_0^x J_1(x) dx = 1 - J_0(x),$$

$$(2) \int_0^x [J_{n-1}(x) - J_{n+1}(x)] dx = 2J_n(x) \quad (n > 0).$$

17. Prove that

$$\lambda \xi^{-\lambda} \int_0^\xi x^{\lambda-1} (1-x^\mu)^\nu dx = F\left(-\nu, \frac{\lambda}{\mu}, \frac{\lambda}{\mu} + 1, \xi^\mu\right)$$

when $F(\alpha, \beta, \gamma, x)$ denotes the hypergeometric series

$$1 + \frac{\alpha \cdot \beta}{1 \cdot \gamma} x + \frac{\alpha \overline{\alpha+1}}{1 \cdot 2} \frac{\beta \overline{\beta+1}}{\gamma \overline{\gamma+1}} x^2 + \frac{\alpha \overline{\alpha+1} \overline{\alpha+2}}{1 \cdot 2 \cdot 3} \cdot \frac{\beta \overline{\beta+1} \overline{\beta+2}}{\gamma \overline{\gamma+1} \overline{\gamma+2}} x^3 + \dots$$

[I. C. S., 1898.]

18. Assuming that the speed of the current in a river at a distance x from the bank follows the law

$$v = v_0 + kx(a-x),$$

where a is the breadth of the river and v_0 and k are constants, find by integration how far down stream a man will be carried who rows 4 miles an hour, pointing the boat's head always straight at the opposite bank, so as to cross in the least time possible: the width of the river being half a mile, the banks being straight and parallel, and the speed of the current being 2 miles an hour near the banks, and 3 miles an hour in mid-stream. [I. C. S., 1905.]

19. Find the moment of inertia of a rectangle of sides $2a$, $2b$ about a line joining the mid-points of the opposite sides of length $2a$.

The section of a ship at the water line is 120 feet long. If the middle line be divided into six equal portions, the ordinates of the boundary of the area at the middle points of the segments are given by the following table:

Distances from end	10	30	50	70	90	110
Ordinates	10	20	20	18	14	8

Draw a figure showing the section, remembering that it is symmetrical on both sides of the middle line. By approximate methods of summation (treating the segments as rectangles) find

the value of Ak^2 , the moment of inertia of the section about the middle line, and the height of the metacentre above the centre of buoyancy if the ship displaces 36,000 cubic feet of water.

$$\left[\text{The height required} = \frac{\text{Moment of Inertia}}{\text{Displacement}}. \right]$$

[I. C. S., 1908.]

20. A substance A transforms into a substance B , the rate of transformation in grammes per second at the time t being equal to ax , where x denotes the number of grammes of A existing at that instant. In like manner B transforms into a third substance C , the rate of transformation being by , where y is the number of grammes of B existing at time t .

Write down the relations between x , y and their differential coefficients with respect to the time, and show that these equations are satisfied by putting

$$x = Pe^{-at}, \quad y = Qe^{-at} + Re^{-bt},$$

provided Q and R are properly determined in terms of P , y being zero when $t = 0$.

Also show that the quantity of B existing will be greatest at $\frac{\log \text{Nap } a - \log \text{Nap } b}{a - b}$ seconds after the zero of time. [I. C. S., 1908.]

21. Prove that the integral of the sum of an infinite series taken over a range within which the series is absolutely convergent, is equal to the sum of the integrals of the terms.

Employ this theorem to find an expansion of $\log(1 + z)$ in ascending powers of z , pointing out the range for which the expansion is valid.

Having given $\log_e 2 = 0.69315$, prove that $\log_e 61 = 4.1109$ to five significant figures. [I. C. S., 1904.]

22. OK is the diameter bounding a semicircle of radius r , P any point on OK , and PQ an ordinate to the diameter OK . If x denote the length OP and z the area which PQ cuts off the semicircle, interpret $\frac{dz}{dx}$ and $\frac{d^2z}{dx^2}$.

Find a curve for which the area bounded by the curve, the axes of x and y and the ordinate at a distance x from the axis of y , is $a^2 \tan \frac{x}{a}$. [I. C. S., 1902.]

23. From the equation

$$\frac{1}{y} \left(ah + \int_h^x y dx \right) = h,$$

where a and h are constants, find y in terms of x .

The value of a being 2 feet, and of h 10 feet, evaluate y when x is 30 feet. [I. C. S., 1910.]

24. Denoting by A the area between the curve $y = f(x)$ and the axis of x , from the value zero to the value a of x , show that, when $f(x)$ is a rational integral algebraic function of the third degree,

$$A = \frac{a}{6}(y_0 + 4y_1 + y_2),$$

where $y_0 = f(0)$, $y_1 = f\left(\frac{a}{2}\right)$, $y_2 = f(a)$.

Compare the result given by this rule with the true value, taken to three places of decimals, for the curve $y = \sin x$, between the values 0 and 0.5 of x reckoned in radians. [I. C. S., 1912.]

25. Verify that the area of the curve

$$y = A + Bx + Cx^2 + Dx^3$$

between the limits $x = h$ and $x = -h$ is equal to the product of h and the sum of the ordinates at

$$x = h/\sqrt{3} \quad \text{and} \quad x = -h/\sqrt{3}.$$

In the case of the curve

$$y = A + Bx + Cx^2 + Dx^3 + Ex^4 + Fx^5 \equiv f(x),$$

verify in like manner that the area between $x = h$ and $x = -h$ is equal to

$$\{5f(h\sqrt{3/5}) + 8f(0) + 5f(-h\sqrt{3/5})\}h/9.$$

[I. C. S., 1913.]

26. Find the differential coefficient of

$$e^{-x}(1 + x + x^2/2! + x^3/3! + x^4/4!);$$

and deduce that the sum of the first five terms of the exponential series is less than e^x by the quantity

$$e^x \int_0^x \frac{a^4 e^{-a}}{4!} da.$$

What would be the corresponding result if the series were taken to n terms instead of to five terms?

[I. C. S., 1913.]

27. Weddle's Rule for finding, approximately, the area bounded by a curve, two ordinates, and a base forming part of the axis of x , is: "Divide the base into six equal parts and draw the ordinates at the points of division, making, with the extreme ordinates, seven in all. Of these ordinates add the first, third, fourth, fifth and seventh, and five times the second, fourth and sixth. Multiply the sum by one-twentieth of the base."

Prove that the rule gives the true result when the limits are 0 and 1 and the curve has any of the forms

$$y = a, \quad y = ax^2, \quad y = ax^4, \quad y = a(x - 1/2)^n,$$

where a is a constant and n is an odd positive integer.

Find, by the rule, the value of $6 \times \int_0^{1/\sqrt{3}} \frac{dx}{1+x^2}$ to seven places of decimals. Check the result by integration.

[I. C. S., 1911.]

28. Show that the work done by a gas in altering its volume from v_1 to v_2 according to the Adiabatic Law

$$pv^\gamma = p_0v_0^\gamma$$

is

$$\frac{p_0v_0^\gamma}{\gamma-1} \left[\frac{1}{v_1^{\gamma-1}} - \frac{1}{v_2^{\gamma-1}} \right].$$

If the law be Isothermal ($\gamma = 1$), show that this becomes

$$p_0v_0 \log \frac{v_2}{v_1}.$$

If a gas expands isothermally from state p_1, v_1 to state p_2, v_2 (Operation I.), then expands adiabatically from state p_2, v_2 to state p_3, v_3 (Operation II.), then contracts isothermally from state p_3, v_3 to state p_4, v_4 (Operation III.), then contracts adiabatically from state p_4, v_4 to state p_1, v_1 (Operation IV.),

- (1) find the amounts of work done by or upon the gas during each of these four operations, drawing a graph of the whole cycle of changes;
- (2) show that the work done in the whole cycle of operations is measured by

$$(p_1 v_1 - p_3 v_3) \log \frac{v_2}{v_1} \text{ (the adiabatic portions cancelling);}$$

- (3) that $v_1 v_3 = v_2 v_4$;

- (4) that, writing

$$\begin{aligned} p_1 v_1 &= p_2 v_2 = \alpha_1, & p_3 v_3 &= p_4 v_4 = \alpha_2, \\ p_2 v_2^\gamma &= p_3 v_3^\gamma = \beta_1, & p_4 v_4^\gamma &= p_1 v_1^\gamma = \beta_2, \end{aligned}$$

the above expression for the work may be written

$$\frac{\alpha_1 - \alpha_2}{\gamma - 1} \log \frac{\beta_1}{\beta_2}.$$

[This cycle of operations is known as a Carnot's cycle for a perfect heat engine.]

29. If dQ be the whole heat absorbed by a body of uniform temperature whilst its temperature changes continuously from θ to $\theta + d\theta$, and if ϕ be a function of the independent variables which define the state of the body and such that

$$d\phi = \frac{dQ}{\theta},$$

ϕ is called the Entropy of the body (Clausius).

Show that if a graph be drawn to represent θ as a function of ϕ , the area between the graph, the ϕ -axis and the ordinates corresponding to the initial and final states represents on some scale the heat absorbed.

In the case of a perfect gas satisfying the law $\frac{pv}{\theta} = \text{const.} = R$, assume the Thermodynamic Equation

$$dQ = C_v d\theta + p dv,$$

where C_v is the specific heat at constant volume, and show that in changing from state θ_1, v_1, ϕ_1 to state θ_2, v_2, ϕ_2 ,

$$\left(\frac{\theta_2}{\theta_1}\right)^{C_v} \left(\frac{v_2}{v_1}\right)^R = \frac{e^{\phi_2}}{e^{\phi_1}}.$$

Taking the temperature as a function of the entropy and simultaneous values of ϕ and θ as given in the following table:

ϕ	1.60	1.70	1.75	1.85
θ	450	400	370	340

and assuming that one unit of area indicates one unit of heat, what is the total heat received during these changes?

[There is a brief sketch of the fundamental formulae of Thermodynamics on pages 56 and 57 of *Solutions of Senate House Problems for 1878* which may be found useful. Students may also read Tait's *Thermodynamics* or Parker's *Thermodynamics* for detailed accounts of the theory; other useful books are Zeuner, *Théorie Mécanique de la Chaleur*; Briot, *Théorie Mécanique de la Chaleur*.]

30. In the case of a saturated vapour, if C' be the specific heat of the vapour, i.e. the heat imparted to one gramme of the saturated vapour to keep it constantly in the saturated state when slowly compressed till the temperature rises one degree Fahrenheit; C that of the liquid from which it is derived at the same pressure and temperature, L the latent heat, then it can be shown that

$$C' = C + \frac{dL}{d\theta} - \frac{L}{\theta},$$

where θ is the absolute temperature.

Let C_p be the specific heat of the liquid at constant pressure, which, as liquids are practically incompressible, is so nearly equal to C that no appreciable error results from regarding them as identical.

Then Regnault has shown experimentally that the sum of the free and latent heat, viz. $L + \int_{273}^{\theta} C_p d\theta$, is not a constant as had been supposed by Watt in his earlier experiments, but is a function of the temperature θ , viz. putting $\theta = 273 + \theta'$ and J being the number of ergs in one calorie (41,539,739.8 ergs or about 3 foot-lbs.), he obtained the equations

$$(1) \quad L + \int_{273}^{\theta} C_p d\theta = J(\alpha + \beta\theta' + \gamma\theta'^2),$$

$$(2) \quad C_p = J(\alpha' + \beta'\theta' + \gamma'\theta'^2),$$

experimentally, determining the constants $\alpha, \beta, \gamma; \alpha', \beta', \gamma'$ for several vapours.

Using these data, prove that

$$(1) \quad \frac{dL}{d\theta} + C_p = J(\beta + 2\gamma\theta'),$$

$$(2) \quad L = J \left[\alpha + (\beta - \alpha')\theta' + (2\gamma - \beta')\frac{\theta'^2}{2} - \gamma'\frac{\theta'^3}{3} \right],$$

$$(3) \quad c' \equiv \frac{C'}{J} = \beta + 2\gamma\theta' - \frac{\alpha + (\beta - \alpha')\theta' + (2\gamma - \beta')\frac{\theta'^2}{2} - \gamma'\frac{\theta'^3}{3}}{273 + \theta'}.$$

31. Show that the integral equivalent of the equation

$$\int_0^x \frac{dx}{1+x+x^2} + \int_0^y \frac{dy}{1+y+y^2} + \int_0^z \frac{dz}{1+z+z^2} = 0$$

is of the form

$$xyz + a(yz + zx + xy) + b(x + y + z) + c = 0,$$

where a, b, c are certain constants.

[Oxf. I. P., 1913.]

32. If the variables x, y, z be so related that

$$yz = F(x), \quad zx = F(y), \quad xy = F(z),$$

show that $\int_{x_1}^x F(x) dx + \int_{y_1}^y F(y) dy + \int_{z_1}^z F(z) dz = xyz - x_1y_1z_1$.

For example, if $x + y + z = 0$, and $yz + zx + xy = -\frac{1}{4}x^2y^2z^2$, show that

$$\int_{x_1}^x \frac{\sqrt{1+x^4}}{x^2} dx + \int_{y_1}^y \frac{\sqrt{1+y^4}}{y^2} dy + \int_{z_1}^z \frac{\sqrt{1+z^4}}{z^2} dz = \frac{3}{4}(xyz - x_1y_1z_1).$$

[Bertrand, Calc. Int., p. 383.]

33. If $y = \int_0^x e^{\sin x} dx$, expand y in powers of x as far as x^5 . [Oxf. I. P., 1911.]

34. Prove that $\int_0^\infty \frac{x^2 + 3x + 3}{(x+1)^3} e^{-x} \sin x \, dx = \frac{1}{2}$. [Oxf. I. P., 1915.]

35. Integrate $\int x^{-2n-2} (1-x)^n (1-cx)^n dx$, where n is a positive integer. [Oxf. I. P., 1917.]

36. Prove that the fifth differential coefficient of

$$x^4 \int \phi(x) \, dx - 4x^3 \int x\phi(x) \, dx + 6x^2 \int x^2\phi(x) \, dx - 4x \int x^3\phi(x) \, dx + \int x^4\phi(x) \, dx$$

is

$$24\phi(x). \text{[Oxf. I. P., 1917.]}$$

37. Integrate $\int \frac{\cos 8\theta - \cos 7\theta}{1 + 2\cos 5\theta} d\theta$.

38. If $f(x)$ and $F(x)$ be two functions continuous and finite between 0 and x , such that

$$F(x) \equiv \int_0^x f(t) \, dt,$$

$$f(x) \equiv 1 - \int_0^x F(t) \, dt,$$

obtain their expansions in ascending powers of x .

[Oxf. I. P., 1915.]

39. Prove that $\int_0^1 \frac{dx}{(1+x)(2+x)} = 0.288$ nearly. [Math. Trip. I., 1916.]

40. If $y^2 = a^2x^2 + c$, express in terms of y the differential coefficients of the functions

$$\log(ax + y), \quad xy$$

with regard to x .

Hence evaluate $\int \frac{dx}{y}$ and $\int y \, dx$, and prove that

$$\int_2^{\frac{5}{2}} \sqrt{x^2 - 4} \, dx = \frac{15}{8} - 2 \log 2. \text{[Math. Trip. I., 1914.]}$$

41. Prove that

$$\{\log(a + \theta h) - \log a\} - \theta \{\log(a + h) - \log a\}$$

can be expressed in the form

$$\theta(1 - \theta) \int_0^h \frac{x \, dx}{(a+x)(a+\theta x)}.$$

Deduce that in calculating a logarithm to base 10 by the method of proportional parts from tables which give the logarithms of all integers from 10^4 to 10^5 , the error is one of defect and cannot amount to $\frac{1}{8}10^{-8}\mu$, where $\mu = \log_{10} e = .43429$. Is this negligible in seven-figure tables? [Math. Trip. Pt. II., 1919.]

42. Integrate $\int \sin \theta \sqrt{\frac{1 + \cos 9\theta}{1 + \cos \theta}} d\theta$, and show that

$$\int \sqrt{\frac{1 + \cos(4n+1)\theta}{1 + \cos \theta}} d\theta = \theta - 2 \left\{ \frac{\sin \theta}{1} - \frac{\sin 2\theta}{2} + \frac{\sin 3\theta}{3} - \cdots - \frac{\sin 2n\theta}{2n} \right\},$$

n being a positive integer.

Chapter III: Change of the Independent Variable — Exercises

Exercises I — Substitution Technique (Arts. 47–58)

1. Evaluate

(i) $\int \frac{3x^2}{1+x^3} dx$. Put $x^3 = z$.

(ii) $\int_0^1 \frac{dx}{x^2+4x+5}$. Put $x+2 = z$.

(iii) $\int_2^3 \frac{dx}{(x-1)\sqrt{x^2-2x}}$. Put $x-1 = z$; $\therefore \frac{dx}{x-1} = \frac{dz}{z}$, etc.

(iv) $\int \frac{dx}{e^x + e^{-x}}$. Put $e^x = z$.

(v) $\int \frac{e^x dx}{2e^{2x} + 2e^x + 1}$. Put $e^x = \frac{z}{1-z}$.

(vi) $\int \tan^2 x \sec^2 x dx$.

(vii) $\int \frac{dx}{\cosh^2 mx}$.

2. Evaluate

(i) $\int_0^a \sqrt{a^2 - x^2} dx$. Put $x = a \sin \theta$.

(ii) $\int_0^{2a} \sqrt{2ax - x^2} dx$. Put $x = a(1 - \cos \theta)$.

Draw graphs to illustrate these two integrations.

3. Find the values of

(i) $\int_0^a x \sqrt{a^2 - x^2} dx$.

(ii) $\int_0^a x^2 \sqrt{a^2 - x^2} dx$.

Interpret the meaning of these integrations.

4. Integrate $\int \frac{(ax^2 - b) dx}{x \sqrt{c^2 x^2 - (ax^2 + b)^2}}$. Put $ax + \frac{b}{x} = z$.

5. Integrate $\int \frac{x^{2n} dx}{(a^2 + x^2)^{n+\frac{3}{2}}}$. Put $x = a \tan \theta$. [St. John's, 1883.]

6. Integrate $\int \sec^{\frac{2}{3}} \theta \operatorname{cosec}^{\frac{2}{3}} \theta d\theta$. Put $\tan \theta = z$. [St. John's, 1883.]

7. Integrate $\int \sqrt{\frac{\sin x}{\cos^5 x}} dx$. Put $\tan x = z$. [Trinity, 1883.]

8. Integrate

- (i) $\int \frac{dx}{x\sqrt{x^4-1}}.$
- (ii) $\int \frac{dx}{x\sqrt{1-x^4}}.$
- (iii) $\int \frac{dx}{x\sqrt{1+x^4}}.$

9. Integrate

- (i) $\int \left(1 - \frac{1}{x^2}\right) e^{x+\frac{1}{x}} dx.$
- (ii) $\int \frac{ax^2-b}{x^3+(ax^2+b)^2} dx.$
- (iii) $\int \frac{ax^2-b}{x^{n+2}}(ax^2+b)^n dx.$
- (iv) $\int \frac{1}{(c+ex)^2} \cos \frac{a+bx}{c+ex} dx.$
- (v) $\int \frac{e^{a \tan^{-1} x}}{1+x^2} dx.$
- (vi) $\int \frac{e^{a \sin^{-1} x}}{\sqrt{1-x^2}} dx.$
- (vii) $\int \frac{\sin x \cos x dx}{a^2 \cos^2 x + b^2 \sin^2 x}.$

10. Integrate

- (i) $\int \{\phi(x)\psi'(x) + \phi'(x)\psi(x)\} dx.$
- (ii) $\int \frac{\phi(x)\psi'(x) - \phi'(x)\psi(x)}{[\phi(x)]^2} dx.$
- (iii) $\int \frac{\phi'(x) dx}{1 + [\phi(x)]^2}.$
- (iv) $\int e^{\phi(x)} \phi'(x) dx.$
- (v) $\int e^{-\psi(x)} \frac{\phi'(x) - \phi(x)\psi'(x) \log \phi(x)}{\phi(x)} dx.$

11. Show that

$$\int (x-a) \sqrt{\frac{x-4a}{x}} dx = a^2(\sinh 2u - 2 \sinh u)$$

where $x = 4a \cosh^2 \frac{u}{2}.$

[Ox. I. Pub., 1899.]

Exercises II — The Hyperbolic Functions (Arts. 59–73)

Establish the following results:

1. $\int \cosh x dx = \sinh x.$

2. $\int \sinh x \, dx = \cosh x.$
3. $\int \operatorname{sech}^2 x \, dx = \tanh x.$
4. $\int \operatorname{cosech}^2 x \, dx = -\coth x.$
5. $\int \frac{\sinh x}{\cosh^2 x} dx = \int \operatorname{sech} x \tanh x \, dx = -\operatorname{sech} x.$
6. $\int \frac{\cosh x}{\sinh^2 x} dx = \int \operatorname{cosech} x \coth x \, dx = -\operatorname{cosech} x.$
7. $\int \operatorname{cg} x \, dx = \operatorname{gd} x.$
8. $\int \operatorname{cg}^2 x \, dx = \operatorname{sg} x.$
9. $\int \frac{dx}{\operatorname{cg} x} = \operatorname{tg} x.$
10. (a) $\operatorname{sg}(u+v) = \frac{\operatorname{sg} u + \operatorname{sg} v}{1 + \operatorname{sg} u \operatorname{sg} v},$
 (b) $\operatorname{cg}(u+v) = \frac{\operatorname{cg} u \operatorname{cg} v}{1 + \operatorname{sg} u \operatorname{sg} v},$
 (c) $\operatorname{sg} 2u = \frac{2 \operatorname{sg} u}{1 + \operatorname{sg}^2 u},$
 (d) $\operatorname{cg} 2u = \frac{\operatorname{cg}^2 u}{1 + \operatorname{sg}^2 u},$
 (e) $\operatorname{tg} 2u = 2 \frac{\operatorname{sg} u}{\operatorname{cg}^2 u}.$

Exercises III — Cosec, Sec & Radical Standard Forms (Arts. 74–89)

Write down the integrals of

1. $\frac{1}{9-x^2}, \frac{1}{9-4x^2}, \frac{1}{x^2-4}, \frac{1}{9x^2-4}, \sqrt{16-9x^2}, \sqrt{3x^2-5}, \sqrt{2+3x^2}.$
2. $\frac{1}{\sqrt{x(x-4)}}, \frac{1}{\sqrt{x(4-x)}}, \frac{1}{\sqrt{x(4+x)}}, \frac{1}{\sqrt{2+2x-x^2}}, \frac{1}{\sqrt{x^2-2x+2}}, \sqrt{x^2+2ax}.$
3. $\frac{x}{\sqrt{9-x^2}}, \frac{x}{\sqrt{x^2-9}}, \frac{x}{\sqrt{9-4x^2}}, \frac{x^2}{\sqrt{1-x^2}}, \frac{x^2}{\sqrt{x^2+1}}.$
4. $x\sqrt{x^2+a^2}, (x+b)\sqrt{x^2+a^2}, \frac{ax+b}{\sqrt{x^2+c^2}}.$
5. $x(x^2+a^2)^{\frac{n}{2}}, (x+a)(x^2+2ax+b)^{\frac{n}{2}}, (ax-b)(ax^2-2bx+c)^{\frac{n}{2}}.$
6. $\frac{x^2+2x+3}{\sqrt{1-x^2}}, \frac{x^2+2x+3}{\sqrt{x^2+1}}, \frac{x^2+2x+3}{\sqrt{x^2+x+1}}, \frac{x^2+ax+b}{\sqrt{x^2+cx+d}}.$
7. $\sqrt{x^2+4x+5}, \sqrt{-x^2+4x+5}, \sqrt{4x^2+4x+5}, \sqrt{-4x^2+4x+5}.$

8. $\sqrt{\frac{x+a}{x-a}}, \sqrt{\frac{a+x}{a-x}}, x\sqrt{\frac{a+x}{a-x}}, (x+a)\sqrt{\frac{x+b}{x-b}}, \frac{(x+a)^{\frac{3}{2}}}{(x-a)^{\frac{1}{2}}}.$
9. $\operatorname{cosec} nx, \operatorname{cosec}(2x+b), \frac{1}{4\cos^3 x - 3\cos x}, \frac{1+\tan^2 x}{1-\tan^2 x}, \frac{1}{2}(\cot x + \tan x).$
10. $\frac{1}{a\sin x + b\cos x}, \frac{1}{\sin 2x + \cos 2x}, \frac{a\sin x + b\cos x}{c\sin x + d\cos x}.$
11. Deduce $\int \operatorname{cosec} x \, dx = \log \tan \frac{x}{2}$ by expressing $\operatorname{cosec} x$ as
- $$\frac{1}{2} \left(\cot \frac{x}{2} + \tan \frac{x}{2} \right).$$

12. Deduce $\int \sec x \, dx = \log \tan \left(\frac{\pi}{4} + \frac{x}{2} \right)$ by

- (i) putting $\sin x = z,$
(ii) putting $\sec x + \tan x = z.$

Show that $\int \sec x \, dx = \cosh^{-1}(\sec x).$

13. Integrate $\int \frac{\cos \theta \, d\theta}{\sin \theta \sqrt{1 - \sin^{2n} \theta}}.$

General Examples

1. If APB be a semicircle, centre O , and PN an ordinate to the diameter AB , and $P'N'$ a contiguous ordinate, show that

$$Lt \sum \frac{NN'}{NP} = \text{circular measure of the angle } OPN,$$

the summation being from the centre to any point N between O and B , and NN' being indefinitely diminished.

2. Find the area in the first quadrant bounded by the axes of coordinates, the ordinate $x = x_1$ and the curve $x^2 = a^2 - \frac{b^4}{y^2}.$

If the range $x = 0$ to $x = a$ on the x -axis be divided into n equal portions of length h and rectangles be inscribed in the Newtonian manner, examine the limit of the area of the last of these rectangles when h is indefinitely diminished. Find the whole area from $x = 0$ to $x = a$.

3. Find the value of $\int \sqrt{e^{2x} + ae^x} \, dx.$ [R. P.]

4. Evaluate

(i) $\int \frac{dx}{\sqrt{1-3x-x^2}}.$ [I. C. S., 1884.]

(ii) $\int \frac{dx}{x\sqrt{x^2+x-6}}.$ (Put $x = \frac{1}{y}.$) [Oxford Second Public Ex., 1880.]

(iii) $\int \frac{(1+x) \, dx}{(2x^2+3x+4)^{\frac{3}{2}}}.$ [Colleges β , 1891.]

(iv) $\int \frac{x-1}{(x^2+2x-1)^{\frac{3}{2}}} dx.$ [Trinity, 1892.]

(v) $\int \frac{3x+4}{(x^2+2x+5)^{\frac{3}{2}}} dx.$ [Math. Trip., 1887.]

5. Show that the result of integrating $\int \frac{dx}{\sqrt{a^2-x^2}}$ may be exhibited as

(i) $2 \cos^{-1} \frac{\sqrt{a+x} - \sqrt{a-x}}{2\sqrt{a}}$ or as

(ii) $2 \sin^{-1} \frac{\sqrt{a+x} + \sqrt{a-x}}{2\sqrt{a}}$

or as (iii) $2 \tan^{-1} \sqrt{\frac{a+R}{a-R}}$, where $R = \sqrt{a^2-x^2}$.

6. If $R \equiv ax^2 + 2bx + c$ and $K = \frac{b^2-ac}{a}$, show that

$$\int \frac{dx}{\sqrt{R}} = \frac{1}{\sqrt{a}} \tanh^{-1} \sqrt{\frac{R}{R+K}}$$

or $= \frac{1}{\sqrt{-a}} \tan^{-1} \sqrt{-\frac{R}{R+K}}$, according as a is positive or negative.

7. Evaluate

(i) $\int \frac{dx}{x\sqrt{1+x^6}}.$ (Put $x^3 = \tan \theta$.) [Oxf. I., 1888.]

(ii) $\int \frac{dx}{(x+1)\sqrt{x^2-1}}.$ (Put $x = \sec \theta$.) [Oxf. I., 1888.]

8. Show that

(i) $\int_0^1 \frac{dx}{(1+x^2)(1-x^2)^{\frac{1}{2}}} = \frac{\pi}{2\sqrt{2}}.$ [Trinity, 1888.]

(ii) $\int \frac{dx}{x^4\sqrt{1+x^2}} = \frac{2x^2-1}{3x^3} \sqrt{1+x^2}.$ [Trinity, 1882.]

9. Integrate $\int \sqrt{1+e^x+e^{2x}} dx.$

10. A sphere of given radius a consists of an infinite number of concentric shells of very small thickness, the density at the surface of any shell varying as the n th power of its radius. Find the mass of the sphere. [Ox. I. Pub., 1903.]

If any diameter AB cut one of the shells at P and the density of the shell varies inversely as (i) $OP\sqrt{AP.PB}$, (ii) $OP^2\sqrt{AP.PB}$, find the mass of the sphere in each case, O being the centre.

11. A triangle ABC is divided into strips by lines parallel to BC ; a point is taken in each strip, and the square of the perpendicular from this point to BC is multiplied by the area of the strip; the same is done with all the strips, and the sum of the products is formed. Express by a definite integral the limiting value of this sum when the breadths of all the strips are diminished indefinitely, and evaluate the integral in terms of the base BC and the distance of A therefrom. [Ox. I. Pub., 1901.]

12. Prove that if $u^2 = x^2 + 2px + q$, an integral of the form

$$\int f(x, u) dx$$

can always be rationalized (provided f is a rational algebraic function) by one of the substitutions

$$\frac{u}{\sqrt{p^2 - q}} = \frac{2y}{1 - y^2} \quad \text{or} \quad \frac{u}{\sqrt{q - p^2}} = \frac{1 + y^2}{1 - y^2}. [\text{Coll. } a, 1890.]$$

13. Find the relation connecting x and y , being given

$$x^2 y^4 \left(\frac{dy}{dx} \right)^2 = a^2 + b^2 y^3. [\text{I. C. S., 1889.}]$$

14. Show that $\int_{\sqrt{3}}^{2\sqrt{3}} z^3 (z^2 - 3)^{\frac{3}{2}} dz = \frac{16038}{35}$. [Ox. I., 1888.]

15. Integrate

$$(i) \int \frac{\sin \theta d\theta}{\sqrt{a \cos^2 \theta + 2b \cos \theta + c}},$$

$$(ii) \int \frac{\cos \theta d\theta}{\sqrt{a \sin^2 \theta + 2b \sin \theta + c}},$$

$$(iii) \int \frac{d\theta}{\cos \theta \sqrt{a \cos^2 \theta + 2b \sin \theta \cos \theta + c \sin^2 \theta}},$$

$$(iv) \int \frac{d\theta}{\sin \theta \sqrt{a \cos^2 \theta + 2b \sin \theta \cos \theta + c \sin^2 \theta}},$$

$$(v) \int \frac{d\theta}{\sin \theta \sqrt{a \cos^2 \theta + b \sin^2 \theta + c}}.$$

[Trin., 1888.]

16. Integrate

$$(i) \int \frac{x^4}{(a^2 - x^2)^{\frac{3}{2}}} dx,$$

$$(ii) \int \frac{x^2 dx}{(a + bx^2)\sqrt{c^2 - x^2}}.$$

[Trin., 1888.]

17. (a) Evaluate $\int_1^7 (x^2 - 6x + 13) dx$, first directly, second by putting $x^2 - 6x + 13 = y$. (Draw a graph and explain fully.)

(b) Evaluate $\int_0^{\frac{2b}{a}} (ax^2 - 2bx + c) dx$, and explain by a graph the result when $2b^2 = 3ac$. Obtain the same result by substituting $ax^2 - 2bx + c = y$, taking $b^2 < ac$. Also obtain $\int_0^{\frac{3b}{a}} (ax^2 - 2bx + c) dx$ by this substitution, explaining your limits for y by means of the graph.

18. Point out the fallacy in the following argument:

$$\int_{-1}^1 \frac{dx}{1+x^2} = [\tan^{-1} x]_{-1}^1 = \frac{\pi}{4} - \left(-\frac{\pi}{4}\right) = \frac{\pi}{2}.$$

But putting $x = \frac{1}{y}$, $dx = -\frac{dy}{y^2}$. When $x = -1$, $y = -1$. When $x = 1$, $y = 1$.

$$\therefore \int_{-1}^1 \frac{dx}{1+x^2} = - \int_{-1}^1 \frac{dy}{1+y^2} = - \int_{-1}^1 \frac{dx}{1+x^2},$$

for, as the result is numerical, the letter used in integration cannot affect the result.

Hence $2 \int_{-1}^1 \frac{dx}{1+x^2} = 0$; but $\int_{-1}^1 \frac{dx}{1+x^2} = \frac{\pi}{2}$; $\therefore \pi = 0$!

19. Point out the fallacy in the following reasoning:

We have, if we put $x = e^t$,

$$\left(x \frac{d}{dx}\right)^a x^k = \left(\frac{d}{dt}\right)^a e^{kt} = k^a e^{kt} = k^a x^k.$$

But when $a = -1$, we have

$$\left(x \frac{d}{dx}\right)^{-1} x^k = \frac{1}{x} \int x^k dx = \frac{x^k}{k+1},$$

and these two results do not agree.

[R.P.]

20. Prove that $\iota \operatorname{gd} \left(\frac{1}{\iota} \operatorname{gd} u \right) = u$,

[Cayley, E.F.]

and show that if $\operatorname{gd} u = a_1 u + a_3 u^3 + a_5 u^5 + \dots$, then will $\operatorname{gd}^{-1} u = a_1 u - a_3 u^3 + a_5 u^5 - \dots$.

21. If $\sec x + \tan x = 1 + S_1 \frac{x}{1!} + S_2 \frac{x^2}{2!} + S_3 \frac{x^3}{3!} + \dots$,

show that $\operatorname{gd}^{-1} x = x + S_3 \frac{x^3}{3!} + S_5 \frac{x^5}{5!} + \dots$,

that $S_{p+1} = S_p + \binom{p}{2} S_{p-2} S_2 + \binom{p}{4} S_{p-4} S_4 + \dots$

and $S_n - \binom{n}{2} S_{n-2} + \binom{n}{4} S_{n-4} - \dots + \cos \frac{n\pi}{2} = \sin \frac{n\pi}{2}$,

and that $S_3 = 2$, $S_5 = 16$, $S_7 = 272$, $S_9 = 7936$, etc.

[Diff. Calc., Art. 573, etc.]

22. Integrate $\int \frac{dx}{(a+x)(c+x)^{\frac{1}{2}}}$ by putting $c+x = (a-c)z^2$ or $(c-a)z^2$, according as $a >$ or $< c$.

Taking the case $a > c$, consider the same integral with $a+da$ replacing a , subtract the original integral, divide by da , and take the limit when da is indefinitely diminished.

Hence obtain $\int \frac{dx}{(a+x)^2(c+x)^{\frac{1}{2}}}$.

Deduce also $\int \frac{dx}{(a+x)(c+x)^{\frac{3}{2}}}$.

23. Evaluate $\int \frac{dx}{(x^2-a^2)(x^2-c^2)^{\frac{1}{2}}}$.

(i) if $a > c$,

(ii) if $a < c$. (Put $x = c \sec \phi$).

[Math. Trip., 1878.]

24. Show that $\int \frac{(x-p)^{2n+1}}{(ax^2+2bx+c)^{n+\frac{3}{2}}} dx = \frac{1}{(b^2-ac)^{n+1}} \int (y^2-q^2)^n dy,$

where $q^2 \equiv ap^2 + 2bp + c$

and $y\sqrt{ax^2+2bx+c} = (ap+b)x + bp + c.$

[Colleges, 1901.]

25. If $F(x) = a_1f(x) + a_2f(2x) + a_3f(3x) + \dots,$

prove that $\frac{a_1}{1^x} + \frac{a_2}{2^x} + \frac{a_3}{3^x} + \dots = \frac{\int_0^\infty t^{x-1} F(t) dt}{\int_0^\infty t^{x-1} f(t) dt}.$

[Ox. J. M. Sch., 1904.]

26. Integrate

(i) $\int \frac{1+x^2}{1-x^2} \frac{dx}{\sqrt{1+x^4}}.$

(ii) $\int \frac{1-x^2}{1+x^2} \frac{dx}{\sqrt{1+x^4}}.$

[Euler.]

27. Show that if $F(x, y)$ be a rational function of x and $y,$

$$\int F\left(x, \sqrt[n]{\frac{\alpha x + \beta}{\gamma x + \delta}}\right) dx$$

can be thrown into rational form by the substitution

$$\frac{\alpha x + \beta}{\gamma x + \delta} = z^n.$$

Hence show that

$$\int \left(\frac{1-2x}{1+2x}\right)^{\frac{3}{2}} dx = 3 \tan^{-1} \sqrt{\frac{1-2x}{1+2x}} - 2 \sqrt{\frac{1-2x}{1+2x}} - \frac{\sqrt{1-4x^2}}{2}.$$

28. Show that if $F(x, y)$ be any rational integral function of x and $y,$

$$\int F(x, \sqrt{ax^2+2bx+c}) dx$$

can be thrown into rational form by any of the substitutions

(1) $\sqrt{ax^2+2bx+c} = \sqrt{a}(x+z),$

(2) $\sqrt{ax^2+2bx+c} = xz + \sqrt{c},$

(3) $x - x_2 = z^2(x - x_1),$

where x_1, x_2 are the roots of $ax^2 + 2bx + c = 0.$

[Bertrand, C.I., p. 39.]

Apply each of these methods to the integration of

$$\int \frac{x dx}{\sqrt{x^2 - 6x + 8}},$$

showing that the result in each case reduces to

$$\sqrt{x^2 - 6x + 8} + 3 \cosh^{-1}(x - 3),$$

as derived by the method of Art. 85.

29. If $x^3 - 3a^2x = a^2y$,

$$\text{show that } \frac{dx}{\sqrt{x^2 - 4a^2}} = \frac{1}{3} \frac{dy}{\sqrt{y^2 - 4a^2}},$$

and hence obtain Cardan's formula for the solution of a cubic.

[J. M. Sch. Ox.]

30. Evaluate $\int_0^{\pi/2} \frac{\cos x \, dx}{1 - \sin^2 a \cos^2 x}$,

and deduce the expansion

$$\frac{2a}{\sin 2a} = 1 + \frac{2}{3} \sin^2 a + \frac{2.4}{3.5} \sin^4 a + \dots, \quad \text{where } \frac{\pi}{2} > a > 0. [\text{Oxf. I. P., 1915.}]$$

31. Show that

$$\int \frac{dx}{(1+x^4)\{(1+x^4)^{\frac{1}{2}} - x^2\}^{\frac{1}{2}}} = \tan^{-1} \frac{x}{\{(1+x^4)^{\frac{1}{2}} - x^2\}^{\frac{1}{2}}}. [\text{Euler, C.I., iv.}]$$

$$\text{Integrate } \int \frac{dx}{(1+x^{2n})\{(1+x^{2n})^{\frac{1}{n}} - x^2\}^{\frac{1}{2}}}.$$

32. Show that the integrals

$$\int \frac{dx}{(1-x^m)(2x^m-1)^{\frac{1}{2m}}} \quad \text{and} \quad \int \frac{x^{m-1}dx}{(1-x^m)(2x^m-1)^{\frac{1}{2m}}}$$

are reduced to the integration of rational fractions by the respective substitutions $2x^m - 1 = u^{2m}x^{2m}$ and $2x^m - 1 = u^{2m}$. [Lexell, Actes de Pétersbourg, 1781, ii.; Lacroix, C.D., ii., p. 65.]

Chapter IV: Integration by Parts. Powers of Sines and Cosines — Exercises

Exercises I — Integration by Parts (Arts. 90–100)

Integrate by parts

1. xe^{3x} , x^2e^{ax} , x^5e^{-x} , $x \cosh x$, $x^2 \sinh x$.
2. $x \cos x$, $x^5 \cos 2x$, $x^2 \cos^2 x$, $x^2 \cos 3x \sin x$, $x \sin x \sin 2x \sin 3x$.
3. $e^x \sin 2x$, $e^x \sin^2 x$, $e^{3x} \sin^3 x \cos x$, $e^{-5x} \cos x \sin^2 x \cos 3x$.
4. $x^3 \log x$, $x^n \log x$, $x^n (\log x)^2$, $x^n (\log x)^3$.
5. $e^{ax} \sin px \sin qx \sin rx$, $e^{ax} \sin px \sin qx \cos rx$.
6. $e^{ax} \sin px \sin qx \cos^2 rx$, $e^{ax} \cos px \cos qx \cos^2 (p+q)x$.

7. Evaluate

$$\int_0^\pi x \sin x \, dx, \quad \int_0^{\pi/2} x^2 \cos x \, dx, \quad \int_0^{\pi/2} x^2 \cos 2x \, dx.$$

8. Integrate

$$\int \sin^{-1} x \, dx, \quad \int x \sin^{-1} x \, dx, \quad \int x^3 \sin^{-1} x \, dx, \quad \int x \tan^{-1} x \, dx.$$

Exercises II — Reduction Formulae for $\int x^m \sin nx \, dx$, $\int x^m \cos nx \, dx$ (Arts. 101–102)

Write down the integrals of

1. $\int x^6 e^x \, dx$, $\int x^5 \sinh x \, dx$, $\int x^5 \cosh^2 x \, dx$.
2. $\int_0^{\pi/2} x^3 \sin x \, dx$, $\int_0^{\pi/2} x^3 \sin^2 x \, dx$, $\int_0^{\pi/2} x^4 \sin x \cos x \, dx$.
3. $\int_0^\pi x^5 \sin x \, dx$, $\int_0^\pi x^5 \cos^2 x \, dx$, $\int_0^1 x^3 \cosh x \, dx$.
4. $\int_0^{\pi/2} x^3 (a^2 \cos^2 x + b^2 \sin^2 x) \, dx$, $\int_1^3 x^3 \log x \, dx$, $\int_0^1 x \tan^{-1} x \, dx$.
5. $\int_0^{\pi/2} e^x \sin x \cos^2 x \, dx$, $\int_0^{\pi/2} x \sin x \sin 2x \sin 3x \, dx$.

Exercises III — Powers and Products of Sines, Cosines, Secants, Cosecants, Tangents and Cotangents (Arts. 103–126)

1. Integrate $\sin^2 x$, $\sin^3 x$, $\sin^4 x$, $\sin^5 x$, $\sin^8 x$, $\sin^9 x$, $\sin^{2n} x$, $\sin^{2n+1} x$, doing those with odd indices in two ways.

2. Integrate $\sin^2 x \cos^2 x$, $\sin^3 x \cos^3 x$, $\sin^4 x \cos^4 x$, $\sin^3 x \cos^6 x$, $\sin^6 x \cos^3 x$, $\sin^6 x \cos^4 x$.
3. Integrate $\frac{\sin^2 x}{\cos^4 x}$, $\cos^2 x \operatorname{cosec}^4 x$, $\sec^2 x \operatorname{cosec}^2 x$, $\frac{1}{\sin^4 x \cos^4 x}$.
4. Evaluate $\int_0^{\pi/4} \sin^2 x \, dx$, $\int_0^{\pi/4} \cos^5 x \, dx$, $\int_0^{\pi/4} \cos^6 x \, dx$.
5. Integrate $\sin ax \cos^2 bx$, $\sin 3x \cos^3 x$, $\sin nx \cos^2 x$.

6. Show that

$$\int \sin x \sin 2x \sin 3x \, dx = -\frac{1}{8} \cos 2x - \frac{1}{16} \cos 4x + \frac{1}{24} \cos 6x.$$

7. Show that

$$(i) \int \sin mx \cos nx \, dx = -\frac{\cos(m+n)x}{2(m+n)} - \frac{\cos(m-n)x}{2(m-n)}.$$

$$(ii) \int \sin mx \sin nx \, dx = \frac{\sin(m-n)x}{2(m-n)} - \frac{\sin(m+n)x}{2(m+n)}.$$

$$(iii) \int \cos mx \cos nx \, dx = \frac{\sin(m-n)x}{2(m-n)} + \frac{\sin(m+n)x}{2(m+n)}.$$

Deduce from (ii) and (iii) the values of

$$\int \sin^2 mx \, dx \quad \text{and} \quad \int \cos^2 mx \, dx,$$

and verify the results by independent integration.

8. Prove that $\int_0^\pi \sin mx \sin nx \, dx$ and $\int_0^\pi \cos mx \cos nx \, dx$ are both zero so long as m and n are integral and unequal. But if m and n are equal integers their values are each equal to $\frac{\pi}{2}$.

General Examples

1. Prove that $\int u \frac{d^2 v}{dx^2} dx = u \frac{dv}{dx} - v \frac{du}{dx} + \int v \frac{d^2 u}{dx^2} dx$.

2. Perform the following integrations:

$$(i) \int \cos^{-1} x \, dx.$$

$$(ii) \int \cos^{-1} \frac{1}{x} \, dx.$$

$$(iii) \int x^3 \tan^{-1} x \, dx.$$

$$(iv) \int x \sec^2 x \, dx.$$

$$(v) \int x \sec x \tan x \, dx.$$

$$(vi) \int (ax+b) \log(cx+d) \, dx.$$

$$(vii) \int \tan^{-1} \sqrt{1-x^2} \, dx.$$

[St. John's, 1886.]

$$(viii) \int \frac{x^4}{1+x^2} \tan^{-1} x \, dx.$$

[Ox. II. P., 1889.]

3. Integrate

$$(i) \int e^{a \sin^{-1} x} \, dx.$$

$$(ii) \int \frac{x \sin^{-1} x}{(1-x^2)^{\frac{1}{2}}} \, dx.$$

$$(iii) \int \frac{x^3 \sin^{-1} x}{(1-x^2)^{\frac{3}{2}}} \, dx.$$

4. Integrate

$$(i) \int \frac{e^{m \tan^{-1} x}}{1+x^2} \, dx.$$

$$(ii) \int \frac{e^{m \tan^{-1} x}}{(1+x^2)^{\frac{3}{2}}} \, dx.$$

$$(iii) \int \frac{e^{m \tan^{-1} x}}{(1+x^2)^2} \, dx.$$

$$(iv) \int \frac{e^{m \tan^{-1} x}}{(1+x^2)^{\frac{5}{2}}} \, dx.$$

$$(v) \int \frac{e^{m \tan^{-1} x}}{(1+x^2)^{\frac{n}{2}+1}} \, dx \quad (n \equiv \text{a positive integer}).$$

5. Integrate

$$(i) \int x e^{bx} \cos ax \, dx.$$

$$(ii) \int x^2 e^{ax} \sin bx \, dx.$$

[a, 1888.]

$$(iii) \int x e^x \sin^2 x \, dx.$$

6. Integrate

$$(i) \int e^{ax} (\sin bx + \cos bx) \, dx.$$

$$(ii) \int e^{ax} (\sinh bx + \cosh bx) \, dx.$$

$$(iii) \int e^{ax} \sinh bx \cosh ax \, dx.$$

$$(iv) \int e^{ax} \cosh ax \sin bx \, dx.$$

$$(v) \int x^2 3^x \sin 4x \, dx.$$

$$(vi) \int \cos \left(b \log \frac{x}{a} \right) \, dx.$$

$$(vii) \int \cosh \left(b \log \frac{x}{a} \right) \, dx.$$

$$(viii) \int_0^\pi \theta \sin \theta \cosh(\cos \theta) d\theta. \quad [a, 1891.]$$

7. Integrate

$$(i) \int \frac{xe^x}{(x+1)^2} dx.$$

$$(ii) \int e^x \frac{1 + \sin x}{1 + \cos x} dx.$$

$$(iii) \int e^x \frac{1 - \sin x}{1 - \cos x} dx.$$

$$(iv) \int \frac{\cosh x + \sinh x \sin x}{1 + \cos x} dx.$$

$$(v) \int \frac{dx}{1 + e^x}. \quad [\text{Mech. Sc. Trip.}]$$

$$(vi) \int \sqrt{1 + e^{nx}} dx.$$

$$(vii) \int e^x \frac{1 + x^2}{(1 + x)^2} dx. \quad [\text{Ox. I. P., 1890.}]$$

8. Integrate

$$(i) \int (\log x)^2 dx. \quad [\text{Ox. I. P., 1888.}]$$

$$(ii) \int \left(x + \frac{1}{x} + \frac{1}{x^2} \right) \log x dx. \quad [\text{Ox. I. P., 1889.}]$$

$$(iii) \int x^{-2} \tan^{-1} x dx. \quad [\text{Ox. II. P., 1887.}]$$

$$(iv) \int \log \left(x + \sqrt{a^2 + x^2} \right) dx. \quad [\text{Math. Trip., 1882.}]$$

$$(v) \int x \log \left(x + \sqrt{a^2 + x^2} \right) dx. \quad [\text{St. John's, 1884.}]$$

$$(vi) \int (a + x) \sqrt{a^2 + x^2} dx. \quad [\text{St. John's, 1888.}]$$

$$(vii) \int (a^2 + x^2) \sqrt{a + x} dx. \quad [\text{St. John's, 1888.}]$$

$$(viii) \int e^{ax} x^2 \sin(bx + c) dx. \quad [\text{Coll., 1892.}]$$

$$(ix) \int x^3 (1 - x^{\frac{2}{3}})^{\frac{3}{2}} dx. \quad [\text{Ox. I. P., 1890.}]$$

9. Integrate

$$(i) \int x e^{ax} \sin bx \sin cx dx.$$

$$(ii) \int x e^{ax} \sin bx \sin^2 cx dx.$$

10. Show that if u be a rational integral function of x ,

$$\int e^{x/a} u dx = a e^{x/a} \left\{ u - a \frac{du}{dx} + a^2 \frac{d^2 u}{dx^2} - a^3 \frac{d^3 u}{dx^3} + \dots \right\},$$

where the series within the brackets is necessarily finite.

[Trin. Coll., 1881.]

11. If $u \equiv \int e^{ax} \cos bx \, dx$, $v \equiv \int e^{ax} \sin bx \, dx$, prove that

$$\tan^{-1} \frac{v}{u} + \tan^{-1} \frac{b}{a} = bx,$$

and that $(a^2 + b^2)(u^2 + v^2) = e^{2ax}$.

12. Evaluate $\int x^2 \log(1 - x^2) \, dx$, and deduce that

$$\frac{1}{1.5} + \frac{1}{2.7} + \frac{1}{3.9} + \cdots = \frac{8}{9} - \frac{2}{3} \log_e 2. [\text{a, 1889.}]$$

13. Integrate $\int \sec^{\frac{5}{3}} \theta \operatorname{cosec}^{\frac{5}{3}} \theta \, d\theta$, $\int \sec^{\frac{5}{3}} \theta \sin \theta \, d\theta$.

14. Find the value of

$$\int \left\{ u \frac{d^n v}{dx^n} - (-1)^n v \frac{d^n u}{dx^n} \right\} dx. [\gamma, 1890.]$$

15. Evaluate

$$\int \left[\frac{d^3 u}{dx^3} \left(\frac{dv}{dx} - \frac{dw}{dx} \right) + \frac{d^3 v}{dx^3} \left(\frac{dw}{dx} - \frac{du}{dx} \right) + \frac{d^3 w}{dx^3} \left(\frac{du}{dx} - \frac{dv}{dx} \right) \right] dx. [\gamma, 1890.]$$

16. Establish the following formulae for integration by parts, u and v being functions of x , and accents denoting differentiations and suffixes integrations with respect to x , and $u^{(n)}$ denoting u with n accents:

$$(i) \int uv \, dx = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \cdots + (-1)^{n-1} u^{(n-1)}v_n$$

$$+ (-1)^n \int u^{(n)} dv_{n+1}.$$

$$(ii) \iint (uv)(dx)^2 = uv_2 - 2u'v_3 + 3u''v_4 - 4u'''v_5 + \cdots + (-1)^{n-1} nu^{(n-1)}v_{n+1}$$

$$+ (-1)^n n \int u^{(n)} v_{n+1} \, dx + (-1)^n \int dx \int u^{(n)} v_n \, dx. [\text{a, 1888.}]$$

17. If u be a function of x , and differentiations and integrations are respectively denoted by accents and suffixes, and (n) means n accents, show that

$$\log u = 1 - \left(\frac{1}{u} \right)' u_1 + \left(\frac{1}{u} \right)'' u_2 - \left(\frac{1}{u} \right)''' u_3 + \cdots + (-1)^n \int \left(\frac{1}{u} \right)^{(n)} du_n. [\text{St. John's, 1889.}]$$

18. If u, v, w be functions of x , and accents and suffixes denote differentiations and integrations respectively, show that

$$2uvw = (vw)'u_1 - (vw)''u_2 + (vw)'''u_3 - \cdots + (-1)^{m-1} \int (vw)^{(m)} du_m$$

$$+ (wu)'v_1 - (wu)''v_2 + (wu)'''v_3 - \cdots + (-1)^{n-1} \int (wu)^{(n)} dv_n$$

$$+ (uv)'w_1 - (uv)''w_2 + (uv)'''w_3 - \cdots + (-1)^{p-1} \int (uv)^{(p)} dw_p. [\text{St. John's, 1889.}]$$

19. Prove that

$$\int_0^1 v^{xv} dv = 1 - \frac{x}{2^2} + \frac{x^2}{3^3} - \frac{x^3}{4^4} + \frac{x^4}{5^5} - \dots \text{ etc. [Math. Trip., 1878.]}$$

20. Find the value of $\int_0^1 x^x dx$ correct to five decimal places. [J. M. Sch. Ox., 1904.]

21. Prove that

$$e^{a^2 x^2} \int_0^x e^{-a^2 x^2} dx = x + \frac{2}{3} a^2 x^3 + \frac{2^2}{3 \cdot 5} a^4 x^5 + \dots$$

$$+ \frac{(2a^2)^n}{3 \cdot 5 \dots (2n-1)} e^{a^2 x^2} \int_0^x x^{2n} e^{-a^2 x^2} dx. [\text{Ox. I. Pub., 1899.}]$$

22. Find the sum of the series, supposed convergent,

$$\frac{x^5}{1 \cdot 3 \cdot 5} - \frac{x^7}{3 \cdot 5 \cdot 7} + \frac{x^9}{5 \cdot 7 \cdot 9} - \text{etc. to } \infty. [\text{Coll., 1881.}]$$

23. If y and z be functions of x , and $u = yz' - zy'$, prove the following:

(i) $\int zu^{-2}(y'z'' - z'y'') dx = -y^{-1}(1 + y'zu^{-1}),$

(ii) the integration of $zy^{-1}u^{-2}(yz'' - zy'')$ can be reduced to that of y^{-2} .

[St. John's, 1886.]

24. Show how the method of integration by parts may be applied to find

$$\int f(x) \frac{d^{n+1}V}{dx^{n+1}} dx,$$

where $f(x)$ is a rational algebraical expression of the n th degree.

Prove that $\int_{-1}^1 f(x) \frac{d^{n+1}(x^2 - 1)^{n+1}}{dx^{n+1}} dx = 0.$ [Coll., 1876.]

25. Prove that $\int (\cos x)^n dx$ may be expressed by the series,

$$\sin x - N_1 \frac{\sin^3 x}{3} + N_2 \frac{\sin^5 x}{5} - N_3 \frac{\sin^7 x}{7} + \dots, \text{ etc.,}$$

$N_1, N_2, N_3, \dots$ being the coefficients of the expansion $(1 + a)^{\frac{n-2}{2}}$, and n having any real value positive or negative. [Smith's Prize, 1876.]

26. Prove that

$$\int x^n e^x \sin x dx = e^x \sum_{r=0}^{r=n} (-1)^r \frac{n!}{(n-r)!} x^{n-r} 2^{-\frac{r+1}{2}} \sin \left\{ x - \frac{(r+1)\pi}{4} \right\}.$$

27. Express the infinite series

$$\frac{1}{2} + \frac{1 \cdot 3}{2 \cdot 4} \cdot \frac{1}{2} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \cdot \frac{1}{3} + \dots$$

as a definite integral, and find its value.

[Ox. II. P., 1902.]

28. Show that

$$2^m \int \cos mx \cos^m x \, dx = A + x + m \frac{\sin 2x}{2} + \frac{m(m-1)}{1.2} \frac{\sin 4x}{4} + \cdots + \frac{\sin 2mx}{2m},$$

where m is an integer and A is independent of x .

[Coll. a, 1885.]

29. Evaluate the integral

$$\int_0^T a_1 \sin \frac{2\pi t}{T} \cdot a_2 \sin \frac{2\pi}{T}(t + \lambda) \, dt,$$

and draw curves showing how its value depends on that of λ .

[Mech. Sc. Trip., 1899.]

30. Prove that if $y = f(x)$ and $x = \phi(y)$ are equivalent relations, then, between any corresponding limits,

$$\int \sqrt{f'(x)} \, dx = \int \sqrt{\phi'(y)} \, dy.$$

Hence, or otherwise, prove that if $\tan \beta = \sqrt{1-c} \tan \alpha$,

$$\int_0^\alpha \frac{dx}{\sqrt{1-c \sin^2 x}} = \int_0^\beta \frac{dx}{\sqrt{1-c \cos^2 x}}. [\text{Ox. II. P., 1886.}]$$

31. Prove that the remainder R in the series

$$\theta = \tan \theta - \frac{1}{3} \tan^3 \theta + \cdots + \frac{(-1)^n}{2n+1} \tan^{2n+1} \theta + R$$

may be written as a definite integral,

$$(-1)^{n+1} \int_0^\theta \tan^{2n+2} \theta \, d\theta. [\text{Coll., 1881.}]$$

32. Show that the integrals $\int_0^x f(z) \, dz$, $\int_0^x z^n f^{(n)}(z) \, dz$ are connected thus:

$$\begin{aligned} \int_0^x f(z) \, dz &= x f(x) - \frac{x^2}{2!} f'(x) + \frac{x^3}{3!} f''(x) - \cdots \\ &+ (-1)^{n-1} \frac{x^n}{n!} f^{(n-1)}(x) + (-1)^n \int_0^x z^n f^{(n)}(z) \, dz, \end{aligned}$$

and that if one can be integrated the other can also be integrated.

[Bernoulli.]

33. Integrate

$$\int \left\{ (2n+1) \cos(2n + \frac{1}{2})\theta + (2n-2) \cos(2n - \frac{3}{2})\theta \right\} (\cos \theta)^{\frac{1}{2}} \, d\theta,$$

and prove that when n is a positive integer,

$$\int_0^{\pi/2} \cos(2n + \frac{1}{2})\theta (\cos \theta)^{\frac{1}{2}} \, d\theta = 0. [\text{Oxford II. Pub., 1913.}]$$

34. Find the sum of the areas included between the axis of x and the arc of the curve $y = x \sin(x/a)$ from the ordinate $x = 0$ to the ordinate $x = n\pi a$, n being any positive integer, odd or even.
[Oxf. I. P., 1911.]

35. Evaluate $\int_0^{2a} \frac{x^n}{\sqrt{2ax - x^2}} \, dx$ when n is any positive integer.

[Oxf. I. P., 1916.]

36. Show that $\int_0^1 x \log \left(1 + \frac{1}{2}x\right) dx = \frac{3}{4} \left(1 - 2 \log \frac{3}{2}\right)$, and prove that this is less than $\int_0^1 \frac{1}{2}x^2 dx$.
[Math. Trip., Part I., 1913.]

37. If $T_n = \int_0^x \tan^n x dx$, show that $(n-1)(T_n + T_{n-2}) = \tan^{n-1} x$.

Given that $\pi = 3.141592\dots$, $\log_e 2 = 0.693147\dots$, show that

$$\int_0^{\pi/4} \tan^5 x dx = 0.09657\dots, \quad \int_0^{\pi/4} \tan^4 x dx = 0.11873\dots [\text{Math. Trip. I., 1915.}]$$

38. Prove that $\int_0^{1/\sqrt{2}} \frac{\sin^{-1} x}{(1-x^2)^{\frac{3}{2}}} dx = \frac{\pi}{4} - \frac{1}{2} \log_e 2$. [Math Trip. I., 1917.]

39. Find the area A between the curve

$$y = a \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x \right)$$

and the axis of x between the limits 0 and π ; and the volume V obtained by rotating this area about the axis of x .

Prove that $4V = \pi^2 a A$.

[Math. Trip. I., 1913.]

40. Show that

$$\int_0^1 x^{2p-1} \log(1+x) dx = \frac{1}{2p} \left\{ \frac{1}{1.2} + \frac{1}{3.4} + \dots + \frac{1}{(2p-1)2p} \right\}. [\text{Math. Trip., Pt. I., 1916.}]$$

Chapter V: Rational Algebraic Fractional Forms — Exercises

Exercises — Partial Fractions and Rational Algebraic Fractional Forms (Arts. 127–169)

Integrate with regard to x the expressions in the following seven groups:

1. Linear unrepeatd factors

(i) $\frac{1}{x(x^2 - 1)}.$

(ii) $\frac{1}{(x - 1)(x - 3)(x - 5)}.$

(iii) $\frac{1 - 3x^2}{3x - x^3}.$

(iv) $\frac{x^2 + x + 1}{(x^2 - 1)(x^2 - 4)}.$

(v) $\frac{(x - 1)(x - 2)}{(x^2 - 9)(x^2 - 16)}.$

(vi) $\frac{(x - a)(x - b)(x - c)}{(x - a_1)(x - b_1)(x - c_1)}.$

(vii) $\frac{(x - a)(x - b)(x - c)}{(x^2 - a_1^2)(x^2 - b_1^2)(x^2 - c_1^2)}.$

(viii) $\frac{x + 1}{x^2 + 10x - 75}.$

(ix) $\frac{x + 1}{x^2 + 10x - 119}.$

(x) $\frac{x + 1}{x^3 - 31x^2 + 311x - 1001}.$

2. Linear repeated factors:

(i) $\frac{1}{(x - 1)^3(x + 1)}.$

(ii) $\frac{1}{(x - 1)^4(x + 1)^4}.$

(iii) $\frac{x + 1}{x^4(x - 1)^4}.$

(iv) $(ax^2 + bx^3)^{-1}.$

(v) $(x^2 - 7x + 12)^{-2}.$

(vi) $\frac{x^3}{(x - a)^2(x - b)}.$

(vii) $\frac{x^2 - 3x + 3}{x^3 - 7x^2 + 16x - 12}.$

[I. C. S., 1900.]

3. Quasi-linear occurrence of factors. Powers of x all even:

$$(i) \int \frac{dx}{(x^2 + a^2)(x^2 + b^2)}.$$

$$(ii) \int \frac{(x^2 + a^2)(x^2 + b^2)}{(x^2 + c^2)(x^2 + d^2)} dx.$$

$$(iii) \int \frac{x^2(x^2 + a^2)}{(x^2 + c^2)} dx.$$

$$(iv) \int \frac{x^2 dx}{(x^2 + 1)(2x^2 + 1)}.$$

$$(v) \int \frac{ax^2 + b}{(cx^2 + d)(ex^2 + f)} dx.$$

$$(vi) \int \frac{ax^2 + b}{x^2(cx^2 + d)(ex^2 + f)(gx^2 + h)} dx.$$

In the last two c, d, e, f, g, h may be considered positive.

4. Quasi-linear factors. Numerator an odd function, Denominator even, or Numerator even, Denominator odd:

$$(i) \int \frac{dx}{x(x^2 + 1)}.$$

$$(ii) \int \frac{x^2 + 2}{x(x^4 - 1)} dx.$$

$$(iii) \int \frac{dx}{x^7 - 6x^5 + 11x^3 - 6x}.$$

$$(iv) \int \frac{x dx}{(ax^2 + bx + c)^2 + (ax^2 - bx + c)^2}.$$

5. Quadratic factors not repeated:

$$(i) \int \frac{dx}{x^4 + x^2 + 1}.$$

$$(ii) \int \frac{(x + 1)^2}{x^4 + x^2 + 1} dx.$$

$$(iii) \int \frac{x^2 + 1}{x^4 + 1} dx.$$

$$(iv) \int \frac{x^2 + 1}{x^4 - x^2 + 1} dx.$$

$$(v) \int (x^2 + a^2)(x^4 + a^2x^2 + a^4)^{-1} dx.$$

$$(vi) \int (x^2 - a^2)(x^4 + a^2x^2 + a^4)^{-1} dx.$$

$$(vii) \int \frac{x^2 + 3x + 1}{x^4 + x^2 + 1} dx.$$

$$(viii) \int \frac{dx}{x^4 + 1}.$$

6. Linear factors repeated. Quadratic factors not repeated.

- (i) $\frac{x^2 dx}{(x-1)^2(x^2-2x+4)}.$
- (ii) $\frac{dx}{(1+x)^2(1+2x+4x^2)}.$
- (iii) $\frac{x^4 dx}{(x-1)^2(x^2+4)}.$
- (iv) $\frac{dx}{(x+1)^2(x^2+1)}.$
- (v) $\frac{dx}{(x-1)^2(x^2+1)}.$
- (vi) $\frac{dx}{x(x-1)^2(x^2+1)}.$
- (vii) $\frac{dx}{x^3(a^2+x^2)}.$
- (viii) $\frac{dx}{x^3(a^2+x^2)(b^2+x^2)}.$
- (ix) $\frac{dx}{(x-1)^3(x^2+x+1)}.$
- (x) $\frac{dx}{(2x-3)^2(4x^2+5)}.$

7. Repeated quadratic factors:

- (i) $\frac{dx}{x(x^2+1)^2}.$
- (ii) $\frac{dx}{(x-1)^2(x^2+1)^2}.$
- (iii) $\frac{(x+1) dx}{(x^2+1)^2}.$
- (iv) $\frac{(x+a)(x+b)}{(x^2+c^2)^3} dx.$

8. Evaluate $\int_0^{\pi/4} \sqrt{\tan \theta} d\theta$ and $\int_0^{\pi/4} \sqrt{\cot \theta} d\theta.$

9. Evaluate

- (i) $\int_0^{\pi/4} \frac{dx}{\cos^4 x - \cos^2 x \sin^2 x + \sin^4 x},$
- (ii) $\int_0^{\pi/4} \frac{dx}{\cos^4 x + \cos^2 x \sin^2 x + \sin^4 x}.$

10. Evaluate $\int_0^{\pi/2} \frac{\cos x dx}{(1+\sin x)(2+\sin x)}.$

11. Show that $\int_0^{\infty} \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)(x^2+c^2)} = \frac{\pi}{2(a+b)(b+c)(c+a)}.$

12. Show that $\int_{-\infty}^{+\infty} \frac{dx}{(x^2 \pm ax + a^2)(x^2 \pm bx + b^2)} = \frac{2\pi}{\sqrt{3}} \frac{a+b}{ab(a^2+ab+b^2)}.$ [γ , 1891.]

13. Show that the sum of the infinite series

$$\frac{1}{a} - \frac{1}{a+b} + \frac{1}{a+2b} - \frac{1}{a+3b} + \dots \quad (a > 0, b > 0)$$

can be expressed as a definite integral, viz.

$$\int_0^1 \frac{t^{a-1}}{1+t^b} dt.$$

And hence prove that

$$\frac{1}{1} - \frac{1}{4} + \frac{1}{7} - \frac{1}{10} + \frac{1}{13} - \frac{1}{16} + \dots = \frac{1}{3} \left(\pi 3^{-\frac{1}{2}} + \log_e 2 \right). [\text{Oxford, 1887.}]$$

14. Integrate:

$$(i) \int \frac{25 dx}{2x^4 + 3x^3 + 3x - 2}. \quad [\text{Colleges, 1882.}]$$

$$(ii) \int \frac{x^5 + 2}{x^5 - x} dx. \quad [\text{St. John's, 1881.}]$$

$$(iii) \int \frac{(1+x^2) dx}{1 - 2x^2 \cos \alpha + x^4}. \quad [\text{Colleges, 1882.}]$$

$$(iv) \int \frac{dx}{1+x^5}. \quad [\text{Colleges a, 1891.}]$$

$$(v) \int_0^1 \frac{1+x^2}{(1-x^2)^2 + a^2 x^2} dx.$$

$$15. \text{ Prove that } \int_0^\infty \frac{dx}{1+x^6} = \frac{\pi}{3}. \quad [\text{St. John's, 1881.}]$$

16. Prove that

$$\begin{aligned} & \int \frac{dx}{(x-a)^p(x-b)^q} \\ &= \sum_{r=0}^{p-2} \frac{Q_r}{(a-b)^{q+r}} \frac{(x-a)^{-p+r+1}}{-p+r+1} + \sum_{r=0}^{q-2} \frac{P_r}{(b-a)^{p+r}} \frac{(x-b)^{-q+r+1}}{-q+r+1} \\ &+ \frac{1}{(a-b)^{p+q-1}} Q_{p-1} \log(x-a) + \frac{1}{(b-a)^{p+q-1}} P_{q-1} \log(x-b), \end{aligned}$$

where P_r and Q_r are the coefficients of z^r in $(1+z)^{-p}$ and $(1+z)^{-q}$ respectively.

17. Integrate

$$(i) \int \frac{dx}{(5x^3 - 3x)^2(x^2 - 1)}. \quad [\text{Math. Trip., 1878.}]$$

$$(ii) \int \frac{dx}{(5x^3 - 3x^2)^2(x^2 - 1)}.$$

$$(iii) \int \frac{\sqrt{x} dx}{(1+x)(2+x)(3+x)}. \quad [\text{Oxford I., 1888.}]$$

18. Prove

$$(i) \int_{-\infty}^{\infty} \frac{2x^2 - x + 1}{(x^2 + x + 1)^3} dx = \frac{10}{9} \pi \sqrt{3}. \quad [\text{Colleges } \beta, 1891.}]$$

$$(ii) \int_0^\infty \frac{\sqrt{x}}{1+x^2} dx = \frac{\pi}{\sqrt{2}}. \quad [\text{Trinity, 1882.}]$$

$$(iii) \int_x^1 \frac{dx}{1+x^4} = \frac{1}{2\sqrt{2}} \left\{ \pi - 2 \tan^{-1} \frac{\sqrt{2}}{x^{-1} - x} \right\}. \quad [\text{Trinity, 1895.}]$$

19. Integrate

$$\int \frac{x dx}{x^3 + 1}.$$

$$\text{Prove that } \frac{1}{2.5} + \frac{1}{8.11} + \frac{1}{14.17} + \dots \text{ to } \infty = \frac{1}{9} \left[\frac{\pi}{\sqrt{3}} - \log 2 \right]. \quad [\text{Colleges, 1896.}]$$

$$20. \text{ Integrate } \int \frac{(\sqrt{\cot x} - \sqrt{\tan x}) dx}{1 + 3 \sin 2x}. \quad [\text{Colleges } \beta, 1890.]$$

21. Integrate

$$(i) \int \tan^{-1} \sqrt{\frac{a^2 x - 1}{b^2 x - 1}} dx.$$

$$(ii) \int \sqrt{a^2 + \sqrt{b^2 + c/x}} dx. \quad [\text{Math. Trip., 1898.}]$$

$$22. \text{ Integrate } \int \frac{(\sqrt{a} - \sqrt{x})^2 dx}{(a^2 + ax + x^2)\sqrt{x}}. \quad [\text{Colleges, 1896.}]$$

$$23. \text{ Integrate } \int \frac{5x^3 + 3x - 1}{(x^3 + 3x + 1)^3} dx. \quad [\text{J. M. Sch., Ox., 1904.}]$$

$$24. \text{ Evaluate } \int_0^{\pi/4} \frac{\sin^4 x}{\cos^5 x} dx. \quad [\text{St. John's, 1892.}]$$

$$25. \text{ Integrate } \int \frac{dx}{x(x-a)^n}, n \text{ being a positive integer.} \quad [\text{St. John's, 1882.}]$$

$$26. \text{ Integrate } \int \left(\frac{x-b}{x-a} \right)^{\frac{n}{2}} dx. \quad [\text{Colleges a, 1885.}]$$

$$27. \text{ Integrate } \int \frac{dx}{1 + 3e^x + 2e^{2x}}. \quad [\text{Math. Trip., 1895.}]$$

28. Sum the series

$$\frac{x^3}{1.3} + \frac{x^5}{3.5} + \frac{x^7}{5.7} + \dots \text{ ad inf.,}$$

assuming it to be convergent.

Deduce that

$$\frac{1}{1.3} \cdot \frac{1}{2^3} + \frac{1}{3.5} \cdot \frac{1}{2^5} + \frac{1}{5.7} \cdot \frac{1}{2^7} + \dots \text{ ad inf.} = \frac{1}{4} - \frac{3}{16} \log 3. [\text{I. C. S., 1899.}]$$

29. Prove that

$$1 - \frac{1}{7} + \frac{1}{9} - \frac{1}{15} + \frac{1}{17} - \dots \text{ ad inf.} = \frac{\pi}{8}(1 + \sqrt{2}). [\text{Colleges } \beta, 1888.]$$

$$30. \text{ Evaluate } \int x^2 \log(1 - x^2) dx,$$

and deduce that

$$\frac{1}{1.5} + \frac{1}{2.7} + \frac{1}{3.9} + \dots = \frac{8}{9} - \frac{2}{3} \log 2. [\text{Colleges a, 1889.}]$$

31. Integrate $\int (a^4 + x^4)^{-1} dx$.

Prove that

$$\frac{1}{1.5} + \frac{1}{9.13} + \frac{1}{17.21} + \frac{1}{25.29} + \cdots = \frac{\sqrt{2}}{32} \left\{ \pi + \log(3 + 2\sqrt{2}) \right\}. [\text{Math. Trip., 1896.}]$$

32. Show that

$$\int \frac{dx}{x(x+1)(x+2)(x+3)\dots(x+n)} = \frac{1}{n!} \sum_{r=0}^{r=n} (-1)^r {}^nC_r \log(x+r).$$

33. Show that

$$\int \frac{(1+x)^n}{(1-2x)^3} dx = \frac{3^n}{2^{n+2}} \frac{1}{(1-2x)^2} - n \frac{3^{n-1}}{2^{n+1}} \frac{1}{1-2x} - \frac{n(n-1)3^{n-2}}{2^{n+2}} \log(1-2x)$$

+ a rational integral algebraic expression of a finite number of terms.

34. Show that if $c < 1$,

$$= x + \frac{1}{2} \frac{x^2}{1-c} + \frac{1}{3} \frac{x^3}{(1-c)(1-c^2)} + \frac{1}{4} \frac{x^4}{(1-c)(1-c^2)(1-c^3)} + \dots \text{to } \infty.$$

35. Show that

$$\begin{aligned} & \int \frac{x^{n+2} dx}{(x-a_1)(x-a_2)(x-a_3)\dots(x-a_n)} \\ &= \frac{x^3}{3} + H_1 \frac{x^2}{2} + H_2 x + \sum \frac{a_1^{n+2}}{(a_1-a_2)(a_1-a_3)\dots(a_1-a_n)} \log(x-a_1), \end{aligned}$$

where H_r is the sum of the homogeneous products r at a time of $a_1, a_2, \dots, a_n$.

36. Show that the part of the indefinite integral

$$\int \frac{1}{x^3} \frac{f(x)}{\phi(x)} dx$$

which becomes infinite when $x = 0$, f and ϕ being rational integral functions of x which do not vanish when $x = 0$, is

$$-\frac{1}{2x^2} \frac{f(0)}{\phi(0)} - \frac{1}{x} \frac{f'(0)\phi(0) - \phi'(0)f(0)}{[\phi(0)]^2} + \frac{1}{2} \log x \frac{f''(0)\{\phi(0)\}^2 - 2f'(0)\phi'(0)\phi(0) - f(0)[\phi(0)\phi''(0) - 2\{\phi'(0)\}^2]}{[\phi(0)]^3}. [\text{Ox. I. P., 1901.}]$$

37. Show that when a rational fraction is decomposed into its simple or “partial” fractions, the decomposition is unique.

38. If $F(x)$ be a function of the $(n-1)$ th degree which assumes the values $u_1, u_2, u_3, \dots, u_n$ when $x = x_0, x_1, x_2, \dots, x_n$ respectively, show that

$$F(x) = u_1 \frac{(x-x_2)(x-x_3)\dots(x-x_n)}{(x_1-x_2)(x_1-x_3)\dots(x_1-x_n)} \\ + u_2 \frac{(x-x_1)(x-x_3)\dots(x-x_n)}{(x_2-x_1)(x_2-x_3)\dots(x_2-x_n)} \\ + \dots \dots \dots \\ + u_n \frac{(x-x_1)(x-x_2)\dots(x-x_{n-1})}{(x_n-x_1)(x_n-x_2)\dots(x_n-x_{n-1})}.$$

39. Prove that if $p < n + 1$,

$$na^{n-p} \int \frac{x^{p-1} dx}{x^n - a^n} = \log(x - a) + \sum_{r=1}^{r=\frac{n-1}{2}} \cos \frac{2rp\pi}{n} \log \left(x^2 - 2ax \cos \frac{2r\pi}{n} + a^2 \right) \\ - 2 \sum_{r=1}^{r=\frac{n-1}{2}} \sin \frac{2rp\pi}{n} \tan^{-1} \frac{x - a \cos \frac{2r\pi}{n}}{a \sin \frac{2r\pi}{n}} \quad \text{if } n \text{ be odd,}$$

and

$$= \log(x - a) + (-1)^p \log(x + a) \\ + \sum_{r=1}^{r=\frac{n-2}{2}} \cos \frac{2rp\pi}{n} \log \left(x^2 - 2ax \cos \frac{2r\pi}{n} + a^2 \right) \\ - 2 \sum_{r=1}^{r=\frac{n-2}{2}} \sin \frac{2rp\pi}{n} \tan^{-1} \frac{x - a \cos \frac{2r\pi}{n}}{a \sin \frac{2r\pi}{n}} \quad \text{if } n \text{ be even.}$$

40. Prove that if $p < n + 1$,

$$na^{n-p} \int \frac{x^{p-1} dx}{x^n + a^n} = (-1)^{p-1} \log(x + a) \\ - \sum_{r=1}^{r=\frac{n-1}{2}} \cos(2r-1) \frac{p\pi}{n} \log \left\{ x^2 - 2ax \cos(2r-1) \frac{\pi}{n} + a^2 \right\} \\ + 2 \sum_{r=1}^{r=\frac{n-1}{2}} \sin(2r-1) \frac{p\pi}{n} \tan^{-1} \frac{x - a \cos(2r-1) \frac{\pi}{n}}{a \sin(2r-1) \frac{\pi}{n}} \quad \text{if } n \text{ be odd,}$$

and

$$= - \sum_{r=1}^{r=\frac{n}{2}} \cos(2r-1) \frac{p\pi}{n} \log \left\{ x^2 - 2ax \cos(2r-1) \frac{\pi}{n} + a^2 \right\} \\ + 2 \sum_{r=1}^{r=\frac{n}{2}} \sin(2r-1) \frac{p\pi}{n} \tan^{-1} \frac{x - a \cos(2r-1) \frac{\pi}{n}}{a \sin(2r-1) \frac{\pi}{n}} \quad \text{if } n \text{ be even.}$$

41. Prove that

$$\int_0^x \frac{dx}{1 - x^{2n}} = \frac{1}{2n} \sum_{r=0}^{r=n-1} \left(\cos \frac{r\pi}{n} \tanh^{-1} \frac{2x \cos \frac{r\pi}{n}}{1 + x^2} + \sin \frac{r\pi}{n} \tan^{-1} \frac{2x \sin \frac{r\pi}{n}}{1 - x^2} \right) \cdot [\text{Math. Trip., 1884.}]$$

42. Show that $\int_0^\infty \frac{t}{t^5 + 1} dt = \frac{4\pi}{5\sqrt{10 + 2\sqrt{5}}}$.

43. (i) Show that the remainder left after dividing the rational integral function $f(x)$ by $(x - c)^2 + b^2$ is

$$\left[f(c) - \frac{b^2}{2!} f^{(2)}(c) + \frac{b^4}{4!} f^{(4)}(c) - \dots + (-1)^r \frac{b^{2r}}{(2r)!} f^{(2r)}(c) + \dots \right] \\ + (x - c) \left[f'(c) - \frac{b^3}{3!} f^{(3)}(c) + \frac{b^5}{5!} f^{(5)}(c) - \dots + (-1)^r \frac{b^{2r+1}}{(2r+1)!} f^{(2r+1)}(c) + \dots \right],$$

where $f^{(s)}(c)$ denotes $\frac{d^s f(c)}{dc^s}$.

- (ii) If $f(x)$ and $\phi(x)$ are rational integral functions of x , and $\phi(x)$ does not contain $(x-c)^2 + b^2$ as a factor, show that it is possible to determine finite values for the constants P and Q in such a manner that

$$f(x) - [P(x-c) + Q]\phi(x)$$

is divisible, without remainder, by $(x-c)^2 + b^2$.

- (iii) Apply the last result to show (or prove in any manner) that

$$\frac{f(x)}{[(x-c)^2 + b^2]^r \phi(x)}$$

can be expressed in the form

$$\frac{\chi(x)}{\phi(x)} + \sum_{n=1}^{n=r} \frac{P_n(x-c) + Q_n}{[(x-c)^2 + b^2]^n},$$

r being a positive integer, $\chi(x)$ a rational integral function of x , and P_n and Q_n constants. [I.C.S., 1892.]

44. If

$$\frac{F(x)}{f(x)} \equiv \phi(x) + \frac{\psi(x)}{f(x)},$$

where F, f, ϕ, ψ are rational polynomials of degrees $m+n, n, m, n-1$ respectively, show that if $a_1, a_2, a_3, \dots, a_n$ be the roots of $f(x) = 0$, considered all different, $\psi(x)$ will be determinable from

$$\begin{vmatrix} \psi(x), & x^{n-1}, & x^{n-2}, & \dots, & x, & 1 \\ F(a_1), & a_1^{n-1}, & a_1^{n-2}, & \dots, & a_1, & 1 \\ F(a_2), & a_2^{n-1}, & a_2^{n-2}, & \dots, & a_2, & 1 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ F(a_n), & a_n^{n-1}, & a_n^{n-2}, & \dots, & a_n, & 1 \end{vmatrix} = 0.$$

Also determine $\psi(x)$ when $f(x) = 0$ has equal roots.

[Oxford I. P., 1913.]

45. Integrate $\iint \left\{ \frac{ax^2 + bx + c}{(x-\alpha)(x-\beta)(x-\gamma)} \right\}^2 dx.$

[Oxford I. P., 1917.]

46. Prove that if n be a positive integer,

$$(i) \int \frac{\cos(n-2p)\theta}{\sin n\theta} d\theta = \frac{1}{n} \sum_{r=1}^{r=n} \cos \frac{2pr\pi}{n} \log \sin \left(\theta - \frac{r\pi}{n} \right),$$

$$(ii) \int \frac{\sin(n-2p)\theta}{\sin n\theta} d\theta = \frac{1}{n} \sum_{r=1}^{r=(n-1)} \sin \frac{2pr\pi}{n} \log \operatorname{cosec} \left(\theta - \frac{r\pi}{n} \right).$$

47. Integrate

$$(i) \int \frac{dx}{1+e^x},$$

$$(ii) \int \frac{dx}{\sqrt{\sin^3 x \sin(x+a)}},$$

and prove that

$$(iii) \int_0^\infty \frac{x dx}{(1+x)(1+x^2)} = \frac{\pi}{4},$$

$$(iv) \int_1^\infty \frac{(x^2+3) dx}{x^6(x^2+1)} = \frac{1}{30}(58-15\pi).$$

[Math. Trip. I., 1917.]

48. Obtain the rational part of $\int \frac{dx}{(x^5 + 1)^3}$. [Math. Trip. II., 1915.]

49. Prove that

$$\frac{2n}{(1+x)^{2n} + (1-x)^{2n}} = \sum_{r=0}^{r=n} \frac{\sin a_r \cos^{2n-2} a_r}{\sin(2n-1)a_r} \frac{1}{x^2 + \tan^2 a_r},$$

where $a_r = (2r+1)\pi/4n$.

[Oxf. II. P., 1899.]

Write down the values of the integrals

$$\int \frac{dx}{(1+x)^{2n} + (1-x)^{2n}}, \quad \int \frac{x dx}{(1+x)^{2n} + (1-x)^{2n}}.$$

50. Show that

$$\int_0^\infty \frac{x dx}{(a+x)^n - (a-x)^n} = \frac{\pi}{2na^{n-2}} \Sigma (-1)^{\lambda-1} \sin \frac{\lambda\pi}{n} \cos^{n-3} \frac{\lambda\pi}{n},$$

the summation extending from $\lambda = 1$ to $\lambda = \frac{n-1}{2}$ or to $\lambda = \frac{n-2}{2}$, according as n is odd or even. [Cf. Wolstenholme's Problems, No. 1912.]

Write down the value of the integral

$$\int \frac{x^2 dx}{(a+x)^n - (a-x)^n}.$$

51. Show that if n be even and $x+y=1$,

$$\begin{aligned} \int \frac{dx}{x^n y^n} &= \frac{1}{n-1} \left[\frac{1}{y^{n-1}} - \frac{1}{x^{n-1}} \right] + \frac{n}{1} \cdot \frac{1}{n-2} \left[\frac{1}{y^{n-2}} - \frac{1}{x^{n-2}} \right] \\ &+ \frac{n(n+1)}{1.2} \cdot \frac{1}{n-3} \left[\frac{1}{y^{n-3}} - \frac{1}{x^{n-3}} \right] + \dots + \frac{n(n+1) \dots (2n-2)}{1.2 \dots (n-1)} \log \frac{x}{y}. \end{aligned}$$

52. Show that if $p < q$,

$$Lt_{n=\infty} \frac{\prod_{r=1}^{r=n} \left(1 - \frac{p^2 x^2}{r^2 \pi^2} \right)}{\prod_{r=1}^{r=n} \left(1 - \frac{q^2 x^2}{r^2 \pi^2} \right)} = \sum_{r=1}^{r=\infty} \frac{2r\pi}{pq} \frac{\sin \left(\frac{p}{q} r\pi \right)}{\cos r\pi} \frac{1}{x^2 - \frac{r^2 \pi^2}{q^2}}. \quad [\text{Todhunter, I.C., p. 38.}]$$

Deduce that if $p < q$,

$$\int \frac{\sin px}{\sin qx} dx = -2 \sum_{r=1}^{r=\infty} \frac{1}{q} \frac{\sin \left(\frac{p}{q} r\pi \right)}{\cos r\pi} \tanh^{-1} \frac{qx}{r\pi} \quad (-\pi < qx < \pi).$$

Chapter VI: Integrals of Forms $\int \frac{dx}{(a + b \cos x + c \sin x)^n}$, etc. — Exercises

Exercises — Various Standard Methods (Arts. 170–208)

1. Integrate

(i) $\int \frac{a \sin \theta + b \cos \theta}{c \sin \theta + e \cos \theta} d\theta.$ [a, 1883.]

(ii) $\int \frac{d\theta}{\cos \theta - \sin \theta}.$ [I.C.S., 1880.]

(iii) $\int \frac{a + \beta \sin \theta}{a + b \cos \theta} d\theta.$ [Trin. H. and Magd.]

(iv) $\int \frac{dx}{\cos a + \cos x}.$ [I.C.S., 1889.]

(v) $\int \frac{dx}{a \cos x + b \sin x}.$ [Coll., 1876.]

(vi) $\int \frac{1 - \tan \theta}{1 + \tan \theta} d\theta.$ [Trin., 1884.]

(vii) $\int \frac{\sec \theta}{1 + \operatorname{cosec} \theta} d\theta.$ [Ox. I. P., 1889.]

(viii) $\int \frac{dx}{3(1 - \sin x) - \cos x}.$ [a, 1881.]

(ix) $\int \frac{\sqrt{2} dx}{2\sqrt{2} + \cos x + \sin x}.$ [Ox. I. P., 1888.]

(x) $\int \frac{dx}{a + b \tan x}.$ [St. John's, 1888.]

(xi) Apply the transformation $t = \tan \frac{1}{2}x$ to the integrals

$$\int \frac{4 dx}{5 + 3 \cos x}, \quad \int \frac{4 dx}{3 + 5 \cos x}.$$

Hence or otherwise, evaluate these integrals to the nearest hundredth, when the limits are $x = 0$ and $\frac{1}{4}\pi$. Prove in any way that the second is the greater of the two integrals, when taken between 0 and $\frac{1}{4}\pi$. [Math. Trip. I., 1913.]

(xii) Prove that

$$\int_0^a \frac{dx}{x + (a^2 - x^2)^{\frac{1}{2}}} = \frac{1}{4}\pi, \quad \int_0^a \frac{a dx}{\{x + (a^2 - x^2)^{\frac{1}{2}}\}^2} = \frac{1}{\sqrt{2}} \log_e(1 + \sqrt{2}),$$

the positive sign being taken for the radical in each of the subjects of integration. [Math. Trip. II., 1913.]

2. Evaluate

(i) $\int_0^\pi \frac{dx}{4 + 5 \sin x}.$ [I. C. S., 1889.]

$$(ii) \int_0^\pi \frac{d\theta}{a + c \cos \theta} \quad (c < a). \quad [\text{I. C. S., 1879.}]$$

$$(iii) \int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}. \quad [\text{St. John's, 1882.}]$$

$$(iv) \int_0^\pi \frac{dx}{1 - 2a \cos x + a^2}. \quad [\text{I. C. S., 1888.}]$$

3. Show that $\int_0^\pi \frac{dx}{1 - \cos a \cos x} = \frac{\pi}{2} \operatorname{cosec} a$, [Ox. II. P., 1889; Trin., 1887.] and integrate

$$\int \frac{\cos a \cos x + 1}{\cos a + \cos x} dx.$$

4. Evaluate

$$(i) \int \frac{dx}{a^2 - \beta^2 \cos^2 x} \quad (a > \beta). \quad [\text{Ox. II. P., 1887.}]$$

$$(ii) \int \frac{dx}{\sin x \cos x + \sin x + \cos x - 1}. \quad [\gamma, 1899.]$$

$$(iii) \int \frac{dx}{\sin x + \sin 2x}. \quad [\beta, 1891.]$$

5. Evaluate

$$\int \frac{dx}{(e^x + e^{-x})^2}. [\text{Ox. I. P., 1889.}]$$

6. Integrate

$$(i) \int \frac{dx}{(4 + 5 \cos x)^2}. \quad [\alpha, 1883.]$$

$$(ii) \int \frac{dx}{(a + b \cos x)^2}. \quad [\text{St. John's, 1884.}]$$

$$(iii) \int \frac{d\theta}{(a \cos \theta + b \sin \theta)^2}. \quad [\alpha, 1881.]$$

$$(iv) \int \frac{dx}{(a + b \cos x + c \sin x)^2}. \quad [\text{Coll., 1892.}]$$

(v) Employ the substitution $\tan \frac{\theta}{2} = \sqrt{\frac{1+e}{1-e}} \tan \frac{u}{2}$ to evaluate the integral $\int_0^\phi \frac{d\theta}{(1 + e \cos \theta)^2}$.
[Math. Trip. I., 1909.]

7. Prove that

$$(i) \int_0^\pi \frac{d\theta}{(1 + \cos a \cos \theta)^2} = \pi \operatorname{cosec}^3 a, \quad [\text{Coll., 1886; St. John's, 1886.}]$$

$$(ii) \int_0^\pi \frac{\sin x \, dx}{(1 + \cos a \sin x)^2} = \frac{2(\sin a - a \cos a)}{\sin^3 a}, \quad [\beta, 1887.]$$

and evaluate

$$\int_0^\pi \frac{x \sin x \, dx}{(1 + \cos a \cos x)^3} \quad \text{and} \quad \int_0^\pi \frac{x \sin x \, dx}{(1 + \cos a \cos x)^4},$$

if a be less than $\frac{\pi}{2}$.

8. Evaluate the integral

$$\int \cos 2\theta \log(1 + \tan \theta) \, d\theta. [\gamma, 1882.]$$

9. Find the values of the following integrals: $\left(a < \frac{\pi}{2}\right)$

(i) $\int_0^\infty \frac{dx}{2 \cos a + e^x + e^{-x}}.$

(ii) $\int_0^\pi \frac{dx}{2 + \cos a(e^x + e^{-x})}.$ [Trin., 1882.]

10. Evaluate

(i) $\int_0^{\frac{\pi}{2}} \frac{d\theta}{a^2 \cos^2 \theta + b^2 \sin^2 \theta}.$ [Ox. I. P., 1889.]

(ii) $\int_0^{\frac{\pi}{2}} \frac{d\theta}{4 + 5 \sin^2 \theta}.$ [I. C. S., 1885.]

(iii) $\int_0^{\frac{\pi}{2}} \frac{a \sin^2 \theta + b \cos^2 \theta}{c \sin^2 \theta + d \cos^2 \theta} d\theta$ (c and d positive). [St. John's.]

(iv) $\int_0^{\frac{\pi}{2}} \frac{d\theta}{(a^2 \cos^2 \theta + \beta^2 \sin^2 \theta)^2}.$ [Ox. II. P., 1887.]

(v) $\int_0^{\frac{\pi}{4}} \frac{\sin 2\theta d\theta}{\sin^4 \theta + \cos^4 \theta},$ [I. C. S., 1891.]

and

(vi) Shew that if $c > a > 0$,

$$\int_0^a \frac{\sqrt{a^2 - x^2}}{c^2 - x^2} dx = \pi(c - \sqrt{c^2 - a^2})/2c. [\text{Math. Trip. I., 1908.}]$$

11. Prove that $\int_0^\pi \frac{d\theta}{a + b \cos \theta} = \frac{\pi}{\sqrt{a^2 - b^2}},$

where $a > b$, and deduce or otherwise obtain the value of

$$\int_0^\pi \frac{d\theta}{(a + b \cos \theta)^3}. [\gamma, 1899.]$$

12. Prove that if $a > b$,

$$\int_0^\pi \frac{dx}{(a + b \cos x)^4} = \frac{\pi}{2} \left\{ \frac{5a^3}{(a^2 - b^2)^{\frac{7}{2}}} - \frac{3a}{(a^2 - b^2)^{\frac{5}{2}}} \right\}. [\gamma, 1888.]$$

Evaluate $\int_0^\pi \frac{dx}{(1 + e \cos x)^4},$ where $e < 1$. [St. John's, 1892.]

13. Evaluate $\int \frac{dx}{a + be^x + ce^{-x}},$ where $a^2 < 4bc$. [I. C. S., 1897.]

14. Prove that if $a < \frac{\pi}{6},$

$$\int_0^a \frac{\sin x}{\cos 3x} dx = \frac{1}{6} \log \frac{\cos^3 a}{\cos 3a}. [\text{C. S., 1896.}]$$

15. Prove that $\int_0^{2\pi} \frac{d\phi}{a + b \cos \phi + c \sin \phi} = \frac{2\pi}{\sqrt{a^2 - r^2}},$

where $r^2 = b^2 + c^2$ and $r < a$.

[C. S., 1900.]

16. Integrate

$$(i) \int \frac{\sqrt{\tan x}}{\sin x \cos x} dx.$$

$$(ii) \int \frac{\sec x \, dx}{a + b \tan x}.$$

$$(iii) \int \frac{\sec^2 x \, dx}{a + b \tan x}.$$

17. Integrate

$$(i) \int \frac{d\theta}{15 \sin^2 \theta - 16 \cos \theta}.$$

$$(ii) \int_0^\pi \frac{x}{1 + \sin x} dx.$$

$$(iii) \int \frac{\cot \theta - 3 \cot 3\theta}{3 \tan 3\theta - \tan \theta} d\theta.$$

[Ox. I. P., 1888.]

$$(iv) \int \frac{\sin 2x \, dx}{(a + b \cos x)^2}.$$

[a, 1889.]

18. Integrate

$$(i) \int \cos 2\theta \log \frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} d\theta.$$

$$(ii) \int \frac{\sin \theta - \cos \theta}{\sqrt{\sin 2\theta}} d\theta.$$

$$(iii) \int \sqrt{\frac{1 - \cos \theta}{\cos \theta (1 + \cos \theta) (2 + \cos \theta)}} d\theta.$$

19. Integrate $\int \sqrt{\frac{1 + \sin x}{1 - \sin x}} \cdot \frac{2 + \sin x}{2 - \sin x} dx.$

20. Integrate

$$\int \frac{\sin \theta - \cos \theta}{(\sin \theta + \cos \theta) \sqrt{\sin \theta \cos \theta + \sin^2 \theta \cos^2 \theta}} d\theta.$$

21. Integrate $\int \frac{\sin^3 \theta \, d\theta}{(1 + \cos^2 \theta) \sqrt{1 + \cos^2 \theta + \cos^4 \theta}}.$

22. Integrate

$$(i) \int \sqrt{1 + \sin x} \, dx.$$

$$(ii) \int \frac{\sin x}{\sqrt{1 + \sin x}} dx.$$

$$(iii) \int \frac{\tan x}{\sqrt{a + b \tan^2 x}} dx.$$

23. Integrate $\int \frac{\sinh x \sin x - \cosh x}{1 - \cos x} dx.$

24. Integrate $\int \frac{x^2 \, dx}{(x \sin x + \cos x)^2}.$

25. Integrate $\int \frac{\sec x \operatorname{cosec} x}{\log \tan x} dx.$

26. Integrate

(i) $\int \sin^{-1} \frac{2x}{1+x^2} dx.$

(ii) $\int \tan^{-1} \frac{3x-x^3}{1-3x^2} dx.$

(iii) $\int \tan^{-1} \frac{-1+\sqrt{1+x^2}}{x} dx.$

27. Integrate $\int \frac{\sqrt{1+x^2}}{1-x^2} dx.$

28. Integrate $\int \frac{\sin x}{\sin 2x} dx, \int \frac{\sin x}{\sin 3x} dx, \int \frac{\sin x}{\sin 4x} dx,$

and prove that

$$5 \int \frac{\sin x}{\sin 5x} dx = \sin \frac{2\pi}{5} \log \left\{ \frac{\sin \left(x - \frac{2\pi}{5} \right)}{\sin \left(x + \frac{2\pi}{5} \right)} \right\} - \sin \frac{\pi}{5} \log \left\{ \frac{\sin \left(x - \frac{\pi}{5} \right)}{\sin \left(x + \frac{\pi}{5} \right)} \right\}. [\text{Trin. Coll., 1892.}]$$

29. Integrate $\int \frac{d\theta}{\sin^2 \theta - \sin^2 a}$ and $\int \frac{\cos \theta}{\sin^2 \theta - \sin^2 a} d\theta.$

Show that $\frac{\sin \theta}{\sin n\theta}$ can be expressed in partial fractions of type

$$\frac{1}{\sin^2 \theta - \sin^2 a} \quad \text{or} \quad \frac{\cos \theta}{\sin^2 \theta - \sin^2 a},$$

according as n is an odd or an even integer and can thereby be integrated.

30. Integrate

(i) $\int_0^{\frac{\pi}{2}} \frac{\sin 3x \, dx}{(a + b \cos x)^2}.$

(ii) $\int_0^{\frac{\pi}{2}} \frac{\sin^3 x \, dx}{a + b \cos x}.$

(iii) $\int_0^{\frac{\pi}{2}} \frac{\sin^3 x \, dx}{(a + b \cos x)^2}.$

(iv) $\int_0^{\frac{\pi}{2}} \frac{\sin 3x \, dx}{(a + b \cos x)^n} \quad (n > 3).$

31. Show how to effect the integration of

$$\int \frac{\cos^p x}{\sin 2nx} dx, \quad \int \frac{\cos^p x}{\cos nx} dx,$$

p and n being integers.

[ϵ , 1883, and Coll., 1879.]

32. Integrate $\int \cot(x-a) \cot(x-\beta) \, dx,$

[γ , 1891.]

and show that

$$\int \cot(x-a) \cot(x-b) \cot(x-c) \, dx = \Sigma \cot(a-b) \cot(a-c) \log \sin(x-a). [\text{Trinity, 1891.}]$$

33. Show that

$$\int \sin x \sec(x-a) \sec(x-\beta) dx = \frac{1}{\sin(\beta-a)} [\cos a \cosh^{-1} \sec(x-a) - \cos \beta \cosh^{-1} \sec(x-\beta)] .[\text{Trinity, 1889.}]$$

34. Prove that

$$\int_0^\beta x \sec x \sec(\beta-x) dx = \beta \operatorname{cosec} \beta \log \sec \beta. [\text{Oxf. II. P., 1901.}]$$

35. Prove that, if a and β be positive quantities,

$$\int_0^{\frac{\pi}{2}} \frac{dx}{(a \cos^2 x + \beta \sin^2 x)^{n+1}} = \frac{\pi}{2} \frac{(-1)^n}{n!} \left(\frac{d}{da} + \frac{d}{d\beta} \right)^n (a\beta)^{-\frac{1}{2}}. [\text{a, 1884.}]$$

36. Prove that, if

$$f(x) = (x-a_1)(x-a_2) \dots (x-a_n)$$

and

$$P = \prod_{r=1}^{r=n} (1 - 2a_r \cos \theta + a_r^2),$$

where $a_1, a_2, \dots a_n$ denote real quantities, then

$$\int_0^\pi \frac{d\theta}{P} = \pi \sum_1^n \epsilon_r / A_r,$$

where $A_r = a_r f(1/a_r) f'(a_r)$, and $\epsilon = -1$, or $+1$, according as a_r is numerically greater or less than 1. [St. John's, 1886.]

37. If c be less than $a \sin \theta$, show that the coefficient of c^m in the expansion of

$$\frac{2c}{\sqrt{c^2 - a^2}} \tan^{-1} \left\{ \sqrt{\frac{c-a}{c+a}} \tan \left(\frac{\pi}{4} - \frac{\theta}{2} \right) \right\}$$

is

$$(-1)^m \frac{1}{a^m} \int \frac{d\theta}{\sin^m \theta} + A_m,$$

where A_m is independent of θ .

[Coll., 1892.]

38. Show that if n be a positive integer,

$$\int \frac{\cos n\theta - \cos na}{\cos \theta - \cos a} d\theta = \theta \frac{\sin na}{\sin a} + \frac{2}{\sin a} \sum_{r=1}^{r=n-1} \frac{\sin r\theta \sin(n-r)a}{r}. [\text{Hermite.}]$$

39. Prove that

$$\int_0^x (1 + \cos x)^n dx = \frac{1}{2^n} \cdot \binom{2n}{n} x + \frac{1}{2^{n-1}} \sum_1^n \binom{2n}{n-r} \frac{\sin rx}{r}.$$

40. Show that

$$\int \prod_1^n \cot(\theta - a_r) d\theta = \theta \cos \frac{n\pi}{2} + \sum_1^n A_r \log \sin(\theta - a_r),$$

where $A_r = \cot(a_r - a_1) \cot(a_r - a_2) \dots \cot(a_r - a_n)$, the factor $\cot(a_r - a_r)$ being omitted. [Hermite.]

41. (1) Show that

$$\int \frac{dx}{a - \sin x} = \frac{2}{\sqrt{a^2 - 1}} \tan^{-1} \left(\frac{a \tan \frac{x}{2} - 1}{\sqrt{a^2 - 1}} \right) \quad (a > 1).$$

(2) Differentiate with regard to x ,

$$\tan^{-1} \sqrt{\frac{a+1}{a-1} \cdot \frac{1-\sin x}{1+\sin x}}.$$

Deduce from (1) and (2) that

$$\tan^{-1} \sqrt{\frac{a+1}{a-1} \cdot \frac{1-\sin x}{1+\sin x}} + \tan^{-1} \frac{a \tan \frac{x}{2} - 1}{\sqrt{a^2 - 1}}$$

is independent of x , and verify your conclusion.

[C. S., 1898.]

42. Integrate $\int \frac{dx}{1 + ab - a \cos x - b \sec x}$,

where $a < 1$, $b > 1$, $ab \neq 1$.

[Oxf. I. P., 1917.]

43. Integrate

(i) $e^x(x \cos x + \sin x)$.

(ii) $(x^2 + 1)^{-\frac{3}{2}}$.

(iii) $(x^4 + 2x^3 + 2x + 1)/(x^2 + x + 1)^3$.

[Oxf. I. P., 1918.]

44. Integrate $\int \frac{\cos^3 x \, dx}{\sin(x-a) \sin(x-\beta) \sin(x-\gamma)}$.

45. Deduce from the identity

$$\int_0^{\frac{\pi}{2}} \cos n\theta \, d\theta = \int_0^{\frac{\pi}{2}} d\theta \left\{ 1 - \frac{n^2}{2!} \sin^2 \theta - \frac{n^2(2^2 - n^2)}{4!} \sin^4 \theta - \frac{n^2(2^2 - n^2)(4^2 - n^2)}{6!} \sin^6 \theta - \dots \right\}$$

the expression for $\sin x$ as an infinite product.

[Oxf. II. P., 1887.]

46. Evaluate the integrals

(i) $\int (x + a \log x)^2 \, dx$.

(ii) $\int \frac{(a+x) \, dx}{x^2 + ax \log x}$.

(iii) $\int \frac{(1 - \log x) \, dx}{(x + a \log x)^2}$.

[Math. Tripos, 1885.]

47. Show that

$$\int_0^x \frac{x^2 + a(a-1)}{(x \sin x + a \cos x)^2} \, dx = \frac{a \sin x - x \cos x}{x \sin x + a \cos x}. [\text{Trin. Coll., 1891.}]$$

48. Evaluate the indefinite integrals

(i) $\int \frac{(\sin x + \cos x)^2}{\{(x-1) \cos x - (x+1) \sin x\}^2} \, dx$.

(ii) $\int \frac{x^2}{\{(x-1) \cos x - (x+1) \sin x\}^2} \, dx$.

[Colleges, 1886.]

49. If $T \equiv x + \sqrt{1+x^2}$, show that

$$\int T^r dx = \frac{1}{2} \left\{ \frac{T^{r-1}}{r-1} + \frac{T^{r+1}}{r+1} \right\}.$$

50. Integrate

$$(i) \int e^x \frac{x^3 - x + 2}{(1+x^2)^2} dx. \quad [\text{Ox. II. P., 1899.}]$$

$$(ii) \int \frac{\log(\cos \theta + \sqrt{\cos 2\theta})}{1 - \cos^2 \theta} d\theta. \quad [\text{Coll. a, 1891.}]$$

51. Prove that, if n be an integer,

$$\int_0^\pi \frac{\cos nx}{1 + \cos a \cos x} dx = \pi \operatorname{cosec} a (\tan a - \sec a)^n,$$

and deduce the value of

$$\int_0^\pi \frac{\cos nx}{(1 + \cos a \cos x)^2} dx. [\text{Colleges } \gamma, 1891.]$$

52. From considering the integral

$$\int_0^\pi \frac{\cos n\theta}{a - \cos \theta} d\theta,$$

show that

$$1 + \frac{n+2}{1} \frac{\cos^2 a}{2^2} + \frac{(n+3)(n+4)}{1.2} \frac{\cos^4 a}{2^4} + \dots = 2^n (\sec a - \tan a)^n \sec^n a \operatorname{cosec} a.$$

53. Prove that, if $0 < a < \frac{\pi}{2}$ and n be a positive integer,

$$\int_0^\pi \sin n\phi \tan^{-1} \left(\frac{\tan \frac{\phi}{2}}{\tan \frac{a}{2}} \right) d\phi = \frac{\pi}{2n} [(\sec a - \tan a)^n - (-1)^n].$$

54. Show that

$$\int \frac{\sin n\theta}{\sin \theta} \frac{d\theta}{\cos n\theta - \cos na} = \frac{1}{n} \sum_{r=0}^{n-1} \operatorname{cosec} \beta_r \log \frac{\sin \frac{1}{2}(\beta_r + \theta)}{\sin \frac{1}{2}(\beta_r - \theta)},$$

where $\beta_r = a + \frac{2r\pi}{n}$.

55. Discuss the integration of

$$(a) \int \frac{\sin p\theta}{\sin q\theta} d\theta, \quad (b) \int \sin \theta \frac{\sin p\theta}{\sin q\theta} d\theta,$$

where p and q are positive integers.

56. With the help of the substitution $x^{-1} = \sqrt{t^2 - 1}$, or otherwise prove that

$$\int_0^\infty \frac{dx}{(9 + 25x^2)\sqrt{1+x^2}} = \frac{1}{12} \tan^{-1} \frac{4}{3}. [\text{Math. Trip., Pt. II., 1920.}]$$

Chapter VII: Further Reduction Formulae — Exercises

Exercises — Integrals of Form $\int x^{m-1}(a+bx^n)^p dx$ (Arts. 211–223)

1. Prove that

$$\int x^{m-1}(a+bx)^p dx = \frac{x^m(a+bx)^p}{m+p} + \frac{ap}{m+p} \int x^{m-1}(a+bx)^{p-1} dx.$$

2.

$$\int \frac{(a+bx)^p}{x^{m+1}} dx = -\frac{(a+bx)^{p+1}}{max^m} + \frac{(p-m+1)b}{m} \frac{1}{a} \int \frac{(a+bx)^p}{x^m} dx.$$

3.

$$\int \frac{dx}{x^m(a+bx)} = -\frac{1}{m-1} \cdot \frac{1}{a} \cdot \frac{1}{x^{m-1}} + \frac{1}{m-2} \cdot \frac{b}{a^2} \cdot \frac{1}{x^{m-2}} - \frac{1}{m-3} \cdot \frac{b^2}{a^3} \cdot \frac{1}{x^{m-3}} + \cdots + (-1)^{m-1} \frac{1}{1} \cdot \frac{b^{m-2}}{a^{m-1}} \cdot \frac{1}{x} + (-1)^m \frac{b^{m-1}}{a^m}.$$

4.

$$\int \frac{(a+bx)^p}{x} dx = \frac{(a+bx)^p}{p} + a \int \frac{(a+bx)^{p-1}}{x} dx. [\text{Bertrand.}]$$

5.

$$\int \frac{dx}{(a+bx^2)^{p+1}} = \frac{x}{2ap(a+bx^2)^p} + \frac{2p-1}{2ap} \int \frac{dx}{(a+bx^2)^p}. [\text{Bertrand.}]$$

6.

$$\int \frac{x^n dx}{(a+bx^2)^{p+1}} = \frac{x^{n+1}}{3ap(a+bx^2)^p} - \frac{n-3p+1}{3ap} \int \frac{x^n}{(a+bx^2)^p} dx, [\text{Bertrand.}]$$

and evaluate

$$\int \frac{x^7}{a+bx^3} dx, \quad \int \frac{x^3}{(a+bx^3)^2} dx, \quad \int \frac{dx}{x^2(a+bx^3)}.$$

7.

$$\int x^n(a+bx^4)^p dx = \frac{x^{n+1}(a+bx^4)^p}{n+1} - \frac{4bp}{n+1} \int x^{n+4}(a+bx^4)^{p-1} dx,$$

$$\int \frac{x^n}{a+bx^4} dx = \frac{x^{n-3}}{(n-3)b} - \frac{a}{b} \int \frac{x^{n-4}}{a+bx^4} dx,$$

$$\int \frac{dx}{(a+bx^4)^{p+1}} = \frac{x}{4ap(a+bx^4)^p} + \frac{4p-1}{4ap} \int \frac{dx}{(a+bx^4)^p}, [\text{Bertrand.}]$$

and evaluate

$$\int \frac{dx}{(a+bx^4)^4}, \quad \int \frac{dx}{x^2(a+bx^4)^3}.$$

Exercises — Reduction Formulae for $\int \sin^p x \cos^q x dx$ (Arts. 224–231)

1. Prove that

$$\int \sin^p \theta \cos^q \theta d\theta = \frac{\sin^{p+1} \theta \cos^{q-1} \theta}{p+q} - \frac{q-1}{(p+q)(p+q-2)} \sin^{p-1} \theta \cos^{q-1} \theta + \frac{(p-1)(q-1)}{(p+q)(p+q-2)} \int \sin^{p-2} \theta \cos^q \theta d\theta$$

the indices being both diminished.

2. Prove that

$$\int \frac{\sin^p \theta}{\cos^q \theta} d\theta = \frac{\sin^{p-1} \theta}{(q-1) \cos^{q-1} \theta} - \frac{p-1}{q-1} \int \frac{\sin^{p-2} \theta}{\cos^{q-2} \theta} d\theta.$$

3. Prove that

$$\int \frac{\cos^p \theta}{\sin^q \theta} d\theta = -\frac{\cos^{p-1} \theta}{(q-1) \sin^{q-1} \theta} - \frac{p-1}{q-1} \int \frac{\cos^{p-2} \theta}{\sin^{q-2} \theta} d\theta.$$

4. Prove that

$$\begin{aligned} \int \frac{d\theta}{\sin^p \theta \cos^q \theta} &= \frac{1}{(q-1) \sin^{p-1} \theta \cos^{q-1} \theta} + \frac{p+q-2}{q-1} \int \frac{d\theta}{\sin^p \theta \cos^{q-2} \theta} \\ &= -\frac{1}{(p-1) \sin^{p-1} \theta \cos^{q-1} \theta} + \frac{p+q-2}{p-1} \int \frac{d\theta}{\sin^{p-2} \theta \cos^q \theta}. \end{aligned}$$

5.

$$\int \frac{\sin^p \theta}{\cos \theta} d\theta = -\frac{\sin^{p-1} \theta}{p-1} + \int \frac{\sin^{p-2} \theta}{\cos \theta} d\theta.$$

6.

$$\int \frac{\cos^p \theta}{\sin \theta} d\theta = \frac{\cos^{p-1} \theta}{p-1} + \int \frac{\cos^{p-2} \theta}{\sin \theta} d\theta.$$

7. (a)

$$\int \sin^{2n} \theta d\theta = -\frac{c}{2n} \left[s^{2n-1} + \frac{2n-1}{2n-2} s^{2n-3} + \frac{(2n-1)(2n-3)}{(2n-2)(2n-4)} s^{2n-5} + \dots + \frac{1.3 \dots (2n-1)}{2.4 \dots (2n-2)} s \right] + \frac{1.3 \dots (2n-1)}{2.4 \dots 2n} c,$$

(b)

$$\int \sin^{2n+1} \theta d\theta = -\frac{c}{2n+1} \left[s^{2n} + \frac{2n}{2n-1} s^{2n-2} + \frac{2n(2n-2)}{(2n-1)(2n-3)} s^{2n-4} + \dots + \frac{2.4 \dots 2n}{1.3 \dots (2n-1)} s \right],$$

where c and s stand respectively for $\cos \theta$ and $\sin \theta$.

[Bertrand.]

8. (a)

$$\int \cos^{2n} \theta d\theta = \frac{s}{2n} \left[c^{2n-1} + \frac{2n-1}{2n-2} c^{2n-3} + \frac{(2n-1)(2n-3)}{(2n-2)(2n-4)} c^{2n-5} + \dots + \frac{1.3 \dots (2n-1)}{2.4 \dots (2n-2)} c \right] + \frac{1.3 \dots (2n-1)}{2.4 \dots 2n} s,$$

(b)

$$\int \cos^{2n+1} \theta d\theta = \frac{s}{2n+1} \left[c^{2n} + \frac{2n}{2n-1} c^{2n-2} + \dots + \frac{2.4 \dots 2n}{1.3 \dots (2n-1)} c \right],$$

c and s being respectively $\cos \theta$ and $\sin \theta$.

[Bertrand.]

9. Prove

(a)

$$\int \operatorname{cosec}^5 \theta d\theta = -\frac{1}{4} \frac{c}{s^4} - \frac{1.3}{2.4} \frac{c}{s^2} + \frac{1.3}{2.4} \log \tan \frac{\theta}{2}.$$

(b)

$$\int \sec^5 \theta d\theta = \frac{1}{4} \frac{s}{c^4} + \frac{1.2}{2.4} \frac{s}{c^2} + \frac{3}{8} \log \tan \left(\frac{\theta}{2} + \frac{\pi}{4} \right),$$

where $c \equiv \cos \theta$, $s \equiv \sin \theta$.

10. Prove that

(a)

$$\int \cos^{2n} \theta d\theta = \frac{1}{2^{2n-1}} \left[\frac{\sin 2n\theta}{2n} + 2n \frac{\sin(2n-2)\theta}{2n-2} + \frac{2n(2n-1)}{1.2} \frac{\sin(2n-4)\theta}{2n-4} + \dots + \frac{2n(2n-1) \dots (n+1)}{1.2 \dots n} \theta \right]$$

(b)

$$\int \cos^{2n+1} \theta d\theta = \frac{1}{2^{2n}} \left[\frac{\sin(2n+1)\theta}{2n+1} + (2n+1) \frac{\sin(2n-1)\theta}{2n-1} + \frac{(2n+1)2n}{1.2} \frac{\sin(2n-3)\theta}{2n-3} + \dots + \frac{(2n+1)2n \dots 2}{1.2 \dots n} \cos \theta \right]$$

(c)

$$\int \sin^{2n} \theta d\theta = \frac{(-1)^n}{2^{2n-1}} \left[\frac{\sin 2n\theta}{2n} - 2n \frac{\sin(2n-2)\theta}{2n-2} + \frac{2n(2n-1)}{1.2} \frac{\sin(2n-4)\theta}{2n-4} - \dots + (-1)^{n-1} \frac{2n(2n-1) \dots (n+1)}{1.2 \dots n} \cos \theta \right]$$

(d)

$$\int \sin^{2n+1} \theta d\theta = \frac{(-1)^{n+1}}{2^{2n}} \left[\frac{\cos(2n+1)\theta}{2n+1} - (2n+1) \frac{\cos(2n-1)\theta}{2n-1} + \dots + (-1)^n \frac{(2n+1) \dots (n+2)}{1.2 \dots n} \cos \theta \right]$$

Exercises — Integrals of Form $\int x^m(a+bx+cx^2)^p dx$ (Art. 247)

1. Taking

$$\int \frac{dx}{X^{p+1}} = \frac{b+2cx}{pkX^p} + \frac{2(2p-1)c}{pk} \int \frac{dx}{X^p},$$

where $X \equiv a+bx+cx^2$ and $k \equiv 4ac-b^2$,

prove

$$\begin{aligned} \int \frac{dx}{X^2} &= \frac{b+2cx}{kX} + \frac{2c}{k} \int \frac{dx}{X}, \\ \int \frac{dx}{X^3} &= \frac{(b+2cx)}{k} \left(\frac{1}{2X^2} + \frac{3c}{kX} \right) + \frac{6c^2}{k^2} \int \frac{dx}{X}, \\ \int \frac{dx}{X^4} &= \frac{b+2cx}{k} \left(\frac{1}{3X^3} + \frac{5c}{3kX^2} + \frac{10c^2}{k^2X} \right) + \frac{20c^3}{k^3} \int \frac{dx}{X}. \text{[Bertrand.]} \end{aligned}$$

2. Show that if $I_n \equiv \int \frac{x^n}{X} dx$, then $cI_n + bI_{n-1} + aI_{n-2} = \frac{x^{n-1}}{n-1}$, and prove

$$\int \frac{x dx}{X} = \frac{1}{2c} \log X - \frac{b}{2c} \int \frac{dx}{X}.$$

Deduce

$$\begin{aligned} \int \frac{x^2 dx}{X} &= \frac{x}{c} - \frac{b}{2c^2} \log X + \frac{b^2-2ac}{2c^2} \int \frac{dx}{X}, \\ \int \frac{x^3 dx}{X} &= \frac{x^2}{2c} - \frac{bx}{c^2} + \frac{b^2-ac}{2c^3} \log X + \frac{3ac-b^2}{2c^3} \int \frac{dx}{X}. \text{[Bertrand.]} \end{aligned}$$

3. Prove

$$\int \frac{dx}{xX} = \frac{1}{2a} \log \frac{x^2}{X} - \frac{b}{2a} \int \frac{dx}{X},$$

and deduce

$$\int \frac{dx}{x^3 X} = \frac{b}{a^2 x} - \frac{1}{2ax^2} + \frac{b^2 - ac}{2a^3} \log \frac{x^2}{X} + \frac{b(3ac - b^2)}{2a^3} \int \frac{dx}{X}. [\text{Bertrand.}]$$

(The value of $\int \frac{dx}{X}$ occurring in each of these results is given in Art. 80.)

4. If $X \equiv a + bx + cx^2$, prove that

$$\int \frac{x^{m+1}}{X^{p+1}} dx = -\frac{x^m}{2cpX^p} + \frac{m}{2cp} \int \frac{x^{m-1} dx}{X^p} - \frac{b}{2c} \int \frac{x^m}{X^{p+1}} dx. [\text{Bertrand.}]$$

5. Prove that if $X \equiv x^2 + ax + a^2$,

$$\int X^{\frac{n}{2}} dx = \frac{2x + a}{2(n+1)} X^{\frac{n}{2}} + \frac{3na^2}{4(n+1)} \int X^{\frac{n}{2}-1} dx. [\text{St. John's, 1889.}]$$

6. Prove that if $X \equiv x^2 + x + 1$,

(a)

$$\int \frac{dx}{x^4 X} = \frac{1}{2} \log \frac{x^2}{X} - \frac{1}{3x^3} + \frac{1}{2x^2} - \frac{1}{\sqrt{3}} \tan^{-1} \frac{2x+1}{\sqrt{3}}.$$

(b)

$$\int \frac{dx}{X^{p+1}} = \frac{1+2x}{3pX^p} + \frac{2(2p-1)}{3p} \int \frac{dx}{X^p}.$$

7. Show that if p be a positive integer and $X \equiv x^2 + x + 1$,

(a)

$$\begin{aligned} \int \frac{dx}{(x^2 + x + 1)^{p+1}} &= \frac{(1+2x)}{3} \left[\frac{1}{pX^p} + \frac{2p-1}{p(p-1)} \left(\frac{2}{3}\right) \frac{1}{X^{p-1}} + \frac{(2p-1)(2p-3)}{p(p-1)(p-2)} \left(\frac{2}{3}\right)^2 \frac{1}{X^{p-2}} + \cdots + \frac{(2p-1)(2p-3)\dots 3.1}{p(p-1)(p-2)\dots 2.1} \frac{2}{\sqrt{3}} \left(\frac{2}{3}\right)^p \tan^{-1} \frac{2x+1}{\sqrt{3}} \right]. \end{aligned}$$

(b) $\int \frac{dx}{(x^2 + x + 1)^{\frac{2n+1}{2}}}$, n being a positive integer,

$$= \frac{1+2x}{3} \left[\frac{1}{2n-1} \frac{2}{X^{\frac{2n-1}{2}}} + \frac{n-1}{(2n-1)(2n-3)} \left(\frac{2}{3}\right) \frac{2^3}{X^{\frac{2n-3}{2}}} + \frac{(n-1)(n-2)}{(2n-1)(2n-3)(2n-5)} \left(\frac{2}{3}\right)^2 \frac{2^5}{X^{\frac{2n-5}{2}}} + \cdots + \right]$$

Exercises — Miscellaneous Reduction Formulae (Arts. 248–274)

1. If $X \equiv ax^n + b$, obtain reduction formulae for the integral $u_{p,q} \equiv \int \frac{x^p}{X^q} dx$ of the forms,

(i) $Au_{p,q} + Bu_{p-n,q} + R = 0,$

(ii) $A'u_{p,q} + B'u_{p,q-1} + R' = 0,$

where A, B, A', B' are constants and R, R' are algebraic functions of x . [Math. Trip., 1896.]

2. Prove that

$$(a) \int \cos^{2n} \phi d\phi = \frac{1}{2n} \tan \phi \cos^{2n} \phi + \left(1 - \frac{1}{2n}\right) \int \cos^{2n-2} \phi d\phi, \quad [\text{Trinity, 1891.}]$$

$$(b) \int \sec^{2n+1} \phi d\phi = \frac{1}{2n} \tan \phi \sec^{2n-1} \phi + \left(1 - \frac{1}{2n}\right) \int \sec^{2n-1} \phi d\phi. \quad [\text{I.C.S., 1886.}]$$

3. Prove that

$$\int (a^2 + x^2)^{\frac{2n+1}{2}} dx = \frac{x}{2n+2} (a^2 + x^2)^{\frac{2n+1}{2}} + \frac{2n+1}{2n+2} a^2 \int (a^2 + x^2)^{\frac{2n-1}{2}} dx. [\text{I.C.S., 1886.}]$$

4. Investigate a formula of reduction applicable to

$$\int x^m (1 + x^2)^{\frac{n}{2}} dx,$$

where m and n are positive integers, and complete the integration if $m = 5, n = 7$. [St. John's Coll., 1881.]

5. If $\phi(n) \equiv a^{2n-1} \int_0^\infty \frac{dx}{(a^2 + x^2)^n}$, prove that $\phi(n) = \frac{2n-3}{2n-2} \phi(n-1)$. [R.P.]

6. Investigate formulae of reduction for

$$(a) \int \frac{dx}{(x^2 + a^2)^{\frac{n}{2}}}.$$

$$(b) \int x^n (a + bx)^{p+\frac{1}{2}} dx.$$

$$(c) \int \frac{x^m}{(a^2 + x^2)^{\frac{1}{2}}} dx.$$

$$(d) \int \frac{x^n}{(a^3 + x^3)^{\frac{n}{3}}} dx.$$

$$(e) \int \frac{x^m}{(x^3 - 1)^{\frac{1}{3}}} dx.$$

$$(f) \int x^{2n} (x^2 + a^2)^{\frac{2p+1}{2}} dx,$$

and obtain the value of $\int x^8 (x^3 - 1)^{-\frac{1}{3}} dx$. [Colleges, Camb.]

7. Investigate a formula of reduction for

$$\int \frac{x^{2n+1}}{(1 - x^2)^{\frac{1}{2}}} dx,$$

and by means of this integral show that

$$\frac{1}{2n+2} + \frac{1}{2} \cdot \frac{1}{2n+4} + \frac{1.3}{2.4} \cdot \frac{1}{2n+6} + \frac{1.3.5}{2.4.6} \cdot \frac{1}{2n+8} + \dots \text{ad inf.} = \frac{2.4.6 \dots 2n}{3.5.7 \dots (2n+1)}.$$

Sum also the series

$$\frac{1}{2n+1} + \frac{1}{2} \cdot \frac{1}{2n+3} + \frac{1.3}{2.4} \cdot \frac{1}{2n+5} + \frac{1.3.5}{2.4.6} \cdot \frac{1}{2n+7} + \dots \text{ad inf.} [\text{Math. Trip., 1897.}]$$

8. Find a reduction formula for

$$\int_1^x \frac{x^n dx}{\sqrt{x-1}}.$$

Show that

$$\frac{2.4.6 \dots 2n}{3.5.7 \dots (2n+1)} \left[1 + \frac{1}{2}x + \frac{1.3}{2.4}x^2 + \dots + \frac{1.3 \dots (2n-1)}{2.4 \dots 2n}x^n \right] = 1 + \frac{a_1}{3}(x-1) + \frac{a_2}{5}(x-1)^2 + \dots + \frac{1}{2n+1}(x-1)^n$$

where $a_1, a_2, \dots$ are the binomial coefficients.

[St. John's, 1886.]

9. Prove that if $u_n \equiv \int_0^{\pi/2} \sin^{2n} x dx$, then

$$u_n = \left(1 - \frac{1}{2n}\right) u_{n-1} - \frac{1}{n2^{n+1}},$$

and deduce

$$\int_0^{\pi/2} \sin^{2n} x dx = -\frac{1}{2^{n+1}} \left\{ \frac{1}{n} + \frac{2n-1}{n(n-1)} + \frac{(2n-1)(2n-3)}{n(n-1)(n-2)} + \dots \right\} + \frac{(2n-1)(2n-3) \dots 3}{2n(2n-2) \dots 4.2} \cdot \frac{\pi}{4}. [\text{Math. Trip.}]$$

10. Prove that

$$\int_0^1 x^{4m+1} \sqrt{\frac{1-x^2}{1+x^2}} dx = \frac{1.3.5 \dots (2m-1)}{2.4.6 \dots 2m} \cdot \frac{\pi}{4} - \frac{2.4.6 \dots 2m}{3.5.7 \dots 2m+1} \cdot \frac{1}{2}.$$

11. Find a reduction formula for

$$\int e^{ax} \cos^n x dx,$$

where n is a positive integer, and evaluate

$$\int e^{ax} \cos^4 x dx. [\text{Oxford, 1889.}]$$

12. Find formulae of reduction for

$$(1) \int x^n \sin x dx,$$

$$(2) \int e^{ax} \sin^n x dx.$$

Deduce from the latter a formula of reduction for

$$\int \cos ax \sin^n x dx. [\text{Colleges } \gamma, 1890.]$$

13. Show that

$$(m+n)(m+n-2) \int \sin^m \theta \cos^n \theta d\theta = (m-1) \sin^{m+1} \theta \cos^{n-1} \theta - (n-1) \sin^{m-1} \theta \cos^{n+1} \theta \\ + (m-1)(n-1) \int \sin^{m-2} \theta \cos^{n-2} \theta d\theta. [\text{Trin. Coll. Camb., 1889.}]$$

14. Show that

$$2^m \int \cos mx \cos^m x dx = C + x + m \cdot \frac{\sin 2x}{2} + \frac{m(m-1)}{1.2} \cdot \frac{\sin 4x}{4} + \dots + \frac{\sin 2mx}{2m},$$

where m is a positive integer.

[Colleges a , 1885.]

15. Show that

$$\int_0^{\pi/2} \sin^{2m} \theta \cos^{2m-1} \theta d\theta = \frac{(2m-2)(2m-4)\dots 4.2}{(4m-1)(4m-3)\dots (2m+1)},$$

m being a positive integer.

[Oxford, 1889.]

16. Evaluate the integral $\int_{-\pi/2}^{\pi/2} e^{-nx} \cos^m x dx$, m being a positive integer.

[Coll., 1886.]

17. Prove that if

$$I_{m,n} \equiv \int \cos^m x \sin nx dx,$$

$$(m+n)I_{m,n} = -\cos^m x \cos nx + mI_{m-1,n-1}$$

and

$$[I_{m,m}]_0^{\pi/2} = \frac{1}{2^{m+1}} \left(\frac{2}{1} + \frac{2^2}{2} + \frac{2^3}{3} + \dots + \frac{2^m}{m} \right). [\text{Bertrand.}]$$

18. If $u_{m,n} \equiv \int_0^{\pi/2} \cos^m x \sin nx dx$, prove that

$$u_{m,n} = \frac{1}{m+n} + \frac{m}{m+n} u_{m-1,n-1}.$$

Hence find the value, when m is a positive integer, of

$$\int_0^{\pi/2} \cos^m x \sin 2mx dx. [\gamma, 1887.]$$

19. If $I_{m,n} \equiv \int \cos^m x \cos nx dx$, prove that

$$I_{m,n} = -\frac{\cos^2 nx}{m^2 - n^2} \frac{d}{dx} \left(\frac{\cos^m x}{\cos nx} \right) + \frac{m(m-1)}{m^2 - n^2} I_{m-2,n},$$

and show that

$$[I_{m,n}]_0^{\pi/2} = \frac{m(m-1)}{m^2 - n^2} [I_{m-2,n}]_0^{\pi/2}.$$

20. Prove that $\int_0^{\pi/2} \cos^n x \cos nx dx = \frac{\pi}{2^{n+1}}$, n being a positive integer.

[Bertrand.]

21. If m and n be positive integers, and if $m+n$ be even, prove that

$$\int_0^{\pi/2} \cos^m \theta \cos n\theta d\theta = \frac{\pi}{2^{m+1}} \frac{m!}{\frac{m+n}{2}! \frac{m-n}{2}!}. [\text{Colleges, 1882.}]$$

22. If $\int_0^{\pi/2} \cos^m x \cos nx dx$ be denoted by $f(m, n)$, show that

$$f(m, n) = \frac{m}{m-n} f(m-1, n+1) = \frac{m}{m+n} f(m-1, n-1). [\text{Oxford, 1890.}]$$

23. Prove that if n be a positive integer, greater than unity,

$$\int_0^{\pi/2} \cos^{n-2} x \sin nx dx = \frac{1}{n-1}. [\text{Oxford I. P., 1889.}]$$

24. If $u_{m,n} \equiv \int x^m \operatorname{cosec}^n x \, dx$, prove that

$$(n-1)(n-2)u_{m,n} = (n-2)^2 u_{m,n-2} + m(m-1)u_{m-2,n-2} - x^{m-1} \{m \sin x + (n-2)x \cos x\} \operatorname{cosec}^{n-1} x. [\text{Math. Trip.,}]$$

25. If $\int_0^\infty e^{-x} x^{n-1} \log x \, dx \equiv \phi(n)$,

prove that $\phi(n+2) - (2n+1)\phi(n+1) + n^2\phi(n) = 0$. [R.P., St. John's Coll., 1881.]

26. Show that if $U_n \equiv \int_0^1 \frac{x^n e^x \, dx}{\sqrt{1-x}}$,

$$2U_{n+1} + U_n(2n-1) - 2nU_{n-1} = 0. [\text{Colleges } \beta, 1887.]$$

27. Prove that if $\phi(m) \equiv \int x^m (x^3 + 3ax + c)^{-\frac{1}{2}} \, dx$,

then

$$(2m-1)\phi(m) + 3a(2m-3)\phi(m-2) + (2m-4)c\phi(m-3) = 2x^{m-2}(x^3 + 3ax + c)^{\frac{1}{2}}. [\text{Trinity, 1886.}]$$

28. If $u_m \equiv \int_0^{\pi/2} e^{-nx} \cos^m x \, dx$,

prove that $(m^2 + n^2)u_m = m(m-1)u_{m-2} + n$.

[Oxford I. P., 1900.]

29. Prove that if $I_m \equiv \int_0^{\pi/2} \frac{\sin^m x \, dx}{(1 - k^2 \sin^2 x)^{\frac{1}{2}}}$,

then $(m-1)k^2 I_m - (m-2)(1+k^2)I_{m-2} + (m-3)I_{m-4} = 0$.

[Trinity, 1890.]

30. Obtain a reduction formula for the integral

$$I_n \equiv \int (a \cos^2 \theta + 2h \sin \theta \cos \theta + b \sin^2 \theta)^{-n} \, d\theta$$

in the form

$$2(n+1)(ab - h^2)I_{n+2} - (2n+1)(a+b)I_{n+1} + 2nI_n = -\frac{1}{2n} \frac{d^2 I_n}{d\theta^2}. [\text{Math. Trip., 1898.}]$$

31. Show that

$$\int_0^{\pi/2} \frac{d\phi}{(1 - e^2 \sin^2 \phi)^3} = \frac{8 - 8e^2 + 3e^4}{(1 - e^2)^{\frac{5}{2}}} \frac{\pi}{16},$$

e being less than unity.

[St. John's Coll., 1885.]

32. If $I_n \equiv \int \frac{\sin nx}{\sin x} \, dx$, prove that $(n-1)(I_n - I_{n-2}) = 2 \sin(n-1)x$,

and hence that

$$\begin{aligned} \int_0^{\pi/2} \frac{\sin nx}{\sin x} &= \frac{\pi}{2}, \text{ if } n \text{ be odd,} \\ &= 2 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots + (-1)^{\frac{n}{2}+1} \frac{1}{n-1} \right), \text{ if } n \text{ be even.} \end{aligned}$$

33. If $X \equiv a + bx^n + cx^{2n}$ and $I_{m,p} \equiv \int x^m X^p dx$,

prove the existence of reduction formulae of the nature of

- (i) $x^{m+1}X^{p+1} = A_1I_{m,p} + B_1I_{m+n,p} + C_1I_{m+2n,p}$;
- (ii) $x^{m-2n+1}X^{p+1} = A_2I_{m,p} + B_2I_{m-n,p} + C_2I_{m-2n,p}$;
- (iii) $x^{m+1}X^p = A_3I_{m,p} + B_3I_{m+n,p-1} + C_3I_{m+2n,p-1}$;
- (iv) $x^{m+1}X^p = A_4I_{m,p} + B_4I_{m,p-1} + C_4I_{m+n,p-1}$;
- (v) $x^{m-n+1}X^{p+1} = A_5I_{m,p} + B_5I_{m-n,p} + C_5I_{m-n,p+1}$;

and find the values of the fifteen constants.

34. Show that

$$\int \frac{dx}{(a + bx^2 + cx^4)^p}$$

can be reduced to the integration of

$$\int \frac{dx}{a + bx^2 + cx^4} \quad \text{and} \quad \int \frac{x^2 dx}{a + bx^2 + cx^4} \quad (b > 0, \quad b^2 > 4ac),$$

and integrate these expressions; p being integral.

[Bertrand.]

35. Show that, if $u_{m,n} \equiv \int \frac{x^m}{(\log x)^n} dx$,

$$(n-1)u_{m,n} = -\frac{x^{m+1}}{(\log x)^{n-1}} + (m+1)u_{m,n-1}. [\text{Oxford, I. P., 1889.}]$$

36. Find reduction formulae for

- (a) $\int \tanh^n x \, dx$;
- (b) $\int \frac{x}{\sin^n x} \, dx$;
- (c) $\int \frac{dx}{(a + be^x + ce^{-x})^n}$.

37. If $I_m \equiv \int \frac{x^m dx}{\sqrt{X}}$, where $X \equiv ax^2 + 2bx + c$, show that

$$amI_m + (2m-1)bI_{m-1} + (m-1)cI_{m-2} = x^{m-1}\sqrt{X}. [\beta, 1891.]$$

38. Establish a reduction formula for

$$I_n \equiv \int \frac{dx}{(Ax^2 + B)^n \sqrt{X}},$$

where $X \equiv ax^4 + bx^2 + c$, in the form

$$\frac{x\sqrt{X}}{(Ax^2 + B)^{n-1}} = \lambda I_n + \mu I_{n-1} + \nu I_{n-2} + \xi I_{n-3},$$

showing that

$$\begin{aligned}\lambda &= (2n-2) \frac{B}{A^2} (A^2c - ABb + B^2a), \\ \mu &= -(2n-3) \frac{1}{A^2} (A^2c - 2ABb + 3B^2a), \\ \nu &= -(2n-4) \frac{1}{A^2} (Ab - 3Ba), \\ \xi &= -(2n-5) \frac{1}{A^2} a.\end{aligned}$$

39. Show that, if

$$I_{m,n} \equiv \int \sin^m \theta \cos n\theta \, d\theta, \quad J_{m,n} \equiv \int \sin^m \theta \sin n\theta \, d\theta,$$

then

$$J_{m-1,n+1} = \left(1 - \frac{n}{m}\right) I_{m,n}, \quad I_{m-1,n+1} = -\left(1 - \frac{n}{m}\right) J_{m,n},$$

where m is a positive integer; and point out how these results can be used to find the values of $I_{m,n}$ and $J_{m,n}$. [C. S., 1896.]

40. If T be a function of x such that

$$\left(\frac{dT}{dx}\right)^2 = A + 3BT + 3CT^2 + DT^3,$$

prove that

$$\frac{d}{dx} \left(\frac{1}{T^{n-1}} \frac{dT}{dx} \right) = -\frac{(n-1)A}{T^n} - \frac{3}{2}(2n-3) \frac{B}{T^{n-1}} - 3 \frac{(n-2)C}{T^{n-2}} - \frac{(2n-5)D}{2T^{n-3}}.$$

Apply the result to investigate a reduction formula for

$$\int \frac{dx}{T^n}.$$

By a consideration of the case where $C = 0$, $D = 0$ (or in any other way), obtain a reduction formula for

$$\int \frac{dx}{(a + 2bx + cx^2)^n}. \text{ [I. C. S., 1897.]}$$

41. Prove that

$$\int_0^\infty e^{-x^2} x^{2n} \, dx = \sqrt{\pi} \frac{1.3.5 \dots (2n-1)}{2.4.6 \dots 2n} \int_0^\infty e^{-x^2} x^{2n+1} \, dx,$$

where n is a positive integer.

[Colleges a , 1890.]

42. If $u_n \equiv \int_a^b x^n \{(b-x)(x-a)\}^{-\frac{1}{2}} \, dx$,

show that $2nu_n = (2n-1)(a+b)u_{n-1} - 2(n-1)abu_{n-2}$.

[Oxford I. Pub., 1912.]

43. By applying the substitution $x = a \cos^2 \theta + b \sin^2 \theta$ (or otherwise), prove that the definite integral

$$\int_a^b \frac{x^n \, dx}{\sqrt{(x-a)(b-x)}}$$

is a rational integral function of a and b when n is an integer; and evaluate it when $n = 3$. [Oxf. I. P., 1913.]

44. If $u_n \equiv \int \frac{\cos 2nx}{\sin^2 x} dx$,

obtain a formula of reduction connecting u_n and u_{n-1} .

Hence, or otherwise, evaluate

$$\int_x^{\pi/2} \frac{\cos 2nx}{\sin^2 x} dx,$$

where n is a positive integer and $\pi/2 > x > 0$.

Consider the case when the lower limit is negative.

[Oxf. I. P., 1915.]

45. By multiplying the inequality $1 \geq 2 \sin x - \sin^2 x$ by $\sin^{2n-1} x$ and by $\sin^{2n} x$, and integrating between 0 and $\frac{1}{2}\pi$, show that

$$\left\{ \frac{(4n+3)(2n+1)}{4n+4} \cdot \frac{\pi}{2} \right\}^{\frac{1}{2}} > \frac{2.4 \dots 2n}{1.3 \dots (2n-1)} > \left\{ \frac{2n(2n+1)}{4n+1} \cdot \pi \right\}^{\frac{1}{2}}. [\text{Math. Trip. I., 1915.}]$$

46. The expression

$$\frac{1-a}{(1-a \sin^2 \theta)^{\frac{1}{2}}} - (1-a \sin^2 \theta)^{\frac{1}{2}},$$

where $1 > a > 0$, is expanded in ascending powers of a , and the coefficient of a^n is denoted by u_n . Prove that

$$\int_0^{\pi/2} u_n d\theta = 0. [\text{Math. Trip. I., 1916.}]$$

47. If $s_n \equiv \int \frac{\sin(2n-1)x}{\sin x} dx$, $v_n \equiv \int \frac{\sin^2 nx}{\sin^2 x} dx$,

prove the reduction formulae

$$n(s_{n+1} - s_n) = \sin 2nx, \quad v_{n+1} - v_n = s_{n+1};$$

and show that if v_n be taken between the limits 0 and $\frac{1}{2}\pi$, its value is $\frac{1}{2}n\pi$, where n is an integer.

[Math. Trip. I., 1914.]

48. If $\lambda^2 \equiv \cos^2 \phi/a^2 + \sin^2 \phi/b^2$, find $\int_0^{\pi/2} \frac{d\phi}{\lambda^2}$,

and prove that

$$\int_0^{\pi/2} \frac{d\phi}{\lambda^6} = \pi ab \{3(a^4 + b^4) + 2a^2 b^2\}/16,$$

and that

$$\int_0^1 \frac{3\mu^2 - 1}{2} \frac{d\mu}{(1 - e^2 \mu^2)^{\frac{3}{2}}} = \frac{e^2}{3(1 - e^2)^{\frac{3}{2}}}. [\epsilon, 1883.]$$

49. If $U_n \equiv \int \sin^m x (a + b \cos x)^{-n} dx$, prove that U_n can be calculated by means of a reduction formula of the nature

$$AU_n + BU_{n-1} + CU_{n-2} = \sin^{m+1} x (a + b \cos x)^{-n+1};$$

and determine the constants A, B, C .

50. Prove that

$$\int_c^\infty \frac{dx}{(a^2 - 2cx + x^2)^{n+1}} = \frac{2n!}{n! n!} \frac{\pi}{\lambda^{n+\frac{1}{2}}},$$

where λ denotes $4(a^2 - c^2)$ and is supposed positive.

[Trin., 1887.]

Chapter VIII: Form $\int F(x, \sqrt{R}) dx$, where R is Quadratic — Exercises

Exercises — Cases I–III: X and Y Linear or Quadratic (Arts. 275–290)

Integrate the following expressions:

1. $\frac{1}{x\sqrt{x+1}}, \frac{1}{(x-1)\sqrt{x+2}}, \frac{x+1}{(x-1)\sqrt{x+2}}, \frac{x^2+x+1}{(x+2)\sqrt{x-1}}.$
2. $\frac{1}{(x^2+1)\sqrt{x}}, \frac{1}{(x^2+2x+2)\sqrt{x+1}}, \frac{x}{(x^2+2x+2)\sqrt{x+1}}, \frac{x^2+1}{(x^2+2x+2)\sqrt{x+1}}.$
3. $\frac{1}{x\sqrt{x^2+1}}, \frac{1}{(x+1)\sqrt{x^2+1}}, \frac{x}{(x+1)\sqrt{x^2+1}}, \frac{x^2+x+1}{(x+1)\sqrt{x^2+2x+3}}.$

4. Prove that

$$\int \frac{dx}{(x+c)\sqrt{x}} = \frac{2}{\sqrt{c}} \tan^{-1} \sqrt{\frac{x}{c}} \quad \text{or} \quad \frac{1}{\sqrt{-c}} \log_e \frac{\sqrt{x} - \sqrt{-c}}{\sqrt{x} + \sqrt{-c}},$$

according as c is positive or negative.

[C. S., 1904.]

5. Integrate $\frac{\sqrt{\sin \theta}}{\cos \theta (\cos \theta + \sin \theta) \sqrt{2 \cos \theta + 3 \sin \theta}}.$
6. Integrate $\frac{\sqrt{\sin \theta} + \sqrt{\cos \theta}}{(a^2 \cos^2 \theta - b^2 \sin^2 \theta) \sqrt{\cos \left(\theta - \frac{\pi}{4} \right)}} \quad (a > b > 0).$
7. Integrate $\frac{\tan 2\theta}{\sqrt{\cos^6 \theta + \sin^6 \theta}}.$ Integrate $\frac{(x^3+1)^2}{(x-1)\sqrt{x^2+1}}.$
8. Integrate

$$\frac{1}{\sqrt{(a-x)(x-b)}}, \quad \frac{1}{(a-x)\sqrt{x-b}}, \quad \frac{1}{(x-b)\sqrt{a-x}}, \quad \frac{1}{(1 \pm x)\sqrt{1 \pm x^2}}.$$

[St. John's, 1890.]

9. Integrate $\frac{1}{x\sqrt{(x-a)(x-b)}}.$

[Colleges, 1876.]

10. Show that

$$\int \frac{x-3}{x\sqrt{x^2-3x+2}} dx = \cosh^{-1}(2x-3) + \frac{3}{\sqrt{2}} \cosh^{-1} \left(\frac{4-3x}{x} \right).$$

[St. John's, 1883.]

11. Integrate $\int \frac{x-\alpha}{x-\beta} \frac{dx}{\sqrt{x^2+2px+q}} \quad (p^2 > q).$

[a, 1887.]

12. Integrate

$$(i) \int \frac{dx}{x\sqrt{x^2+x+1}},$$

[Coll., 1879.]

$$(ii) \int \frac{dx}{x\sqrt{x^2 - a^2}}, \quad [\epsilon, 1883.]$$

$$(iii) \int \frac{dx}{(1+x^2)\sqrt{1-x^2}}. \quad [I. C. S., 1888.]$$

13. Integrate

$$(i) \int \frac{dx}{(1-x)\sqrt{1-x^2}} \text{ (in two ways)}. \quad [\text{Trinity}, 1890.]$$

$$(ii) \int \frac{dx}{(1-2x)\sqrt{1+4x}}. \quad [\text{Coll.}, 1892.]$$

14. Integrate

$$\int \frac{dx}{(x-\lambda)\sqrt{x-\mu}}, \quad \int \frac{x^3 dx}{x^2 + \lambda^2}, \quad \int \frac{dx}{(x+1)(x+2)(x+3)}. \quad [\text{Math. Trip.}, \text{Pt. I.}, 1920.]$$

General Examples

1. Obtain the following integrals:

$$(i) \int (1+x)^{-1} x^{-\frac{1}{2}} dx.$$

$$(ii) \int (1+x)^{-1} (1+2x)^{-\frac{1}{2}} dx.$$

$$(iii) \int x^{-1} (2-3x+x^2)^{-\frac{1}{2}} dx.$$

$$(iv) \int (1+x)^{-1} (1+x+x^2)^{-\frac{1}{2}} dx.$$

$$(v) \int \frac{\sqrt{1+x+x^2}}{1+x} dx.$$

$$(vi) \int \frac{x^2+x-1}{(x+1)\sqrt{x^2-1}} dx.$$

$$(vii) \int \frac{dx}{x\sqrt{a^n+x^n}}.$$

$$(viii) \int \frac{1-x}{1+x} \frac{1}{\sqrt{x+x^2+x^3}} dx.$$

2. Integrate

$$(i) \int \frac{dx}{(x+2)\sqrt{x^2-1}}.$$

$$(ii) \int \frac{\sqrt{x^2-1}}{x+2} dx. \quad [\text{Barnes Schol.}, 1887.]$$

3. Show that

$$\int \frac{dx}{(x-p)\sqrt{a+2bx+cx^2}} = \frac{1}{\{-(a+2bp+cp^2)\}^{\frac{1}{2}}} \sin^{-1} \left\{ \frac{(a+bx)+p(b+cx)}{(x-p)\sqrt{b^2-ac}} \right\}$$

where p lies between the roots of $a+2bx+cx^2=0$, supposed real. [Trinity, 1886 and 1891.]

4. Show that

$$\int \frac{dx}{(x^2 + a^2)\sqrt{x^2 + b^2}} = \frac{1}{a\sqrt{b^2 - a^2}} \cos^{-1} \frac{a}{b} \sqrt{\frac{x^2 + b^2}{x^2 + a^2}}, \quad \text{if } a < b,$$

$$\text{and} \quad = \frac{1}{a\sqrt{a^2 - b^2}} \cosh^{-1} \frac{a}{b} \sqrt{\frac{x^2 + b^2}{x^2 + a^2}}, \quad \text{if } a > b.$$

5. Prove that

$$\int \frac{(x+1) dx}{(2x^2 - 2x + 1)\sqrt{3x^2 - 2x + 1}} = \cosh^{-1} \sqrt{\frac{3x^2 - 2x + 1}{2x^2 - 2x + 1}} + 2 \cos^{-1} \frac{1}{\sqrt{2}} \sqrt{\frac{3x^2 - 2x + 1}{2x^2 - 2x + 1}}.$$

6. Integrate

$$(i) \int \frac{(3x+4) dx}{(5x^2+8x)\sqrt{4x^2-2x+1}},$$

$$(ii) \int \frac{dx}{(x^2+2ax+b^2)\sqrt{x^2+2ax+c^2}},$$

where $a < b < c$.

7. Integrate

$$(i) \int \frac{x dx}{(a^2 + b^2 - x^2)\sqrt{(a^2 - x^2)(x^2 - b^2)}}. \quad [\text{St. John's, 1888.}]$$

$$(ii) \int \frac{(x+b) dx}{(x^2 + a^2)\sqrt{x^2 + c^2}} \quad (a > c). \quad [\text{St. John's, 1889.}]$$

$$(iii) \int \frac{d\theta}{\sin \theta \sqrt{a \cos^2 \theta + b \sin^2 \theta + c}}. \quad [\text{Trinity, 1888.}]$$

8. Find the values of

$$(i) \int \frac{\sin x dx}{(\cos x + \cos \alpha)\sqrt{(\cos x + \cos \beta)(\cos x + \cos \gamma)}}. \quad [\gamma, 1890.]$$

$$(ii) \int \frac{dx}{\cos(x+\alpha)\sqrt{\cos(x+\beta)\cos(x+\gamma)}}. \quad [\gamma, 1890.]$$

9. Integrate

$$\int \frac{a^2 - x^2}{(a^2 - ax + x^2)(a^4 + a^2x^2 + x^4)^{\frac{1}{2}}} dx,$$

transforming by the substitution

$$x^2 + ax + a^2 = y^2(x^2 - ax + a^2). [a, 1884.]$$

10. Integrate

$$(i) \int \frac{(8x-13) dx}{(3x^2-10x+9)\sqrt{-x^2+10x-13}}.$$

$$(ii) \int \frac{dx}{(x-1)(x-2)\sqrt{(x-3)(x-4)}}. \quad [\text{Coll., 1892.}]$$

$$(iii) \int \frac{(x+9) dx}{(x^2-5x+4)\sqrt{x^2-2x+2}}.$$

$$(iv) \int \frac{(x-a) dx}{(x-b)(x-c)(x-d)\sqrt{x-e}}.$$

$$(v) \int \frac{(x+3) dx}{(x^2+x+1)\sqrt{x^2+x+2}}.$$

11. Integrate

$$(i) \int \frac{x^4-1}{x^2\sqrt{x^4+x^2+1}} dx;$$

[Coll., 1901.]

$$(ii) \int \frac{x^{-1}(x^2-1)}{\sqrt{x^4+x^2+1}} dx.$$

12. Show that

$$\int \sqrt{\frac{\sin(x-a)}{\sin(x+a)}} dx = \cos a \cos^{-1} \left(\frac{\cos x}{\cos a} \right) - \sin a \cosh^{-1} \left(\frac{\sin x}{\sin a} \right).$$

[Coll., 1901.]

13. Integrate

$$(i) \int \frac{1-2x^2}{1+2x^2} \sqrt{\frac{1+x^2}{1-x^2}} x dx.$$

[R. P.]

$$(ii) \int \frac{dx}{(1+x^4)\sqrt{(\sqrt{1+x^4}-x^2)}}.$$

[R. P.; Euler, C.I., vol. iv.]

$$(iii) \int \frac{dx}{(a^2-ax-x^2)\sqrt{ax+x^2}}.$$

[Oxf. I. P., 1900.]

14. Evaluate the integral

$$\int \frac{dx}{(a^2-\tan^2 x)(b^2-\tan^2 x)^{\frac{1}{2}}}.$$

[Math. Trip., 1886.]

15. Prove that

$$\int \frac{dx}{\sin x \sin^{\frac{1}{2}}(2x+a)} = -\frac{1}{\sqrt{\sin a}} \cosh^{-1} \frac{\sin(x+a)}{\sin x}.$$

[Coll., 1892.]

16. Show how to integrate

$$\int (fx+g)(ax^2+bx+c)^{\frac{n}{2}} dx,$$

where n is any positive or negative integer.

[a, 1890.]

$$17. \text{ Show that } \int \frac{(a+bx) dx}{(a+2bx+cx^2)^{\frac{3}{2}}} = \frac{x}{(a+2bx+cx^2)^{\frac{1}{2}}}.$$

[Trinity, 1889.]

18. Prove by the substitution

$$y^2 = (ax^2+2bx+c)/(Ax^2+2Bx+C),$$

where A and $AC-B^2$ are positive, that the integral

$$\int \frac{(Mx+N) dx}{(Ax^2+2Bx+C)\sqrt{ax^2+2bx+c}}$$

becomes of the form

$$P_1 \int \frac{dy}{\sqrt{\lambda_1-y^2}} + P_2 \int \frac{dy}{\sqrt{y^2-\lambda_2}},$$

where λ_1 and λ_2 are the roots of the quadratic

$$(a - \lambda A)(c - \lambda C) - (b - \lambda B)^2 = 0,$$

and P_1, P_2 are definite constants.

Integrate completely the function

$$\frac{x+3}{(2x^2-10x+17)\sqrt{4x^2-26x+49}}. [\text{Math. Trip., 1891.}]$$

19. Prove $\int_0^a \sqrt{\tanh^2 a - \tanh^2 x} dx = \frac{\pi}{2}(1 - \operatorname{sech} a).$ [β , 1889.]

20. Integrate $\int (x^2 + b^2)^{-1}(x^2 + a^2 + b^2)^{-\frac{1}{2}} dx.$ [Ox. II. P., 1902.]

21. Integrate

$$\int \frac{\sin^3 x dx}{(1 + \cos^2 x)\sqrt{1 + \cos^2 x + \cos^4 x}}, [\text{St. John's, 1882.}]$$

and evaluate

$$\int_{-1}^1 \frac{dx}{(a^2 + c^2 x^2)\sqrt{1 - x^2}}.$$

22. Show that $\int \frac{dx}{(x-a)\sqrt{Ax^2+2Bx+C}}$ is transcendental unless $Aa^2 + 2Ba + C = 0.$ [J. M. Sch. Ox., 1904.]

Establish the results

$$(i) \int \frac{dx}{(x-1)\sqrt{x^2-4x+5}} = \frac{1}{\sqrt{2}} \cosh^{-1} \frac{\sqrt{2(x^2-4x+5)}}{x-1}$$

and

$$(ii) \int \frac{dx}{(x-2)\sqrt{3+2x-2x^2}} = \sin^{-1} \frac{\sqrt{3+2x-2x^2}}{\sqrt{7}(x-2)}.$$

[Coll. a, 1890.]

23. Show that

$$\int_{-1}^{+1} \frac{(1-ax)(1-bx)}{(1-2ax+a^2)(1-2bx+b^2)} \frac{dx}{\sqrt{1-x^2}} = \frac{\pi}{2} \frac{2-ab}{1-ab}. [\delta, 1884.]$$

24. Describe the steps whereby the integral of a rational function of a single variable, x , can be obtained.

Prove that if the sign of summation refer to the suffixes 1, 2, 3 in cyclical order, the integral

$$\int dx \sqrt{(x-a)(x-b)} \Sigma(c_2 - c_3) \left[\frac{2(c_1 - a)(c_1 - b)}{(x - c_1)^2} - \frac{2c_1 - a - b}{x - c_1} \right]$$

is a certain constant multiple of

$$(x-a)^{\frac{3}{2}}(x-b)^{\frac{3}{2}}(x-c_1)^{-1}(x-c_2)^{-1}(x-c_3)^{-1}. [\text{Math. Trip., 1896.}]$$

25. Determine the degenerate form of the elliptic integral

$$\int \frac{ds}{\sqrt{4(s-s_1)(s-s_2)(s-s_3)}}, \quad s_1 > s_2 > s_3,$$

when s_2 is made to coincide with s_1 or with s_3 .

[Int. Arts, London.]

26. Prove, by means of the substitution $\frac{x-\beta}{x-\alpha} = y^2$, that

$$\int \frac{dx}{(x-\gamma)\sqrt{(x-\alpha)(x-\beta)}} = \frac{-2}{\sqrt{(\alpha-\gamma)(\gamma-\beta)}} \tan^{-1} \sqrt{\frac{\alpha-\gamma}{\gamma-\beta} \cdot \frac{x-\beta}{x-\alpha}}$$

or $= \frac{2}{\sqrt{(\alpha-\gamma)(\beta-\gamma)}} \tanh^{-1} \sqrt{\frac{\alpha-\gamma}{\beta-\gamma} \cdot \frac{x-\beta}{x-\alpha}}$. [Int. Arts, London.]

27. Prove that

$$\int_0^1 \frac{dx}{(1+x)(2+x)\sqrt{x(1-x)}} = \pi \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{6}} \right). \text{ [Math. Trip. I., 1912.]}$$

28. Show that the integral

$$\int \frac{dx}{x\sqrt{3x^2+2x-1}}$$

is rationalized by the assumption $x = (1+y^2)/(3-y^2)$, and hence, or otherwise, find its value.

Prove that if m be a positive proper fraction, the value of the above integral when taken between limits $\frac{1}{3}$ and $\frac{1}{2+m}$ is the same as when taken between limits $\frac{1}{2+m}$ and $\frac{1}{m(2+m)}$. [Math. Trip. I., 1910.]

29. Prove by means of the substitution

$$\frac{a-x}{x-b} = \frac{a-d}{b-c} \cdot \frac{c-y}{y-d},$$

that, if m be any positive quantity, and $a > b > c > d$,

$$\int_b^a \frac{\{(a-x)(x-c)\}^{m-1}}{\left\{ \frac{(a-x)(x-d)}{a-d} + \frac{(x-b)(x-c)}{b-c} \right\}^m} dx = \int_d^c \frac{\{(a-x)(c-x)\}^{m-1}}{\left\{ \frac{(a-x)(x-d)}{a-d} + \frac{(b-x)(c-x)}{b-c} \right\}^m} dx.$$

[Math. Trip., 1878.]

[See Wolstenholme's *Mathematical Problems*, Numbers 1900–1903 for a group of similar examples.]

30. By the transformation $px + \frac{q}{x} = \sqrt{2pq}/z$, integrate

$$\int \frac{px^2 - q}{px^2 + q} \frac{dx}{\sqrt{p^2x^4 + q^2}}. \text{ [Cf. Euler, C.I., iv., p. 22.]}$$

31. Apply the transformation $x^2 + \frac{1}{x^2} = \frac{2}{z^2}$ to integrate

(i) $\int \frac{\sqrt{1+x^4}}{1-x^4} dx,$

(ii) $\int \frac{x^2 dx}{(1-x^4)\sqrt{1+x^4}}.$

[Euler, C.I., iv.]

32. Show that

$$\int \frac{2 d\theta}{\sin \theta \sqrt[4]{\cos 2\theta}} = \tanh^{-1} \frac{\cos \theta}{\sqrt[4]{\cos 2\theta}} + \tan^{-1} \frac{\cos \theta}{\sqrt[4]{\cos 2\theta}}.$$

33. Show that the transformation

$$\{(a + bx^n)^\lambda - b^\lambda x^{\lambda n}\} = \left(\frac{x}{u}\right)^{\lambda n}$$

will reduce the integration

$$\int \frac{x^{m-1} dx}{(a + bx^n)\{(a + bx^n)^\lambda - b^\lambda x^{\lambda n}\}^{\frac{m}{\lambda n}}} \quad \text{to the form} \quad \frac{1}{a} \int \frac{u^{m-1} du}{1 + b^\lambda u^{\lambda n}}.$$

[Euler, C.I., iv., 53 and 56; Peacock, p. 305.]

34. (i) Show that

$$\int \frac{e^x(2-x^2)}{(1-x)\sqrt{1-x^2}} dx = e^x \sqrt{\frac{1+x}{1-x}}, [\text{Peacock, p. 309.}]$$

and (ii) integrate

$$\int e^x \frac{1 + nx^{n-1} - x^{2n}}{(1-x^n)\sqrt{1-x^{2n}}} dx.$$

34. Integrate

$$(i) \int \frac{2-3x}{2+3x} \sqrt{\frac{1+x}{1-x}} dx.$$

$$(ii) \int \frac{\sin^5 \theta + 2 \cos^5 \theta}{\cos \theta \sin 4\theta} d\theta.$$

[St. John's, 1881.]

35. Show that

$$(i) \int \sqrt{\frac{a^2 - c^2 x^2}{a^2 - x^2}} \frac{dx}{x} = \frac{1}{2} \log \left\{ \frac{y-1}{y+1} \frac{(y+c)^c}{(y-c)^c} \right\},$$

$$\text{where } y = \sqrt{\frac{a^2 - c^2 x^2}{a^2 - x^2}}.$$

$$(ii) \int \frac{x^2 + 1}{x^2 - 1} \frac{dx}{\sqrt{1 - ax^2 + x^4}} = \frac{1}{\sqrt{a-2}} \cos^{-1} \frac{x\sqrt{a-2}}{x^2 - 1}.$$

[Hall, I.C., p. 325.]

36. If $F(x, y)$ be a rational algebraic function of x and y , show that

$$\int F\left(x, \sqrt{1+x^2}\right) \left(x + \sqrt{1+x^2}\right)^{\frac{p}{q}} dx$$

may be integrated by the transformation $x = \sinh(q \log z)$.

37. Show that

$$(i) \int (\cos 2\theta)^{\frac{3}{2}} \cos \theta d\theta = \frac{1}{8} \sin \theta (3 + 2 \cos 2\theta) \sqrt{\cos 2\theta} + \frac{3}{8\sqrt{2}} \sin^{-1}(\sqrt{2} \sin \theta).$$

$$(ii) \int_{\alpha}^{\theta} \frac{\sin \theta}{\sqrt{\sin^2 \alpha - \sin^2 \theta}} d\theta = \log \left(\frac{\cos \alpha}{\cos \theta + \sqrt{\sin^2 \alpha - \sin^2 \theta}} \right).$$

$$(iii) \int_0^{\pi} e^{2x \cos \theta} d\theta = \pi \left[1 + \frac{x^2}{(1!)^2} + \frac{x^4}{(2!)^2} + \frac{x^6}{(3!)^2} + \dots \right].$$

38. Show that

$$(i) \int_0^1 x^{a \pm cx} dx = \frac{1}{a+1} \mp \frac{c}{(a+2)^2} + \frac{c^2}{(a+3)^3} \mp \frac{c^3}{(a+4)^4} + \dots$$

$$(ii) \int_0^1 x^{\pm x^2} dx = 1 \pm \frac{1}{3^2} + \frac{1}{5^3} \mp \frac{1}{7^4} + \dots$$

[Anglin.]

39. If $\phi(x) = a_0 + \frac{1}{2}a_2x^2 + \frac{1 \cdot 3}{2 \cdot 4}a_4x^4 + \dots$, show that

$$(i) \frac{4}{\pi} \int_0^{\pi/2} \cos^2 \theta \phi(\sin \theta) d\theta = 1^2 \cdot a_0 + \frac{1}{3} \left(\frac{1}{2}\right)^2 a_2 + \frac{1}{3} \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2 a_4 + \dots$$

$$(ii) \frac{20}{9\pi} = \frac{1}{2} \cdot 1^2 + \frac{1}{3} \left(\frac{1}{2}\right)^2 + \frac{1}{4} \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2 + \frac{1}{5} \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right)^2 + \dots$$

$$(iii) \frac{4}{\pi} - 1 = \frac{1}{1^2} \left(\frac{1}{2}\right)^2 + \frac{1}{3^2} \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2 + \frac{1}{5^2} \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right)^2 + \dots$$

[Anglin.]

40. Integrate

$$(i) \int \frac{\sin 2\theta d\theta}{\sqrt{\sin^4 \theta + 4 \sin^2 \theta \cos^2 \theta + 2 \cos^4 \theta}}.$$

$$(ii) \int \frac{(z^2 - 1) dz}{(z^2 + z + 1)\sqrt{z^4 + 4z^3 + 4z^2 + 4z + 1}}.$$

41. If J be the Jacobian of two quadratic functions of x , viz.

$$u_1 \equiv a_1x^2 + 2b_1x + c_1, \quad u_2 \equiv a_2x^2 + 2b_2x + c_2,$$

$$J \equiv \begin{vmatrix} 2(a_1x + b_1), & 2(b_1x + c_1) \\ 2(a_2x + b_2), & 2(b_2x + c_2) \end{vmatrix},$$

show that if $u_1 = 0$, $u_2 = 0$ have no positive roots, then

$$\int_0^\infty \frac{J}{u_1 u_2} dx = 2 \log \frac{a_1 c_2}{a_2 c_1}.$$

42. By means of the identity

$$\int_0^{\pi/2} (a + \sin^2 x)^n \cos x dx = \int_0^{\pi/2} (1 + a - \sin^2 x)^n \sin x dx,$$

prove that

$$a^n + {}^nC_1 \frac{a^{n-1}}{3} + {}^nC_2 \frac{a^{n-2}}{5} + {}^nC_3 \frac{a^{n-3}}{7} + \dots$$

$$= (1+a)^n - 2 \frac{n}{3} (1+a)^{n-1} + 2^2 \frac{n(n-1)}{3 \cdot 5} (1+a)^{n-2}$$

$$- 2^3 \frac{n(n-1)(n-2)}{3 \cdot 5 \cdot 7} (1+a)^{n-3} + \dots$$

[Wolstenholme, *Problems*, No. 1929; Wiggins, E. Times, No. 13323.]

43. Show that

(i)

$$\begin{aligned}
& a^n + {}^nC_1 \frac{1}{p+2} a^{n-1} + {}^nC_2 \frac{1 \cdot 3}{(p+2)(p+4)} a^{n-2} \\
& \quad + {}^nC_3 \frac{1 \cdot 3 \cdot 5}{(p+2)(p+4)(p+6)} a^{n-3} + \dots \\
& = (1+a)^n - {}^nC_1 \frac{p+1}{p+2} (1+a)^{n-1} + {}^nC_2 \frac{(p+1)(p+3)}{(p+2)(p+4)} (1+a)^{n-2} \\
& \quad - {}^nC_3 \frac{(p+1)(p+3)(p+5)}{(p+2)(p+4)(p+6)} (1+a)^{n-3} + \dots
\end{aligned}$$

(ii)

$$\begin{aligned}
& a^n + {}^nC_1 \frac{1}{p+2} a^{n-1} b + {}^nC_2 \frac{1 \cdot 3}{(p+2)(p+4)} a^{n-2} b^2 \\
& \quad + {}^nC_3 \frac{1 \cdot 3 \cdot 5}{(p+2)(p+4)(p+6)} a^{n-3} b^3 + \dots \\
& = (a+b)^n - {}^nC_1 \frac{p+1}{p+2} (a+b)^{n-1} b + {}^nC_2 \frac{(p+1)(p+3)}{(p+2)(p+4)} (a+b)^{n-2} b^2 \\
& \quad - {}^nC_3 \frac{(p+1)(p+3)(p+5)}{(p+2)(p+4)(p+6)} (a+b)^{n-3} b^3 + \dots
\end{aligned}$$

44. (i) Integrate

$$\int \frac{2x^3 - 1}{x^6 + 2x^3 - x^2 + 1} dx. [\text{Ox. I. P., 1903.}]$$

(ii) Integrate

$$\int \frac{(3x^4 - 1) dx}{x^8 + 2x^4 - 16x^2 + 1}.$$

(iii) Prove that

$$\int_{-\pi/4}^{\pi/4} \frac{(1 - \sin \theta \cos \theta + \sin^2 \theta) \cos^2 \theta d\theta}{(1 + \sin \theta \cos \theta)^3} = \frac{10\pi}{9} \sqrt{3}.$$

45. (i) Show that

$$\int_0^a x^{n-1} (1-a+x)^{-n-1} dx = \frac{1}{n} \frac{a^n}{1-a}.$$

(ii) Evaluate

$$\int_a^{a+h} \frac{(a+h-x)^{n-1}}{(n-1)!} \frac{d^n}{dx^n} f(x) dx.$$

How could your result be applied to the summation of series?

[a, 1886.]

46. Discuss the integration of

$$\int f \left[x, \left(\frac{a+bx}{a'+b'x} \right)^{\frac{m}{n}}, \left(\frac{a+bx}{a'+b'x} \right)^{\frac{p}{q}}, \left(\frac{a+bx}{a'+b'x} \right)^{\frac{r}{s}} \right] dx,$$

where f denotes a rational integral algebraic function of the quantities indicated.

[Lacroix, C.I., ii., p. 35.]

47. If $F(x)$ be a rational integral algebraic function of x , show that $\int_{-1}^{+1} \frac{F(x)}{\sqrt{1-x^2}} = \pi k$, where k is the coefficient of $\frac{1}{a}$ in the product

$$F(a) \left[\frac{1}{a} + \frac{1}{2} \cdot \frac{1}{a^3} + \frac{1 \cdot 3}{2 \cdot 4} \cdot \frac{1}{a^5} + \dots \right]; [\text{St. John's, 1891.}]$$

or where k is the constant term in the expansion of $\frac{x F(x)}{(x^2 - 1)^{\frac{1}{2}}}$. [Coll., 1892.]

48. If $f(x)$ be an arbitrary algebraic polynomial of degree $n - 1$, and

$$P_n(x) \equiv A \frac{d^n}{dx^n} (x - a)^n (x - b)^n,$$

where A is a constant, then

$$\int_a^b f(x) P_n(x) dx = 0. [\text{Lond. Univ.}]$$

49. Prove that

$$\int_0^a \frac{x dx}{\cos x \cos(a - x)} = \frac{a}{\sin a} \log \sec a. [\text{Coll., 1896.}]$$

50. Show that if a be less than unity,

$$\int_0^\pi \frac{x \sin x dx}{1 + a^2 \cos^2 x} = \pi \frac{\tan^{-1} a}{a}. [a, 1891.]$$

51. Integrate

$$(i) \int \frac{b^4 x + x^5}{(b^4 - x^4)^2} \sin^{-1} \frac{x^2}{b^2} dx. [\delta, 1881.]$$

$$(ii) \int \frac{d\phi}{\cos \phi} \sqrt{1 - \frac{1}{2} \sin^2 \phi}. [\text{St. John's, 1885.}]$$

52. From the definition of a Bessel's function, viz.

$$J_n(x) = \frac{x^n}{2^n \Gamma(n+1)} \left[1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2 \cdot 4(2n+2)(2n+4)} - \dots \right],$$

derive the results

$$\frac{\sin x}{x} = \int_0^{\pi/2} J_0(x \sin \theta) \sin \theta d\theta,$$

$$\frac{1 - \cos x}{x} = \int_0^{\pi/2} J_1(x \sin \theta) d\theta. [\text{Coll., 1896.}]$$

53. Integrate

$$(i) \frac{x^3}{(x+1)^3(x^3-1)}.$$

$$(ii) \frac{1}{\sqrt{1+3\sin x \cos x + 2\sin^2 x \cos^2 x}}.$$

$$(iii) \frac{1}{\sqrt{(1+\sin x)(2+\sin x)}}. [\text{Math. Trip., 1897.}]$$

54. Show that

$$\int \sin^m x \sin nx \, dx = \sin^2 nx \frac{d}{dx} \left\{ \frac{\phi(\sin x)}{\sin nx} \right\},$$

where the form of the function ϕ is defined by the relation

$$\begin{aligned} \phi(z) = \frac{z^m}{n^2 - m^2} - \frac{m(m-1)}{(n^2 - m^2)\{n^2 - (m-2)^2\}} z^{m-2} \\ + \frac{m(m-1)(m-2)(m-3)}{(n^2 - m^2)\{n^2 - (m-2)^2\}\{n^2 - (m-4)^2\}} z^{m-4} - \dots, \end{aligned}$$

m being a positive integer and n not being of the form $\pm(m-2r)$, where r is a positive integer not greater than $\frac{m}{2}$. [Math. Trip., 1897.]

55. Draw graphs of the transformation formula

$$(a_2x^2 + 2b_2x + c_2)y^2 = a_1x^2 + 2b_1x + c_1$$

corresponding to those of Arts. 301 and 309 for

$$(a_2x^2 + 2b_2x + c_2)y = a_1x^2 + 2b_1x + c_1.$$

Chapter IX: General Theorems — Exercises

Exercises — Limiting Forms Expressed as Definite Integrals (Arts. 319–320)

1. Determine by integration the limiting values of the sums of the following series when n is infinitely great:

(i) $\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \cdots + \frac{1}{n+n}$. [a, 1884.]

(ii) $\frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \frac{n}{n^2+3^2} + \cdots + \frac{n}{n^2+n^2}$. [Oxford, 1888.]

(iii) $\frac{1}{\sqrt{2n-1^2}} + \frac{1}{\sqrt{4n-2^2}} + \frac{1}{\sqrt{6n-3^2}} + \cdots + \frac{1}{\sqrt{2n^2-n^2}}$. [Clare, etc., 1882.]

(iv) $\frac{1}{n} \left\{ \sin^{2k} \frac{\pi}{2n} + \sin^{2k} \frac{2\pi}{2n} + \sin^{2k} \frac{3\pi}{2n} + \cdots + \sin^{2k} \frac{\pi}{2} \right\}$, k being a positive integer. [St. John's, 1886.]

2. Show that the limit when n is increased indefinitely of

$$\frac{(n-m)^{\frac{1}{2}}}{n} + \frac{(2^2n-m)^{\frac{1}{2}}}{2n} + \frac{(3^2n-m)^{\frac{1}{2}}}{3n} + \cdots + \frac{(n^3-m)^{\frac{1}{2}}}{n^2} \quad \text{is} \quad \frac{3}{2}.$$

[Colleges, 1892.]

3. Find the limit when n is indefinitely great of the series

$$\frac{\sqrt{n-1}}{n} + \frac{\sqrt{2n-1}}{2n} + \frac{\sqrt{3n-1}}{3n} + \cdots + \frac{\sqrt{n^2-1}}{n^2}.$$

[Colleges, 1890.]

4. Evaluate

$$Lt_{n=\infty} \left[\frac{1}{\sqrt{2a^2n-1}} + \frac{1}{\sqrt{4a^2n-1}} + \frac{1}{\sqrt{6a^2n-1}} + \cdots + \frac{1}{\sqrt{2a^2n^2-1}} \right].$$

5. Evaluate

$$Lt_{n=\infty} \left[\frac{n^2}{(n^2+1^2)^{\frac{3}{2}}} + \frac{n^2}{(n^2+2^2)^{\frac{3}{2}}} + \cdots + \frac{n^2}{\{n^2+(n-1)^2\}^{\frac{3}{2}}} \right].$$

[C. S., 1901.]

Miscellaneous Examples (Arts. 321–353)

1. Integrate

(i) $\int \frac{\sin^2 x \, dx}{(x \cos x - \sin x)^2}$; [L.]

(ii) $\int \frac{\log x \, dx}{x^2(1 - \log x)^2}$. [L.]

2. Prove that

$$\int_{-c}^c \frac{x(c^2 - x^2) dx}{(b^2 + c^2 - 2bx)^{\frac{5}{2}}} = \frac{4}{5} \frac{c^5}{b^4} \quad (b > c). \text{[L.]}$$

3. If $X = a + 2bx + cx^2$, show that $\int \frac{dx}{X^n}$ can be made to depend upon $\int \frac{dx}{X^{n-1}}$.

Find a reduction formula for $\int \cos mx \sin^n x dx$, and apply it to the case $n = 4$. [L.]

4. Evaluate $\int \frac{2x - 3}{5x^2 - 16x + 14} \frac{dx}{\sqrt{3x^2 - 10x + 9}}$. [L.]

5. Prove that $\int_0^b u \frac{d^n v}{dx^n} dx$ can be made to depend upon $\int_0^b v \frac{d^n u}{dx^n} dx$.

Hence show that if $f(x)$ be an arbitrary polynomial of degree $n - 1$, and

$$P_n(x) = \frac{d^n(Ax^2 + Bx + C)^n}{dx^n},$$

then

$$\int_{\alpha}^{\beta} f(x) P_n(x) dx = 0,$$

where α, β are the roots, considered real, of the quadratic $Ax^2 + Bx + C = 0$.

6. Prove that the effect of the operation $p \frac{d}{dt} + q$ on a periodic function $a \cos(nt + \epsilon)$ is to multiply the amplitude a by $\sqrt{p^2 n^2 + q^2}$, and to increase the angle $nt + \epsilon$ by $\tan^{-1} \frac{pn}{q}$.

Write down the effect of the operation

$$\left(p \frac{d}{dt} + q\right) / \left(P \frac{d}{dt} + Q\right),$$

and generally, of the operation

$$\left(\alpha + \beta \frac{d}{dt} + \gamma \frac{d^2}{dt^2} + \dots\right) / \left(A + B \frac{d}{dt} + C \frac{d^2}{dt^2} + \dots\right)$$

on the same periodic function.

[Int. Arts, London.]

7. When $y^2 \equiv ax^2 + 2bx + c$, prove that

$$\int \frac{dx}{y} = \frac{1}{\sqrt{a}} \cosh^{-1} \frac{y\sqrt{a}}{\sqrt{ac - b^2}}, \quad \frac{1}{\sqrt{a}} \sinh^{-1} \frac{y\sqrt{a}}{\sqrt{b^2 - ac}} \quad \text{or} \quad \frac{1}{\sqrt{-a}} \sin^{-1} \frac{y\sqrt{-a}}{\sqrt{b^2 - ac}},$$

the real form to be chosen, and deduce the value of the integral in the degenerate case when $a = 0$. [Int. Arts, London.]

8. Find the limiting value of $(n!)^{\frac{1}{n}}/n$, when n is infinite.

9. Find the limiting value when n is infinite of the n th part of the sum of the n quantities

$$\frac{n+1}{n}, \frac{n+2}{n}, \frac{n+3}{n}, \dots, \frac{n+n}{n},$$

and show that it bears to the limiting value of the n th root of the product of the same quantities the ratio $3e : 8$, where e is the base of the Napierian logarithms. [Oxford 1886, and I. P., 1911.]

10. If na is always equal to unity, and n is indefinitely great, show that the limiting value of the product

$$(1 + a^4)\{1 + (2a)^4\}^{\frac{1}{2}}\{1 + (3a)^4\}^{\frac{1}{3}}\{1 + (4a)^4\}^{\frac{1}{4}} \dots \{1 + (na)^4\}^{\frac{1}{n}}$$

is $e^{\pi^2/48}$.

[Oxford, 1888.]

11. Show that the limit of the sum of n terms of the series

$$\frac{n^3}{(n^2 + 1^2)(n^2 + 2 \cdot 1^2)} + \frac{n^3}{(n^2 + 2^2)(n^2 + 2 \cdot 2^2)} + \dots + \frac{n^3}{(n^2 + n^2)(n^2 + 2n^2)},$$

when n is infinite, is $\sqrt{2} \tan^{-1} \sqrt{2} - \frac{\pi}{4}$.

[γ , 1901.]

12. Find

$$Lt_{n=\infty} \left[\frac{\sqrt{n-a}}{n-c} + \frac{\sqrt{2n-a}}{2n-c} + \frac{\sqrt{3n-a}}{3n-c} + \dots + \frac{\sqrt{n^2-a}}{n^2-c} \right].$$

13. Find the limiting value, when n is infinite, of

$$\left\{ \tan \frac{\pi}{2n} \cdot \tan \frac{2\pi}{2n} \cdot \tan \frac{3\pi}{2n} \dots \tan \frac{n\pi}{2n} \right\}^{\frac{1}{n}}. [\text{Oxford I. P., 1903.}]$$

14. Show that the limit of the product

$$\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right)^{\frac{1}{2}} \left(1 + \frac{3}{n}\right)^{\frac{1}{3}} \dots \left(1 + \frac{n}{n}\right)^{\frac{1}{n}},$$

when n is increased indefinitely, is $e^{\pi^2/12}$.

[Colleges, 1896.]

15. Find the limit, when n is indefinitely increased, of

$$\frac{1}{n} \left\{ \sec \frac{x}{n} + \sec \frac{2x}{n} + \dots + \sec \frac{(n-1)x}{n} \right\},$$

where x is $< \frac{\pi}{2}$.

Examine the case when $x > \frac{\pi}{2}$.

16. Find the limiting value of

$$2 \log n - \log \left[(1 + n^2)^{\frac{1}{n}} (2^2 + n^2)^{\frac{1}{n}} \dots (2n^2)^{\frac{1}{n}} \right],$$

when n is indefinitely increased.

[Oxford I. P., 1900.]

17. Show from elementary considerations that when n increases indefinitely,

$$1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} - \log n$$

approaches a finite limit intermediate between $\frac{1}{2}$ and 1.

[St. John's, 1884.]

18. If $f(x) = f(a+x)$, show that

$$\int_a^{na} f(x) dx = (n-1) \int_0^a f(x) dx,$$

and illustrate geometrically.

[Oxford I. P., 1888.]

19. Prove that $\int_0^a \phi(x) dx = \int_0^a \phi(a-x) dx$,

and show that

$$(1) \int_0^\pi \frac{x \sin^n x}{1 + \cos^2 x} dx = \frac{\pi}{2} \int_0^\pi \frac{\sin^n x}{1 + \cos^2 x} dx,$$

and evaluate this integral when $n = 1$ and when $n = 3$.

$$(2) \int_0^{\pi/2} \log \frac{(1 + \sin^2 x)^2}{1 + \frac{1}{4} \sin^2 2x} dx = \frac{\pi}{2} \log 2.$$

20. If $\phi(x) = -\phi(2a-x)$, show that

$$\int_b^{2a} \phi(x) dx = - \int_0^b \phi(x) dx. [\text{Colleges, 1886.}]$$

21. Prove that

$$\int_b^c \frac{\phi(x-b)}{\phi(c-x)} dx = \int_b^c \frac{\phi(c-x)}{\phi(x-b)} dx,$$

provided $\phi(x)$ remains finite when x vanishes.

[St. John's, 1883.]

22. Prove that if $\phi(x)$, $\psi(x)$, $\phi'(x)$, $\psi'(x)$ be continuous and finite from $x = a$ to $x = b$,

$$\int_a^b \phi'(x) \psi'(x) dx = \phi\{a + \theta(b-a)\}[\psi(b) - \psi(a)],$$

where θ is a positive proper fraction.

23. Prove that $\int_a^{\pi-a} x f(\sin x) dx = \frac{\pi}{2} \int_a^{\pi-a} f(\sin x) dx$.

[St. John's, 1883.]

24. Show that

$$\int_a^b f^n(x) \phi(c-x) dx - \int_a^b f(x) \phi^n(c-x) dx = \sum_{r=1}^n [f^{r-1}(x) \phi^{n-r}(c-x)]_a^b,$$

where $f^n(x)$ means the n th differential coefficient of $f(x)$.

[γ , 1893.]

25. Show that, if $\psi(x) = \int_0^x \phi(x) \phi'(2a-x) dx$,

then $\psi(2a) - 2\psi(a) = [\phi(a)]^2 - \phi(0)\phi(2a)$.

[Trinity, 1895.]

26. If $f(x, y)$ is symmetrical in x and y , prove that

$$\int_{1-b}^b x f(x, 1-x) dx = \frac{1}{2} \int_{1-b}^b f(x, 1-x) dx. [\text{Colleges } a, 1889.]$$

27. Examine under what limitations the formula

$$\int_b^a \phi(x) dx = \int_b^c \phi(x) dx + \int_c^a \phi(x) dx$$

holds good.

Show that

$$\int_{-\infty}^{\infty} \left(x + \frac{1}{x}\right) \phi\left(x - \frac{1}{x}\right) \frac{dx}{x} = 2 \int_{-\infty}^{\infty} \phi(x) dx. [\text{Math. Tripos, 1884.}]$$

28. If

$$A_n = 1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n+1}, \quad B_m = \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2m},$$

show that when n and m are both infinite and the ratio $n : m$ tends to a limit k^2 ,

$$A_n - B_m = \log 2 + \log k. [\text{Colleges } a, 1888.]$$

29. Show that

$$\frac{1}{D^n} e^{ax} \frac{\sin}{\cos}(bx + \alpha) = e^{ax} \frac{\frac{\sin}{\cos}\left(bx + \alpha - n \tan^{-1} \frac{b}{a}\right)}{(a^2 + b^2)^{\frac{n}{2}}} + A_0 + A_1x + A_2x^2 + \cdots + A_{n-1}x^{n-1},$$

A_0, A_1 , etc., being arbitrary constants, and also that it may be written

$$e^{ax} \frac{1}{(D+a)^n} \frac{\sin}{\cos}(bx + \alpha) + A_0 + A_1x + \cdots + A_{n-1}x^{n-1},$$

and explain how the latter operation is to be conducted.

30. If $I_1 = \int_0^{\pi/2} \log(1 + a_1 \sin^2 \theta) d\theta$,

show that $I_1 = \frac{\pi}{4} \log(1 + a_1) + \frac{1}{2} I_2$,

where $I_2 = \int_0^{\pi/2} \log(1 + a_2 \sin^2 \theta) d\theta$

and $4(1 + a_2)(1 + a_1) = (2 + a_1)^2$.

Hence show that

$$I_1 = \frac{\pi}{4} \log \left[(1 + a_1)(1 + a_2)^{\frac{1}{2}}(1 + a_3)^{\frac{1}{4}} \dots \right],$$

where $4(1 + a_{r+1})(1 + a_r) = (2 + a_r)^2$.

31. Show that if $n > 1$,

$$\int_0^1 \tanh \frac{1}{nx} dx < \frac{1}{n} (1 + \log n). [\text{Oxford I. P., 1911.}]$$

32. How is the equation

$$\int_a^b f'(x) dx = f(b) - f(a)$$

to be interpreted when $f(x)$ is not a single-valued function?

Illustrate your answer by evaluating

$$\int_0^{\frac{1}{2}n\pi} \frac{d\theta}{a^2 \cos^2 \theta + b^2 \sin^2 \theta},$$

where a and b are real and n is a positive integer.

[Oxford I. P., 1912.]

33. Remembering that $\int_0^\infty$ means the limit tended to by $\int_\epsilon^\eta$ as the first of the two positive quantities ϵ, η tends to zero, and the second to infinity, prove that if $a > 1$, the value of

$$\int_0^\infty (a^n e^{-ax} - e^{-x}) x^{n-1} dx$$

is zero if $n > 0$, but not if $n = 0$.

[Oxford I. P., 1917.]

34. If $f(x)$ be any function of x which can be put into partial fractions of the form $\frac{A}{a^2 - x^2}$, then will

$$\int_0^\infty \frac{f(x)}{1+x^2} dx = \frac{\pi}{2} f(\sqrt{-1}). [\text{R. P.}]$$

35. If $0 < b < a$, $a_1 = \frac{1}{2}(a+b)$, $b_1 = (ab)^{\frac{1}{2}}$,

prove that $b < b_1 < a_1 < a$, $a_1 - b_1 < \frac{1}{2}(a-b)$.

Show that if $(a+b) \tan \theta \cot \phi = a - b \tan^2 \theta$,

then

$$\int_0^{\pi/2} (a^2 \cos^2 \theta + b^2 \sin^2 \theta)^{-\frac{1}{2}} d\theta = \int_0^{\pi/2} (a_1^2 \cos^2 \phi + b_1^2 \sin^2 \phi)^{-\frac{1}{2}} d\phi. [\text{Math. Trip., Part II., 1915.}]$$

36. Show that

$$\int_0^{\pi/2} \sin \theta \tan^{-1}(\sin \theta) d\theta = \frac{\pi}{2}(\sqrt{2} - 1). [\text{Math. Trip., 1882.}]$$

37. Show how to evaluate $\int R(x, y) dx$, where $R(x, y)$ denotes any rational algebraic function of the coordinates x, y of a point on a conic. [St. John's, 1891.]

38. Show that if a be greater than unity,

$$\int_0^\pi \frac{x dx}{a^2 - \cos^2 x} = \frac{\pi^2}{2a\sqrt{a^2 - 1}}. [\text{Oxf. I. P., 1890.}]$$

39. Prove that

$$\int_0^\infty \phi(x) dx = \int_0^1 \left\{ \phi(x) + \frac{1}{x^2} \phi\left(\frac{1}{x}\right) \right\} dx. [\text{St. John's Coll., 1882.}]$$

40. Prove that $\int_0^\pi \frac{x \sin x dx}{2 + \cos 2x} = \frac{\pi}{\sqrt{2}} \tan^{-1} \sqrt{2}$. [Oxf. I. P., 1889.]

41. Integrate $\int_a^b \frac{dx}{c^2 - x^2}$, when c lies between a and b . [R. P.]

42. Prove that

$$\int_0^1 x^n (2-x)^n dx = 2^{2n} \int_0^1 x^n (1-x)^n dx. [\text{Oxf. II. P., 1886.}]$$

43. Prove that

$$\int_a^{\pi/2-a} 2 \cos^2 x \phi(\sin 2x) dx = \int_{2a}^{\pi/2} \phi(\sin x) dx. [\text{St. John's.}]$$

44. Show that

$$\theta = \int_{a-c}^r \frac{(a^2 - c^2 + r^2)}{r \sqrt{2(a^2 + c^2)r^2 - (a^2 - c^2)^2 - r^4}} dr$$

is the equation of a circle.

[Math. Trip., 1882.]

45. Find the integrals

$$(a) \int \left\{ \left(\frac{x}{e} \right)^x + \left(\frac{e}{x} \right)^x \right\} \log x dx;$$

(b) $\int \frac{dx}{x} \sqrt{x^2(x-3)^2 - 4x};$

(c) $\int \frac{dx}{\sin^3 x \tan^n \frac{x}{2}}.$ [St. John's, 1887.]

46. Prove that

$$\int_0^\alpha \log(1 + \tan \alpha \tan x) dx = \alpha \log \sec \alpha. [\text{Colls., 1896.}]$$

47. Evaluate

(i) $\int \frac{xe^x}{(x+1)^2} dx.$ [Trin., 1891.]

(ii) $\int e^x \frac{x^2 + 1}{(x+1)^2} dx.$ [Hall, I.C.]

(iii) $\int \frac{e^{x\sqrt{2}}}{1 - x\sqrt{2}} \cdot \frac{1 - x^2}{\sqrt{1 - 2x^2}} dx.$

(iv) $\int \frac{xe^x dx}{(e^x - 1)^3}.$ [Hall, I.C.]

(v) $\int \frac{dx}{(1 + x \tan x)^2}.$ [Trin., 1891.]

(vi) $\int_e^{e^2} \frac{\sec x \csc x}{\log \tan x} dx.$ [Trin., 1884.]

(vii) $\int \frac{\log(1 + x^2)}{\sqrt{1 - x}} dx.$

48. If $I_n \equiv \int_{-1}^1 (1 - x^2)^n \cos ax dx,$

show that $a^2 I_n = 2n(2n - 1)I_{n-1} - 4n(n - 1)I_{n-2}$, provided $n > 1$.

Show also that

$$I_n = \frac{n!}{a^{2n+1}} \{f(a) \sin a + g(a) \cos a\},$$

where $f(a)$ and $g(a)$ are algebraic functions of a , of degrees $\leq n$, with integral coefficients. [Trin., 1892.]

49. Show that

(i) $\int x^3 \sqrt{\frac{1+x^2}{1-x^2}} dx = \frac{1}{2} \tan^{-1} \sqrt{\frac{1+x^2}{1-x^2}} - \frac{1}{4} (2+x^2) \sqrt{1-x^4}.$

(ii) $\int \frac{x^2 + 1}{x^2 - 1} \frac{dx}{\sqrt{1 - ax^2 + x^4}} = \frac{1}{\sqrt{a-2}} \cos^{-1} \frac{x\sqrt{a-2}}{x^2 - 1} \quad (a > 2).$

[Hall, I.C., p. 325 and p. 346.]

50. Show that

$$\int_0^1 x^{cx^a} dx = 1 - \frac{c}{(a+1)^2} + \frac{c^2}{(2a+1)^3} - \frac{c^3}{(3a+1)^4} + \dots [\text{Anglin.}]$$

51. Prove that

(i) $\frac{2}{\pi} \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 + \sin^2 \theta}} = 1^2 - \left(\frac{1}{2}\right)^2 + \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2 - \dots$

$$(ii) \quad 1 + \frac{1}{2^2} \left(\frac{1}{2} \right)^2 + \frac{1}{3^2} \left(\frac{1 \cdot 3}{2 \cdot 4} \right)^2 + \dots = \frac{11}{\pi} - 4.$$

[Anglin.]

52. If $\phi(x) = a_1x + \frac{2}{3}a_3x^3 + \frac{2 \cdot 4}{3 \cdot 5}a_5x^5 + \dots$,

prove that

$$(i) \quad \int_0^{\pi/2} \cos^2 \theta \phi(\sin \theta) d\theta = \frac{1}{3}a_1 + \frac{1}{5} \left(\frac{2}{3} \right)^2 a_3 + \frac{1}{7} \left(\frac{2 \cdot 4}{3 \cdot 5} \right)^2 a_5 + \dots$$

$$(ii) \quad \frac{\pi}{2} = 1 + \frac{1}{3} \cdot 1^2 + \frac{1}{5} \left(\frac{2}{3} \right)^2 + \frac{1}{7} \left(\frac{2 \cdot 4}{3 \cdot 5} \right)^2 + \dots$$

$$(iii) \quad \pi - 3 = \frac{1}{3^2} \cdot 1^2 + \frac{1}{5^2} \left(\frac{2}{3} \right)^2 + \frac{1}{7^2} \left(\frac{2 \cdot 4}{3 \cdot 5} \right)^2 + \dots$$

[Anglin.]

Chapter X: Differentiation, etc., under an Integration Sign — Exercises

General Examples (Arts. 354–366)

1. Prove that

$$\frac{d}{da} \int_{a^q}^{a^p} a^m x^n dx = a^{m-1} \left[\left(\frac{m}{n+1} + p \right) a^{(n+1)p} - \left(\frac{m}{n+1} + q \right) a^{(n+1)q} \right],$$

and verify the result by performing the integration first.

2. If A be the area bounded by a parabola and its latus rectum ($4a$), prove

(1) by differentiating the integral $4 \int_0^a \sqrt{ax} dx$ with regard to a ,

(2) by first integrating and then differentiating with regard to a ,

that

$$\frac{dA}{da} = \frac{16a}{3}.$$

3. Apply the method of Art. 355 to prove that

$$\frac{d}{dc} \int_{\frac{c}{2}}^{\frac{c\sqrt{3}}{2}} \sqrt{c^2 - x^2} dx = \frac{1}{6}\pi c,$$

and explain geometrically each step of the process.

Obtain the same result by first integrating and then differentiating the result with regard to c ; and also geometrically.

4. Show that if $u = \int_0^\infty e^{-ax^3 - bx^2} dx$,

then

$$3ab \frac{\partial^2 u}{\partial b^2} - 3a \frac{\partial u}{\partial b} - 2b^2 \frac{\partial u}{\partial a} = 1,$$

provided a be positive.

[Trinity, 1888.]

5. Show that $\frac{d^n}{dc^n} \int_{-c}^c f(x+c) dx = 2^n f^{(n-1)}(2c)$.

[a, 1883.]

6. If $f(x+c) = f(x)$ for all values of x , show that

$$\int_0^c f(y+az) dy$$

is independent of z .

[a, 1887.]

7. Prove that

$$\int_0^{\pi/2} \frac{dx}{(a^2 \cos^2 x + \beta^2 \sin^2 x)^{n+1}} = \frac{\pi}{2^{n+1}} \frac{(-1)^n}{n!} \left(\frac{1}{a} \frac{\partial}{\partial a} + \frac{1}{\beta} \frac{\partial}{\partial \beta} \right)^n (a\beta)^{-1}.$$

8. Prove that if $u = (a^n + b^n) \int_{\beta}^{\alpha} F\{(a-b)x\} dx$,

where F denotes any function, β and α being independent of a and b , and n being a positive integer, then

$$\left\{ (a^n + b^n) \left(\frac{\partial}{\partial a} + \frac{\partial}{\partial b} \right)^n - 2 \cdot n! \right\} u = 0.$$

[Oxford, 1886.]

9. If $u = \int_0^c \phi(x, t) dt$,

where c is a function of u and x , prove that

$$\frac{du}{dx} = \frac{\int_0^c \frac{\partial \phi}{\partial x} dt + \frac{\partial c}{\partial x} \phi(x, c)}{1 - \frac{\partial c}{\partial u} \phi(x, c)}.$$

[δ , 1885.]

10. If $u = \int_{\alpha}^{\beta} \phi(x, y) dy$,

where α and β are functions of x and u , prove that

$$\frac{du}{dx} = \frac{\int_{\alpha}^{\beta} \frac{\partial \phi}{\partial x} dy + \phi(x, \beta) \frac{\partial \beta}{\partial x} - \phi(x, \alpha) \frac{\partial \alpha}{\partial x}}{1 - \phi(x, \beta) \frac{\partial \beta}{\partial u} + \phi(x, \alpha) \frac{\partial \alpha}{\partial u}}.$$

11. Comment upon the application of the rule of Art. 355 to the case

$$\frac{d}{da} \int_{-a}^a \frac{\phi(x) dx}{\sqrt{a^2 - x^2}}.$$

Prove that in this case the true result is

$$\frac{1}{a} \int_{-a}^a \frac{x \phi'(x) dx}{\sqrt{a^2 - x^2}}.$$

12. If

$$u = \int_{f(a)}^{F(a)} \phi(\theta, a) d\theta,$$

we have

$$\frac{du}{da} = \int_{f(a)}^{F(a)} \frac{\partial \phi(\theta, a)}{\partial a} d\theta + \phi\{F(a), a\} \frac{dF}{da} - \phi\{f(a), a\} \frac{df}{da}.$$

Do you consider that this formula fails in the case in which

$$F(a) = a, \quad f(a) = 0 \quad \text{and} \quad \phi(\theta, a) = \frac{1}{\sqrt{\cos \theta - \cos a}} ?$$

If so, to what extent and in what respect?

Prove that in this case

$$\frac{du}{da} = \frac{\sin a}{2\sqrt{2}} \int_0^{\pi/2} \frac{\sin^2 \theta d\theta}{\left(1 - \sin^2 \frac{a}{2} \sin^2 \theta\right)^{3/2}}.$$

Make any remarks that occur to you as to the reasons for the peculiar form which the general formula assumes in this case.

[e, 1884.]

13. Show that the equation

$$\frac{d}{dx} \int_0^1 x^3 e^{x^2(1-\frac{1}{t})} \frac{dt}{t^2} = \int_0^1 \frac{\partial}{\partial x} \left[x^3 e^{x^2(1-\frac{1}{t})} \frac{1}{t^2} \right] dt$$

ceases to hold for $x = 0$.

[Math. Tripos, 1897.]

14. Find a curve in which the abscissa of the centroid of the area of that portion bounded by the curve, the coordinate axes and an ordinate is proportional to the abscissa of the bounding ordinate.
[Colleges, 1878.]

15. A vessel in the form of a right circular cylinder with vertical axis and a flat horizontal base is filled to varying depths with liquid of varying density. If the depth of the centre of gravity of the liquid be always $\frac{1}{n}$ of the immersed portion of the axis, show that the density varies as (depth) $^{\frac{2-n}{n-1}}$.

16. Find the general equation of all solids of revolution for which the distance from the vertex of the centroid of a segment made by a plane perpendicular to the axis, is proportional to the height of the segment.
[Todhunter, *Integral Calculus*, p. 198.]

17. Find the form of the curve for which the area bounded by the curve, the coordinate axes and an ordinate is such that the moments of inertia of this area about the coordinate axes are in a constant ratio.

18. A body moves from rest at a distance a towards a centre of attraction varying inversely as the distance. Show that the time of describing the space between βa and $\beta^n a$ will be a maximum when

$$\beta n^{\frac{1}{2(n-1)}} = 1$$

$$\left[\text{It may be assumed that } \left(\frac{dx}{dt} \right)^2 \propto \log \frac{a}{x} \right]$$

[Tait and Steele, *Dynamics of a Particle*.]

19. Find the density of a parabolic plate as a function of the abscissa in order that the distance of the centroid from the vertex may vary as the square root of the length of the plate. [a , 1881.]

20. Find the equation of the curve such that the area included by the ordinate at any point, the axis of x and the curve is in a constant ratio to the area included by the ordinate, the axis of x and the tangent.
[Math. Tripos, 1882.]

21. Prove that

$$\frac{d}{da} \int_0^a \frac{F(x) dx}{\sqrt{a-x}} = \int_0^a \frac{F(x) + 2x \frac{dF(x)}{dx}}{2a\sqrt{a-x}} dx.$$

Under what circumstances will $\int_0^a \frac{F(x) dx}{\sqrt{a-x}}$ be independent of a ? [Todhunter, *Int. Calc.*]

22. If $\tan \frac{\pi}{8} \sin \phi = x \sqrt{\frac{1-x^2}{1+x^2}} = \sin \psi$,

verify that

$$\int_0^x \frac{dx}{\sqrt{1-x^8}} = \frac{1}{2\sqrt{2}} \int_0^\phi \frac{d\phi}{\sqrt{1 - \tan^2 \frac{\pi}{8} \sin^2 \phi}} + \sin^2 \frac{\pi}{8} \int_0^\psi \frac{d\psi}{\sqrt{1 - \tan^2 \frac{\pi}{8} \sin^2 \psi}}.$$

[Math. Tripos, 1896.]

23. Prove that

$$b^2 \int_0^\infty \frac{\cos bx}{a^2 + x^2} dx + 6 \int_0^\infty \frac{\cos bx}{(a^2 + x^2)^2} dx - 8a^2 \int_0^\infty \frac{\cos bx}{(a^2 + x^2)^3} dx = 0.$$

[ϵ , 1884.]

24. Verify that

$$y = A \int_0^1 v^{\beta-1} (1-v)^{\gamma-\beta-1} (1-xv)^{-\alpha} dv$$

satisfies the differential equation of the hypergeometric series, viz.

$$x(1-x) \frac{d^2 y}{dx^2} + \{\gamma - (\alpha + \beta + 1)x\} \frac{dy}{dx} - \alpha\beta y = 0,$$

when $\beta > 0$ and $\gamma > \beta$.

25. If $u = \int_0^\pi e^{qx \cos \theta} \{A + B \log(x \sin^2 \theta)\} d\theta$,

verify that

$$x \frac{d^2 u}{dx^2} + \frac{du}{dx} - q^2 xu = 0.$$

26. Prove that $y = \frac{1}{\pi} \int_0^\pi (x + \cos \phi \sqrt{x^2 - 1})^n d\phi$

satisfies the equation

$$\frac{d}{dx} \left[(x^2 - 1) \frac{dy}{dx} \right] = n(n+1)y.$$

[St. John's, 1892.]

27. Show that the differential equation

$$\frac{d^2 u}{dx^2} + u = \frac{a}{x} + \frac{b}{x^2}$$

is satisfied by

$$u = \int_0^\infty \frac{a \sin z + b \cos z}{x + z} dz.$$

Write down the complete solution.

[St. John's, 1883.]

28. If $y = \int_0^\pi \sqrt{x} e^{nx \cos \theta} \cos\{\sqrt{7} \log(\sqrt{x} \sin \theta) + \alpha\} d\theta$,

prove that

$$\frac{d^2 y}{dx^2} = \left(n^2 - \frac{2}{x^2} \right) y.$$

[St. John's, 1889.]

29. Prove that

$$\int_0^\pi (\cosh x - \sinh x \cos \phi)^{\frac{2n-1}{2}} d\phi = \int_0^\pi \frac{d\phi}{(\cosh x - \sinh x \cos \phi)^{\frac{2n+1}{2}}}.$$

Prove also that if

$$P = \int_0^\pi (\cosh x - \sinh x \cos \phi)^{\frac{1}{2}} d\phi,$$

$$\frac{dP}{dx} = \frac{1}{2} \int_0^\pi \frac{\cos \phi d\phi}{(\cosh x - \sinh x \cos \phi)^{\frac{3}{2}}}.$$

[a, 1886.]

30. If $\frac{y}{x} = \int_0^{\pi/2} \cos(mx^n \sin \phi) \cos^{\frac{1}{n}} \phi d\phi$, prove that y satisfies the equation

$$\frac{d^2 y}{dx^2} + m^2 n^2 x^{2n-2} y = 0.$$

[a, 1886.]

31. Verify that

$$y = \int e^{ux} V[A + B \log\{U_1(a_2 + b_2 x)\}] du,$$

where $U_1 = b_2 u^2 + b_1 u + b_0$, $\log VU_1 = \int \frac{a_2 u^2 + a_1 u + a_0}{U_1} du$,

and the limits are given by $e^{ux} VU_1 = 0$, satisfies the equation

$$(a_2 + b_2 x) \frac{d^2 y}{dx^2} + (a_1 + b_1 x) \frac{dy}{dx} + (a_0 + b_0 x) y = 0,$$

provided $a_1 b_2 - a_2 b_1 = b_2^2$.

[Spitzer, *Crelle*, vol. liv.]

32. Verify that if x be positive

$$u = C_1 \int_{-q}^q e^{xt} (t^2 - q^2)^{\frac{a}{2}-1} dt + C_2 \int_{-\infty}^{-q} e^{xt} (t^2 - q^2)^{\frac{a}{2}-1} dt,$$

and if x be negative

$$u = C_1 \int_{-q}^q e^{xt} (t^2 - q^2)^{\frac{a}{2}-1} dt + C_2 \int_q^{\infty} e^{xt} (t^2 - q^2)^{\frac{a}{2}-1} dt$$

solves the differential equation

$$x \frac{d^2 u}{dx^2} + a \frac{du}{dx} - q^2 x u = 0.$$

[Petzval.]

33. Prove that

$$\int_{\frac{\pi}{2}-a}^{\frac{\pi}{2}} \sin \theta \arccos \frac{\cos a}{\sin \theta} d\theta = \frac{\pi}{2} (1 - \cos a).$$

[Trinity, 1886.]

34. Prove that

$$\int_0^1 \log \frac{1+ax}{1-ax} \frac{dx}{x\sqrt{1-x^2}} = \pi \sin^{-1} a.$$

[Oxford, 1888.]

35. Establish the known result

$$\int_0^{\infty} \frac{\log x}{(a+x)^2} dx = \frac{\log a}{a},$$

and hence prove that when n is a positive integer

(a)

$$\int_0^{\infty} \frac{\log x}{(a+x)^{n+2}} dx = \frac{1}{(n+1)a^{n+1}} \left\{ \log a - \frac{1}{1} - \frac{1}{2} - \frac{1}{3} - \cdots - \frac{1}{n} \right\},$$

(β)

$$\int_0^{\infty} \frac{(\log x)^2}{(a+x)^{n+2}} dx = \frac{1}{(n+1)a^{n+1}} \left\{ \left(\frac{\pi^2}{3} - \frac{1}{1^2} - \frac{1}{2^2} - \frac{1}{3^2} - \cdots - \frac{1}{n^2} \right) + \left(\log a - \frac{1}{1} - \frac{1}{2} - \frac{1}{3} - \cdots - \frac{1}{n} \right)^2 \right\}$$

[Math. Tripos, 1883.]

36. If the operator Δ , applied to a function of a , has the effect of changing a to $a+1$, and subtracting the original function, show that

$$\Delta \int_a^b \phi(x, a) dx = \int_a^b \Delta \phi(x, a) dx,$$

where a and b are independent of a .

Prove that

$$\int_0^\infty e^{-ax}(e^{-x} - 1)^n dx = \frac{(-1)^n}{a(a+1)\dots(a+n)}.$$

[Bertrand, *C.I.*, p. 183.]

37. Given $u = \int_0^\infty \frac{\cos ax}{1+x^2} dx$, differentiating twice we have

$$\frac{d^2 u}{da^2} = - \int_0^\infty \frac{x^2 \cos ax}{1+x^2} dx.$$

But this is indeterminate when x is infinite. Discuss the validity of the differentiation. [Bertrand, *Cal. Int.*, p. 181.]

38. Is it true that

$$\int_0^1 \left[\int_0^1 \frac{a^2 - x^2}{(a^2 + x^2)^2} dx \right] da = \int_0^1 \left[\int_0^1 \frac{a^2 - x^2}{(a^2 + x^2)^2} da \right] dx ?$$

If not, why not? [See Art. 1899, Vol. II.]

Evaluate each side separately and compare the results.

[Bertrand, *Cal. Int.*, p. 187.]

39. If $P + \iota Q = \phi(x + \iota y)$, show that in general

$$\int_a^b \int_\alpha^\beta \frac{\partial Q}{\partial y} dx dy = \int_\alpha^\beta \int_a^b \frac{\partial P}{\partial x} dy dx$$

and

$$\int_a^b \int_\alpha^\beta \frac{\partial P}{\partial x} dx dy = - \int_\alpha^\beta \int_a^b \frac{\partial Q}{\partial y} dy dx.$$

Examine the case $\phi(x + \iota y) = e^{-(x+\iota y)^2}$, taking $a = 0$, $\alpha = 0$ and $b = \infty$.

[Bertrand.]

40. If $f(x) = (2-x)^{\frac{1}{2}} \int_0^{\pi/2} (1 - x \sin^2 \theta)^{-\frac{1}{2}} d\theta$, show that

$$\frac{df(x)}{dx} = \frac{1}{2}(2-x)^{-\frac{1}{2}} \int_0^{\pi/2} \cos 2\theta \left[(1 - x \cos^2 \theta)^{-\frac{3}{2}} - (1 - x \sin^2 \theta)^{-\frac{3}{2}} \right] d\theta.$$

Hence show that as x increases from 0 to 1, $f(x)$ increases from $\frac{\pi}{\sqrt{2}}$ to ∞ .

[C. S., 1898.]

41. Prove that

$$\int_0^u du \int_0^u du \int_0^u du \dots \int_0^u du f(u) = \frac{1}{(n-1)!} \int_0^u (u-z)^{n-1} f(z) dz,$$

there being n integration signs in the left member of the equality.

[R. P.]

42. Show that

$$\frac{d^2}{dc^2} \left\{ \int_0^c \int_0^c \phi(x+y+c) dx dy \right\} = 9\phi(3c) - 8\phi(2c) + \phi(c).$$

[Oxf. II. P., 1890.]

43. Show that the quartic function

$$Q \equiv ax^4 + 4bx^3 + 6cx^2 + 4dx + e$$

can, in general, be expressed in three different ways as the sum of two squares $P^2 + R^2$, where

$$P \equiv a^{-\frac{3}{2}} [(ax + b)^2 + 3(ac - b^2) - 2\lambda]$$

and

$$R \equiv a^{-\frac{5}{2}} \lambda^{-\frac{1}{2}} [2(ax + b)\lambda + a^2d - 3abc + 2b^3],$$

λ having any one of three determinate values $\lambda_1, \lambda_2, \lambda_3$.

Verify the evaluation of the integral $\int \frac{dx}{Q}$ in the form

$$\frac{a^2}{4} \left[(\lambda_2 - \lambda_3) \lambda_1^{\frac{1}{2}} \tan^{-1} \frac{R_1}{P_1} + (\lambda_3 - \lambda_1) \lambda_2^{\frac{1}{2}} \tan^{-1} \frac{R_2}{P_2} + (\lambda_1 - \lambda_2) \lambda_3^{\frac{1}{2}} \tan^{-1} \frac{R_3}{P_3} \right] / \Lambda,$$

where $\Lambda = (\lambda_2 - \lambda_3)(\lambda_3 - \lambda_1)(\lambda_1 - \lambda_2)$.

[Math. Trip., 1897.]

44. Show that

$$\int_0^a x^{n-1} (1 - a + x)^{-n-r} dx = \frac{(n-1)!}{(n+r-1)!} \left(\frac{d}{da} \right)^{r-1} \frac{a^{n+r-1}}{1-a}$$

[I. C. S., 1892.]

Chapter XI: Preliminary to Integration of $\int \frac{M}{N} \frac{dx}{\sqrt{Q}}$.

Definitions of Elliptic Functions — Exercises

Exercises — Legendre's Formulae (Art. 391)

Prove the following twelve results:

$$1. \int_0^\theta \frac{d\theta}{\Delta^3} = \frac{1}{k'^2} E(\theta, k) - \frac{k^2 \sin \theta \cos \theta}{k'^2 \Delta}.$$

[Putting $P = \frac{\sin \theta \cos \theta}{\Delta}$ and differentiating, we obtain, after a little reduction, $k^2 \frac{dP}{d\theta} = \Delta - \frac{k'^2}{\Delta^3}$, then integrating we obtain the result stated.]

$$2. \int_0^\theta \frac{\sin^2 \theta d\theta}{\Delta} = \frac{1}{k^2} (F - E).$$

$$3. \int_0^\theta \frac{\cos^2 \theta d\theta}{\Delta} = \frac{1}{k^2} (E - k'^2 F).$$

$$4. \int_0^\theta \frac{\sec^2 \theta d\theta}{\Delta} = \frac{1}{k'^2} (\Delta \tan \theta + k'^2 F - E).$$

$$5. \int_0^\theta \frac{\tan^2 \theta d\theta}{\Delta} = \frac{1}{k'^2} (\Delta \tan \theta - E).$$

$$6. \int_0^\theta \frac{\tan^2 \frac{\theta}{2} d\theta}{\Delta} = 2\Delta \tan \frac{\theta}{2} + F - 2E.$$

$$7. \int_0^\theta \frac{\sec^2 \frac{\theta}{2} d\theta}{\Delta} = 2\Delta \tan \frac{\theta}{2} + 2F - 2E.$$

$$8. \int_0^\theta \Delta \sec^2 \theta d\theta = \Delta \tan \theta + F - E.$$

$$9. \int_0^\theta \Delta \tan^2 \theta d\theta = \Delta \tan \theta + F - 2E.$$

$$10. \int_0^\theta \Delta^3 d\theta = \frac{k^2}{3} \Delta \sin \theta \cos \theta + \frac{4 - 2k^2}{3} E - \frac{k'^2}{3} F.$$

$$11. \int_0^\theta \Delta \sin^2 \theta d\theta = -\frac{1}{3} \Delta \sin \theta \cos \theta + \frac{2k^2 - 1}{3k^2} E + \frac{k'^2}{3k^2} F.$$

$$12. \int_0^\theta \Delta \cos^2 \theta d\theta = \frac{1}{3} \Delta \sin \theta \cos \theta + \frac{1 + k^2}{3k^2} E - \frac{k'^2}{3k^2} F.$$

General Examples (Arts. 367–392)

-
1. By putting $x = \frac{1 - \sin \theta}{1 + \sin \theta}$, shew that

$$u = \int_x^1 \frac{dx}{\sqrt{x(1+6x+x^2)}} = \frac{1}{\sqrt{2}} \int_0^\theta \frac{d\theta}{\sqrt{1 - \frac{1}{2} \sin^2 \theta}} = \frac{1}{\sqrt{2}} F\left(\theta, \frac{1}{\sqrt{2}}\right);$$

and that

$$x = \frac{1 - \operatorname{sn}(u\sqrt{2})}{1 + \operatorname{sn}(u\sqrt{2})}, \quad \left(\operatorname{mod.} \frac{1}{\sqrt{2}}\right); \quad \text{i.e. } u = \frac{1}{\sqrt{2}} \operatorname{sn}^{-1}\left(\frac{1-x}{1+x}, \frac{1}{\sqrt{2}}\right).$$

2. Prove that

$$F_1(\theta, k) = \frac{\pi}{2} \left(1 + \frac{1^2}{2^2} k^2 + \frac{1^2 \cdot 3^2}{2^2 \cdot 4^2} k^4 + \frac{1^2 \cdot 3^2 \cdot 5^2}{2^2 \cdot 4^2 \cdot 6^2} k^6 + \dots\right);$$

and that $F_1\left(\theta, \frac{1}{10}\right) = 1.574745$ very nearly.

3. Prove that

$$E_1(\theta, k) = \frac{\pi}{2} \left(1 - \frac{1^2}{2^2} k^2 - \frac{1^2 \cdot 3^2}{2^2 \cdot 4^2} k^4 - \frac{1^2 \cdot 3^2 \cdot 5^2}{2^2 \cdot 4^2 \cdot 6^2} k^6 - \dots\right).$$

4. Prove that

$$\Pi_1(\theta, k, n) = \frac{\pi}{2} \left[1 + \left(\frac{1}{2} k^2 - n\right) \frac{1}{2} + \left(\frac{1 \cdot 3}{2 \cdot 4} k^4 - \frac{1}{2} k^2 n + n^2\right) \frac{1 \cdot 3}{2 \cdot 4} + \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} k^6 - \frac{1 \cdot 3}{2 \cdot 4} k^4 n + \frac{1}{2} k^2 n^2 - n^3\right) \frac{1}{2} + \dots\right]$$

5. Establish the truth of

$$(a) \quad \left(\operatorname{sn} u + \frac{1}{\operatorname{cn} u}\right)^2 + \left(\operatorname{cn} u + \frac{1}{\operatorname{sn} u}\right)^2 = \left(1 + \frac{1}{\operatorname{sn} u \operatorname{cn} u}\right)^2.$$

$$(b) \quad \frac{\frac{\operatorname{cn} u}{\operatorname{sn} u} - \frac{\operatorname{sn} u}{\operatorname{cn} u}}{\operatorname{cn} u + \operatorname{sn} u} = \frac{1}{\operatorname{sn} u} - \frac{1}{\operatorname{cn} u}.$$

$$(c) \quad \left(\frac{1}{\operatorname{sn} u} - \operatorname{sn} u\right) \left(\frac{1}{\operatorname{cn} u} - \operatorname{cn} u\right) \left(\frac{\operatorname{sn} u}{\operatorname{cn} u} + \frac{\operatorname{cn} u}{\operatorname{sn} u}\right) = 1.$$

$$(d) \quad \frac{1 - \operatorname{cn} u}{1 + \operatorname{cn} u} = \left(\frac{1}{\operatorname{sn} u} - \frac{\operatorname{cn} u}{\operatorname{sn} u}\right)^2.$$

6. Prove that

$$(1) \quad \operatorname{dn}^2 u - k^2 \operatorname{cn}^2 u = k'^2,$$

$$(2) \quad \frac{1}{\operatorname{cn}^2 u} = 1 + \operatorname{tn}^2 u,$$

$$(3) \quad \frac{1}{\operatorname{sn}^2 u} = 1 + \frac{1}{\operatorname{tn}^2 u}.$$

7. Prove that

$$(1) \quad \frac{d}{du} \operatorname{sn}^2 u = 2 \operatorname{sn} u \operatorname{cn} u \operatorname{dn} u,$$

$$(2) \quad \int \operatorname{sn}^p u \operatorname{cn} u \operatorname{dn} u \, du = \frac{\operatorname{sn}^{p+1} u}{p+1},$$

$$(3) \quad \int \frac{\operatorname{dn} u}{a + b \operatorname{cn} u} \, du = \frac{1}{\sqrt{a^2 - b^2}} \cos^{-1} \left(\frac{a \operatorname{cn} u + b}{a + b \operatorname{cn} u} \right), \quad a > b.$$

8. Prove

$$\begin{aligned} 2 \operatorname{sn} u \operatorname{cn} v &= \sin(\operatorname{am} u + \operatorname{am} v) + \sin(\operatorname{am} u - \operatorname{am} v), \\ 2 \operatorname{cn} u \operatorname{cn} v &= \cos(\operatorname{am} u + \operatorname{am} v) + \cos(\operatorname{am} u - \operatorname{am} v). \end{aligned}$$

9. By putting $x = a \cos \theta$, show that

$$\int_x^a \frac{dx}{\sqrt{a^4 - x^4}} = \frac{1}{a\sqrt{2}} \operatorname{cn}^{-1} \left(\frac{x}{a}, \frac{1}{\sqrt{2}} \right).$$

10. Prove $\int_a^x \frac{dx}{\sqrt{x^4 - a^4}} = \frac{1}{a\sqrt{2}} \operatorname{cn}^{-1} \left(\frac{a}{x}, \frac{1}{\sqrt{2}} \right).$

11. By putting $x = a\sqrt{\frac{1+z}{1-z}}$, show that

$$\int_x^\infty \frac{dx}{\sqrt{a^4 + x^4}} = \frac{1}{2a} \operatorname{cn}^{-1} \left(\frac{x^2 - a^2}{x^2 + a^2}, \frac{1}{\sqrt{2}} \right).$$

12. Prove that

$$\int_x^\infty \frac{dx}{\sqrt{a^4 + 2a^2x^2 \cos 2\alpha + x^4}} = \frac{1}{2a} \operatorname{cn}^{-1} \left(\frac{x^2 - a^2}{x^2 + a^2}, \sin \alpha \right).$$

13. Prove that

$$\operatorname{sn} K = 1, \quad \operatorname{cn} K = 0, \quad \operatorname{dn} K = k', \quad \operatorname{tn} K = \infty.$$

14. Prove that

$$\begin{aligned} (1) \quad \frac{d}{du} (\operatorname{sn} u + \operatorname{cn} u)^n &= n(\operatorname{sn} u + \operatorname{cn} u)^{n-1} (\operatorname{cn} u - \operatorname{sn} u) \operatorname{dn} u, \\ (2) \quad \int (k^2 \operatorname{sn} u + \operatorname{cn} u)^n (k^2 \operatorname{cn} u - \operatorname{sn} u) \operatorname{dn} u \, du &= \frac{(k^2 \operatorname{sn} u + \operatorname{cn} u)^{n+1}}{n+1}. \end{aligned}$$

15. Draw graphs of $y = \Delta\theta$ and $y = \frac{1}{\Delta\theta}$, showing that the former consists of an undulating curve lying entirely below the line $y = 1$ and the other of an undulating line lying entirely above the line $y = 1$. Take the cases $k^2 = \frac{1}{2}$ and $k^2 = \frac{1}{4}$.

Show that the areas bounded by these curves, the x -axis, the y -axis and any ordinate at a point whose abscissa is θ represent $E(\theta)$ and $F(\theta)$ completely. Examine what happens in the limiting cases $k = 0$ and $k = 1$.

16. Show that the complete elliptic integrals of the First and Second Species may be expressed as

$$F_1 = \frac{\pi}{2} f \left(\frac{1}{2}, \frac{1}{2}, 1, k^2 \right),$$

$$E_1 = \frac{\pi}{2} f \left(-\frac{1}{2}, \frac{1}{2}, 1, k^2 \right),$$

where $f(a, b, c, x)$ is the hypergeometric series

$$1 + \frac{a \cdot b}{1 \cdot c} x + \frac{a \overline{a+1} \, b \overline{b+1}}{1 \cdot 2 \, c \overline{c+1}} x^2 + \dots$$

17. Show by differentiating $F(\theta, k)$ and $E(\theta, k)$ with regard to k

$$(1) \quad \frac{dE}{dk} = \frac{1}{k} (E - F),$$

$$(2) \quad \frac{dF}{dk} = \frac{1}{kk'^2}(E - k'^2 F) - \frac{k}{k'^2} \frac{\sin \theta \cos \theta}{\Delta}.$$

Hence, eliminating E and F alternately, show that

$$(1 - k^2) \frac{d^2 F}{dk^2} + \frac{1 - 3k^2}{k} \frac{dF}{dk} - F + \frac{\sin \theta \cos \theta}{\Delta^3} = 0,$$

$$(1 - k^2) \frac{d^2 E}{dk^2} + \frac{1 - k^2}{k} \frac{dE}{dk} + E - \frac{\sin \theta \cos \theta}{\Delta} = 0,$$

and for the complete functions F_1, E_1

$$(1 - k^2) \frac{d^2 F_1}{dk^2} + \frac{1 - 3k^2}{k} \frac{dF_1}{dk} - F_1 = 0,$$

$$(1 - k^2) \frac{d^2 E_1}{dk^2} + \frac{1 - k^2}{k} \frac{dE_1}{dk} + E_1 = 0.$$

Chapter XII: Quadrature (I). Plane Surfaces, Cartesian and Polar Equations — Exercises

Exercises — Cartesian Equations (Arts. 393–406)

1. Obtain the area bounded by a parabola and its latus rectum. A series of ordinates are drawn between the vertex and the latus rectum, parallel to the latter, viz. $x = \left(\frac{r}{n}\right)^{\frac{2}{3}} a$, where $r = 1, 2, 3, \dots, n-1$. Show that they divide the aforementioned area into n equal parts.
2. Obtain the areas bounded by the curve, the x -axis, and the specified ordinates in the following cases:
 - (a) The catenary $y = c \cosh \frac{x}{c}$, from $x = 0$ to $x = h$.
 - (b) The logarithmic curve $y = e^x$, from $x = 0$ to $x = h$.
 - (c) The logarithmic curve $y = \log_e x$, from $x = 1$ to $x = h$ ($h > 1$).
 - (d) The ellipse $y = \frac{b}{a} \sqrt{a^2 - x^2}$, from $x = \sqrt{a^2 - b^2}$ to $x = a$.
 - (e) The hyperbola $xy = k^2$, from $x = a$ to $x = b$, a and b both > 0 ; first, if the hyperbola be rectangular, second, if the angle between the asymptotes be ω .
 - (f) The curve $y = xe^{x^2}$, from $x = 0$ to $x = h$.
3. Obtain the area (1) bounded by the parabolas $y^2 = 4ax$, $x^2 = 4ay$;
(2) bounded by the parabolas $y^2 = 4ax$, $x^2 = 4by$.
In what ratio is this area divided by the common chord in each case?
4. Find the areas of the portions into which the ellipse $x^2/a^2 + y^2/b^2 = 1$ is divided (1) by the straight line $y = c$;
(2) by the two straight lines $y = c$, $x = d$, supposed to cut within the ellipse.
5. Trace the curve $x^2 y^2 = a^2 (y^2 - x^2)$, and find the whole area included between the curve and its asymptotes.
6. Find the area between the curve $y^2(a+x) = (a-x)^3$ and its asymptote.
7. Find the area of the loop of the curve

$$y^2 x + (x+a)^2 (x+2a) = 0.$$

8. Two curves in which $y \propto x^m$ and two in which $y \propto x^n$ form a quadrilateral; show that its area is

$$\frac{m-n}{(m+1)(n+1)} (x_1 y_1 - x_2 y_2 + x_3 y_3 - x_4 y_4),$$

where (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , (x_4, y_4) are the coordinates of the corners taken in order. [Trinity, 1891.]

9. By means of the integral $\int y dx$ taken round the contour of the triangle formed by the intersecting lines,

$$y = a_1 x + b_1,$$

$$y = a_2x + b_2,$$

$$y = a_3x + b_3,$$

show that they enclose the area

$$\frac{(b_1 - b_3)^2}{2(a_1 - a_3)} + \frac{(b_2 - b_1)^2}{2(a_2 - a_1)} + \frac{(b_3 - b_2)^2}{2(a_3 - a_2)}.$$

[Smith's Prize, 1876.]

10. A four-sided figure is formed by the three parabolas,

$$y^2 - 9ax + 81a^2 = 0,$$

$$y^2 - 4ax + 16a^2 = 0,$$

$$y^2 - ax + a^2 = 0,$$

and the axis of x . Prove that its area is $12a^2$, and is equal to the area enclosed by the chords of the area.

[Colleges α , 1886.]

11. Find the curvilinear area enclosed between the parabola $y^2 = 4ax$ and its evolute. [Oxf. I. P., 1889.]
12. Show that the area cut off from a semi-cubical parabola by a tangent is divided by the tangent at the cusp in the ratio 64 : 17. [Oxford II. P., 1889.]
13. (i) Find the area of a loop of the curve

$$ay^2 = x^2(a - x).$$

[I. C. S., 1882.]

- (ii) Find the whole area of the curve

$$a^2y^2 = a^2x^2 - x^4.$$

[I. C. S., 1881.]

14. Trace the curve $a^2x^2 = y^3(2a - y)$, and prove that its area is equal to that of the circle whose radius is a . [I. C. S., 1887 and 1890.]
15. Trace the curve $a^4y^2 = x^5(2a - x)$, and prove that its area is to that of the circle of radius a as 5 to 4.
16. Find the area of the curve

$$y(x^2 + 1) = x^3 - 1$$

from $x = 0$ to $x = 1$.

[St. John's, 1881.]

17. (i) Find the area between $y^2 = \frac{x^3}{a - x}$ and its asymptote.

- (ii) Show that the whole area between

$$y^2(x - a)(b - x) = c^2x^2$$

and its asymptote is $\pi c(a + b)$.

[Ox. II. P., 1903.]

- (iii) Show that the area between the curve

$$y^2x = a^3 - a^2x$$

and its asymptote is that of a circle of radius a .

[St. John's, 1889.]

18. Find the area between the axis of x , the hyperbola $x^2/a^2 - y^2/b^2 = 1$, and the line $y = x \tan \alpha$, where

$$\frac{b}{a} > \tan \alpha > 0.$$

[Ox. I. P., 1901.]

If A be the vertex, O the centre, and P any point on the hyperbola $x^2/a^2 - y^2/b^2 = 1$, prove that

$$x = a \cosh \frac{2S}{ab}, \quad y = b \sinh \frac{2S}{ab},$$

where S is the sectorial area AOP .

[Math. Tripos, 1885.]

19. Find by integration the area lying on the same side of the axis of x as the positive part of the axis of y , and which is contained by the lines

$$\begin{aligned} y^2 &= 4ax, \\ x^2 + y^2 &= 2ax, \\ x &= y + 2a. \end{aligned}$$

Express the area both when x is the independent variable and when y is the independent variable.

[Colleges, 1882.]

20. Prove that the area of the loop of

$$a(x - y)(x - 2y) = y^3 \quad \text{is} \quad \frac{a^2}{60}.$$

[Coll. β , 1891.]

21. Find the areas of the two regions of space bounded by the straight line $y = c$, and the curves whose equations are

$$\begin{aligned} x^2 + y^2 &= c^2, \\ y^3 + 4x^2 &= 4c^3. \end{aligned}$$

[I. C. S., 1891.]

22. Prove that the area contained between the curve

$$(x + 3a)(x^2 + y^2) = 4a^3$$

and its asymptote is $3a^2\sqrt{3}$.

[Oxf. I. P., 1901.]

23. Prove that the area of the curve

$$x^4 - 3ax^3 + a^2(2x^2 + y^2) = 0$$

is $\frac{3}{8}\pi a^2$.

[Math. Trip., 1893.]

24. Find the area of one loop of the curve

$$y^4 - y^2 + x^2 = 0.$$

[Colleges α , 1885.]

25. Through the cusp of the evolute of a parabola, a line is drawn perpendicular to the axis. Show that it divides the area between the parabola and the evolute in the ratio 17 : 5. [C. S., 1896.]

26. Show that the ordinate $x = a$ divides the area between $y^2(2a - x) = x^3$ and its asymptote into two parts in the ratio

$$3\pi - 8 : 3\pi + 8.$$

[Math. Trip. I., 1912.]

Exercises — Polar Coordinates (Arts. 407–414)

Find the areas bounded by

1. $r^2 = a^2 \cos^2 \theta - b^2 \sin^2 \theta$, the central pedal of a hyperbola.
2. One loop of $r = a \sin 4\theta$. Also state the total area.
3. One loop of $r = a \sin 5\theta$. Also state the total area.
4. One loop of $r = a \sin n\theta$.

Give the total area in the cases, (i) n even; (ii) n odd.

5. The portion of $r = ae^{\theta \cot \alpha}$ bounded by the radii vectores

$$\theta = \beta, \quad \theta = \beta + \gamma \quad (\gamma < 2\pi).$$

6. Any sector of $r^{\frac{1}{2}}\theta = a^{\frac{1}{2}}$ ($\theta = \alpha$ to $\theta = \beta$).
7. Any sector of the reciprocal spiral $r\theta = a$ ($\theta = \alpha$ to $\theta = \beta$).
8. The cardioide $r = a(1 - \cos \theta)$.
9. The Limaçon $r = a + b \cos \theta$, (i) if $a > b$; (ii) if $a < b$ obtain the two areas of outer and inner portions.
10. Find the area included between the two loops of the curve

$$r = a(2 \cos \theta + \sqrt{3}).$$

[Oxf. I. P., 1889.]

11. Prove that the area in the positive quadrant of the curve

$$(x^2 + y^2)^5 = (a^2 x^3 + b^2 y^3)^2 \quad \text{is} \quad \frac{1}{3}(a^2 + b^2).$$

[γ, 1899.]

12. Find the area of the closed part of the Folium

$$r = \frac{3a \sin \theta \cos \theta}{\sin^3 \theta + \cos^3 \theta}.$$

[I. C. S., 1884.]

13. Show that the area of a loop of the curve

$$ax^{2n+1} - b^2 x^n y^n + cy^{2n+1} = 0$$

is $\frac{1}{2(2n+1)} \frac{b^4}{ac}$, a and c being positive.

[Colleges, 1881.]

14. Trace the curve whose equation is

$$r^4 = a^4 \sec \theta \tan \theta,$$

and find the area between the curve and any pair of radii vectores drawn from the pole. [Trinity, 1882.]

15. Trace the lemniscate $r^2 = a^2 \cos 2\theta$ and its first positive pedal, and show that the area of a loop of the latter is double the area of a loop of the former.

Find the areas of each of the two small lozenge-shaped portions common to the two loops of the pedal.

16. Show that the area contained between the curve

$$r = a \cos 5\theta$$

and the circle $r = a$ is three-fourths of the area of the circle. [Oxf. I. P., 1888.]

17. Find the area between the curve $r = a(\sec \theta + \cos \theta)$ and its asymptote. [St. John's, 1881.]

18. Prove that the area of the curve

$$r^2(2c^2 \cos^2 \theta - 2ac \sin \theta \cos \theta + a^2 \sin^2 \theta) = a^2 c^2$$

is equal to πac . [I. C. S., 1879.]

19. Find the area of the curve

$$r = 3a \cos \theta + a \cos 3\theta.$$

[Math. Trip., 1882.]

20. Find the area of the loop of the curve

$$r^2 = a^2 \theta \cos \theta$$

between $\theta = 0$ and $\theta = \frac{\pi}{2}$.

General Problems on Quadrature (Cartesians and Polars)

1. Find the area bounded by

$$x^2 + y^2 = 4a^2, \quad x^2 + y^2 = 2ay \quad \text{and} \quad x = a.$$

[H. C. S.]

Also the area of the loop of the curve

$$by^2 = x^2(a - x)$$

(a and b both positive). [I. C. S., 1882.]

2. Find the whole area of the curve

$$y^2 = x^2 \frac{a^2 - x^2}{a^2 + x^2}.$$

[I. C. S., 1885; Colleges, 1892.]

3. A parabola $y^2 = ax$ cuts the hyperbola $x^2 - y^2 = 2a^2$ at the points P, Q ; and the tangent at P to the hyperbola cuts the parabola again at R . Find the area of the curvilinear triangle PQR .

4. Find the area included between one of the branches of the curve $x^2 y^2 = a^2(x^2 + y^2)$ and its asymptotes.

Find the whole area of the curve

$$x^4 + y^4 = a^2(x^2 + y^2).$$

[Colleges α , 1887.]

5. Trace the curve $a^2 y^2 = x^3(2a - x)$, and prove that its area is equal to that of the circle whose radius is a . [I. C. S., 1887.]

6. Prove that the whole area of

$$(x^2 + a^2)y^2 + 3a^3y + 2a^4 = 0$$

is

$$(3 - 2\sqrt{2})\pi a^2.$$

[Colleges β , 1891.]

7. Find the area of the loops of the curve

$$y^4 - x^4 - a^2y^2 + b^2x^2 = 0 \quad \text{when } b^2 > a^2.$$

[Oxford I. P., 1902.]

8. Find the area bounded by the cycloid

$$x = a(\theta + \sin \theta),$$

$$y = a(1 - \cos \theta),$$

and the straight line joining two consecutive cusps.

9. Show that the coordinates of a point P on the Folium of Descartes $x^3 + y^3 = axy$ can be expressed as

$$x = \frac{at}{1+t^3}, \quad y = \frac{at^2}{1+t^3}.$$

Show that as t varies from 0 to ∞ P traces out a closed loop, and that its area is $\frac{a^2}{6}$. [Colleges, 1896.]

10. Prove that the area of either loop of the curve

$$x^5 + y^5 - 5a^2x^2y = 0$$

is

$$\frac{2\pi a^2}{\sqrt{10 + 2\sqrt{5}}}.$$

[γ , 1893.]

11. Show that in that part of the curve $(x + y - 3c)xy + c^3 = 0$ for which x is positive, the area between the curve, the axis of x , and the ordinate which touches the curve is $\frac{1}{2}c^2$. [St. John's, 1886.]

12. Trace the curve $y^4 + x^3y = a^2x^2$,

and show that the area of the segment which lies between the axis of y and the straight line whose equation is $y = x$ is $\frac{1}{6}a^2 \log 2$. [Colleges ϵ , 1883.]

13. Pairs of ordinates of the hyperbola $xy = a^2$ are determined by the condition that the area included by any pair, the curve, and the x -axis is constant; show that the lengths of any such pair are in a constant ratio. [Oxford I. P., 1888.]

14. Show that the area between the curve

$$x(x^2 + y^2 - a^2) + \frac{2}{9}a^3\sqrt{3} = 0$$

and its asymptote is πa^2 .

[St. John's, 1892.]

15. Show that the area between the inner branch of the curve

$$(x^2 + y^2 - a^2)^2 = \frac{1}{4}x^2(x^2 + y^2)$$

and the positive parts of the two axes is $\pi a^2/3\sqrt{3}$. [St. John's, 1888.]

16. Prove that the whole area of the epicycloid generated by a point on a circle of radius $\frac{a}{4}$ rolling on a fixed circle of radius a is to the area of the fixed circle in the ratio of 15 to 8.

17. Find the whole area of the curve whose equation is

$$(x^2 + y^2)(x + y + a)(x + y - a) + x^2y^2 = 0.$$

[Colleges, 1886.]

18. Find the area of a loop of the curve

$$x^4 + y^4 = 2a^2xy.$$

[Oxford I. P., 1888.]

19. Find the area cut off from an ellipse by a focal chord.

[Colleges α , 1883.]

20. Prove that the areas cut off by the equiangular spiral $r = ae^{\theta \cot \alpha}$ from the space bounded by any two fixed lines through the pole are in geometrical progression. [Oxford I. P., 1900.]

21. Find the area of the curve $r = a\theta e^{b\theta}$ enclosed between two given radii vectores and two successive branches of the curve. [Trinity, 1881.]

22. Find the area of the loop of the curve $r = a\theta \cos \theta$ between $\theta = 0$ and $\theta = \frac{\pi}{2}$. [Oxford II. P., 1890.]

23. Find the area of the curve

$$(r - a \cos \theta)^2 = a^2 \cos 2\theta.$$

[Colleges α , 1887.]

24. Show that the area of the loop of the folium $x^3 + y^3 = 3axy$ is divided by the parabola $y^2 = ax$ in the ratio 5 : 4.

In what ratio does the line $x + y = 2a$ cut the loop in the above folium. [Oxford I. P., 1889.]

25. Find the area included between the axis of y and the curve

$$y^2 + 2y - 2x(y + 1) = x^4 - 3x^3 + 3,$$

the curve being supposed to stop at the node. [St. John's, 1884.]

26. Determine by integration the area of the ellipse

$$x^2 + xy + y^2 = 1.$$

27. (i) Find the whole area enclosed by the hypocycloid

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}.$$

[Oxford I. P., 1888.]

- (ii) Prove that the area of the locus of intersection of pairs of tangents at right angles for this curve is $\frac{1}{4}\pi a^2$. [Math. Tripos, 1888.]

28. Prove that the locus of the points of bisection of the intercepts on the normals of a cycloid between the cycloid and its base divides the area between the cycloid and its base into two parts in the ratio 7 : 5. [Oxford II. P., 1886.]

29. Trace the curve $x^{2n+1} + y^{2n+1} = (2n+1)ax^ny^n$, when n is even, and when n is odd, n being a positive integer; and prove that the area of the loop is $(2n+1)\frac{a^2}{2}$. Prove that this is also the area between the infinite branches of the curve and the asymptote. [St. John's, 1882.]

30. Find the whole area contained between the curve

$$x^2(x^2 + y^2) = a^2(y^2 - x^2)$$

and its asymptotes.

[Oxford I. P., 1887.]

31. Find the area bounded by the circle $x = a \cos \theta$, $y = a \sin \theta$ and the hyperbola $x = b \cosh u$, $y = b \sinh u$; that area being taken which lies within the circle and on the convex side of the hyperbola, and b being less than a . [Trinity, 1888.]

32. (a) Show that in the Archimedean Spiral $r = a\theta$, if $A_1, A_2, A_3, A_4, \dots$ be the areas of the inner loop and the successive heart-shaped figures formed by the convolutions of the curve

$$A_1 = \frac{\pi^2 a^2}{4}, \quad A_{n+1} = 2n\pi^2 a^2.$$

- (b) In the Reciprocal Spiral $r\theta = a$, if $A_1, A_2, A_3 \dots$ be the areas of the successive closed loops,

$$A_n = \frac{4a^2}{\pi} \frac{1}{4n^2 - 1}.$$

33. Find the area of the loop of the curve

$$(x+y)(x^2+y^2) = 2axy.$$

[Oxford I. P., 1890.]

34. At all points of the first negative pedal of the curve $r = \cosh(m\theta \cot \alpha)$ lines are drawn making a constant angle α with the tangent. Show that the area bounded by any pair of such lines, the curve enveloped and the first negative pedal is

$$A\{1 + (m^2 - 1) \cos^2 \alpha\},$$

where A is the area of the corresponding portion of the first negative pedal bounded by radii vectores from the pole. [Colleges α , 1891.]

35. Find the area of that portion of the loop of the curve

$$r^2 = p \cos \theta + q \sin \theta,$$

which is not enclosed by the curve

$$r^2 = b + a \cos \theta.$$

If a family of such curves be taken (by varying p and q), such that this area is constant, show that the envelope of the system is a curve whose equation is

$$r^2 = c + a \cos \theta.$$

[Colleges β , 1889.]

36. Show that the whole area enclosed by the outer line of the curve $r^{\frac{2}{3}} = a^{\frac{2}{3}} \cos \frac{2}{3}\theta$ is $\frac{9}{8}a^2\sqrt{3}$.
[Colleges, 1876.]

37. In a hyperbola, C is the centre, A the end of the transverse axis and P any point (x, y) on the same branch of the curve as A ; prove that twice the area of the sector CAP is

$$ab \log \left(\frac{x}{a} + \frac{y}{b} \right).$$

38. Show that the area contained between a hyperbola, any tangent and a line parallel to the asymptote which bisects the part of the tangent intercepted between the curve and the asymptote

$$= \frac{ab}{2} \left(\log 2 - \frac{5}{8} \right),$$

and is constant.

[Trinity, 1886.]

39. Prove that the area of the curve

$$x = \frac{ap}{(1+p^2)^2}, \quad y = \frac{1}{2} \frac{ap^2(1-p^2)}{(1+p^2)^2}$$

is $\frac{1}{32}\pi a^2$.

[Math. Tripos, 1882.]

40. Show that the area cut off from the ellipse

$$ax^2 + 2hxy + by^2 = 1$$

by the line $lx + my = 1$ is

$$\alpha\beta(\theta - \sin\theta \cos\theta),$$

where α, β are the semiaxes of the ellipse and

$$\cos\theta = \frac{\sqrt{ab - h^2}}{\sqrt{am^2 + bl^2 - 2hlm}}.$$

[Colleges, 1892.]

41. Trace the curve whose equation is

$$x(x^{2n} + y^{2n}) = a^2 y^{2n-1},$$

and prove that the area between the curve, the axis of x and a tangent parallel to the axis of y is

$$\frac{a^2}{4n}(2n - 1 - \log 2n).$$

[St. John's, 1885.]

42. Show that in the curve

$$r^2 = \sec 2\theta \log(2 \cos^2 \theta)$$

the area between the curve and the lines $\theta = \pm \frac{1}{4}\pi$ is $\left(\frac{1}{4}\pi\right)^2$.

[St. John's, 1886.]

43. Find in integral form, and completely, the area enclosed between two confocal conics and two given radii from the centre.

[Trinity, 1881.]

44. Prove that the area of each of the two equal and similar pieces of the ellipse $x^2/a^2 + y^2/b^2 = 1$ which are cut off by the hyperbola $x^2/\alpha^2 - y^2/\beta^2 = 1$ ($\alpha < a$) is

$$ab \sin^{-1} \frac{\beta(a^2 - \alpha^2)^{\frac{1}{2}}}{(a^2\beta^2 + \alpha^2b^2)^{\frac{1}{2}}} - \alpha\beta \sinh^{-1} \frac{b(a^2 - \alpha^2)^{\frac{1}{2}}}{(\alpha^2\beta^2 + a^2b^2)^{\frac{1}{2}}}.$$

[St. John's, 1887.]

45. Prove that the areas of the two loops of the curve

$$r^2 - 2ar \cos \theta - 8ar + 9a^2 = 0$$

are

$$(32\pi + 24\sqrt{3})a^2 \quad \text{and} \quad (16\pi - 24\sqrt{3})a^2.$$

[Math. Tripos, 1875.]

46. The area between two tangents to the same convolution of an equiangular spiral at right angles to one another, and the curve, is

$$pp' + \frac{1}{2}(p^2 - p'^2) \cot 2\gamma,$$

where p, p' are the perpendiculars from the pole on the tangents and γ is the angle of the spiral. [Colleges, 1882.]

47. A circle with centre at the origin cuts the loop of the Folium $x^3 + y^3 - 3axy = 0$. If the angle subtended at the origin by the common chord equals

$$2 \tan^{-1} \frac{2^{\frac{1}{3}} - 1}{2^{\frac{1}{3}} + 1},$$

prove that the area between the loop and the circle is

$$\frac{a^2}{2} \left[1 - (2^{\frac{2}{3}} + 2^{\frac{4}{3}}) \tan^{-1} \frac{2^{\frac{1}{3}} - 1}{2^{\frac{2}{3}}} \right].$$

[Colleges, 1885.]

48. The centre of a circle of constant radius a moves along a fixed straight line AB in its plane, and from A a fixed point in the line a tangent AP is drawn to the circle. Show that the area included between the locus of P and its asymptotes is πa^2 . [Math. Tripos, 1882.]

49. Show that the curve

$$r = a \left(\frac{1}{2}\sqrt{3} + \cos \frac{\theta}{2} \right)$$

has three loops, whose areas are

$$a^2 \left(\frac{5}{4}\pi + 2\sqrt{3} \right), \quad a^2 \left(\frac{5}{6}\pi - \frac{5}{4}\sqrt{3} \right), \quad a^2 \left(\frac{5}{12}\pi - \frac{3}{4}\sqrt{3} \right)$$

respectively.

[Colleges, 1892.]

50. Show that the area of the Cassinian

$$r^4 - 2a^2r^2 \cos 2\theta + a^4 = b^4$$

is

$$2 \int_0^{\pi/2} \sqrt{b^4 - a^4 \sin^2 \phi} d\phi, \quad \text{provided } b > a,$$

but is

$$2 \int_0^{\pi/2} \frac{b^4 \cos^2 \phi d\phi}{\sqrt{a^4 - b^4 \sin^2 \phi}}, \quad \text{when } a > b.$$

[Math. Tripos, 1883.]

51. Prove that the area of the first negative pedal of an ellipse with respect to the focus is

$$\frac{\pi a^2(2 - 3e^2)}{2\sqrt{1 - e^2}} \quad (e < \tfrac{1}{2}),$$

where a and e are the semi major axis and the eccentricity of the ellipse. [Colleges, 1892.]

How do you interpret this result if $e < \frac{1}{2}$?

52. Find the area of the curve whose Cartesian equation is

$$a^2(y - x)^2 = (a + x)^3(a - x).$$

[Math. Tripos, 1896.]

53. Find the value of $\int_0^1 v_x dx$, v_x being the real root of the cubic

$$v_x^3 + v_x^2 v_1 + v_x v_1^2 - \frac{c}{x} = 0.$$

[Colleges, 1872; R. P.]

54. Find the area in the first quadrant bounded by the axes of coordinates and the curve

$$\sinh^{-1} \frac{x}{a} + \sinh^{-1} \frac{y}{b} = c,$$

taking a, b, c all positive.

[I. C. S., 1897.]

55. Trace the whole curve

$$x^2 y^2 = c^2(a - x)(x - b),$$

where $0 < b < a$, and find its whole area.

[I. C. S., 1898.]

56. It is given that the abscissa ON and ordinate NP of a point on any arch of a cycloidal arc are $a(\theta - \sin \theta)$ and $a(1 - \cos \theta)$. NP is produced to K so that $NK = 2a$, and the rectangle $ONKA$ is completed. Prove that the area included by ON , NP and the arc OP never differs from three-fourths of $ONKA$ by more than

$$\frac{3a^2}{8}\sqrt{3};$$

and find for what positions of P the difference vanishes.

[I. C. S., 1912.]

57. Trace on squared centimetre paper the curves

$$x^4 + y^4 = 4a^2 xy,$$

$$x^4 + y^4 = 4ax^2 y,$$

taking $a = 10$ cm., and estimate the area of a loop of each curve.

Prove that

$$\int_0^\infty \frac{t^2}{(1 + t^4)^2} dt = \frac{1}{4} \int_0^\infty \frac{t^2}{1 + t^4} dt = \frac{\pi}{8\sqrt{2}},$$

and hence calculate the area of a loop of the second curve. Find also the area of a loop of the first curve. Give each area to the nearest square centimetre when a is 10 centimetres. [C. S., 1913.]

58. Obtain the area contained between the two curves

$$r^2 \cos 2\theta = 4a^2 \cos^4 \theta \quad \text{and} \quad r^2 \cos 2\theta = a^2.$$

[Oxf. I. P., 1912.]

59. Show that the area of the loop of the curve

$$x^7 + y^7 = ax^3y^3$$

is equal to $a^2/14$.

[Oxf. I. P., 1914.]

60. Prove by any method that the area of the ellipse

$$\{a(x-2) + 3y\}^2 + 4(x+1)(x-2) = 0$$

is independent of a , and find the area.

Prove also that the straight line $y = x$ divides the ellipse $x^2 + 3y^2 = 6y$ into two areas which are in the ratio

$$4\pi - 3\sqrt{3} : 8\pi + 3\sqrt{3}.$$

[Oxf. I. P., 1916.]

61. Trace the curve

$$r \cos \theta = a \sin 3\theta,$$

and show that the area of a loop is

$$\frac{1}{8}a^2(9\sqrt{3} - 4\pi).$$

[Math. Trip. I., 1919.]

62. Show that the curve $r = a(2 \cos \theta + \cos 3\theta)$ has three loops, the area of the larger loop being $\frac{10\pi + 9\sqrt{3}}{12}a^2$, and the areas of the two smaller loops being $\frac{5\pi - 9\sqrt{3}}{24}a^2$. [Math. Trip. I., 1916.]

63. Show that the coordinates of any point on the curve

$$y^2(a+x) = x^2(3a-x)$$

may be taken as

$$x = a \sin 3\theta / \sin \theta, \quad y = a \sin 3\theta / \cos \theta,$$

and prove that the area of the loop and the area between the curve and its asymptote are both equal to $3\sqrt{3}a^2$. [Math. Trip. I., 1915.]

64. Show that the area of the loop of the curve

$$(x^2 + y^2)^2 - 4axy^2 = 0$$

in the positive quadrant is $\frac{1}{4}\pi a^2$.

[Math. Trip. I., 1920.]

65. Having established Simpson's Rule, that if

$$y = y(x) \equiv a_0 + a_1x + a_2x^2 + a_3x^3,$$

then

$$\int_0^1 y \, dx = \frac{1}{6} \{y(0) + y(1) + 4y(\frac{1}{2})\},$$

prove that if $y(x)$ also contains a term a_4x^4 the error in still using Simpson's Rule is

$$\frac{1}{120}a_4.$$

[Math. Trip. I., 1920.]

Chapter XIII: Quadrature (II). Tangential Polars, Pedal Equations and Pedal Curves, Intrinsic Equations, etc. — Exercises

Problems on Quadrature (Arts. 416–450)

1. Interpret geometrically $\int_{p_0}^{p_1} \sqrt{r^2 - p^2} dp$ in the case of the curve $r = f(p)$.

Prove that the value of $\int \sqrt{r^2 - p^2} dp$, taken all round an ellipse whose semiaxes are a, b , and whose centre is the pole, is $\pi(a - b)^2$. [Oxford I. P., 1903.]

2. Use the pedal equation of an ellipse, viz. $\frac{a^2 b^2}{p^2} = a^2 + b^2 - r^2$, to show that the area of the portion of an ellipse included between the curve, the semi-major axis and a central radius vector r , is

$$\frac{ab}{2} \tan^{-1} \sqrt{\frac{a^2 - r^2}{r^2 - b^2}},$$

a, b being the semiaxes of the ellipse.

[Colleges, 1882.]

3. Find the area of the part of the ellipse $p^2(2a - r) = b^2 r$ included between two focal radii vectores drawn, one to an extremity of the minor axis and the other to the nearer extremity of the major axis. [Oxford I. P., 1889.]
4. Find the area included between an ellipse and its evolute and bounding radii of curvature, the one coinciding with the major axis and the other inclined at an angle of $\frac{\pi}{4}$ to it. [Colleges, 1884, and β , 1888.]
5. Through every point of an ellipse a line is drawn outwards normal to the ellipse and equal to the radius of curvature at the point. Show that the area of the curve thus obtained is

$$\pi \frac{9a^4 + 14a^2 b^2 + 9b^4}{2ab}.$$

[Colleges α , 1891.]

6. Show that the area of that part of the evolute of an ellipse (eccentricity $> \frac{1}{\sqrt{2}}$) which lies outside the ellipse is

$$a^4 b^4 \int_{b^2}^{\frac{a^2+b^2}{3}} \frac{(a^2 + b^2 - 3\rho)^2}{\rho^3 (a^2 + b^2 - \rho)^2} \frac{d\rho}{\sqrt{(\rho - b^2)(a^2 - \rho)}}.$$

[Colleges, 1882.]

7. Find the area of the pedal of the curve

$$(ax)^{\frac{2}{3}} + (by)^{\frac{2}{3}} = (a^2 - b^2)^{\frac{2}{3}},$$

the origin being taken at $x = \sqrt{a^2 - b^2}$, $y = 0$.

[Oxford I. P., 1888.]

8. Show that the area of the space between the epicycloid $p = A \sin B\psi$ and its pedal curve taken from cusp to cusp is $\frac{1}{4}\pi A^2 B$. [Colleges, 1878.]

9. Show that the area between an epicycloid and the arc of the fixed circle included between two consecutive cusps is

$$\frac{\pi b^2}{a}(3a + 2b),$$

where a and b are the radii of the fixed and rolling circles respectively. [Colleges α , 1884.]

Show also that the area of the corresponding sector of the fixed circle is that of an ellipse with semi-axes the radii of the two circles. [Oxford I. P., 1913.]

10. Show that the p - ψ equation to a cycloid when one of the cusps is taken as origin is

$$p = 2a(\sin \psi - \psi \cos \psi),$$

where a is the radius of the generating circle; and find the area between the curve from cusp to cusp and the corresponding arc of the pedal with regard to a cusp. [Oxford II. P., 1903.]

11. Show that the area bounded by that portion of the cardioide $r^{\frac{1}{2}} = a^{\frac{1}{2}} \sin \frac{1}{2}\theta$, which lies in the first quadrant, the terminal tangents, and the corresponding portion of the locus of the extremity of the polar subtangent, is

$$3a^2(10 - 3\pi)/16.$$

[Math. Tripos, 1896.]

12. Show that in the curve in which the area bounded by the curve and the radii vectores from a certain fixed point varies as the square of the length of the bounding arc, the radius of curvature varies as the projection of the radius vector on the tangent. [Colleges α , 1891.]

13. The pedal of a cycloid with regard to any point on its axis meets the cycloid at the vertex A and cuts the tangent at the cusp in Q ; find the area between it and the chord AQ ; and prove that this area is least when the origin is the middle point of the axis. [St. John's, 1883.]

14. An elliptic wire is pushed in one plane through a very short straight tube; find the equation to the locus of the centre, and prove that the area of each loop is $\frac{\pi}{2}(a - b)^2$, where a and b are the semi-axes. [Colleges, 1886.]

15. A point Q is taken on the normal drawn outward at a point P of a catenary, the parameter of which is c . Prove that if PQ is equal to the length of the arc of the catenary measured from the vertex to P , the area between the locus of Q and the catenary, and bounded by the normal at the vertex and by another normal inclined at an angle ψ to this, is

$$\frac{c^2}{2}(\tan^2 \psi + \tan \psi - \psi).$$

[Colleges γ , 1882.]

16. Prove that the pedal of the cardioide $r = a \cos^2 \frac{\theta}{2}$ with respect to the cusp consists of two closed regions of areas A and B , A consisting of the inner loop and B being external to A and bounded by the outer line of the curve and such that

$$2A + B = \frac{15\pi a^2}{32}.$$

[Colleges γ , 1899.]

17. Prove that the area of the pedal of the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ with respect to the point $(a, 0)$ is five times as great as the area of its pedal with respect to the origin. [Oxford II. P., 1899.]

18. The tangent at a point P of a lemniscate cuts the curve again at Q, R . Prove that the middle point of QR is at the same distance from the nodal point as P ; and that the equation to its locus is

$$a^{10}(x^2 - y^2) = r^4\{a^8 + 4(a^4 - r^4)(a^4 - 4r^4)\},$$

where $r^2 \equiv x^2 + y^2$.

Show that it can be written $r^2 = a^2 \cos \frac{2}{5}\theta$.

Trace the curve completely, and prove that the portion corresponding to the upper half of one branch of the lemniscate divides the other branch into two parts whose areas are in the ratio of

$$6 - 3\sqrt{3} : 3\sqrt{3} - 4.$$

[St. John's, 1884.]

19. Show that the area of a loop of the curve

$$(x^2 - a^2)^2 + (y^2 - 3a^2)^2 = a^4$$

is

$$a^2\sqrt{2} \left(\frac{\pi}{3} - \log_e \frac{\sqrt{3}+1}{2} \right).$$

[Math. Tripos, 1882.]

20. The tangent at every point P of a closed finite curve is produced to Q so that PQ is constant. Find the area between the locus of Q and the original curve. How is the result to be explained, (i) if the curvature of the first curve is sometimes in one direction, sometimes in the opposite direction; (ii) if the curve cuts itself a given number of times. [St. John's Coll., 1881.]

21. A straight line of constant length c is drawn from each point of a closed oval curve making a given angle α with the normal at that point. Prove that the area of the curve traced out by the end of the line is

$$S + \pi c^2 \pm lc \cos \alpha,$$

where S is the area of the given oval curve and l is its length.

[Coll. γ , 1893.]

22. Show that the area of the polar reciprocal of a curve whose equation is given in rectangular coordinates is

$$\frac{1}{2}k^4 \int \frac{\frac{d^2y}{dx^2}}{\left(y - x \frac{dy}{dx}\right)^2} dx,$$

x, y being the coordinates of a point on the original curve.

Apply this to find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

[Colleges, 1886.]

23. The area of a given closed oval curve is A ; the bisectors of the internal and external angles between tangents to it which meet at a given constant angle 2α envelop curves whose areas are A_1 and A_2 ; show that

$$A_1 \cos^2 \alpha + A_2 \sin^2 \alpha = A.$$

[Colleges γ , 1888.]

24. Prove that for any closed curve which has a centre, the area of the locus of intersection of tangents at right angles, and the area of the locus of intersection of normals at right angles differ by twice the area of the curve.

[Math. Tripos, 1888.]

25. O being a fixed point, OP a radius vector of any curve, OP is produced to Q so that $OP.PQ = a^2$, and A is the area between the locus of Q and the given curve. If A' be the area of the inverse of the curve with respect to O , the constant of inversion being a , show that $A' - A$ is independent of the form of the curve.

If the given curve be a circle, and O a point on its circumference, find the area of any part bounded by the locus of Q , the circle and two radii vectores from O . [St. John's, 1891.]

26. A circle rolls on the outside of an oval curve, the pedals of the curve, of the locus of the centre of the circle and of the envelope of the circle are of areas A_0 , A_1 , A_2 , respectively; prove that $A_2 - 2A_1 + A_0$ depends only on the rolling circle.

Show that if the area of the oval curve, of the locus of the centre of the circle and of the envelope of the circle be S_0 , S_1 , S_2 respectively,

$$A_2 - 2A_1 + A_0 = S_2 - 2S_1 + S_0.$$

[Trinity, 1878.]

27. One of the curves given by the equation

$$y = a^2 \frac{d}{ds} \left\{ \frac{dy}{dx} + \frac{1}{3} \left(\frac{dy}{dx} \right)^3 \right\}$$

cuts the axis of x twice at the angle α . Prove that the area between the curve and the axis is

$$a^2 \{ \tan \alpha \sec \alpha + \log(\sec \alpha + \tan \alpha) \}.$$

[Oxf. I. P., 1912.]

28. A curve concave to the axis of x is such that the product of the ordinate and radius of curvature at any point is constant and equal to c^2 (The *Elastica*, or *Bent Bow*). Prove that the maximum value of the ordinate is $2c \sin \frac{\alpha}{2}$, where α is the angle at which the curve crosses the axis of x . [Ox. I. P., 1903.]

Show that the area which lies between the bow and the bow-string is $2c^2 \sin \alpha$.

29. Show that the area of a closed curve, which is the envelope of the line $x \cos \psi + y \sin \psi = p$, is the value of the integral

$$-\frac{1}{2} \int \left(\frac{dp}{d\psi} + p \right) \left(\frac{dp}{d\psi} - p \right) d\psi$$

taken completely round the curve.

[Math. Trip., 1898.]

30. The integral $-\frac{1}{2} \int \left(\frac{dp}{d\psi} + np \right)^2 d\psi$ is taken round a closed curve, n being taken equal to $\tan \psi$ or to $-\cot \psi$, according as the one or the other is numerically less than unity. Show that the value of the integral differs from the area of the curve by the sum of the squares of the perpendiculars from the origin upon the tangents at the points where the integral changes form. [Math. Trip., 1898.]

31. In the cycloid prove that the conic locus of points with regard to which the area of the pedal is constant, is in general a circle, and find the point for which the area of the pedal is a minimum. [Ox. I. P., 1900.]

32. In a catenary, A is the vertex, P any point on the curve, AO , PN perpendiculars upon the directrix, PY a tangent and NY perpendicular to it. Show that the area of the figure $ONPA$ is double that of the triangle YNP .

33. Show that the area of the first positive pedal of the curve $p = f(r)$ may be obtained by the formula

$$\frac{1}{2} \int \frac{p^2}{\sqrt{r^2 - p^2}} \frac{dp}{dr} dr,$$

where the letters p and r are the pedal coordinates of a point on the original curve.

Apply this method to find the area of the cardioide, which is the first positive pedal of the circle $r^2 = ap$.

34. Employ the formula

$$\frac{1}{2} \int \frac{pr}{\sqrt{r^2 - p^2}} dr$$

to find the area of

$$r^2 + a^2 = b^2 + 2ap \quad (a > b).$$

To what curve does this pedal equation belong?

35. In the epicycloid

$$p^2 = a^2 \frac{r^2 - a^2}{c^2 - a^2},$$

where a and $\frac{c-a}{2}$ are the radii of the fixed and rolling circles respectively, obtain a formula for the area of any sectorial portion with centre of the sector at the origin. Hence show that the area between one foil of the curve and the fixed circle is

$$\pi(c-a)^2(c+2a)/4a.$$

36. When $a < b$ the conchoid of Nicomedes, viz.

$$x^2 y^2 = (a+y)^2 (b^2 - y^2) \quad \text{or} \quad r = a \csc \theta \pm b$$

has a loop. Find its area.

37. Let S be the focus of a parabola, SP_1 , SP_2 two focal radii vectores of lengths r_1 , r_2 . The latus rectum is $4a$ and $P_1P_2 = c$. Prove Lambert's expression for the sectorial area SP_1P_2 , viz.

$$\frac{\sqrt{a}}{3} \left[s^{\frac{3}{2}} - (s-c)^{\frac{3}{2}} \right],$$

where $2s = r_1 + r_2 + c$.

Show that the segment cut off by a focal chord of length c is

$$\frac{1}{3} a^{\frac{1}{2}} c^{\frac{3}{2}}.$$

38. In the case of the Cotes's spirals, whose equations are of the form

$$\frac{1}{p^2} = \frac{A}{r^2} + B,$$

show that the area of the sectorial portion bounded by the curve and the radii vectores r_1 and r_2 is

$$\frac{1}{2B} \left\{ (Br_1^2 + A - 1)^{\frac{1}{2}} \sim (Br_2^2 + A - 1)^{\frac{1}{2}} \right\}, \quad B \neq 0.$$

Examine in detail the particular cases of

- (i) the equiangular spiral;

(ii) the reciprocal spiral;

(iii), (iv) and (v) the cases which reduce to the polar forms,

$$u = a \sinh n\theta, \quad u = a \cosh n\theta, \quad u = a \sin n\theta, \quad \text{respectively.}$$

39. Riccati's *Syntractory* is generated as follows. The tractory is an involute of a common catenary of parameter c , starting from the vertex. PT is a tangent at any point P of the tractory, cutting the directrix of the catenary at T . Q is a point on PT or PT produced such that $QT = c'$. The locus of Q is the syntractory.

Show that the areas between the two branches and the directrix are

$$\frac{\pi}{2}c'(2c \pm c').$$

40. If A be the area of the 'Helmet,'

$$(k+1)\{(x^2 + ka^2)y^2 - 2ay(a^2 - x^2)\} + (a^2 - x^2)^2 = 0, \quad (k \neq -1),$$

and V the volume formed by its revolution about the y -axis, prove that

$$A = \frac{\pi a^2}{\sqrt{k(k+1)}} \left[2(k+1)^{\frac{3}{2}} - (2k+3)k^{\frac{1}{2}} \right],$$

$$V = \frac{2\pi a^3}{3\sqrt{k+1}} \left[3(k+1)^{\frac{3}{2}} \log \frac{\sqrt{k+1} + 1}{\sqrt{k+1} - 1} - 2(3k+4) \right].$$

[For the first part of the example, and for several others of similar character, see Wolstenholme's *Problems*, Nos. 1866 to 1870.]

Chapter XIV: Quadrature, etc. (III). Surface Integrals, Arels, Corresponding Curves — Exercises

Exercises — Surface Integrals, Centroids and Moments of Inertia (Arts. 451–459)

1. In the first quadrant of the circle $x^2 + y^2 = a^2$ the surface density varies at each point as xy .
Find (i) the mass of the quadrant,
(ii) its centroid,
(iii) its moment of inertia about the y -axis.
2. Work out the corresponding results for the portion of the parabola $y^2 = 4ax$ bounded by the axis and the latus rectum, the surface density varying as $x^p y^q$.
3. Find the centroid of a fine rod of uniform sectional area and of which the line-density varies as the n th power of the distance from one end. Also its moment of inertia about that end, about the other end, and about the middle point.
4. Find the centroid of the triangle bounded by the lines $y = mx$, $x = a$ and the x axis when the surface density at each point varies as the square of the distance from the origin. Also find the moment of inertia about the y -axis.

5. Find the centroid of

- (i) either of the areas bounded by the circle $(x - a)^2 + y^2 = a^2$ and the parabola $y^2 = ax$;
- (ii) the centroid of the area bounded by the parabolas

$$y^2 = 4ax, \quad x^2 = 4by;$$

- (iii) the centroid of the area bounded by

$$y^2 = 4ax, \quad y = 2x,$$

the surface density being uniform in each case.

6. Find the moment of inertia of a triangle of uniform surface density
 - (i) about one of its sides;
 - (ii) about an axis perpendicular to its plane through an angular point.
7. Find the area remote from the pole between the circles

$$r = a, \quad r = 2a \cos \theta;$$

and assuming a surface density varying inversely as the distance from the pole, find

- (1) the centroid;
 - (2) the moment of inertia about a line through the pole perpendicular to the plane.
8. Find for the area included between the curves

$$y^2 = 4ax, \quad x^2 = 4ay,$$

- (i) the moment of inertia about the x -axis;
- (ii) the moment of inertia about an axis through the origin and at right angles to the plane of the area.
9. Find the coordinates of the centroid of the area bounded by the catenary $y = c \cosh \frac{x}{c}$, an ordinate, and the coordinate axes.
10. If the density at any point of a circular disc whose radius is a vary directly as the distance from the centre and a circle described on a radius as diameter be cut out, prove that the centroid of the remainder will be at a distance $\frac{6a}{5(3\pi - 2)}$ from the centre. [Math. Trip., 1875.]

Exercises — Corresponding Points and Areas (Arts. 460–465)

1. Find the whole area of a loop of each of the curves

(i) $x(x^2 + y^2) = a(x^2 - y^2)$.

(ii) $(m^2x^2 + n^2y^2)^2 = a^2x^2 - b^2y^2$.

[St. John's, 1887.]

2. Trace the shape of the following curves, and find their areas:

(i) $(x^2 + y^2)^3 = axy^4$.

(ii) $(x^2 + 2y^2)^3 = axy^4$.

[Barnes Scholarships, 1887.]

3. Prove that the area of

$$\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{c^2} \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right)^2 \quad \text{is} \quad \frac{\pi c^2}{2ab} (a^2 + b^2).$$

4. Prove that the area in the positive quadrant of the curve

$$(a^2x^2 + b^2y^2)^{\frac{3}{2}} = mx^3 + ny^3 \quad \text{is} \quad \frac{1}{3ab} \left(\frac{m}{a^3} + \frac{n}{b^3} \right).$$

[a, 1890.]

5. Prove that the area of the curve

$$(a^2x^2 + b^2y^2)^2 = c^6(x^2 - y^2) \quad \text{is} \quad \frac{c^6}{a^3b^3} \left\{ ab + (b^2 - a^2) \tan^{-1} \frac{b}{a} \right\}.$$

[St. John's, 1883.]

6. Show that the area of the loop of the curve

$$\frac{x^5}{a^5} + \frac{y^5}{b^5} = 5 \frac{x^2y^2}{a^2b^2} \quad \text{is} \quad \frac{5}{2} ab.$$

7. Find the area of the curve

$$\frac{l}{\sqrt{r^2 - c^2}} = 1 + e \cos \theta \quad (e < 1).$$

8. Show that the area bounded by

$$(x^2 + y^2 - c^2)(x^2 + y^2) = 4a^2x^2 \quad \text{is} \quad (2a^2 + c^2)\pi.$$

9. Find the area included within the curve whose equation is

$$\left(\frac{x}{a}\right)^{\frac{2}{3}} + \left(\frac{y}{b}\right)^{\frac{2}{3}} = 1.$$

[Colleges, 1885.]

10. Trace the curve

$$\left(\frac{x}{a} + \frac{y}{b}\right)^{\frac{2}{3}} + \left(\frac{x}{a} - \frac{y}{b}\right)^{\frac{2}{3}} = 2,$$

and show that its area is half as great again as that of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

[Math. Tripos, 1884.]

11. Prove that the area of the curve

$$(x^2 + y^2)^5 = a^2 y(x^7 + y^7) \quad \text{is} \quad \frac{105}{512} \pi a^2 + \frac{125}{384} a^2.$$

[St. John's, 1889.]

12. Prove that the area of the curve

$$(a^2 x^2 + b^2 y^2)^5 = 8a^4 b^4 xy(a^4 x^6 + b^4 y^6) \quad \text{is} \quad a^2 + b^2.$$

[St. John's, 1889.]

13. Show that the area in the first quadrant of the curve

$$c^4 \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right)^5 = \left(\frac{x^3}{a} + \frac{y^3}{b} \right)^2 \quad \text{is} \quad \frac{ab(a^2 + b^2)}{3c^2}.$$

14. Trace the curve $4(x^2 + 2y^2 - 2ay)^2 = x^2(x^2 + 2y^2)$, proving that the area of a loop is $4\pi(2 - \sqrt{3})a^2/\sqrt{3}$, and that the area included between the loops is

$$8a^2(2\pi - 3\sqrt{3})/3\sqrt{3}.$$

[Trinity, 1896.]

15. Find the whole area of the curve

$$\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right)^2 = \frac{2xy}{ab},$$

[Oxford I. P., 1890.]

and of a loop of the curve

$$\frac{x^4}{a^4} + \frac{y^4}{b^4} = \frac{2xy}{ab}.$$

[Oxford II. P., 1900.]

16. Show that the area of either oval of

$$x^2\{x^2/a^2 + y^2/b^2 - 1\} + c^2 = 0 \quad \text{is} \quad \frac{1}{2}\pi b(a - 2c).$$

[St. John's, 1890.]

17. If $f(x, y) = 0$ be a closed curve, show that its area is mn times the area of the closed curve $f(mx, ny) = 0$. Trace the curve $(4x^2 + 9y^2)^4 = axy^6$, and find its area. [Oxford II. P., 1890.]

18. Trace the curve $\frac{x^3}{a^3} + \frac{y^3}{b^3} = \frac{3xy}{ab}$, and show that the area of its loop is $\frac{3}{2}ab$.

19. A curve is defined by the equations

$$x = 6a \sin^2 \phi, \quad y = 6a \sin^2 \phi \tan \phi,$$

where ϕ is a variable parameter. Show that the centroid of the portion enclosed between the infinite branches and the asymptote is situated on the x -axis at a distance $5a$ from the origin. [Oxford II. P., 1889.]

20. (i) In an involute of a circle, show that the area swept out by the radius vector drawn from the centre of the circle to a point on the curve varies as the cube of the central perpendicular upon the tangent, the initial line being the radius to the point where the involute meets the circle.

(ii) In the Conchoid of Nicomedes $r = a \sec \theta - b$ in the case when $a < b$, show that the area of the loop is

$$a^2(a \sec^2 \alpha - 2 \sec \alpha \cosh^{-1} \sec \alpha + \tan \alpha),$$

and that the distance of the centroid of the loop from the node is

$$\frac{2}{3}a \frac{3a \sec \alpha - 3 \cosh^{-1} \sec \alpha - \sin \alpha \tan^2 \alpha}{a \sec \alpha - 2 \cosh^{-1} \sec \alpha + \sin \alpha},$$

where $\alpha = \cos^{-1} a/b$.

21. Prove that the area contained by the curve

$$x^4 + 2x^2y^2 + 4ax^2y + 2a^2(y^2 - x^2 - 2ay) + a^4 = 0 \quad \text{is} \quad \pi a^2(4 - 5/\sqrt{2}).$$

Find also the distance from the axis of y of the centre of gravity of that portion of the area which lies in the first quadrant. [Colleges β , 1890.]

22. Show that the area included between the curve

$$s = a \tan \psi,$$

its tangent at $\psi = 0$ and its tangent at $\psi = \phi$ is

$$\frac{1}{2}a^2 \tan \phi + a^2 \tan \frac{1}{2}\phi - a^2 \log(\sec \phi + \tan \phi).$$

[Trinity, 1892.]

23. Show that an expression for the element of area in trilinear coordinates is

$$\csc C \, d\alpha \, d\beta.$$

Show that the area of the conic whose trilinear equation is

$$a^{-1}\beta\gamma + b^{-1}\gamma\alpha + c^{-1}\alpha\beta = 0$$

is to that of the triangle of reference as

$$4\pi : 3\sqrt{3}.$$

[Oxford II. P., 1890.]

24. Show that the coordinates of the centroid of the area bounded by half the cycloid $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$, the line of cusps and the y -axis are given by

$$\frac{9\pi\bar{x}}{(3\pi - 4)(3\pi + 4)} = \frac{3\bar{y}}{7} = \frac{a}{2}.$$

[Wallis.]

25. OB and OC are any two semi-diameters of an ellipse conjugate to each other; find the locus of the intersection of the normals at B and C , and show that the area of the curve is

$$\frac{\pi(a^2 - b^2)^2}{4ab}$$

[R. P.]

26. Tangents to a system of similar and similarly situated concentric ellipses are drawn such that the distance of each from the centre is the same. Find the area of the curve formed by the points of contact. [Trinity, 1885.]

27. Show that the moment of inertia of the portion of a uniform parabolic lamina cut off by the latus rectum about the tangent at an extremity of the latus rectum, is equal to $\frac{12Ma^2}{7}$, $4a$ being the latus rectum and M the mass of the lamina. [Oxf. I. P., 1914.]

28. Prove by integration that the moment of inertia of a uniform triangular lamina ABC of mass M about a perpendicular axis at A is

$$\frac{1}{12}M(3b^2 + 3c^2 - a^2).$$

[Ox. I. P., 1915.]

Chapter XV: Quadrature, etc. (IV). Miscellaneous Theorems, Connexion of a Line-Integral and a Surface-Integral, Mechanical Integration, etc. — Exercises

Examples (Arts. 466–509)

1. Ox, Oy being perpendicular axes, A, B fixed points on Oy and $AMBA$ any closed region of area S lying in the positive quadrant, show that the integral

$$\int [\{\phi(y)e^x - my\} dx + \{\phi'(y)e^x - m\} dy],$$

taken round the curve from A to B , is equal to

$$m(S + a - b) + \phi(b) - \phi(a),$$

$\phi(y), \phi'(y)$ being finite and continuous, m a constant and $OA = a, OB = b$. [J. Math. Schol. Oxford, 1904.]

2. P_1, P_2 are points on a closed oval of area A , such that P_1, P_2 subtends a right angle at a fixed point O . Show that the area of the curve traced out by the middle point of P_1P_2 is equal to

$$\frac{1}{2}A + \frac{1}{8} \int_0^{2\pi} \left(r_1 \frac{dr_2}{d\theta_1} - r_2 \frac{dr_1}{d\theta_2} \right) d\theta_1,$$

where $\theta_2 = \theta_1 + \frac{\pi}{2}$ and $OP_1 = r_1, OP_2 = r_2$. [Colleges β , 1889.]

3. A fixed point O is taken on a central oval which is such that through any point inside it other than the centre one and only one chord can be drawn which is bisected at that point; prove that the locus of the middle point of the chord PQ for a constant sum 2σ of the arcs OP, OQ cuts at right angles the same locus for a constant difference $2\sigma'$ of these arcs; and deduce that the area of the oval is

$$\frac{1}{2} \int_0^l d\sigma \int_0^{\frac{1}{2}l} d\sigma' \sin \theta,$$

where l is the length of the oval, and θ is the angle between the tangents at P and Q . [Math. Tripos, 1889.]

4. A bar AB carries at a point of its length a small wheel having AB for axis and which turns about AB : the end A is constrained to move in a given straight line; show that if the end B is carried round any closed curve without singular points and which does not cut the straight line on which A moves, the area of the curve is measured by the product of AB into the whole length registered by the revolving wheel. [Colleges, 1892.]

[This is the principle of construction of Coffin's Planimeter. A full description will be found on p. 159, *Practical Electrical Engineering*, by Briggs and others. It is the case when the rod OP of Fig. 102 is of infinite length, so that P describes a straight line instead of a circle.]

5. A straight line of given length moves with its extremities on the arcs of two closed curves of given areas, and a point is attached to the moving line.

Prove that when the area traced by this attached point has a minimum value for different positions of the point on the line, the difference of the areas of the circles whose radii are the segments into which the point divides the line is equal to the difference of the areas of the given curves. [St. John's, 1882.]

6. Show that the path of the mid-point of a rod of constant length $2c$, whose ends lie upon an ellipse, is an oval of area $\pi(ab - c^2)$.

If, instead of both ends being on the ellipse, one end lies on the ellipse and the other on the major axis, or if one end lies on the ellipse and the other on the auxiliary circle, find the areas of the paths described by the centre of the rod in both cases.

7. A rigid cyclic quadrilateral $ABCD$ moves in its plane so as to return to its original position after turning through four right angles. Show that if (A) , etc., denote the areas of the curves described by A , etc., and if S_1, S_2 , etc., denote the areas of the triangles BCD, CDA , etc., then

$$S_1(A) + S_3(C) = S_2(B) + S_4(D).$$

Find also the equation connecting the areas described by any three vertices with that described by the centre of the circumcircle of the triangle. [I. C. S., 1909.]

8. Two bars OP, RPQ , of lengths $OP = c, RPQ = b + a$, respectively turn round a fixed pin at O and a joint at P . dS_1, dS_2 denote the polar elements of area about O of the curves traced by P and Q respectively; prove that

$$dS_2 - dS_1 = a d\zeta + a \left(\frac{1}{2}a + b \right) d\theta - \frac{1}{2}a dp,$$

where $PQ = a, RP = b, p$ is the perpendicular from O on RPQ , $d\zeta$ is the displacement of R perpendicular to RPQ and θ is the inclination of RPQ to a fixed line OA . [Math. Trip., Pt. I., 1914.]

Chapter XVI: Rectification (I). Elementary — Exercises

Exercises — Cartesian Rectification (Arts. 511–519)

1. Find the length of the arc of the curve $y^2(2a - x) = x^3$, the cissoid of Diocles. [Huygens, 1625–1695.]
2. Find the curve for which the length of the arc measured from the origin varies as the square root of the ordinate.
3. The major axis of an ellipse is 1 foot in length, and its eccentricity is $\frac{1}{10}$. Prove that its circumference is 3.1337 feet nearly. [Trinity, 1883.]
4. Find the length of any arc of the curve

$$x^{\frac{2}{3}} - y^{\frac{2}{3}} = a^{\frac{2}{3}}.$$

5. Show that in the “catenary of equal strength,” $y = a \log \sec \frac{x}{a}$,

$$s = a \log \tan \left(\frac{x}{2a} + \frac{\pi}{4} \right),$$

and that the intrinsic equation of the curve is $s = a \operatorname{gd}^{-1} \psi$.

6. Show that in the common catenary, or chainette, $y = c \cosh \frac{x}{c}$,

$$s = \sqrt{y^2 - c^2}, \quad s = c \tan \psi, \quad s^2 = c(\rho - c), \quad s = c \sinh \frac{x}{c}.$$

The area bounded by the curve, the directrix, the y -axis and an ordinate is $A = cs$.

The centroid of the arc has coordinates

$$\begin{aligned} \bar{x} &= x - c \tan \frac{\psi}{2} & \bar{y} &= \frac{1}{2}(y + x \cot \psi) \\ &= x - (y - c) \frac{c}{s}, & &= \frac{1}{2} \left(y + \frac{cx}{s} \right). \end{aligned}$$

The centroid of the area bounded by the curve, the directrix, the y -axis and an ordinate is given by

$$\bar{x} = x - (y - c) \frac{c}{s}, \quad \bar{y} = \frac{1}{4} \left(y + \frac{cx}{s} \right),$$

and that both centroids lie on the ordinate through the intersection of the terminal tangents.

7. Show that the length of the curve $y = \log \coth \frac{x}{2}$ from the point (x_1, y_1) to the point (x_2, y_2) is $\log \frac{\sinh x_2}{\sinh x_1}$.
8. Show that in the epi- or hypo-cycloid

$$x = (a + b) \cos \theta - b \cos \frac{a+b}{b} \theta,$$

$$y = (a + b) \sin \theta - b \sin \frac{a+b}{b} \theta.$$

$$(i) s = \frac{4b}{a}(a+b) \cos \frac{a\theta}{2b}, \quad (ii) s = \frac{4b}{a}(a+b) \cos \frac{a}{a+2b}\psi,$$

s being measured from the point where $\theta = \frac{\pi b}{a}$, i.e. a vertex.

9. For the four-cusped hypocycloid

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}},$$

show (i) that $s = \frac{3a}{4} \cos 2\psi$, s being measured from a vertex;

(ii) the whole length of the curve is $6a$;

(iii) $s^3 \propto x^2$, s being measured from the cusp which lies on the y -axis.

10. In the tractrix

$$x = \sqrt{c^2 - y^2} + \frac{c}{2} \log \frac{c - \sqrt{c^2 - y^2}}{c + \sqrt{c^2 - y^2}},$$

show that $s = c \log \frac{c}{y}$.

11. Show that the distance from the vertex of the centroid of a wire in the form of portion of a cycloid, of which the vertex is the middle point, is $\frac{1}{3}$ of the greatest ordinate of the arc.
12. Show that the arc of a parabola of latus rectum $4a$ measured from the vertex, and the radius vector from the focus, are expressible in terms of a parameter t in the respective forms

$$\frac{s}{a} = \frac{t}{1-t^2} + \frac{1}{2} \log \frac{1+t}{1-t}, \quad \frac{r}{a} = \frac{1}{1-t^2}.$$

[Math. Trip. Pt. II., 1915.]

Prove also that $s = \sqrt{r(r-a)} + a \tanh^{-1} \sqrt{\frac{r-a}{r}}$.

Exercises — Polar Rectification (Arts. 520–526)

1. Find the length of any arc of the curve from the formula

$$s = \int \sqrt{r^2 + r'^2} d\theta$$

for the following cases:

(i) $r = a \cos \theta$ (circle). (ii) $r = ae^{m\theta}$ (equiang. spiral).

(iii) $r = a \sin^2 \frac{\theta}{2}$ (cardioid). (iv) $\frac{2a}{r} = 1 + \cos \theta$ (parabola).

(v) $r = a \frac{\sin^2 \theta}{\cos \theta}$ (cissoid). (vi) $r = \frac{3a \sin^2 \theta}{2 \cos^3 \theta}$ (semicub. parab.).

2. Show that the length of the arc of that part of the cardioid

$$r = a(1 + \cos \theta),$$

which lies on the side of the line $4r = 3a \sec \theta$ remote from the pole, is equal to $4a$. [Oxford.]

3. Show that the whole length of the limaon $r = a \cos \theta + b$ is equal to that of an ellipse whose semi-axes are equal in length to the maximum and minimum radii vectores of the limaon. Hence show how to divide the arc of the limaon into four equal parts. [Colleges a , 1888.]

4. Prove that the length of the n th pedal of a loop of the curve

$$r^m = a^m \sin m\theta$$

is

$$a(mn + 1) \int_0^{\pi/m} \sin m\theta \, d\theta, \quad \text{where } m(k - n + 1) = 1.$$

5. Show that the length of a loop of the curve

$$\begin{aligned} 3x^2y - y^3 &= (x^2 + y^2)^3 \\ &= 2 \int_0^1 \frac{d\xi}{\sqrt{1 - \xi^6}}. \end{aligned}$$

[St. John's, 1881.]

6. Show that the rectification of the curve $r^n = a^n \sin n\theta$ is given by the integral

$$s = a \int_0^\xi \frac{d\xi}{\sqrt{1 - \xi^{2n}}}.$$

[Math. Trip., 1896.]

7. Two radii vectores OP , OQ of the curve

$$r = 2a \cos^3 \left(\frac{\pi}{4} + \frac{\theta}{3} \right)$$

are drawn equally inclined to the initial line; prove that the length of the intercepted arc is $a\alpha$, where α is the circular measure of the angle POQ . [Asparagus, Educ. Times.]

8. Show that the centroid of a wire bent into the form of a cardioide $r = a(1 + \cos \theta)$, and with a line density $k \sec \frac{\theta}{2}$, k being a constant, is on the axis of the cardioide at distance $\frac{a}{2}$ from the cusp.

Exercises — The Converse Problem (Arts. 527–529)

1. Find the curves in which

(i) $s = a \sin^{-1} \frac{y}{a}$. (ii) $s = \sqrt{y^2 - c^2}$.

(iii) $s = \frac{r^2}{2a}$. (iv) $y = ce^{-\frac{s}{c}}$.

(v) $s \propto r$. (vi) $s \propto \sqrt{x}$.

(vii) $s = 2\sqrt{2ar}$.

2. Show that the equation

$$nx \frac{d^2s}{dx^2} + \frac{ds}{dx} = 0$$

leads to a cycloid or a four-cusped hypocycloid according as $n = 2$ or $n = 3$.

Exercises — Length of the Arc of an Evolute (Arts. 530–535)

1. Show that in the parabola $y^2 = 4ax$, the length of the arc of the evolute intercepted within the parabola is

$$4a(3\sqrt{3} - 1).$$

2. Find the whole length of the evolute of the cardioide

$$r = a(1 + \cos \theta).$$

3. Show that the length of the evolute of the portion of the Folium of Descartes $x^3 + y^3 = 3axy$, which corresponds to the loop, is $\frac{3a}{4}(4 - \sqrt{2})$.

Problems (Arts. 536–565)

1. Show that the whole length of the curve

$$y^2(a^2 - y^2) = 8a^2x^2 \quad \text{is} \quad \pi a\sqrt{2}.$$

[Oxford I. P., 1890.]

2. Find the whole length of the loop of the curve

$$3ay^2 = x(x - a)^2.$$

[Oxford I. P., 1889.]

3. Show that the arcs of an equiangular spiral, measured from the pole to the different points of its intersection with another equiangular spiral having the same pole but a different angle, will form a series in geometrical progression. [Trinity, 1884.]

4. Show that the length of an arc of the curve $y^n = x^{m+n}$ can be found in finite terms in the cases when $\frac{n}{2m}$ or $\frac{n}{2m} + \frac{1}{2}$ is an integer.

5. Evaluate the expressions,

$$(i) \int y \frac{dx}{ds} ds, \quad (ii) \int x \frac{dy}{ds} ds, \quad (iii) \int \left(\frac{x}{r^2} \frac{dy}{ds} - \frac{y}{r^2} \frac{dx}{ds} \right) ds,$$

wherein the line-integrals are taken round the perimeter of a closed curve. [St. John's, 1890.]

6. If s be the length of the curve

$$r = a \tanh \frac{\theta}{2}$$

between the origin and $\theta = 2\pi$, and A be the area between the same points, show that $A = a(s - a\pi)$. [Oxford I., 1888.]

7. Show that if the arc of the curve

$$r = a \tanh^n \frac{\theta}{2n} \quad (n \text{ being integral}),$$

measured from the origin, be called s , and if A be the corresponding area swept out by the radius vector from the origin,

$$A = \frac{a^2\theta}{2} - na^2 \left[\left(\frac{r}{a} \right)^{\frac{1}{n}} + \frac{1}{3} \left(\frac{r}{a} \right)^{\frac{3}{n}} + \cdots + \frac{1}{2n-1} \left(\frac{r}{a} \right)^{\frac{2n-1}{n}} \right],$$

$$s + r = a\theta - 2na \left[\left(\frac{r}{a}\right)^{\frac{1}{n}} + \frac{1}{3} \left(\frac{r}{a}\right)^{\frac{3}{n}} + \cdots + \frac{1}{n-2} \left(\frac{r}{a}\right)^{\frac{n-2}{n}} \right], \quad \text{if } n \text{ be odd and } > 2,$$

$$s + r = 2na \left[\log \cosh \frac{\theta}{2n} - \left\{ \frac{1}{2} \left(\frac{r}{a}\right)^{\frac{2}{n}} + \frac{1}{4} \left(\frac{r}{a}\right)^{\frac{4}{n}} + \cdots + \frac{1}{n-2} \left(\frac{r}{a}\right)^{\frac{n-2}{n}} \right\} \right], \quad \text{if } n \text{ be even,}$$

the results for $r = a \tanh \frac{\theta}{2}$ giving $2A = a(s - r) = a(2s - a\theta)$.

8. Show that the length of an arc of the Cissoid of Diocles

$$r = a \frac{\sin^2 \theta}{\cos \theta}$$

is $a\sqrt{3}(z - \tanh^{-1} z)$ taken between limits θ_1 and θ_2 where

$$3z^2 \cos \theta = 1 + 3 \cos^2 \theta.$$

9. Show that the intrinsic equation of the semicubical parabola

$$3ay^2 = 2x^3 \quad \text{is} \quad 9s = 4a(\sec^3 \psi - 1).$$

10. In a certain curve

$$x = e^\theta \sin \theta, \quad y = e^\theta \cos \theta.$$

Show that $s = e^\theta \sqrt{2} + C$.

Also that for the curve

$$x = e^{a\theta} \sin b\theta, \quad y = e^{a\theta} \cos b\theta, \quad s = e^{a\theta} \sqrt{a^2 + b^2} + C.$$

Name these curves.

11. Show that the length of an arc of the curve

$$x \sin \theta + y \cos \theta = f'(\theta),$$

$$x \cos \theta - y \sin \theta = f''(\theta),$$

is given by $s = f(\theta) + f''(\theta) + C$.

12. Trace the curve $y^2 = \frac{x}{3a}(a - x)^2$, and find the length of that part of the evolute which corresponds to the loop. [St. John's, 1881 and 1891.]

13. Show that the curve whose pedal equation is $p^2 = r^2 - a^2$ has for its intrinsic equation $s = a \frac{\psi^2}{2}$.

What curve is this?

14. The coordinates of a point on a plane curve are given by the relations

$$x = a[(1 - \theta^2) \cos \theta + 2\theta \sin \theta - 1],$$

$$y = a[(1 - \theta^2) \sin \theta - 2\theta \cos \theta];$$

prove that $3sa^{\frac{1}{3}} = (2a + \rho)(\rho - a)^{\frac{1}{3}}$,

ρ being the radius of curvature at the point and s the arcual distance from the origin. [Oxford II. P., 1888.]

15. The evolute of a parabola whose vertex is A meets the axis in C , and the parabola in Q . Find the perimeter of the figure bounded by AC , the parabolic arc AQ , and the arc of the evolute CQ . [Oxford I. P., 1889.]

16. Prove that the length of the first negative pedal, taken with respect to the origin, of the loop of the folium of Descartes

$$x^3 + y^3 - 3axy = 0$$

is equal to $6a - a\{\pi - \sqrt{2}\log(\sqrt{2} + 1)\}$.

17. Find the length of the arc between two consecutive cusps of the curve

$$(c^2 - a^2)p^2 = c^2(r^2 - a^2).$$

[Oxford I. P., 1889.]

18. Show that the length of the arc of the hyperbola $xy = a^2$ between the limits $x = b$ and $x = c$ is equal to the arc of the curve $p^2(a^4 + r^4) = a^4r^2$, between the limits $r = b$, $r = c$. [Oxford I. P., 1888.]

19. By means of the formula $s = \int \frac{r dr}{\sqrt{r^2 - p^2}}$, find the length of the curve $r = a \sin^2 \frac{\theta}{2}$. [Colleges a , 1887.]

20. If s be the arc of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ measured from the end of the major axis to a point whose eccentric angle is ϕ , prove that

$$s + ae^2 \cos \phi \sin \theta = \int_0^\theta \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} d\theta,$$

where $\theta = \tan^{-1} \left(\frac{a}{b} \tan \phi \right)$.

[Colleges a , 1883.]

21. Show that the circumference of an ellipse can be expressed either as

$$4a \int_0^{\pi/2} (1 - e^2 \sin^2 \theta)^{\frac{1}{2}} d\theta$$

or as

$$4a(1 - e^2) \int_0^{\pi/2} (1 - e^2 \sin^2 \theta)^{-\frac{3}{2}} d\theta,$$

where a is the semi-major axis, e the eccentricity.

[Trinity, 1887.]

22. Show that the three-cusped hypocycloid has equations of the forms

(i) $p = b \cos 3\psi,$

(ii) $r^4 + 8br^3 \cos 3\theta + 18b^2r^2 = 27b^4.$

Show that the length of an arc of the inverse of this curve with respect to the centre is proportional to

$$\tan^{-1}(2\sqrt{2} \sin 3\psi).$$

[St. John's, 1887.]

23. Prove that the intrinsic equation of the curve

$$r^{\frac{1}{5}} = a^{\frac{1}{5}} \cos \frac{1}{5}\theta \quad \text{is} \quad \frac{s}{a} = \frac{5}{16}\psi + \frac{5}{4} \sin \frac{\psi}{3} + \frac{5}{32} \sin \frac{2\psi}{3},$$

where s and ψ are measured from the point $a, 0$, and the tangent at that point. [St. John's, 1889.]

24. A circle of perimeter S and area A rolls externally in its plane entirely round an oval curve of perimeter S and area B . Prove that its centre describes an oval of perimeter $2S$ and area $3A + B$. [Oxford I. P., 1918.]

25. Find the centroid of a sector bounded by two radii vectores, and an arc of the curve whose polar equation is

$$r^2 = a^2(1 - \sin 2\theta)(1 + \sin 2\theta)^{-1},$$

and show that an arc of this curve is expressible as

$$\frac{5a}{2} \int \frac{\cos^2 \chi d\chi}{(1 + \sqrt{5} \sin \chi) \sqrt{1 - 5 \sin^2 \chi}}.$$

[Matz, Educ. Times.]

26. A rod moves always to pass through a fixed point and have one extremity on a straight line distant h from the point. Show that the arc of the curve traced out by its centre of instantaneous rotation, as the rod moves from the perpendicular position to one inclined at 45° to the line, is

$$\frac{1}{4} \{ \log(\sqrt{5} + 2) + 2\sqrt{5} \} h.$$

[Math. Tripos, 1883.]

27. On the tangent at any point P of a curve, PT is taken equal to the radius vector of P ; show how to find the length of any arc of the locus of T . For example, take the equiangular spiral and verify the result geometrically. [St. John's, 1884.]

28. Find the arc of the curve enveloped by the line

$$x \cos \phi + y \sin \phi = (a \cos^2 \phi + b \sin^2 \phi)^{-3}$$

between the points corresponding to $\phi = 0$, $\phi = \frac{\pi}{2}$.

[St. John's, 1891.]

29. Find the whole area of the curve

$$x = a \sin \theta - b \sin 2\theta,$$

$$y = a \cos \theta - b \cos 2\theta,$$

and show that the whole length of its perimeter is equal to that of an ellipse whose semiaxes are $a + 2b$, $a - 2b$. [Colleges a , 1885.]

30. Prove that if s be the arc of the curve

$$r = a \sec \alpha,$$

$$\theta = \tan \alpha - \alpha,$$

where α is a variable parameter, measured from the initial line to a point P on the curve, and if A be the area bounded by the curve, the initial line, and the radius vector to P , then

$$9A^2 = 2as^3.$$

Find the area swept out in any portion of its progress by the intercept of the tangent to the curve between the curve and the first positive pedal with regard to the origin. [Trinity, 1890.]

31. If a curve be given by $r^{2n} = \sin^2 \phi + m^2 \cos^2 \phi$, where r is the radius vector and ϕ the angle it makes with the tangent, show that

$$m \tan \frac{1}{m} (\phi - n\theta) = \tan \phi,$$

θ being the angle the radius vector makes with the initial line (which is to be appropriately chosen).

Obtain also a formula for the rectification of the curve. (The result is not obtainable in finite terms.) [I. C. S., 1898.]

32. Consider the nature of the curves

$$(i) s\psi^2 = a, (ii) \frac{\psi}{2\pi} = \sin \frac{s}{a}, (iii) s = l \sin m\psi,$$

when $m < 1$ and when $m > 1$.

33. Given a closed oval of continuous curvature without any singularities: a series of parallel curves is drawn. Prove that if A denote the area of any one of them and l its perimeter, then

$$4\pi A - l^2$$

is the same for all.

[I. C. S., 1895.]

34. In the equation of the curve $r = a + \epsilon u$, a and ϵ are constants, the latter being small; and u is a function of θ finite for all values of θ and periodic, with a period 2π . Show that if A denote the area of the curve, then its length is $2\sqrt{\pi A}$ accurately as far as small quantities of the first order inclusive.

[I. C. S., 1896.]

35. The area of an ellipse differs from that of its auxiliary circle by 10 per cent. of the area of the latter. Show that the perimeter of the ellipse differs from that of the auxiliary circle by 4.93 per cent. approximately of the perimeter of the latter.

[I. C. S., 1910.]

36. Assuming that for the catenary formed by a hanging elastic wire

$$\frac{x}{c} = u + k \sinh u, \quad \frac{y}{c} = \cosh u + \frac{1}{2}k \cosh^2 u,$$

prove that

$$\frac{s}{c} = \frac{1}{2}ku + \sinh u + \frac{1}{4}k \sinh 2u,$$

reducing to the common catenary when $k = 0$ and approximating to a parabola when k is large. [B.A. Hon. Lond., 1899.]

37. In the cycloid $y = \frac{s^2}{8a}$. Show that the only curve for which both x and y are finite integral functions of s is a straight line.

[Oxf. I. P., 1913.]

38. Find the Cartesian equation (choosing convenient axes of coordinates) of the curve in which

$$\rho^2/a^2 = (d\rho/ds)^2 + 1.$$

[Oxf. I. P., 1917.]

39. Find the intrinsic equation of the curve $27ay^2 = 4x^3$. Prove that the involutes of the curve $27ay^2 = 4x^3$ are given by the equations

$$x = a \tan^2 \psi + c \cos \psi - 2a,$$

$$y = -2a \tan \psi + c \sin \psi,$$

c being an arbitrary constant.

What happens when $c = 0$?

[Oxf. I. P., 1915.]

40. Show that the length of a quadrant of the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ is equal to $\frac{3a}{2}$, and find the length of one quadrant of the curve

$$\left(\frac{x}{a}\right)^{\frac{2}{3}} + \left(\frac{y}{b}\right)^{\frac{2}{3}} = 1.$$

[Math. Tripos, Part I., 1910.]

41. Trace the form of the curve

$$x(1+t^2) = 1-t^2, \quad y(1+t^2) = 2t,$$

as t increases from $-\infty$ to ∞ , and show that its area is π .

Find also the length of any arc of the curve in terms of t . [Math. Trip., Part I., 1910.]

42. Show that the length of the arc of the parabola $y^2 = 4ax$ which is intercepted between the point of intersection of the parabola and $3y = 8x$ is

$$a \left(\log 2 + 1\frac{5}{16} \right).$$

[Math. Trip. I., 1908.]

43. Prove that the perimeter of an ellipse of small eccentricity e and semiaxes a, b is equal to

$$\frac{1}{2}\pi\{3(a+b) - 2\sqrt{ab}\},$$

neglecting e^7 and higher powers.

[Math. Trip. I., 1917.]

44. Prove that the length of an ellipse may be expressed by

$$\iint \frac{dS}{\rho}$$

taken over the area, where dS is an element of the area of the ellipse and ρ the radius of curvature of the similar, similarly situated and concentric ellipse passing through the element dS . [Colleges, 1892.]

45. Find the intrinsic equations of a circle, a catenary, and a cycloid, and trace the curves $s = a\phi^3$, $s\phi = a$ and $s^2 = a^2\phi$.

At any point P of a cycloid the tangent is produced to a length PT equal to the arc measured from the vertex, and at T a perpendicular is drawn equal to the radius of curvature at P . Prove that the locus of the extremity of this perpendicular is the same cycloid moved parallel to its axis through a distance equal to twice the diameter of the generating circle. [St. John's College, 1882.]

Chapter XVII: Rectification (II). Central Conic, Limaon, Lemniscate, Trochoids, etc. Application of Elliptic Functions — Exercises

Exercises — Fagnano's Theorem (Arts. 580–581)

1. Show that if coaxial ellipses be drawn with a given centre such that the areas enclosed between them and their respective director circles is constant, the locus of the Fagnano points is a circle of the same area.
2. Show that the locus of the Fagnano points for similar and similarly situated concentric ellipses is a pair of straight lines.
3. Show that the locus of the Fagnano points which lie on confocal ellipses is

$$(x^{\frac{2}{3}} + y^{\frac{2}{3}})^2(x^{\frac{2}{3}} - y^{\frac{2}{3}}) = c^2,$$

$2c$ being the distance between the foci.

4. Show that if F be the Fagnano point on an ellipse of semiaxes $OA = a$, $OB = b$,

$$2 \operatorname{arc} BF = aE_1 + a - b,$$

$$2 \operatorname{arc} AF = aE_1 - a + b,$$

where E_1 is the complete elliptic integral of the second kind

$$\int_0^{\pi/2} \sqrt{1 - e^2 \sin^2 \phi} d\phi.$$

5. Show that the central perpendicular upon the tangent at a Fagnano point is a geometric mean between the semiaxes, and equal to the semi-diameter conjugate to the radius to the Fagnano point. Further, that the radius of curvature at this point is also equal to the perpendicular, and that the normals at the corresponding point on the evolute pass through the centre. Finally, that the arc of the evolute is at such a point divided in the ratio

$$b^{\frac{3}{2}} : a^{\frac{3}{2}}.$$

6. Show that if a straight rod LM of length $a + b$ slides with its ends on two axes Ox , Oy at right angles and carries a point F whose distances from L and M are respectively a and b , which thus describes an ellipse, then at the instant when LM is tangential to the path of F , F is a Fagnano point on the described ellipse, and the circle on LM for diameter passes through the point on the normal at F where that normal touches the evolute.
7. Show that the tangents at the points $P(x_1, y_1)$, $Q(x_2, y_2)$ on an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, which are related to each other so that $\frac{x_1 x_2}{a^2} = \frac{y_1 y_2}{b^2}$, intersect on a confocal hyperbola which passes through the Fagnano points.

[Many properties of these points will be found in Greenhill's *Elliptic Functions*, pages 182, 183.]

Exercises — Central Conics (Arts. 583–589)

1. In the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, put $a = b \tan \alpha$, $\Delta = \sqrt{1 - \sin^2 \alpha \sin^2 \phi}$, and show that we may take $x = b \tan \alpha \sec \phi \Delta$, $y = b \cos \alpha \tan \phi$, and that

$$\frac{ds}{d\phi} = \frac{b \cos \alpha}{\Delta \cos^2 \phi}, \quad t = \frac{b}{\cos \alpha} \Delta \tan \phi,$$

and

$$s = b \sec \alpha \tan \phi \Delta + b \cos \alpha F(\phi, \sin \alpha) - b \sec \alpha E(\phi, \sin \alpha).$$

2. From the polar equation $r^2 = a^2 \sec 2\theta$ deduce the rectification of the rectangular hyperbola, viz.

$$s = a\sqrt{2}[\Delta \tan \omega + \frac{1}{2}F - E].$$

3. If PQ be a chord of one branch of a hyperbola, touching a confocal ellipse at F , and the confocal cutting that branch of the hyperbola at A and B , and if PR , QS be the other tangents from P and Q to the ellipse, show that the elliptic arcs AR , BS exceed the elliptic arc AFB by the excess of the tangents PR , QS over the chord PQ , i.e. that

$$\text{arc } AR + \text{arc } BS - \text{arc } AFB$$

is rectifiable in terms of known lines.

In particular, examine what happens:

- (1) When F is the vertex of the confocal ellipse.
- (2) When F is at B .
- (3) When PR and QS are at right angles to PQ and F the vertex of the ellipse.

Exercises — The Lemniscate (Art. 592)

1. Find the length of the arc of a lemniscate $r^2 = a^2 \cos 2\theta$ from $\theta = 0$ to $\theta = \frac{\pi}{6}$.
2. Find the area of the curve $y^2 = \frac{1}{1-x^4}$ for the portion in the first quadrant. What connection is there between this problem and the evaluation of the perimeter of the lemniscate?
3. Draw a careful polar graph of the lemniscate $r^2 = 25 \cos 2\theta$, taking one inch as unit of length, and deduce a Cartesian graph of

$$y = 5 \operatorname{cn} \frac{x\sqrt{2}}{5} \quad \left(\operatorname{mod.} \frac{1}{\sqrt{2}} \right).$$

4. Show that the difference between the lengths of the asymptote and the infinite arc of the hyperbola $x^2/a^2 - y^2/b^2 = 1$ in the first quadrant is

$$t - s = \frac{\pi a}{2} \left[\frac{1}{2} \cdot \frac{1}{e} + \frac{1 \cdot 1^2}{2^2 \cdot 4} \cdot \frac{1}{e^3} + \frac{1 \cdot 1^2 \cdot 3^2}{2^2 \cdot 4^2 \cdot 6} \cdot \frac{1}{e^5} + \frac{1 \cdot 1^2 \cdot 3^2 \cdot 5^2}{2^2 \cdot 4^2 \cdot 6^2 \cdot 8} \cdot \frac{1}{e^7} + \dots \right].$$

Miscellaneous Problems

1. Prove that the three equations

$$x = c \log \sec \psi, \quad y = c(\tan \psi - \psi), \quad s = c(\sec \psi - 1),$$

represent one and the same curve.

[I. C. S., 1893.]

2. Find the area of the curve

$$r_1 r_2 = b^2,$$

considering all cases which may arise.

3. Prove that the value of the integral

$$\int pxy \left(\frac{x}{a^3} + \frac{y}{b^3} \right)^2 ds,$$

taken round the ellipse $x^2/a^2 + y^2/b^2 = 1$, is $\frac{\pi}{2}$, p denoting the central perpendicular on the tangent at (x, y) and ds an element of arc.

[I. C. S., 1912.]

4. If the point x, y lies on the curve

$$y^2 = x^2 + 2px + q,$$

prove that

$$\frac{dx}{y} = \frac{dy}{x+p} = \frac{dx+dy}{x+y+p},$$

and hence obtain the integral of $\frac{dx}{\sqrt{x^2 + 2px + q}}$.

If, however, the point (x, y) lie on the circle $x^2 + y^2 = a^2$, show that the corresponding relation is

$$\frac{dx}{y} = -\frac{dy}{x} = \pm \frac{ds}{a},$$

where s is the length of the arc measured to the point (x, y) .

Deduce the known formula for the integral of $\frac{dx}{\sqrt{a^2 - x^2}}$.

[I. C. S., 1908.]

5. Show that if δ stands for $\frac{d}{dx}$,

$$\delta^{-n}(uv) = v\delta^{-n}u - n\frac{dv}{dx}\delta^{-n-1}u + \frac{n(n+1)}{1.2}\frac{d^2v}{dx^2}\delta^{-n-2}u - \dots$$

6. If $\frac{1}{R} \frac{dR}{dx} = \frac{fx+g}{ax^2+bx+c}$,

and if $b^2 - 4ac$ be positive and the roots of $ax^2 + bx + c = 0$ be λ and μ , prove that $R = (x - \lambda)^p(x - \mu)^q$, where

$$\frac{p}{q} = \frac{f}{2a} \pm \frac{2ag - bf}{2a^2(\lambda - \mu)}.$$

And if $a = -1$, $b = 0$, $c = 1$,

$$R = (1 - x^2)^{-\frac{f}{2}} \left(\frac{1+x}{1-x} \right)^{\frac{g}{2}}.$$

If $b^2 - 4ac$ be negative,

$$R = (ax^2 + bx + c)^{\frac{f}{2a}} e^{\frac{2ag-bf}{2a\sqrt{4ac-b^2}} \tan^{-1} \frac{2ax+b}{\sqrt{4ac-b^2}}}.$$

If $b^2 - 4ac = 0$,

$$R = \left(\frac{2ax + b}{2a} \right)^{\frac{f}{a}} e^{-\frac{2ag-bf}{a(2ax+b)}}.$$

[E. J. Routh, Proc. L.M.S., vol. xvi., p. 250.]

7. Show that

$$\iiint \dots \int (x^2 - 1)^{-k-1} \left(\frac{x-1}{x+1} \right)^l dx dx \dots dx,$$

there being $2k+1$ integrations, $2k+1$ being an integer, though k may be a fraction, is equal to

$$\frac{1}{2^{2k+1}M} (x^2 - 1)^k \left(\frac{x-1}{x+1} \right)^l,$$

where $M = (k+l)(k+l-1) \dots$ to $2k+1$ factors. [Cf. Routh, Proc. L.M.S., vol. xvi., p. 249.]

8. ABC is a triangle with the corner A fixed and with sides AC, CB respectively $\sqrt{n}$ and $\sqrt{n+1}$, given lengths.

The side AB ($=r$) makes an angle $\theta = nA - (n+1)B$ with a fixed straight line AX .

Show (1) that the path of B is rectifiable by the formula

$$s = \sqrt{n} \int_0^A \frac{d\theta}{\sqrt{1 - \frac{n}{n+1} \sin^2 \theta}} = \sqrt{n} \operatorname{am}^{-1} A, \quad \text{mod. } \sqrt{\frac{n}{n+1}}.$$

(2) When $n = 1$ the rectification is the same as that of a Bernoulli's Lemniscate.

(3) The inclination of the normal to the radius vector is $A + B$.

(4) The area of the triangle is equal to the area of a sector of the curve starting from the axis AX . [M. Serret's Problem, Calc. Int., p. 269.]

9. C is a point of maximum curvature on the Limaon

$$r = a \cos \theta + b, \quad b > a;$$

A and A' are the two vertices. Prove that the difference between the arcs $AC, A'C$ is $4a$. [St. John's, 1891.]

10. If $y = x^3 - 3a^2x$, prove that

$$\frac{dx}{\sqrt{x^2 - 4a^2}} = \frac{dy}{3\sqrt{y^2 - 4a^6}},$$

and by integration express x explicitly in terms of y .

[Oxford I. P., 1916.]

Apply this method to solve the cubic

$$x^3 - 3x^2 - 45x - 473 = 0.$$

11. Prove that

$$\int_0^{\pi/2} e^{-\sin 2x} \cos x dx = e^{-1} \left\{ 1 + \frac{1}{3.1!} + \frac{1}{5.2!} + \frac{1}{7.3!} + \dots \right\}.$$

[Oxford I. P., 1916.]

12. Prove that if n be an odd positive integer greater than 3,

$$n \int_0^{\pi/2} \sin^n x \, dx = (2 - \sqrt{3}) \frac{n-1}{n-2} \cdot \frac{n-3}{n-4} \cdots \frac{4}{3} - \frac{\sqrt{3}}{8} \left[\frac{1}{2^{n-3}} + \frac{n-1}{n-2} \frac{1}{2^{n-5}} + \cdots + \frac{n-1}{n-2} \cdot \frac{n-3}{n-4} \cdots \frac{4}{3} \right].$$

[Oxford I. P., 1916.]

13. The parameters t_1, t_2 of two points A, B of the unicursal curve

$$x/(1-t^2) = y/(t-t^3) = a/(1+t^2)$$

are equal to $\tan \alpha, \tan \beta$, where

$$-\frac{1}{2}\pi < \alpha < -\frac{1}{4}\pi, \quad \frac{1}{4}\pi < \beta < \frac{1}{2}\pi.$$

Prove that the area of the curvilinear triangle AOB , where O is the double point, is

$$a^2 \left[2 - \frac{\pi}{2} + \beta - \alpha - \sec \beta \sec \alpha \sin(\beta - \alpha) + \frac{1}{2} \tan \beta \tan \alpha \sin 2(\beta - \alpha) \right].$$

[Oxford I. P., 1916.]

14. If n be a positive integer, show that

$$\int_0^\pi x \sin^2 nx \csc^2 x \, dx = \frac{1}{2}n\pi^2.$$

[Oxford I. P., 1912.]

15. By assuming

$$\int \frac{x^3}{\sqrt{1+x^4}} \, dx \quad \text{to be of the form} \quad \frac{a+bx+cx^2+dx^3+ex^4}{\sqrt{1+x^4}},$$

obtain the integration by differentiation and equating coefficients, also obtain the result directly by putting $x^4 = z$.

16. Given a rational integral relation between x and y of the form

$$y^n + A_1 y^{n-1} + A_2 y^{n-2} + \cdots + A_n = 0 = F(x, y), \text{ say,}$$

where $A_1, A_2, \dots, A_n$ are rational functions of x , prove that when $\int y \, dx$ can be expressed algebraically in terms of x , then

$$\int y \, dx = B_0 + B_1 y + B_2 y^2 + B_3 y^3 + \cdots + B_{n-1} y^{n-1},$$

where $B_0, B_1, B_2, \dots$ are rational functions of x .

[Abel.]

17. Assuming X to be a rational function of x , and $y^m = X$, and that $\int y \, dx$ is integrable in algebraic form and expressible as

$$\int y \, dx = P_0 + P_1 y + P_2 y^2 + \cdots + P_{m-1} y^{m-1},$$

where $P_0, P_1, \dots, P_{m-1}$ are rational functions of x , show that

$$P_0 = P_2 = P_3 = \cdots = P_{m-1} = 0,$$

that is the integration must contain one term only, and that $\frac{1}{y} \int y \, dx$ is a rational algebraic function of x .

[Liouville.]

18. If M and T be two rational polynomials, then, provided $\int \frac{M}{\sqrt[m]{T}} dx$ can be integrated in algebraic form at all, the form of the integral is $\frac{\theta}{\sqrt[m]{T}}$, where θ is a function of x .

Show also that

- (1) θ is a rational function of x .
- (2) That $MT = T \frac{d\theta}{dx} - \frac{1}{m} \theta \frac{dT}{dx}$.
- (3) That θ is an integral polynomial expression and not of such form as $\frac{U}{V}$, where U and V are complete polynomials, i.e. not such that V contains x .
- (4) That the degree of the polynomial θ is greater by unity than the degree of M .

Use these facts to show that $\int \frac{x^2 dx}{\sqrt{1+x^4}}$ is not expressible algebraically. [Bertrand, C. I., p. 94.]

19. P, Q, R being any rational algebraic polynomials, and assuming that when $\int \frac{P}{Q} \frac{dx}{\sqrt{R}}$ is integrable by means of the ordinary elementary functions, the integral must be of the form

$$\int \frac{P}{Q} \frac{dx}{\sqrt{R}} = \eta + \frac{\theta}{\sqrt{R}} + A \log(\alpha_1 + \beta_1 \sqrt{R}) + B \log(\alpha_2 + \beta_2 \sqrt{R}) + \dots,$$

where $\eta, \theta, \alpha_1, \beta_1, \alpha_2, \dots$ etc., are rational functions of x , a result established by Abel, show that when the integration can be reduced to one term, the general type of the result is either algebraic of form $\theta/\sqrt{R}$ or may be written as

$$\int \frac{P}{Q} \frac{dx}{\sqrt{R}} = A \tanh^{-1} \frac{\beta \sqrt{R}}{\alpha}.$$

In the latter case show that

- (1) $\alpha^2 - \beta^2 R = Q$.
- (2) $\alpha \beta R \left(\frac{\beta'}{\beta} - \frac{\alpha'}{\alpha} + \frac{1}{2} \frac{R'}{R} \right) = \frac{P}{A}$.
- (3) $2\alpha'Q - Q'\alpha = \frac{2P}{A}\beta$,

where accents denote differentiation with regard to x .

20. Show that

$$\int \frac{x^6(11x^2 + 37x + 28)}{x^{14} - x^3 - 4x^2 - 5x - 2} \frac{dx}{\sqrt{x+2}} = -2 \tanh^{-1} \frac{(x+1)\sqrt{x+2}}{x^7}.$$

21. Prove that

$$\int \frac{x dx}{\sqrt{(x^2 + 2x - 5)(x^2 + 4x - 8)}} = \frac{1}{2} \tanh^{-1} \frac{x+4}{x+5} \sqrt{\frac{x^2 + 4x - 8}{x^2 + 2x - 5}}.$$

22. Prove that

- (1) $\int \frac{5x^2 + 3x + 1}{2x + 1} \frac{dx}{\sqrt{x^4 + 2x^2 + 2x + 1}} = \tanh^{-1} \frac{x\sqrt{x^4 + 2x^2 + 2x + 1}}{x^3 + x + 1}.$
- (2) $\int \frac{3 \tan^2 \theta + 2}{2 \tan^4 \theta + 5 \tan^2 \theta + 4} \sec \theta d\theta = \tan^{-1} \frac{\sin \theta}{1 + \cos^2 \theta}.$

$$(3) \int \frac{\sinh 2x \, dx}{\sqrt{\cosh^2 x + 1} \sqrt{\cosh^2 x + 2}} = 2 \tanh^{-1} \sqrt{\frac{\cosh^2 x + 1}{\cosh^2 x + 2}}.$$

23. Show that

$$(i) \frac{2n+1}{2} a^2 \int \sqrt{\frac{a^{2n-1} + x^{2n-1}}{a^{2n+1} + x^{2n+1}}} \frac{dx}{a^2 - x^2} - \frac{2n-1}{2} \int \sqrt{\frac{a^{2n+1} + x^{2n+1}}{a^{2n-1} + x^{2n-1}}} \frac{dx}{a^2 - x^2} \\ = \tanh^{-1} x \sqrt{\frac{x^{2n-1} + a^{2n-1}}{x^{2n+1} + a^{2n+1}}}.$$

$$(ii) \frac{2n+1}{2} \int \sqrt{\frac{1 + \sin^{2n-1} \theta}{1 + \sin^{2n+1} \theta}} \sec \theta \, d\theta - \frac{2n-1}{2} \int \sqrt{\frac{1 + \sin^{2n+1} \theta}{1 + \sin^{2n-1} \theta}} \sec \theta \, d\theta = \tanh^{-1} \left(\sin \theta \sqrt{\frac{1 + \sin^{2n-1} \theta}{1 + \sin^{2n+1} \theta}} \right).$$

24. Integrate the following:

$$(i) \int \frac{(2x+1)dx}{\sqrt{x^4 + 2x^3 - 3x^2 - 4x + 3}}. \quad (ii) \int \frac{x \, dx}{\sqrt{x^4 + 2x^3 - 3x^2 - ax + a}}. \quad [\text{Abel.}]$$

$$(iii) \int \frac{5x^2 + 15x + 12}{5x^2 + 15x + 9} \frac{dx}{\sqrt{(x+1)(x+2)}}.$$

$$(iv) \int \frac{(1+8x) \, dx}{\sqrt{1+6x+4x^2} \sqrt{1-2x+4x^2}}.$$

$$(v) \int \frac{(2x+a) \, dx}{\sqrt{x^4 + 2ax^3 + 3a^2x^2 + 2a^3x - a^4}}.$$

$$(vi) \int \frac{3x^4 - 2x^3 + 1}{2x^3 - x^2 + 1} \frac{dx}{\sqrt{x^4 + 1}}. \quad (vii) \int \frac{3x+1}{1-x-2x^2-x^3} \frac{dx}{\sqrt{x}}.$$

$$(viii) \int \frac{3x^4 + 1}{1-x^2-x^6} \frac{dx}{\sqrt{x^4 + 1}}. \quad (ix) \int \frac{x^2 - (a+b)x - ab}{\sqrt{x^2 + a^2} \sqrt{x^2 + b^2}} dx.$$

$$(x) \int \frac{1+x-x^2}{1-x+x^2+x^3} \frac{dx}{\sqrt{x^2-1}}. \quad (xi) \int \frac{1+x^4}{1+x^2-x^4} \frac{dx}{\sqrt{x^4-1}}.$$

$$(xii) \int \frac{1+3x^4+2x^5}{1+2x-x^6} \frac{dx}{\sqrt{1+x^4}}.$$

25. Show that the whole perimeter and area of a single loop of the curve $r = 2a \cos n\theta$ ($n > 1$) are respectively equal to the whole perimeter and area of the ellipse $x^2 + n^2 y^2 = a^2$. [Oxf. I. P., 1911.]

26. If an element ds of a curve lie at distance r from the origin, and subtends an angle $d\theta$ there, it is known that unit electric current flowing along ds produces a magnetic force at the origin at right angles to the plane of the curve proportional to $\frac{d\theta}{r}$.

Show that if unit current flows through a thin endless wire of given length in the form of an ellipse, the magnetic force due to the current at the centre of the ellipse is inversely proportional to the area of the ellipse. [Oxford II. P., 1913.]

27. A current of electricity is flowing round a fine wire $ABCD \dots KA$ bent into a plane polygon. O is any point within the polygon, and perpendiculars $OP, OQ, OR, \dots$ are drawn to the sides KA, AB, BC , etc., respectively, and again perpendiculars whose lengths are $\alpha, \beta, \gamma, \dots$ from O upon the sides $PQ, QR, RS, \dots$ of the inscribed polygon $PQRS \dots$. Show that the magnetic force on unit particle situated at O is

$$\iota \sum \frac{\sin A}{\alpha},$$

where ι is the current strength.

28. Show that the perimeter of an ellipse of axes $2a$, $2b$ and small eccentricity e is approximately equal to the perimeter of a circle of diameter $a + b$, with an error which is only about 0.0025 per cent. when e is as great as 0.2. [Math. Trip. Part II., 1913.]

Chapter XVIII: Rectification (III). Miscellaneous Theorems — Exercises

Miscellaneous Problems

1. Show that any point on the Lemniscate $r^2 = a^2 \cos 2\theta$ may be represented by

$$x = a \frac{z + z^3}{1 + z^4}, \quad y = a \frac{z - z^3}{1 + z^4},$$

and hence obtain the rectification of the curve.

[Serret.]

Show that the integral obtained for s reduces to the standard Legendrian form by the further substitution

$$\cos \phi = \frac{z\sqrt{2}}{\sqrt{1 + z^4}}.$$

[Cayley, Ell. Functions, Art. 63.]

2. By the transformation $\frac{z - \iota}{z + \iota} = \frac{a - \iota}{a + \iota} u$,

show that the equation

$$x + \iota y = C e^{\iota \omega} \int \frac{(z - a)^{n+1} (z - \iota)^m}{(z - \alpha)^{n+1} (z + \iota)^{m+2}} dz$$

takes the form

$$x + \iota y = A \int \frac{u^m (u - 1)^{n+1}}{(u - \xi)^{n+1}} du,$$

where

$$\xi = \frac{(a + \iota)(\alpha - \iota)}{(\alpha - \iota)(a + \iota)}, \quad A = \frac{C}{2\iota} e^{\iota \omega} \left(\frac{a + \iota}{\alpha + \iota} \right)^{n+1} \left(\frac{a - \iota}{a + \iota} \right)^{m+1}$$

Hence show that the condition that $x + \iota y$ should be purely algebraic is

$$\frac{d^n}{d\xi^n} \xi^m (\xi - 1)^{n+1} = 0,$$

a and α being supposed conjugate, and m, n positive integers.

Discuss the roots of this equation.

[Serret, Calc. Intg., p. 254.]

3. In Bernoulli's Lemniscate $r^2 = 2a^2 \cos 2\theta$,

show that if $x + \iota y = u$ and $x - \iota y = v$,

the equation of the curve may be written

$$(u^2 - a^2)(v^2 - a^2) = a^4.$$

Further, expressing u^2 and v^2 as $a^2(1 + t^2)$ and $a^2 \left(1 + \frac{1}{t^2}\right)$ respectively, show that the tangent of the angle which the tangent at any point makes with the x -axis is

$$\iota \frac{1 + t^3}{1 - t^3}.$$

Hence, putting the coordinates of two points at which the tangents are parallel, as $\omega\mu$, $\omega^2\mu$ where $\omega^3 = 1$, show that the locus of the mid-points of chords joining such points is

$$\begin{aligned} & [16u^2v^2 - 8a^2(u^2 + v^2) + 3a^4]^2 \\ &= 4a^4[16\{u^4 + v^4 - u^2v^2\} - 12a^2(u^2 + v^2) + 9a^4], \end{aligned}$$

i.e. a curve of the eighth degree.

[R. A. Roberts, Proc. L.M. Soc., vol. xviii.]

4. Obtain an integral for the rectification of the inverse of the parabola $y^2 = 4ax$, with regard to a point on the axis whose coordinates are $(h, 0)$.

If $h = -3a$, show that

$$s = \frac{1}{6a\sqrt{2}} \log \frac{3 + 2\sqrt{2}\sin\omega}{3 - 2\sqrt{2}\sin\omega},$$

where $a \tan^2 \omega$, $2a \tan \omega$ are taken as the current coordinates of a point on the parabola, and the arc of the inverse is measured from the point corresponding to the vertex of the parabola. [Mr. Roberts, loc. cit.]

Show that the semiperimeter is bisected at the point $\omega = \sin^{-1} \frac{3}{4}$.

5. Show that the tangents to the parabola $y^2 = 4a(x + a)$ at the points

$$\{a \sinh^2(u \pm v) - a, 2a \sinh(u \pm v)\},$$

where u is variable but v is a constant, intersect on a confocal parabola; and that if T be a point on this second parabola, and TP_1 , TP_2 the tangents to the first, then

$$TP_1 + TP_2 - \text{arc } P_1P_2 = a(\sinh 2v - 2v),$$

and is constant.

[Oxford I. P., 1911.]

6. Show that $\iint \frac{dA}{r}$, taken over the area cut from a parabola of latus rectum $4a$ by an ordinate distant c from the vertex ($c < a$), where r denotes the distance from the focus, is equal to

$$4\sqrt{ac} - 2(a - c) \log \frac{\sqrt{a} + \sqrt{c}}{\sqrt{a} - \sqrt{c}}.$$

[Oxford I. P., 1911.]

7. Show that $\int_0^{\pi/4} \frac{\sin 15\theta}{\sin \theta} d\theta = \frac{\pi}{4} + \frac{1}{1} - \frac{1}{3} + \frac{1}{5} - \frac{1}{7}$.

8. If

$$u = e^{-\int \phi dx} \int e^{\int \phi dx} (c_0 + c_1x + c_2x^2 + \cdots + c_nx^n) f dx + Ce^{-\int \phi dx},$$

where $c_0, c_1, c_2, \dots, c_n, C$ are $(n + 2)$ arbitrary constants, and

$$\frac{\phi}{f} = a_0 + a_1x + a_2x^2 + \cdots + a_mx^m,$$

where $a_0, a_1, \dots, a_m$ are $(m + 1)$ given constants, show that if m be not greater than n , $\frac{du}{dx}$, obtained by the direct differentiation of u with regard to x , contains only $(n + 1)$ arbitrary constants. [Math. Tripos, 1878.]

9. If $f(m, n) = \int_0^\infty x^m (\cosh x)^{-n} dx$, where m and n are positive integers, each greater than 2, prove that

$$(n - 1)(n - 2)f(m, n) = (n - 2)^2 f(m, n - 2) - m(m - 1)f(m - 2, n - 2).$$

[Oxford I. P., 1914.]

10. Given that a and c are positive, show that the limit when $m \rightarrow \infty$ and $n \rightarrow \infty$ of

$$\frac{1}{n} \left[\frac{1}{a^r} + \frac{1}{\left(a + \frac{c}{n}\right)^r} + \frac{1}{\left(a + \frac{2c}{n}\right)^r} + \frac{1}{\left(a + \frac{3c}{n}\right)^r} + \cdots + \frac{1}{\left(a + mn\frac{c}{n}\right)^r} \right]$$

is finite when $r > 1$; and find this limit.

[Oxf. I. P., 1914.]

11. The increase dS in a man's satisfaction S by an increased expenditure dx on a certain commodity, is expressed by the law $dS = \frac{\lambda}{x-a} dx$. Similar laws, viz.

$$dS = \frac{\mu}{y-b} dy, \quad dS = \frac{\nu}{z-c} dz,$$

hold for two other commodities, where $\lambda, \mu, \nu, a, b, c$ are all positive. Find how the man should expend a given sum E ($> a + b + c$) so that his total satisfaction is greatest. [Oxford I. P., 1914.]

Show that the maximum satisfaction is measured by

$$S = \log \frac{\lambda^\lambda \mu^\mu \nu^\nu (E - a - b - c)^{\lambda+\mu+\nu}}{(\lambda + \mu + \nu)^{\lambda+\mu+\nu}}.$$

12. Evaluate

$$\int_{e^u}^{\infty} \frac{x-1}{x^2 - 2x \cos v + 1} \frac{dx}{\sqrt{x^2 - 2x \cosh u + 1}}.$$

[Oxford II. P., 1914.]

13. Show that the tangent to the curve

$$3a^2(y-x) + x^3 = 0,$$

at the point whose abscissa is h , cuts the curve again at the point whose abscissa is $-2h$, and that the area included between the curve and the tangent is $9h^4/4a^2$. [Oxf. I. P., 1918.]

14. If $f_1(x)$ and $f_2(x)$ are both polynomials in x , show that the integral of $f_1(x)/f_2(x)$ with respect to x can always be written in the form

$$\phi_1(x)/\phi_2(x) + \log \phi_3(x)/\phi_4(x),$$

$\phi_1, \phi_2, \phi_3, \phi_4$ also denote polynomials, not necessarily real.

Find the general form of the integral with respect to x of

$$f_1(x + \sqrt{x^2 - 1})/f_2(x - \sqrt{x^2 - 1}).$$

[Oxf. I. P., 1918.]

15. Show that the area bounded by the curve

$$x = \frac{3at^2}{1+t^3}, \quad y = \frac{3at}{1+t^3},$$

its real asymptote $x + y + a = 0$, and by two lines at right angles to this asymptote through the points $t = -a, t = 0$ of the curve, is

$$\frac{3a^2}{4} \left\{ 1 + \frac{u^4 - 1}{(u^2 + u + 1)^2} \right\},$$

and find the whole area between the curve and its real asymptote.

[Oxf. I. P., 1917.]

16. If $\phi(z)$ be a rational function of z without singularities in the range $0 \leq z \leq 1$, prove that

$$\begin{aligned} \int_0^{\pi/2} \phi(\sin 2x) \cos^2 x \cos 2x \, dx &= \int_0^{\pi/2} \phi(\sin 2x) \cos^2 x \cos^2 2x \, dx \\ &= \int_0^{\pi/2} \phi(\sin 2x) \cos^4 x \cos 2x \, dx. \end{aligned}$$

[Oxford I. P., 1907.]

17. Integrate

$$(i) \int \frac{x(x-a)^{b-1} dx}{(x-b)\{(x-b)^{2b} - (x-a)^{2a}\}^{\frac{1}{2}}},$$

$$(ii) \int \frac{x^{q-1}\{px^{p+q} - qa^{p+q}\} dx}{(x^{p+q} + a^{p+q})^2 + x^{2q}a^{2q}}.$$

18. In the curve $\frac{15x}{a} = \left(\frac{2y}{a}\right)^{\frac{2}{3}} - \left(\frac{2y}{a}\right)^{\frac{5}{3}}$, show that $s^2 = x^2 + \frac{16}{15}y^2$, s being measured from the origin.

Show that the curve is a quintic of which the y -axis is an axis of symmetry, and that the area of the loop $= \left(\frac{8}{189}\right) \left(\frac{5}{3}\right)^{\frac{1}{3}} a^2$.

19. If 2ϕ be the eccentric angle of the point r, θ on the ellipse $c = r(1 - e \cos \theta)$, prove that

$$\{(1+e)^2 - 4e \sin^2 \phi\} \left(\frac{d\theta}{d\phi}\right)^2 = 4(1 - e^2 \cos^2 \theta).$$

Use the fact that

$$\int_0^\pi F(\cos^2 \theta) \, d\theta = 2 \int_0^{\pi/2} F(\cos^2 \theta) \, d\theta$$

and the above to obtain a value of α , such that

$$\int_0^{\pi/2} \frac{d\phi}{\sqrt{(1+e)^2 - 4e \sin^2 \phi}} = 2 \int_0^\alpha \frac{d\phi}{\sqrt{(1+e)^2 - 4e \sin^2 \phi}}.$$

[Oxford I. P., 1917.]

20. A uniform rod of mass M has its extremities at the points $x_1, y_1; x_2, y_2$. Show that the product of inertia of the rod with respect to the axes is given by

$$M \int_0^1 \{tx_2 + (1-t)x_1\} \{ty_2 + (1-t)y_1\} \, dt.$$

Hence show that the product of inertia of the rod is the same as that of three particles of masses

$$\frac{M}{6}, \quad \frac{M}{6}, \quad \frac{2M}{3},$$

placed at the extremities and the middle point of the rod respectively. [Oxford I. P., 1913.]

21. Show that the coordinates of any point on the curve whose intrinsic equation is $s = a(\sec^n \psi - 1)$, where n is an odd integer greater than unity, can be expressed rationally in terms of $\tan \psi$, and show that when $x = 0$ the curve is a cubic with a cusp. [Oxf. I. P., 1911.]

22. Show how to evaluate the integral $\int f(x, y) dx$, where

$$y^2 = ax^2 + 2bx + c$$

and $f(x, y)$ is a rational function of x and y .

Prove that

$$(i) \int_0^a \frac{dx}{x + \sqrt{a^2 - x^2}} = \frac{1}{2}\pi,$$

$$(ii) \int_0^a \frac{a dx}{(x + \sqrt{a^2 - x^2})^2} = \frac{1}{\sqrt{2}} \log(1 + \sqrt{2}),$$

the positive sign being taken for the radical in each of the subjects of integration. [Math. Trip., Part II., 1913.]

23. Show by means of the transformation $y = \frac{(x^2 + 1)^{\frac{1}{2}}}{x + 1}$ that

$$\int_0^\infty \frac{dx}{(x + 1)(x^2 + 1)^{\frac{1}{2}}} = 2 \int_{1/\sqrt{2}}^1 \frac{dy}{(2y^2 - 1)^{\frac{1}{2}}} = \sqrt{2} \log(\sqrt{2} + 1),$$

and verify the result in an independent manner.

[Math. Trip., Part II., 1914.]

24. Integrate $\int \frac{\sin x}{\sin(x - a)} dx$.

[Math. Trip., Part II., 1914.]

25. Evaluate

$$\int \frac{x + 2}{(x + 1)^2(x^2 + 4)}, \quad \int \frac{dx}{(x^2 + 1)^4}, \quad \int \frac{dx}{(5 - 3 \cos x)^2},$$

and the corresponding definite integrals taken between the limits $(0, \infty)$, $(0, \infty)$ and $(0, \pi)$ respectively.

[Math. Trip., Part II., 1914.]

26. Show that

$$(i) \int \frac{\sin 4x}{\sin 6x} dx = \frac{\sqrt{3}}{6} \tanh^{-1} \left(\frac{\sin 2x}{\cos \frac{\pi}{6}} \right).$$

$$(ii) \int \frac{\sin 3x}{\sin 5x} dx = \frac{1}{5} \left[\sin \frac{\pi}{5} \log \frac{\sin \left(\frac{2\pi}{5} + x \right)}{\sin \left(\frac{2\pi}{5} - x \right)} + \sin \frac{2\pi}{5} \log \frac{\sin \left(\frac{\pi}{5} + x \right)}{\sin \left(\frac{\pi}{5} - x \right)} \right].$$

27. Prove that

$$\int_0^{\pi/2} \frac{d\theta}{(1 + \sin^2 \theta)(2 + \sin^2 \theta)} = \frac{\pi}{\sqrt{3}} \sin \frac{\pi}{12}.$$

28. Prove that

$$\int \frac{2 \cos \theta + \sin \theta}{(1 + \sin \theta \cos \theta)^{\frac{3}{2}}} d\theta = \frac{2 \sin \theta}{(1 + \sin \theta \cos \theta)^{\frac{1}{2}}}.$$

Chapter XIX: Moving Curves. Quadrature and Rectification of Loci of Carried Points and Envelopes of Carried Lines — Exercises

Problems

1. A circle of radius a rolls round the circumference of an equal circle. Prove that the area of the epitrochoid, described by a point carried with the rolling circle and distant c from its centre, is

$$(4a^2 + 2c^2)\pi.$$

[Oxf. I. P., 1918.]

2. If a circle roll on the convex side of a parabola from one extremity of the latus rectum to the other, and can just pass between the vertex and the directrix, prove that four times the area traced out by that radius of this circle, which always passes through the point of contact, exceeds the area of the circle by half the rectangle contained by the latus rectum and a line equal to the arc it cuts off.

[R. P.]

3. An equiangular spiral rolls upon a straight line from a point P_1 to a point P_2 of the spiral. O , the pole of the spiral, traces out the path O_1O_2 . From O_1O_2 are drawn perpendiculars O_1N_1 , O_2N_2 on the straight line. Find the area of $O_1N_1N_2O_2$.

[Colleges a , 1881.]

4. A closed oval curve rolls upon a fixed curve. Find an expression for the area of the roulette traced out by any carried point.

In a complete revolution of the closed oval curve, prove that the sum of the areas of the envelopes of two carried lines at right angles to one another which pass through a point fixed to the rolling curve is constant. Prove also that this sum exceeds the area generated by the point, by the area of the rolling curve.

[Colleges γ , 1887.]

5. If a closed oval curve roll with angular velocity ω on a straight line, while a point moves along its evolute with relative velocity $\omega\rho'$, prove that the area included in any portion of a revolution between the straight line, the curve generated by the moving point, and the perpendiculars to the former drawn through the extremities of the latter, is double the corresponding portion of the area between the curve and its evolute, bounded by the initial and final radii of curvature, provided the moving point is initially at the centre of curvature of the point of contact; ρ' being the radius of curvature of the evolute at the point corresponding to the point of the rolling curve in contact with the straight line.

[Colleges δ , 1883.]

6. The cardioide $r = a(1 - \cos \theta)$ rolls on a straight line; prove that the intrinsic equation of the roulette of the cusp is

$$2s = 3a(2\psi - \sin 2\psi),$$

measuring from the point of contact of the cusp.

Prove also that its Cartesian equation is

$$\frac{4a - x}{2a} = \left\{ 2 + \left(\frac{y}{2a} \right)^{\frac{2}{3}} \right\} \sqrt{1 - \left(\frac{y}{2a} \right)^{\frac{2}{3}}},$$

that its area is $\frac{15}{4}\pi a^2$, and that the radius of curvature of the roulette of the cusp at the point of contact is three times its distance from the point of contact.

[Trinity, 1888.]

Find the evolute of the roulette of the pole and the intrinsic equation of the envelope of the axis.

7. A closed curve is moving in any manner in its own plane. Show that if ρ be the radius of curvature of the envelope of the tangent at any point of the curve, then

$$\int \rho ds$$

is equal to twice the area of the curve, the integral being taken all round the curve, ds being an element of arc of the moving curve. [Colleges, 1879.]

8. A plane lamina moves in any given manner on a fixed plane: O is a fixed point on the fixed plane, P a point attached to the moving lamina and fixed upon it. If the area described by P about O be given, show that the locus of all points (P) in the moving plane for which the area is the same, is a circle, and that for different values of the area the corresponding circles are concentric. [St. John's, 1881.]
9. Examine the isoperimetric correspondence between the parabola $y^2 = 4ax$ and the Archimedean spiral $r = 2a\theta$, showing that the spiral can be made to roll upon the parabola in such manner that the pole of the spiral travels along the axis of the parabola.
10. Show that the reciprocal spiral $r\theta = a$ and the exponential curve $y = ae^{-x/a}$ are isoperimetric companions, both curves being rectifiable and corresponding arcs equal, and interpret the result by reference to the locus of the pole of the spiral when suitably started rolling.
11. Establish isoperimetric companionship between the curve

$$\frac{x}{a} = \log \tan \left(\frac{\pi}{4} + \frac{\phi}{2} \right) - \sin \phi, \quad \frac{y}{a} = \frac{\sin^2 \phi}{\cos \phi}$$

and the cissoid $r = a \frac{\sin^2 \theta}{\cos \theta}$.

12. Establish isoperimetric companionship between the semi-cubical parabola $ay^2 = x^3$ and the spiral $8au + \theta^3 = 0$.
13. Show that the curve

$$r = a \log \sec t, \quad \theta = \int \frac{dt}{\log \sec t}$$

is rectifiable and in isoperimetric companionship with the catenary of equal strength $y = a \log \sec \frac{x}{a}$.

14. Show that the curves

$$4x = a \left(\cos \phi - 9 \cos \frac{\phi}{3} \right), \quad 4y = a \left(3 \sin \frac{\phi}{3} - \sin \phi \right)$$

and $r^{\frac{1}{3}} = a^{\frac{1}{3}} \sin \frac{1}{3}\theta$

are rectifiable and isoperimetric companions.

15. Show that the curve

$$2\theta + 6 \frac{\sqrt{a^{\frac{2}{3}} + r^{\frac{2}{3}}}}{r^{\frac{1}{3}}} = 3 \log \frac{\sqrt{a^{\frac{2}{3}} + r^{\frac{2}{3}}} + r^{\frac{1}{3}}}{\sqrt{a^{\frac{2}{3}} + r^{\frac{2}{3}}} - r^{\frac{1}{3}}}$$

is rectifiable and in isoperimetric companionship with

$$x^{\frac{2}{3}} - y^{\frac{2}{3}} = a^{\frac{2}{3}}.$$

16. Show that the curve

$$r = 4a \sin \frac{t}{2} \cos^3 \frac{t}{2}, \quad \theta = \tan \frac{t}{2} - 2t$$

is rectifiable and in isoperimetric companionship with the cardioide $r = a(1 + \cos \theta)$, its pole travelling along the axis of the cardioide as it rolls within the cardioide, the two poles being initially coincident.

17. Show, by taking $r = a\theta$ and $\chi = n\theta$ in Art. 702, that

$$x = \frac{a}{n^2} [n\theta \cos n\theta + (n-1) \sin n\theta],$$

$$y = \frac{a}{n^2} [n\theta \sin n\theta - (n-1)(\cos n\theta - 1)]$$

is an isoperimetric companion of the Archimedean spiral $r = a\theta$.

Hence show

(1) that $x^2 = 2ay$ is isoperimetric with $r = a\theta$;

(2) that $r = a \csc \left\{ \frac{\sqrt{r^2 - a^2}}{a} - \theta \right\}$ is rectifiable and in isoperimetric companionship with $r = 4a\theta$.

18. Show that an ellipse of semiminor axis b and eccentricity e can be made to roll upon the curve

$$\frac{b}{y} = \operatorname{dn} \frac{x}{b} \quad (\text{mod. } e),$$

so that the path of the centre of the ellipse is the x axis.

Show that if the origin be taken at the point for which the end of the major axis a is in contact with the curve, this may be reduced to the form

$$\frac{y}{a} = \operatorname{dn} \frac{x}{b}.$$

[Write $Kb - x$ for x and reduce, see Ch. XXXI., Art. 1352. See also Greenhill, *Elliptic Functions*, p. 72.]

19. Show that the perimeter of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is equal to twice the perimeter of one outer foil of the curve

$$r = b \cos \frac{b\theta}{a},$$

and that the area of the ellipse is equal to four times the area of one outer foil of the same curve.

Show further, that if the vertex of the foil be placed in contact with the inner side of the ellipse at the end of the minor axis, and the foil roll without sliding upon the ellipse, the pole of the rolling foil will traverse the major axis of the ellipse.

Deduce a well-known proposition as to a circle rolling in the interior of another circle of double its radius.

20. An involute of a circle is made to slide, touching the rectangular axes Ox , Oy . Show that the locus of the instantaneous centre on the plane x, y is a straight line. What is the locus of the instantaneous centre relatively to the curve.

Show that the glissettes of carried points are cycloids and trochoids, and the envelopes of carried straight lines are either cycloids or involutes of cycloids. [Besant, *Roulettes and Glissettes*.]

21. A cycloid rolls along a straight line. Show that the intrinsic equations of the envelopes of (1) the axis, (2) the line of cusps, (3) the tangent at the vertex are respectively

$$(1) s = a\psi^2 + 3a \sin^2 \psi,$$

$$(2) s = 3a(\psi + \frac{1}{2} \sin 2\psi),$$

$$(3) s = a(\psi + \frac{3}{2} \sin 2\psi),$$

measuring s in each case from the point on its locus for which $\psi = 0$.

Trace each of these curves, supposing the cycloid to be continued both ways, and the rolling to continue with successive arches of the cycloid, and find the positions of their cusps.

Show that the whole perimeter of the last of these curves is

$$8a\sqrt{2} + 8a \sin^{-1} \sqrt{\frac{2}{3}} - 2\pi a,$$

$$\text{and its area} = \frac{2}{3}\pi a^2.$$

Show that the first evolutes of the second and third curves, and the second evolute of the first are four-cusped hypocycloids.

22. A parabola rolls on a straight line; show that

(1) the locus of the focus is a catenary (Art. 517),

(2) the envelope of the directrix is an equal catenary,

(3) the tangent at the vertex and the latus-rectum envelop parallels to a catenary,

(4) the intrinsic equation of the envelope of the axis is

$$s = a(2 \log \sec \psi + \tan^2 \psi).$$

23. If the cardioide $r = a(1 - \cos \theta)$ move so as to touch a straight line always at the same point, show that the locus of the pole is

$$r = 2a \sin^2 \theta,$$

and that the intrinsic equation of the envelope of the axis is

$$\frac{3s}{a} = 12 \sin^2 \frac{\psi}{3} - 7 \sin^4 \frac{\psi}{3}.$$

24. If an ellipse slide in contact with a given straight line at a given point, the glissette of the foci is

$$(x^2 + y^2)(b^2 + y^2)^2 = 4a^2 y^4,$$

and that of the centre is $x^2 y^2 = (a^2 - y^2)(y^2 - b^2)$.

25. A lamina moves in such manner that a certain point in it describes the path

$$\xi = c \sin \psi - c \cos \psi \log(\sec \psi + \tan \psi),$$

$$\eta = c \cos \psi + c \sin \psi \log(\sec \psi + \tan \psi) - c,$$

referred to fixed axes OX, OY in its plane, whilst a straight line through this point attached to the lamina makes an angle ψ with the Y -axis.

Reduce this motion to rolling. Also show that the difference of the curvatures of the loci of the instantaneous centre on the lamina and on the fixed plane is $\frac{\cos^2 \psi}{c}$.

Show further that the intrinsic equation of the envelope of the line attached to the lamina is

$$\frac{ds}{d\psi} = c \sec \psi \tan \psi + c \log(\sec \psi + \tan \psi).$$

26. A lamina moves in its own plane, so that a point O' upon it traces out a cissoid,

$$\xi = -2a \cos^2 \frac{\theta}{2}, \quad \eta = 2a \frac{\cos^3 \frac{\theta}{2}}{\sin \frac{\theta}{2}}, \quad \text{i.e. } \eta^2(2a + \xi) + \xi^3 = 0,$$

upon a fixed plane with reference to a pair of fixed rectangular axes OX, OY in that plane, whilst a straight line $O'x$ attached to the moving lamina rotates, making an angle θ with OX . Show that the motion is that of rolling of one parabola upon another equal parabola, and deduce from the formula of Art. 660, for the difference of curvature of the I -loci, the radius of curvature of a parabola.

27. A catenary moves in its own plane so as always to touch a given straight line at a given point. Show that the tangential polar equation of the envelope of the axis is

$$\frac{p}{c} = \text{gd}^{-1} \psi,$$

where c is the parameter of the catenary.

28. The centre of a circular disc of radius a travels along a parabolic path $y^2 = 2ax$, spinning at an angular velocity ω in a clockwise direction, the centre receding from the axis with a velocity $a\omega$. Show that the motion thus produced is that of the rolling of an involute of the circle upon the axis of the parabola, and that the velocity of the point of contact is the same as the velocity with which the centre of the circle recedes from the tangent at the vertex.

29. A Bernoulli's lemniscate moves so as to touch a fixed axis at a given point. Show that the tangential polar equation of the envelope of the axis is

$$p^2 = a^2 \sin^2 \frac{\psi}{3} \cos \frac{2\psi}{3},$$

and that the glissette of the pole is

$$r^2 = a^2 \sin \theta.$$

30. A circle rolls on an equal circle and carries with it a fixed tangent. Find the intrinsic equation of the envelope of the carried tangent. [Oxford II. P., 1887.]

31. A triangle of area Δ moves so that two of its sides (a, b) touch an oval of perimeter l at points where the radii of curvature are ρ, ρ' ; prove that the radius of curvature and the perimeter of the envelope of the third side c are

$$\frac{1}{c}(2\Delta - a\rho - b\rho') \quad \text{and} \quad \frac{1}{c}\{4\pi\Delta - (a+b)l\}.$$

[St. John's, 1883.]

32. An ellipse rolls on a fixed horizontal straight line (the axis of x). Show that the locus of the highest point of the ellipse will be

$$x = \int \frac{y^4 - 8a^2b^2}{y^2 \sqrt{(4a^2 - y^2)(y^2 - 4b^2)}} dy,$$

and reduce the integral to the standard form.

[St. John's Coll., 1881.]

33. Prove that the intrinsic equation of the envelope of the directrix of a catenary of parameter c , rolling on a circle of radius c , will be found by eliminating α between the equations

$$\frac{s}{c} = \frac{1}{2} \tan \alpha \sec \alpha + \frac{1}{4} \log \frac{1 + \sin \alpha}{1 - \sin \alpha}$$

and

$$\psi = \alpha + \tan \alpha.$$

[St. John's, 1886.]

34. A given right-angled triangle is made to slide round the outside of a fixed oval curve with the point P on the curve, the side PR touching it and the side PQ normal to it. If s be the perimeter of the oval, prove that the length of the curve enveloped by QR is equal to $(s + 2\pi PQ) \sin PQR$. [St. John's, 1889.]
35. When a curve rolls on a straight line, show how to find the locus of the centre of curvature at the point of contact, and prove that, in the case of a cardioide, the locus is an ellipse. [St. John's, 1889.]
36. When a curve rolls on a fixed curve, prove that the locus of the centre of curvature is inclined to the common tangent at the angle

$$\tan^{-1}\{\rho d\rho' / (\rho + \rho') ds\},$$

where ρ, ρ' are the radii of curvature of the fixed and rolling curves at the point of contact. [St. John's, 1889.]

37. A cardioide $r = a(1 - \cos \theta)$ rolls upon an equal cardioide, the vertices coinciding during the roll. Show that the roulette of the pole of the rolling curve is

$$r = 4a \sin^2 \left(\frac{\pi}{6} + \frac{\theta}{2} \right),$$

that the tangential polar equation of the envelope of the axis is

$$p = 4a \sin \psi \sin^3 \frac{\pi + \psi}{6},$$

and that the area of the roulette of the pole is

$$\frac{a^2}{2}(3\pi + 4\sqrt{3}).$$

38. A cardioide of perimeter $8a$ rolls on the outer side of a cycloid of equal perimeter from cusp to cusp, the vertices coinciding during the roll. Show that the area of the roulette of the cardioide between the roulette and the cycloid $= \frac{9}{2}\pi a^2$.

Show also that the arc of any portion of the roulette of the cusp measured from the vertex of the curve is double the distance of the point of contact of the two curves from the axis of the cycloid.

Show further that the tangential polar equation of the envelope of the axis of the cardioide is

$$p = 2a \sin \psi (\psi + 2 \cos^3 \psi),$$

where p is drawn from the vertex of the cycloid and ψ is measured from its axis.

39. A cycloid of length $8a$ rolls on the outside of a cardioide of equal length, a cusp of the cycloid starting from the cusp of the cardioide. Show that the intrinsic equation of the envelope of the line joining the cusps of the cycloid is

$$2s = 3a\psi + 6a \sin \frac{\psi}{2},$$

ψ being measured from the tangent at the vertex of the cardioide.

[Oxf. II. P., 1913.]

Chapter XX: Rectification of Twisted Curves — Exercises

Problems

1. Show that the equations of a Rhumb line on a sphere of radius r may be written as

$$\begin{aligned} x^2 + y^2 + z^2 &= r^2, \\ (x^2 + y^2)^{\frac{1}{2}} \cosh \left(n \tan^{-1} \frac{y}{x} \right) &= r. \end{aligned}$$

2. Show that the curve of intersection of the cylinders

$$y^2 = 8ax, \quad x = ae^{z/a},$$

is given by

$$s = x + z + \text{const.}$$

3. A sphere of diameter K touches the plane of an ellipse of principal axes a, b at its centre C . A is the other end of the diameter of the sphere through C . The ellipse is projected on to the sphere by lines through A . Show that the length of the curve so described will be

$$4 \int_0^{\pi/2} \frac{K^2 \sqrt{a^2 \sin^2 \phi + b^2 \cos^2 \phi}}{K^2 + a^2 \cos^2 \phi + b^2 \sin^2 \phi} d\phi.$$

[St. John's, 1884.]

4. A curve is drawn upon the surface of a sphere such that

$$\phi \sin \theta = \text{const.},$$

ϕ being the longitude and θ the co-latitude of any point.

Show that $s = a \log \left(\frac{\tan \theta_1/2}{\tan \theta_2/2} \right)$ is the length of the arc between points where $\theta = \theta_1$ and $\theta = \theta_2$, and a is the radius of the sphere.

Give a sketch showing the nature of the curve $\phi \sin \theta = 1$ upon the sphere $r = a$.

5. Show that the line of intersection of the sphere

$$r = c \cos \theta$$

and the cone

$$\tan \theta = \frac{a}{c} e^{\phi \cot \alpha}$$

is rectifiable, and that

$$s = c\theta \sec \alpha.$$

Also show that the conical projection of this curve on the sphere upon the tangent plane at the end of the diameter remote from the origin, the origin being the pole of projection, is an equiangular spiral. Hence deduce the same result by inversion.

6. Show that the curve of intersection of the sphere

$$x^2 + y^2 + z^2 - 2az = 0$$

and the cone

$$(2x^2 + 2y^2 + zx)^2 = z^2(x^2 + y^2)$$

projects conically from the origin into a cardioide upon the plane $z = 2a$. Hence obtain the rectification of the twisted curve.

7. Show that the length of the arc of intersection of the cylinders

$$x^2 = 2y, \quad x^3 = 6z,$$

measured from the origin to any point x, y, z , is $x + z$.

8. Show that for the curve

$$\frac{60x}{-45 - 40t^2 - 12t^4} = \frac{3y}{3 - t^2} = \frac{z}{t} = t,$$

the arc measured from the origin, is given by

$$s + x = \frac{1}{2}\sqrt{2} \cdot z + \text{const.}$$

9. In the curve for which

$$\frac{dx}{dt} = (1-t)(1-t^2), \quad \frac{dy}{dt} = -t(1-t)^2, \quad \frac{dz}{dt} = t(1-t)^2(1+t),$$

show that

$$s = x + y + z + C.$$

10. Show that in the curve of intersection of

$$r = a \cos \theta \quad \text{and} \quad \cos 2\theta = \tan^2 \phi,$$

$$s = \frac{a}{2\sqrt{2}} \left[2E\left(\chi, \frac{1}{\sqrt{2}}\right) - \frac{\sin \chi \cos \chi}{\sqrt{1 - \frac{1}{2} \sin^2 \chi}} \right],$$

where $\sin \chi = \sqrt{2} \sin \phi$.

Show that the inverse of this curve with regard to the origin is a lemniscate, the constant of inversion being a .

11. Show that the rectification of the line of intersection of

$$x^2 + y^2 + z^2 = cz \quad \text{and} \quad y^2 = 4cx$$

is given by

$$s = \frac{c}{2^{\frac{1}{4}} 3^{\frac{1}{2}}} \left\{ \frac{1}{\sqrt{\cos \frac{\pi}{12}}} \tan^{-1} \left(\frac{\sin \theta}{2^{\frac{1}{4}} \sqrt{\cos \frac{\pi}{12}}} \right) + \frac{1}{\sqrt{\sin \frac{\pi}{12}}} \left(\tanh^{-1} \frac{\sin \theta}{2^{\frac{1}{4}} \sqrt{\sin \frac{\pi}{12}}} \right) \right\}$$

where $\tan \theta = \sqrt{\frac{x}{c}}$,

and show that this curve can be inverted into a parabola lying upon a tangent plane to the sphere.

12. A Loxodrome is drawn on a sphere to cut all the meridians at the same constant angle α ; show that the area of the surface of the sphere, included between any arc of this curve and the two meridians through its ends, is

$$a^2 \tan \alpha \log \frac{1 + \sin \psi_1}{1 + \sin \psi_2},$$

where ψ_1 and ψ_2 are the latitudes of the ends of the arc and a is the radius of the sphere. [Ox. II. P., 1900.]

Chapter XXI: Volumes and Surfaces of Solids of Revolution, and their Centroids — Exercises

Exercises I — Volumes of Revolution (Problems 1–9)

1. Prove that the volume of the solid generated by the revolution of an ellipse round its minor axis is a mean proportional between those generated by the revolution of the ellipse and of the auxiliary circle round the major axis. [I. C. S., 1881.]

2. Find the volume of the solid formed by the revolution of a cycloid round a tangent at the vertex.

3. The loop of the curve

$$2ay^2 = x(x - a)^2$$

revolves about the straight line $y = a$; find the volume of the solid generated. [Oxford I. P., 1890.]

4. Show that the volume of the solid formed by the revolution of the cissoid

$$y^2(2a - x) = x^3$$

about its asymptote is equal to $2\pi^2 a^3$. [Trinity, 1886.]

5. Find the volume of the solid produced by the revolution of the loop of the curve

$$y^2 = x^2 \frac{a + x}{a - x}$$

about the axis of x . [I. C. S., 1882.]

6. Prove that the areas of the oblate and prolate spheroids formed by rotating an ellipse of major axis $2a$ and eccentricity e about its principal axes are

$$2\pi a^2 \left(1 + \frac{1 - e^2}{2e} \log \frac{1 + e}{1 - e} \right)$$

and

$$2\pi a^2 \left(1 - e^2 + \frac{\sqrt{1 - e^2}}{e} \sin^{-1} e \right).$$

[Oxford II. P., 1914.]

Prove also that of all prolate spheroids formed by the revolution of an ellipse of given area, the sphere has the greatest surface. [I. C. S., 1891.]

Find the surface of any zone of an ellipsoid of revolution cut off by planes perpendicular to the axis of revolution. [Colleges α , 1888.]

7. If the evolute of a catenary revolve about the directrix of the catenary, show that the area of any portion of the surface generated, cut off by two planes perpendicular to the directrix, varies as the difference of the cubes of the radii of its bounding circles. [Colleges α , 1892.]

8. Find the volume of the solid formed by the revolution about the prime radius of the loop of the curve

$$r^3 = a^3 \theta \cos \theta$$

between $\theta = 0$ and $\theta = \frac{\pi}{2}$. [Oxford II. P., 1890.]

9. If the cardioid $r = a(1 - \cos \theta)$ revolve round the line

$$\rho = r \cos(\theta - \gamma),$$

prove that the volume generated is

$$3\pi^2 a^2 + \frac{5}{2}\pi a^3 \cos \gamma,$$

assuming that the line does not cut the cardioid.

[St. John's, 1882.]

Exercises II — Surfaces of Revolution and Related Volumes (Problems 10–19)

10. Prove that the area of the surface generated by the revolution of a portion of the arc of a cycloid about the normal at one extremity is equal to the area of the cycloid multiplied by

$$\frac{16}{3} \left[(\beta - \gamma) \sin \beta \cos \gamma + \frac{1}{2} \cos(\beta + \gamma) - \frac{1}{2} \cos(3\beta - \gamma) - \frac{1}{2} \cos 2\gamma \right],$$

where γ and β are the angles of inclination of the axis of revolution, and of the normal at the other extremity of the arc, to the axis of the cycloid.

Deduce the areas of the surfaces generated by the revolution of the whole cycloid about its axis and about its base.

[Colleges ϵ , 1884.]

11. Find the volume of the solid formed by revolving a loop of a lemniscate of Bernoulli about the straight line in its plane which passes through the pole and is perpendicular to the axis. [Oxford I. P., 1901.]

12. The lemniscate

$$r^2 = a^2 \cos 2\theta$$

revolves about a tangent at the pole. Show that the volume and surface of the solid generated are respectively

$$\frac{\pi^2 a^3}{4} \quad \text{and} \quad 4\pi a^2.$$

13. A surface is the locus of points which have their distances from a fixed plane inversely proportional to the fifth power of their distances from a fixed point O in that plane. Prove that its volume equals twice that of the sphere which, with its centre at O , touches the surface. [Oxford II. P., 1880.]

14. Find the volume of the solid formed by the revolution of the curve

$$(a - x)y^2 = a^2 x$$

about its asymptote.

[I. C. S., 1883.]

15. Show that the rate of increase of the volume of an anchor ring when the radius of the generating circle is increased while its centre remains at a constant distance a from the axis of revolution is

$$2\pi^2 ad,$$

the diameter of the generating circle being d , increasing at unit rate.

[Trinity Coll., 1881.]

16. A loop of the curve

$$r = a \sin n\theta$$

revolves about the initial line. Find the volume of the solid thus generated, and verify the result by deducing the volume of the ring formed by the revolution of a circle about a tangent. [Colleges α , 1889.]

17. If the curve

$$r = a + b \cos \theta$$

revolve about the initial line, show that the volume generated is

$$\frac{4}{3}\pi a(a^2 + b^2),$$

provided $a < b$.

[Colleges α , 1884.]

18. The curve

$$r = a(1 - \epsilon \cos \theta),$$

when ϵ is very small, revolves about a tangent parallel to the initial line; prove that the volume of the solid thus generated is approximately

$$2\pi^2 a^3(1 + \epsilon^2).$$

[I. C. S., 1892.]

19. The curve

$$r^3 = a^3 \cos 3\theta$$

revolves about $\theta = 0$. Prove that the loop in the third quadrant generates a volume

$$\frac{3\pi a^3}{8}.$$

[Oxford I. P., 1902.]

Exercises III — Loxodromes, Plane Curves, and General Theorems (Problems 20–30)

20. A loxodrome is drawn from a point A on the earth's surface to a point B . If θ_1, ϕ_1 be the longitude and co-latitude of B , and θ_2, ϕ_2 the corresponding quantities for A , show that the area contained between the meridians of A and B and the loxodrome is

$$\frac{2(\theta_1 - \theta_2) \log \left(\cos \frac{\phi_1}{2} + \cos \frac{\phi_2}{2} \right)}{\log \left(\tan \frac{\phi_1}{2} + \tan \frac{\phi_2}{2} \right)},$$

the radius of the earth being taken as unity.

[St. John's, 1884.]

21. Prove that the whole area bounded by the curve

$$x^4 + y^4 = 2axy^2$$

is

$$\frac{\pi a^2}{2\sqrt{2}}.$$

Also show that if the area revolves about the x -axis, either loop generates a solid whose volume is $\frac{2}{3}\pi a^3$. When the area revolves about the y -axis, the whole volume generated is

$$\frac{\pi^2 a^3}{4}\sqrt{2}.$$

22. Determine the curve which generates, by revolving about the axis of x , a volume proportional to the length cut off from the axis by the terminal bounding planes. [Trin. Hall and Magd., 1886.]

23. The axes of two cylinders of radius a intersect at an angle α ; show that the whole volume common to the two is

$$\frac{16}{3}a^3 \csc \alpha.$$

[Trin. H. and Magd., 1886.]

24. Evaluate

$$\iint \frac{dS}{p^n},$$

taken over the surface of a sphere of radius a , p being the perpendicular on the tangent plane from a fixed point within the sphere at a distance b from the centre; showing that

$$\iint \frac{dS}{p^n} = \frac{2\pi}{n-2} \frac{a}{b} [(c-a)^{2-n} - (c+a)^{2-n}]$$

if $n \neq 2$, and

$$2\pi \frac{a}{c} \log \frac{c-a}{c+a}$$

if $n = 2$.

[Oxford II. P., 1892.]

25. Show that the volume traced out by the part of the area of the curve

$$r = f(\theta)$$

which lies between $\theta = \beta$ and $\theta = \alpha$, when the curve revolves about the line $\theta = \gamma$, taking $\alpha > \beta > \gamma$, is

$$\frac{2\pi}{3} \int_{\beta}^{\alpha} [f(\theta)]^3 \sin(\theta - \gamma) d\theta.$$

[Oxford I. P., 1902.]

26. In the case of any portion of a surface revolving about an axis, prove that the volume generated is the sum with the proper signs of the corresponding volumes generated by the projections of the surface on any two planes at right angles to one another through the axis of rotation. [I, 1900.]

27. A point O is taken on a diameter of a sphere (centre C , radius a) so that $OC = c$ ($c < a$); the radius vector of length r drawn from O to any point P on the surface makes an angle θ with OC , and the radius CP makes an angle θ' with OC produced, dS is an element of area of the surface containing the point P ; evaluate the integral

$$\int \frac{\cos \theta \cos \theta'}{r^2} dS$$

taken over the larger of the portions into which the surface is divided by a plane, through O , at right angles to OC . [Oxford I. P., 1901.]

28. Prove the formula

$$\frac{2\pi}{3} \int r^3 \sin \theta d\theta$$

for the volume of the surface formed by the revolution of a closed plane curve about the initial line.

The outer loop of

$$r^3 = a^3 \cos \frac{1}{3}\theta$$

revolves about the initial line. Show that the volume of the surface generated is

$$\frac{1}{15}\pi a^3 \left(7 + \frac{1}{21}\right).$$

[Oxford I. P., 1911.]

29. Find the area of the finite portion of the surface

$$2z = x^2 + y^2$$

cut off by the plane $z = h$.

[Oxford I. P., 1913.]

30. Show how to find the volume of the solid formed by revolving the curve

$$r = f(\theta)$$

about the line $\theta = \alpha$, it being assumed that the curve passes through the origin.

Prove that the volume of the solid formed by revolving one loop of the curve

$$r^2 = a^2 \cos 2\theta$$

about one of the inflexional tangents is

$$\frac{1}{8}\pi^2 a^3.$$

Show also that the distance of the centroid of this solid from the origin is

$$\frac{4a}{3\pi}.$$

[Oxford I. P., 1915.]

Chapter XXII: Surfaces and Volumes in General, and their Centroids, etc. Double and Triple Integration — Exercises

Exercises I — Volumes and Slices (Problems 1–6)

1. Find by integration the volume of a frustum of

- (1) a pyramid on a triangular base,
- (2) a pyramid on a square base,
- (3) a cone.

2. Find the volume of the portion of a sphere bounded by planes through the centre which cut the sphere in the sides of a given spherical triangle ABC .

3. Show that the volume cut off from the paraboloid

$$x^2 + y^2 = 4az$$

by the plane

$$x + y + z = a$$

is

$$18\pi a^3.$$

4. Show that the volume of the solid bounded by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^{2n}}{c^{2n}} = 1$$

is

$$\frac{4n}{2n+1} \pi abc.$$

5. Show that the volume bounded by the surface

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2 \left(\frac{x}{a} + \frac{y}{b} \right) \frac{z^n}{c^n}$$

and the planes

$$z = 0, \quad z = h$$

is

$$\frac{2\pi abh}{2n+1} \left(\frac{h}{c} \right)^{2n}.$$

6. Show that the volume of a slice of the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1,$$

bounded by the parallel planes

$$lx + my + nz = \delta_1, \quad lx + my + nz = \delta_2,$$

is

$$\frac{\pi abc}{3p^3} (\delta_1 - \delta_2) (\delta_1^2 - \delta_1 \delta_2 - \delta_2^2),$$

where p is the central perpendicular upon a tangent plane parallel to the faces of the slice.

Exercises II — Surface Areas and Ellipsoid Integrals (Problems 7–17)

7. If A be the area of a central section of an ellipsoid parallel to the tangent plane at the elementary area δS , show that

$$\iint \frac{dS}{A} = 4,$$

the integration being taken over the surface of the ellipsoid.

8. Prove that over an ellipsoid of semiaxes a, b, c ,

$$\iint p \, dS = 4\pi abc,$$

$$\iint \frac{dS}{p} = \frac{4\pi}{3} \left(\frac{bc}{a} + \frac{ca}{b} + \frac{ab}{c} \right),$$

$$\iint \frac{dS}{p^2} = \frac{2\pi}{3} \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right),$$

dS being an element of surface, and p the central perpendicular upon the tangent plane.

Investigate also the value of

$$\iint \frac{dS}{p^3}.$$

9. Apply the formula

$$V = \frac{1}{3} \iint (lx + my + nz) \, dS$$

to find the volume of an ellipsoid, x, y, z being the coordinates of any point on the surface, and l, m, n the direction cosines of the normal there. [Colleges α , 1881.]

10. If the ellipsoid of semiaxes a, b, c be very nearly spherical, then its area is, to the first order (inclusive) of the small quantities, represented by the difference of the axes

$$4\pi a^{\frac{1}{3}} b^{\frac{1}{3}} c^{\frac{1}{3}}.$$

[Trinity, 1891.]

11. Show that a portion of a spherical surface (radius unity) may be bent into the surface of revolution defined by the equations

$$x = k \cos p \cos \frac{q}{k}, \quad y = k \cos p \sin \frac{q}{k},$$

$$z = E(p, k) - \int_0^p \sqrt{1 - k^2 \sin^2 p} \, dp;$$

and explain the geometrical theory, distinguishing the two cases

$$k < 1, \quad k > 1.$$

[Math. Tripos, 1887.]

12. The curve $z = f(x)$, $y = 0$ revolves about the axis of x , and the surface thus formed is intersected by the right cylinder $y = \phi(x)$, which is symmetrical with respect to the axis of x ; prove that the cylinder cuts off from the first surface a portion the area of which can be determined by the evaluation of the integral

$$\int z \sqrt{1 + \left(\frac{dz}{dx} \right)^2} \sin^{-1} \frac{y}{z} \, dx$$

between proper limits.

[Oxford II. P., 1888.]

13. Show that the cylinder

$$(x - c)^2 + y^2 = (a - c)^2$$

cuts off from the sphere

$$x^2 + y^2 + z^2 = a^2$$

a portion of which the area is

$$8a \left\{ a \cos^{-1} \left(\frac{c^2}{a^2} \right) - c \sqrt{a^2 - c^2} \right\},$$

a being supposed greater than c .

[Oxford II. P., 1888.]

14. Prove that the volume cut off from the paraboloid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{2z}{c}$$

by the plane

$$z = px + qy + r$$

is

$$\frac{\pi abc}{4} \left(\frac{a^2 p^2}{c^2} + \frac{b^2 q^2}{c^2} + \frac{2r}{c} \right)^2.$$

[Oxford II. P., 1902.]

15. Show that the volume enclosed between the surface

$$z^2 \{ (x^2 + y^2 + c^2)^2 - 4c^2 x^2 \} = c^4 y^2$$

and the cylinder

$$x^2 + y^2 = c^2$$

is

$$(\pi - 2)c^3.$$

[Oxford II. P., 1886.]

16. By application of the formulae

$$V = \frac{1}{3} \iint p \, dS, \quad V = \iint z \cos \gamma \, dS$$

to the evaluation of the volume of an ellipsoid, establish the results

(1)

$$\int_0^c \int_0^\mu \frac{(\mu^2 - \nu^2) \, d\mu \, d\nu}{\sqrt{(b^2 - \nu^2)(c^2 - \nu^2)(c^2 - \mu^2)(\mu^2 - b^2)}} = \frac{\pi}{2},$$

(2)

$$\int_0^c \int_0^\mu \frac{(\mu^2 - \nu^2) \sqrt{(c^2 - \mu^2)(c^2 - \nu^2)}}{\sqrt{(b^2 - \nu^2)(\mu^2 - b^2)}} \, d\mu \, d\nu = \frac{\pi}{6} c^2 (c^2 - b^2).$$

(See Art. 820 for the notation.) [Todhunter, *Functions of Laplace, Lam and Bessel*, pages 216, 217; Bertrand, *Calc. Int.*, pages 424, 426.]

17. Show that the volume bounded by the surface

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2 \left(\frac{x}{a} + \frac{y}{b} \right) \phi(z)$$

and the planes

$$z = 0, \quad z = z_1$$

is

$$2\pi ab \int_0^{z_1} [\phi(z)]^2 \, dz.$$

Exercises III — Applications of Multiple Integration (Problems 18–28)

18. A cavity is just large enough to allow of the complete revolution of a circular disc of radius c , whose centre describes a circle of the same radius c , while the plane of the disc is constantly parallel to a fixed plane, and perpendicular to that in which the centre moves. Show that the volume of the cavity is

$$\frac{2c^3}{3}(3\pi + 8).$$

19. If O be a point without a sphere of radius a and centre C , and r the distance of any point of the sphere from O , show that, integrating $\frac{1}{r^n}$ over the surface, we have

$$\iint \frac{dS}{r^n} = \frac{2\pi}{n-2} \frac{a}{c} [(c-a)^{2-n} - (c+a)^{2-n}] \quad \text{if } n \neq 2,$$

and

$$2\pi \frac{a}{c} \log \frac{c-a}{c+a} \quad \text{if } n = 2.$$

What will be the results if O lies within the sphere?

20. A surface is obtained by making the diameter $2a$ of a semicircle move parallel to itself, the path of the centre being perpendicular to the initial plane of the semicircle, whilst the plane of the semicircle rotates round the diameter; and when the plane has moved through an angle θ the distance which the diameter has moved is $c \sin \theta$. Prove that the volume of the whole surface so generated is

$$\frac{4}{3}\pi a^3 + \frac{1}{2}\pi^2 c a^2.$$

[Trinity, 1890.]

21. Use the theorem

$$V = \iiint dx dy dz = \iiint J du dv dw$$

to find the volume of the parallelepiped enclosed by the planes

$$\begin{aligned} ax + by + cz = 0, & \quad a_1x + b_1y + c_1z = 0, & \quad a_2x + b_2y + c_2z = 0, \\ ax + by + cz = d, & \quad a_1x + b_1y + c_1z = d_1, & \quad a_2x + b_2y + c_2z = d_2. \end{aligned}$$

22. Prove that the area of that portion of the surface

$$(m^2 - 1)(x^2 + y^2) = z^2,$$

which is cut out by the surface

$$z = a^{-1}x^2 + b^{-1}y^2,$$

where a and b are positive, is

$$\frac{\pi}{2} m(m^2 - 1) a^{\frac{1}{2}} b^{\frac{1}{2}} (a + b).$$

[Oxford II. P., 1890.]

Exercises IV — General Volume, Surface, and Mass Problems (Problems 23–39)

23. Show that when $f(x)$ is a slowly changing function,

$$\int_a^b f(x) dx$$

is approximately equal to

$$\frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right].$$

Prove that this formula may be used to calculate exactly the volume cut from a hyperboloid of one sheet by parallel planes meeting it in elliptic sections. [Colleges α , 1881.]

24. Prove that the volume included in the positive octant between the surface

$$z^2(x^2 + y^2 + a^2)^{2n+2}(x^2 + 2a^2)^{2n} = a^{8n+4}y^2$$

and the planes

$$x = 0, \quad x = \infty, \quad y = a, \quad y = \infty$$

is

$$\frac{\pi a^3}{2^{2n+1}n} \frac{1 \cdot 3 \cdot 5 \cdots (4n-3)}{2 \cdot 4 \cdot 6 \cdots (4n-2)},$$

n being a positive integer.

25. Show that the area of that part of the sphere $r = 1$, enclosed by the cone

$$\tan \frac{\theta}{2} = \sqrt{3} \cos \phi,$$

is π .

[Colleges α , 1881.]

26. Show that the volume of the solid, the equation to the surface of which is

$$z^2 + \alpha x^2 + 2\delta xy + \beta y^2 = 2\mu z,$$

is

$$\frac{4\pi}{3} \frac{\mu^3}{\sqrt{\alpha\beta - \delta^2}}.$$

[Colleges, 1882.]

27. If in the tangent plane at the vertex of a paraboloid two ellipses be described whose axes are in the principal sections and proportional to their parameters, the cylinders whose bases are these ellipses, and whose generators are parallel to the axis of the paraboloid, will intercept on the surface a portion whose area is proportional to the difference between the radii of curvature of either of the principal sections at the points where it intersects the bounding curve. [Colleges, 1892.]

28. If the density of a tetrahedron at any point vary as the n^{th} power of the sum of the distances of the point from the faces of the tetrahedron, show that the mass of the tetrahedron

$$= kV \frac{1 \cdot 2 \cdot 3}{(r+1)(r+2)(r+3)} \sum \frac{p_1^{r+3}}{(p_1-p_2)(p_1-p_3)(p_1-p_4)},$$

where V is the volume; p_1, p_2, p_3, p_4 are the perpendiculars from the corners upon the opposite faces, and k the density at the centroid of the volume.

Examine what happens in the case of a regular tetrahedron.

29. Find the volume contained between any two planes perpendicular to the axis of x and the surface whose equation is

$$(y^2 + z^2)^2 = (a^2 + \beta x)y^2 + (a'^2 + \beta' x)z^2.$$

[St. John's, 1884.]

30. Show that the mass contained between a paraboloid of revolution and a sphere, with centre at the vertex and diameter $2a$, equal to the latus rectum of the paraboloid, where the density at any point varies as the square of the latus rectum of the paraboloid containing it and having the same vertex and axis as the bounding paraboloid, is

$$\frac{\pi}{15}(7 - 4\sqrt{2})a^3\rho,$$

where ρ is the density at the external surface of the paraboloid.

[Colleges β , 1883.]

31. Find the volume between the surfaces

$$\begin{aligned} z(y^2 + z^2)^\gamma &= a_1, & y^2 + z^2 &= 4\beta_1 z, & y &= b_1 z, \\ z(y^2 + z^2)^\gamma &= a_2, & y^2 + z^2 &= 4\beta_2 z, & y &= b_2 z. \end{aligned}$$

[Colleges δ , 1881.]

32. Prove that if a, b, c be any positive quantities in descending order of magnitude, the solid angle of that part of the cone

$$ax^2z^2 + (by^2 - cz^2)(x^2 + y^2) = 0$$

which lies on the positive side of the plane xy is equal to

$$4 \sin^{-1} \left(\frac{c}{a} \right)^{\frac{1}{2}} - 4 \left(\frac{b}{a+b} \right)^{\frac{1}{2}} \sin^{-1} \left\{ \frac{c(a+b)}{a(c+b)} \right\}^{\frac{1}{2}}.$$

[Colleges β , 1891.]

33. Prove that the volume common to a sphere and a circular cylinder which touches it, and also passes through the centre, is

$$\left(\frac{1}{2} - \frac{2}{3\pi} \right)$$

of the volume of the sphere. Also show that the sum of the two spherical caps cut off by the cylinder forms

$$\frac{1}{2} - \frac{1}{\pi}$$

of the area of the sphere.

[St. John's, 1891.]

34. A sphere of radius a is cut by two diametral planes so as to form a lune of angle α , which is itself cut in two by a plane inclined at an angle β to its edge and passing through one end of it, and equally inclined to the two faces of the lune; show that the volume of the pointed part is

$$\frac{2}{3}a^3 \sin \beta \left\{ (2 + \cos^2 \beta) \tan^{-1} \left(\sin \beta \tan \frac{\alpha}{2} \right) + \frac{\sin \beta \cos^2 \beta \tan \frac{\alpha}{2}}{1 + \sin^2 \beta \tan^2 \frac{\alpha}{2}} \right\}.$$

[St. John's, 1881.]

35. Prove that the moment of inertia about the axis of z of the part of the paraboloid

$$2z = ax^2 + by^2,$$

cut off by the plane

$$lx + my + nz = p,$$

is

$$\frac{\pi}{24n^6(ab)^{\frac{1}{2}}}(bl^2 + am^2 + 2pnab)^2 \{bl^2(a + 7b) + am^2(7a + b) + 2pnab(a + b)\},$$

the density being taken as unity.

[Math. Tripos, 1890.]

36. If $A + B + C = 0$ and the coordinate axes be rectangular, prove that

$$\iint \{(A, B, C, D, E, F \mid x, y, z)^2 (A', B', C', D', E', F' \mid x, y, z)^2\} d\omega = \frac{8\pi}{15} (AA' + BB' + CC' + 2DD' + 2EE' + 2FF')$$

where the integration extends over the whole surface of a sphere of unit radius whose centre is the origin of coordinates.

Also show that the unconditional result is

$$\frac{4\pi}{15} [A'(3A + B + C) + B'(3B + C + A) + C'(3C + A + B) + 4DD' + 4EE' + 4FF'] .$$

[Colleges, 1892.]

37. A flexible envelope is in the form of an oblate spheroid, such that e is the eccentricity of a meridian section: the part between two meridians, the planes of which are inclined to each other at the angle $2\pi(1 - e)$, is cut away, and the edges are then sewn together. Prove that the meridian curve of the new surface is the “curve of sines,” and that the volume enclosed is changed in the ratio

$$3\pi e^2 : 8.$$

[St. John's, 1889.]

38. A surface is such that $ABCD$ being any rectangle in the plane of x, y , with its sides parallel to Ox, Oy , and AP, BQ, CR, DS being drawn parallel to Oz to meet the surface in P, Q, R, S , the volume of the solid $ABCDPQRS$ is equal to the base $ABCD$, multiplied by the arithmetic mean of AP, BQ, CR, DS . Prove that the surface is a hyperbolic paraboloid. [Math. Tripos, 1876.]

39. Show that the integral

$$\iiint e^{\frac{k(lx+my+nz)}{\sqrt{a^2+b^2+c^2}}} dx dy dz$$

taken over the volume of the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

is

$$\frac{4\pi abc}{k^3} (k \cosh k - \sinh k),$$

and find the values of

$$\iiint e^x dx dy dz; \quad \iiint e^{x+y} dx dy dz; \quad \iiint e^{x+y+z} dx dy dz$$

through the same space.

Exercises V — Further Applications of Multiple Integration (Problems 40–53)

40. On a closed oval surface of volume V and surface S , whose curvature is everywhere finite, rolls a sphere of radius a ; the surface of the envelope of the sphere is S' . Prove that the volume of the envelope is

$$V + a(S' + S) - \frac{4}{3}\pi a^3.$$

[Math. Tripos, 1886.]

41. Show that the volume of the pedal of an ellipsoid taken with the centre as origin is less than that taken with regard to any other origin; and that the sum of the volumes of the pedals, taken with regard to the extremities of three semi-conjugate diameters, is six times that taken with regard to the centre.

[Math. Tripos, 1887.]

42. Show that the moment of inertia of the ellipsoid

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 1$$

about the axis of x is

$$\frac{1}{2}M(ca - g^2 + ab - h^2)(abc + 2fgh - af^2 - bg^2 - ch^2)^{-1},$$

where M is the mass of the ellipsoid.

[Trinity, 1890.]

43. Find the envelope of the conics

$$x^2 \sec^3 \theta - y^2 \tan^3 \theta = a^2,$$

where θ is the variable parameter. Show that in addition to certain lines it consists of a curve whose asymptotes are $x = \pm a$. Also, if the area between the axis of x , an asymptote, and the corresponding branch of the curve be A , and the volume generated by the revolution of this branch about the axis of x be V , prove that

$$V = \pi a A = \frac{1}{2}\pi a^3 \int_0^{\pi/2} (\sin \phi)^4 d\phi.$$

[Colleges β , 1890.]

44. Show that the value of

$$\iiint \frac{xyz \, dx \, dy \, dz}{\sqrt{x^2 + y^2 + z^2}}$$

taken throughout the positive octant of the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

is

$$\frac{4\pi abc}{15} \frac{bc + ca + ab}{(b + c)(c + a)(a + b)}.$$

[Oxford II. P., 1888.]

45. Prove that the mass of a sphere of radius a , whose density at any point P is

$$\frac{k}{AP},$$

where k is a constant and A is a fixed point distant $f(> a)$ from the centre of the sphere, is equal to

$$\frac{4\pi k a^3}{3f}.$$

[Oxford I. P., 1914.]

46. Prove that the volume which lies within the sphere

$$x^2 + y^2 + z^2 = a^2$$

and the ellipsoid

$$x^2 \sin^2 \alpha \csc^2 \beta + y^2 \cos^2 \alpha \sec^2 \beta + z^2 = a^2,$$

where $0 < \alpha < \beta < \frac{\pi}{2}$, is

$$\frac{4}{3}a^3 (\pi - 2\beta + 2\alpha \sin 2\beta \csc 2\alpha).$$

[Oxford I. P., 1916.]

47. P is a point of abscissa x (> 0) on the parabola

$$x^2 = 2ay, \quad z = 0,$$

and Sa^2 is the area of the segment bounded by the arc OP and the radius vector OP ; the straight line PQ of length $2Sa$ is drawn parallel to Oz . The locus of Q being a curve which passes through the origin, prove that

(1) the length of the arc OQ is

$$x + \frac{x^3}{6a^2};$$

(2) the cylindrical area bounded by the arcs OP , OQ and the straight line PQ is

$$\frac{a^2}{45} + \frac{(3x^2 - 2a^2)(x^2 + a^2)^{\frac{3}{2}}}{90a^3}.$$

[Oxford I. P., 1916.]

48. Show that the two cylinders

$$\frac{x^2}{a^2} + \frac{z^2}{c^2} = 1 \quad \text{and} \quad y^2 = 2b(c - z)$$

intersect on the plane $z = k$ (where $k^2 < c^2$), a rectangle of area

$$4a \left(1 - \frac{k}{c}\right) \sqrt{2b(c + k)}.$$

Show that the volume cut off from the cylinder

$$\frac{x^2}{a^2} + \frac{z^2}{c^2} = 1$$

by the cylinder

$$y^2 = 2b(c - z)$$

is

$$\frac{128}{105}ac\sqrt{bc}.$$

[Oxford I. P., 1917.]

49. The sphere

$$x^2 + y^2 + z^2 = a^2$$

is intersected by the cylinder

$$x^2 + y^2 = ax.$$

Prove that the ratio of the spherical area cut off by the cylinder to the cylindrical area cut off by the sphere is

$$\pi - 2 : 2.$$

[Oxford I. P., 1915.]

50. Integrate

$$\int_0^1 \left[\int_0^{x^2} \frac{dy}{\sqrt{x^2 - y^2}} \right] dx.$$

[Oxford I. P., 1915.]

51. Find the value of

$$\iint \frac{dx dy}{(a^2 + x^2 + y^2)^p}$$

taken all over the plane x, y ; p being greater than unity.

[Oxford I. P., 1915.]

52. Find the four points where any line parallel to the axis of z intersects the surface

$$(x^2 + y^2 + z^2)^3 = 4(a^2 z^2 + x^2 y^2).$$

Prove that the volume enclosed by that part of the surface which lies above the plane $z = 0$ is

$$\frac{1}{3}a^3.$$

[Oxford II. P., 1915.]

53. If the coordinates of a point on a certain surface be expressed as

$$x = a \sin u, \quad y = a \sin v, \quad z = a \cos u + a \cos v,$$

prove that the area of the portion of the surface bounded by

$$u = 0, \quad u = \frac{\pi}{2}, \quad v = 0, \quad v = \frac{\pi}{2},$$

is

$$\frac{\pi a^2}{4} \left(1 - \frac{c_1^3}{1} - \frac{c_3^3}{3} - \frac{c_5^3}{5} - \frac{c_7^3}{7} - \dots \right),$$

where

$$c_r = \frac{(2r-1)(2r-3) \cdots 1}{2r(2r-2) \cdots 2}.$$

[Oxford II. P., 1915.]